
User Guide

Release 2.3.300

ARM CE-OSS, Infineon

Aug 25, 2026

OVERVIEW

INTRODUCTION

1.1 Trusted Firmware M

Trusted Firmware-M (TF-M) implements the Secure Processing Environment (SPE) for Armv8-M, Armv8.1-M architectures (e.g. the [Cortex-M33](#), [Cortex-M23](#), [Cortex-M55](#), [Cortex-M85](#) processors) and dual-core platforms. It is the platform security architecture reference implementation aligning with PSA Certified guidelines, enabling chips, Real Time Operating Systems and devices to become PSA Certified.

TF-M relies on an isolation boundary between the Non-secure Processing Environment (NSPE) and the Secure Processing Environment (SPE). It can but is not limited to using the [Arm TrustZone technology](#) on Armv8-M and Armv8.1-M architectures. In pre-Armv8-M architectures physical core isolation is required.

TF-M consists of:

- Secure Boot to authenticate NSPE and SPE images
- TF-M Core for controlling the isolation, communication and execution within SPE and with NSPE
- Crypto, Internal Trusted Storage (ITS), Protected Storage (PS), Firmware Update and Attestation secure services

TF-M implements [PSA-FF-M](#) defined IPC and SFN mechanisms to allow communication between isolated firmware partitions. TF-M is highly configurable allowing users to only include the required secure services and features. Project provides *Base Configuration* build with just TF-M core and platform drivers and 4 predefined configurations known as *TF-M Profiles*. TF-M Profiles or TF-M base can be configured to include required services and features as described in the *Configuration* section.

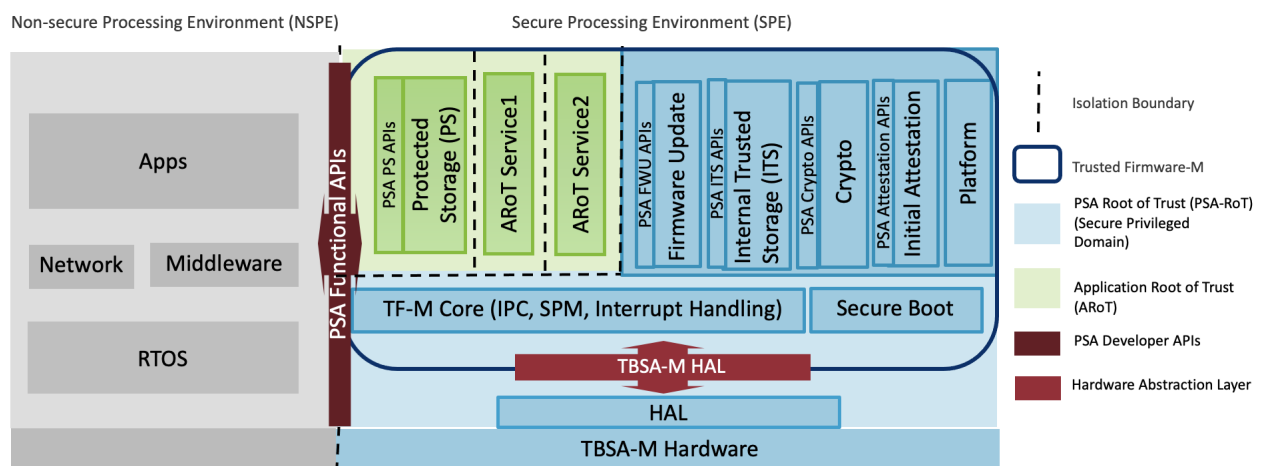


Figure 1:: FF-M compliant design with TF-M

Applications and Libraries in the Non-secure Processing Environment can utilize these secure services with a standardized set of PSA Functional APIs. Applications running on Cortex-M devices can leverage TF-M services to ensure secure connection with edge gateways and IoT cloud services. It also protects the critical security assets such as sensitive data, keys and certificates on the platform. TF-M is supported on several Cortex-M based *Microcontrollers* and Real Time Operating Systems (RTOS).

Terms TFM and TF-M are commonly used in documents and code and both refer to **Trusted Firmware M**. *Glossary* has the list of terms and abbreviations.

1.1.1 Repositories

TF-M is comprised of multiple repositories that supplement each other in making the project both customisable and maintainable.

Table 1:: TF-M Repositories

Repository	Description
trusted-firmware-m	Software implementation of TF-M with documentation and essential tools
tf-m-tests	Tests that focus on the functionalities of TF-M components
tf-m-tools	Non essential tools used for testing and verification of TF-M
tf-m-extras	Extension of the main repository to host examples, demonstrations, third-party modules etc

1.2 License

The software is provided under a BSD-3-Clause *License*. Contributions to this project are accepted under the same license with developer sign-off as described in the *Contributing Process*.

This project contains code from other projects as listed below. The code from external projects is limited to `b12`, `lib` and `platform` folders. The original license text is included in those source files.

- The `b12` folder contains files imported from MCUBoot project and the files have Apache 2.0 license.
- The `lib/ext` folder may contain 3rd party projects and files with diverse licenses. Here are some that are different from the BSD-3-Clause and may be a part of the runtime image. The source code for these projects is fetched from upstream at build time only.
 - `CMSIS_5` - Apache 2.0 license
 - `tf-psa-crypto` - [Apache 2.0 license TF-PSA-Crypto](#)
 - `mcuboot` - [Apache 2.0 license MCUBoot](#)
 - `qcbor` - [Modified BSD-3-Clause license](#)
 - `tf-m-extras` - Set of additional components. Please check individually in [tf-m-extras repository](#)
- The `platform` folder currently contains platforms support imported from the external project and the files may have different licenses.

The document *Supported Platforms* lists the details.

1.3 Release Notes and Process

The Release Cadence and Process provides release cadence and process information.

The Releases provides details of major features of the release and platforms supported.

1.4 Feedback and Support

For this release, feedback is requested via email to tf-m@lists.trustedfirmware.org.

A bi-weekly technical forum is available for discussion on any technical topics online. Welcome to join [TF-M Forum](#).

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GETTING STARTED GUIDES

2.1 First Things First

2.1.1 Prerequisite

Trusted Firmware M provides a reference implementation of the Platform Security Architecture (PSA) specifications, aligning with PSA Certified guidelines. It is assumed that the reader is familiar with the specifications that can be found [here](#).

The current TF-M implementation on Armv8-M leverages TrustZone for Armv8-M so a good understanding of the v8-M architecture is also necessary. A good place to get started with Armv8-M is developer.arm.com.

2.1.2 Build and run instructions

Trusted Firmware M source code is available on git.trustedfirmware.org.

To build & run TF-M:

- Follow the this guide to set up and check your environment.
- Follow the *Build instructions* to compile and build the TF-M source.
- Follow the Run TF-M examples on Arm platforms for information on running the example.

To port TF-M to a another system or OS, follow the *OS Integration Guide*

Contributing Process contains guidance on how to contribute to this project.

2.2 Set up build environments

TF-M officially supports a limited set of build environments and setups. In this context, official support means that the environments listed below are actively used by team members and active developers, hence users should be able to recreate the same configurations by following the instructions described below. In case of problems, the TF-M team provides support only for these environments, but building in other environments can still be possible.

The following environments are supported:

Linux

1. version supported:
Ubuntu 22.04 x64+
2. install dependencies:

```
sudo apt-get install -y git curl wget build-essential libssl-dev python3 \
python3-pip cmake make
```

3. verify CMake version:

```
cmake --version
```

Note

Please download CMake 3.27.7 or later version from <https://cmake.org/download/>.

4. add CMake path into environment:

```
export PATH=<CMake path>/bin:$PATH
```

Windows

1. version supported:

Windows 10 x64

2. install dependencies:

- Git client latest version (<https://git-scm.com/download/win>)
- CMake version 3.21 or later (native Windows version)
- GNU make (<http://gnuwin32.sourceforge.net/packages/make.htm>)
- Python3 (native Windows version) and the pip package manager (from Python 3.4 it's included)

3. add CMake and Python path into environment:

```
set PATH=<CMake_Path>\bin;%PATH%  
set PATH=<python_launcher_path>;%PATH%
```

2.3 Install python dependencies

Clone the TF-M source code, and then install the TF-M's additional Python dependencies.

Linux

1. get the TF-M source code:

```
git clone https://git.trustedfirmware.org/TF-M/trusted-firmware-m.git
```

2. TF-M recommends installing dependencies in a venv

```
# Setup python venv for the project  
python3 -m venv .venv  
  
# NOTE: If your system python install version is <3.10 you can use `uv <https://docs.  
↪astral.sh/uv/getting-started/installation/#standalone-installer>` to setup your .venv  
uv venv --python 3.12  
  
source .venv/bin/activate  
cd </trusted-firmware-m>  
# `-e` installs modules and scripts in editable/development mode  
pip install -e .
```

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```
# NOTE: If you've used `uv` to setup your `.venv`, prepend the `pip` commands with `uv`
# `-e` installs modules and scripts in editable/development mode
uv pip install -e .
```

Windows

1. get the TF-M source code:

```
git clone https://git.trustedfirmware.org/TF-M/trusted-firmware-m.git
```

2. TF-M recommends installing dependencies in a venv

```
# Setup python venv for the project
python3 -m venv .venv

# NOTE: If your system python install version is <3.10 you can use `uv <https://docs.
↪astral.sh/uv/getting-started/installation/#standalone-installer>` to setup your .venv
uv venv --python 3.12

.venv\Scripts\activate.bat
cd <\trusted-firmware-m>
# `-e` installs modules and scripts in editable/development mode
pip install -e .

# NOTE: If you've used `uv` to setup your `.venv`, prepend the `pip` commands with `uv`
# `-e` installs modules and scripts in editable/development mode
uv pip install -e .
```

2.4 Install a toolchain

To compile TF-M code, at least one of the supported compiler toolchains have to be available in the build environment. The currently supported compiler versions are:

- Arm Compiler v6.22+

Linux

- Download the standalone packages from [here](#).
- Add Arm Compiler into environment:

```
export PATH=<ARM_CLANG_PATH>/bin:$PATH
export ARM_PRODUCT_PATH=<ARM_CLANG_PATH>/sw/mappings
```

- Configure proper tool variant and license.

Windows

- Download the standalone packages from [here](#).
- Add Arm Compiler into environment:

```
set PATH=<ARM_CLANG_PATH>\bin;%PATH%
set ARM_PRODUCT_PATH=<ARM_CLANG_PATH>\sw\mappings
```

- Configure proper tool variant and license.

Note

When compiling for a Cortex-M52 target, a Cortex-M85 target, or with an `-mcpu=<name>` option that includes the **+pacbti** feature modifier, and when configured without *User based Licensing*, the compiler could incorrectly report one of these errors:

- Cortex-M52 is not available with the current toolkit edition and license
- Cortex-M85 is not available with the current toolkit edition and license

The use of `-target-feature +pacbti` is disallowed in this variant of Armclang. Please use Armclang version 6.24+ which does not have this issue.

- GNU Arm compiler v14.2.1+

Linux

- Download the GNU Arm compiler from [here](#).
- Add GNU Arm into environment:

```
export PATH=<GNU_ARM_PATH>/bin:$PATH
```

Windows

- Download the GNU Arm compiler from [here](#).
- Add GNU Arm into environment:

```
set PATH=<GNU_ARM_PATH>\bin;%PATH%
```

- IAR Arm compiler v9.50.2+

Linux

- Download IAR build tools from [here](#).
- Add IAR Arm compiler into environment:

```
export PATH=<IAR_COMPILER_PATH>/bin:$PATH
```

Windows

- Download IAR build tools from [here](#).
- Add IAR Arm compiler into environment:

```
set PATH=<IAR_COMPILER_PATH>\bin;%PATH%
```

- Arm Toolchain for Embedded (ATfE) v20.1.0+

Linux

- Download the Arm Toolchain for Embedded from [here](#).
- Add ATfE into environment:

```
export PATH=<ATFE_PATH>/bin:$PATH
```

Windows

- Download the Arm Toolchain for Embedded from [here](#).
- Add ATfE into environment:

```
set PATH=<ATFE_PATH>\bin;%PATH%
```

Note

Not all platforms support this toolchain. Please refer to a platform documentation or check with the platform owner.

2.5 Build AN521 regression sample

Here, we take building TF-M for AN521 platform with regression tests using GCC as an example:

Linux

Get the TF-M tests source code:

```
git clone https://git.trustedfirmware.org/TF-M/tf-m-tests.git
```

Build SPE and NSPE.

```
cd </tf-m-tests/tests_reg>
cmake -S spe -B build_spe -DTFM_PLATFORM=arm/mps2/an521 -DCONFIG_TFM_SOURCE_PATH=<TF-M_
↳source dir absolute path> \
    -DCMAKE_BUILD_TYPE=Debug -DTFM_TOOLCHAIN_FILE=<TF-M source dir absolute path>/
↳toolchain_GNUARM.cmake \
    -DTEST_S=ON -DTEST_NS=ON
cmake --build build_spe -- install

cmake -S . -B build_test -DCONFIG_SPE_PATH=<tf-m-tests absolute path>/tests_reg/build_
↳spe/api_ns \
    -DCMAKE_BUILD_TYPE=Debug -DTFM_TOOLCHAIN_FILE=<tf-m-tests absolute path>/tests_reg/
↳build_spe/api_ns/cmake/toolchain_ns_GNUARM.cmake
cmake --build build_test
```

Windows

Important

Use “/” instead of “\” when assigning Windows paths to CMAKE variables, for example, use “c:/build” instead of “c:\build”.

Get the TF-M tests source code:

```
git clone https://git.trustedfirmware.org/TF-M/tf-m-tests.git
```

Build SPE and NSPE.

```
cd <\tf-m-tests\tests_reg>
cmake -G"Unix Makefiles" -S spe -B build_spe -DTFM_PLATFORM=arm/mps2/an521 -DCONFIG_TFM_
↳SOURCE_PATH=<TF-M source dir absolute path> \
    -DCMAKE_BUILD_TYPE=Debug -DTFM_TOOLCHAIN_FILE=<TF-M source dir absolute path>/
↳toolchain_GNUARM.cmake \
```

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```

-DTEST_S=ON -DTEST_NS=ON
cmake --build build_spe -- install

cmake -G"Unix Makefiles" -S . -B build_test -DCONFIG_SPE_PATH=<tf-m-tests absolute path>/
↳ tests_reg/build_spe/api_ns \
-DMAKE_BUILD_TYPE=Debug -DTFM_TOOLCHAIN_FILE=<tf-m-tests absolute path>/tests_reg/
↳ build_spe/api_ns/cmake/toolchain_ns_GNUARM.cmake
cmake --build build_test

```

Note

The latest Windows support long paths, but if you are less lucky then you can reduce paths by moving the build directory closer to the root by changing the -B option of the commands, for example, to C:\build_spe and C:\build_test folders.

2.6 Run AN521 regression sample

Run the sample code on SSE-200 Fast-Model, using FVP_MPS2_AEMv8M provided by Arm Development Studio.

Note

Arm Development Studio is not essential to develop TF-M, you can skip this section if don't want to try on Arm develop boards.

Linux

1. install Arm Development Studio to get the fast-model.
Download Arm Development Studio from [here](#).
2. Add bl2.axf and tfm_s_ns_signed.bin to symbol files in Debug Configuration menu.

```

<DS_PATH>/sw/models/bin/FVP_MPS2_AEMv8M \
--parameter fvp_mps2.platform_type=2 \
--parameter cpu0.baseline=0 \
--parameter cpu0.INITVTOR_S=0x10000000 \
--parameter cpu0.semihosting-enable=0 \
--parameter fvp_mps2.DISABLE_GATING=0 \
--parameter fvp_mps2.telnetterminal0.start_telnet=1 \
--parameter fvp_mps2.telnetterminal1.start_telnet=0 \
--parameter fvp_mps2.telnetterminal2.start_telnet=0 \
--parameter fvp_mps2.telnetterminal0.quiet=0 \
--parameter fvp_mps2.telnetterminal1.quiet=1 \
--parameter fvp_mps2.telnetterminal2.quiet=1 \
--application cpu0=<build_spe>/api_ns/bin/bl2.axf \
--data cpu0=<build_test>/tfm_s_ns_signed.bin@0x10080000

```

Note

The log is output to telnet by default. It can be also redirected to stdout by adding the following parameter.

```
--parameter fvp_mps2.UART0.out_file=/dev/stdout
```

To automatically terminate the fast-model when it finishes running, you can add the following parameters:

```
--parameter fvp_mps2.UART0.shutdown_on_eot=1
```

Windows

1. install Arm Development Studio to get the fast-model.

Download Arm Development Studio from [here](#).

2. Add bl2.axf and tfm_s_ns_signed.bin to symbol files in Debug Configuration menu.

```
<DS_PATH>\sw\models\bin\FVP_MPS2_AEMv8M \
--parameter fvp_mps2.platform_type=2 \
--parameter cpu0.baseline=0 \
--parameter cpu0.INITVTOR_S=0x10000000 \
--parameter cpu0.semihosting-enable=0 \
--parameter fvp_mps2.DISABLE_GATING=0 \
--parameter fvp_mps2.telnetterminal0.start_telnet=1 \
--parameter fvp_mps2.telnetterminal1.start_telnet=0 \
--parameter fvp_mps2.telnetterminal2.start_telnet=0 \
--parameter fvp_mps2.telnetterminal0.quiet=0 \
--parameter fvp_mps2.telnetterminal1.quiet=1 \
--parameter fvp_mps2.telnetterminal2.quiet=1 \
--application cpu0=<build_spe>/api_ns/bin/bl2.axf \
--data cpu0=<build_test>/tfm_s_ns_signed.bin@0x10080000
```

Note

To automatically terminate the fast-model when it finishes running, you can add the following parameters:

```
--parameter fvp_mps2.UART0.shutdown_on_eot=1
```

After completing the procedure you should see the following messages on the DAPLink UART (baud 115200 8n1):

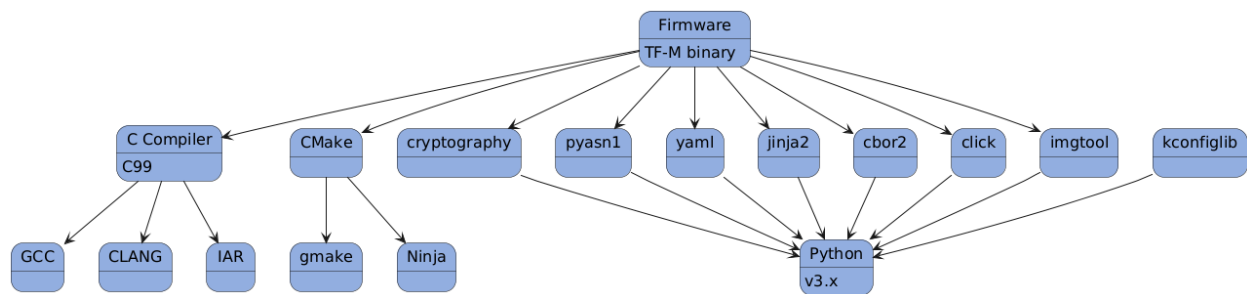
```
...
#### Execute test suites for the Secure area ####
Running Test Suite PSA protected storage S interface tests (TFM_S_PS_TEST_1XXX)...
> Executing 'TFM_S_PS_TEST_1001'
Description: 'Set interface'
TEST: TFM_S_PS_TEST_1001 - PASSED!
> Executing 'TFM_S_PS_TEST_1002'
Description: 'Set interface with create flags'
TEST: TFM_S_PS_TEST_1002 - PASSED!
> Executing 'TFM_S_PS_TEST_1003'
Description: 'Set interface with NULL data pointer'
TEST: TFM_S_PS_TEST_1003 - PASSED!
> Executing 'TFM_S_PS_TEST_1005'
Description: 'Set interface with write once UID'
TEST: TFM_S_PS_TEST_1005 - PASSED!
....
```

2.7 Tool & Dependency overview

To build the TF-M firmware the following tools are needed:

- C compiler of supported toolchains
- CMake version 3.27.7 or later
- Git
- gmake, aka GNU Make
- Python $\geq$ v3.11
- [Optionally] `uv` <<https://docs.astral.sh/uv/getting-started/installation/#standalone-installer>>

2.7.1 Dependency chain



Next steps

Here are some next steps for exploring TF-M:

- Detailed *Build instructions*.
- *IAR Build instructions*.
- Try other Samples and Demos.
- *Documentation generation*.

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3.1 Threat Models

3.1.1 Trusted Firmware-M Generic Threat Model

Introduction

This document introduces a generic threat model of Trusted Firmware-M (TF-M). This generic threat model provides an overall analysis of TF-M implementation and identifies general threats and mitigations.

There is also a dedicated document for physical attack mitigations which can be found *here*.

Note

If you think a security vulnerability is found, please follow Trustedfirmware.org [?] to contact TF-M security team.

Scope

TF-M supports diverse models and topologies. It also implements multiple isolation levels. Each case may focus on different target of evaluation (TOE) and identify different assets and threats. TF-M implementation consists of several secure services, defined as Root of Trust (RoT) service. Those RoT services belong to diverse RoT (Application RoT or PSA RoT) and access different assets and hardware. Therefore each RoT service may require a dedicated threat model.

The analysis on specific models, topologies or RoT services may be covered in dedicated threat model documents. Those threat models are out of the scope of this document.

Methodology

The threat modeling in this document follows the process listed below to build up the threat model.

- Target of Evaluation (TOE)
- Assets identification
- Data Flow Diagram (DFD)
- Threats Prioritization
- Threats identification

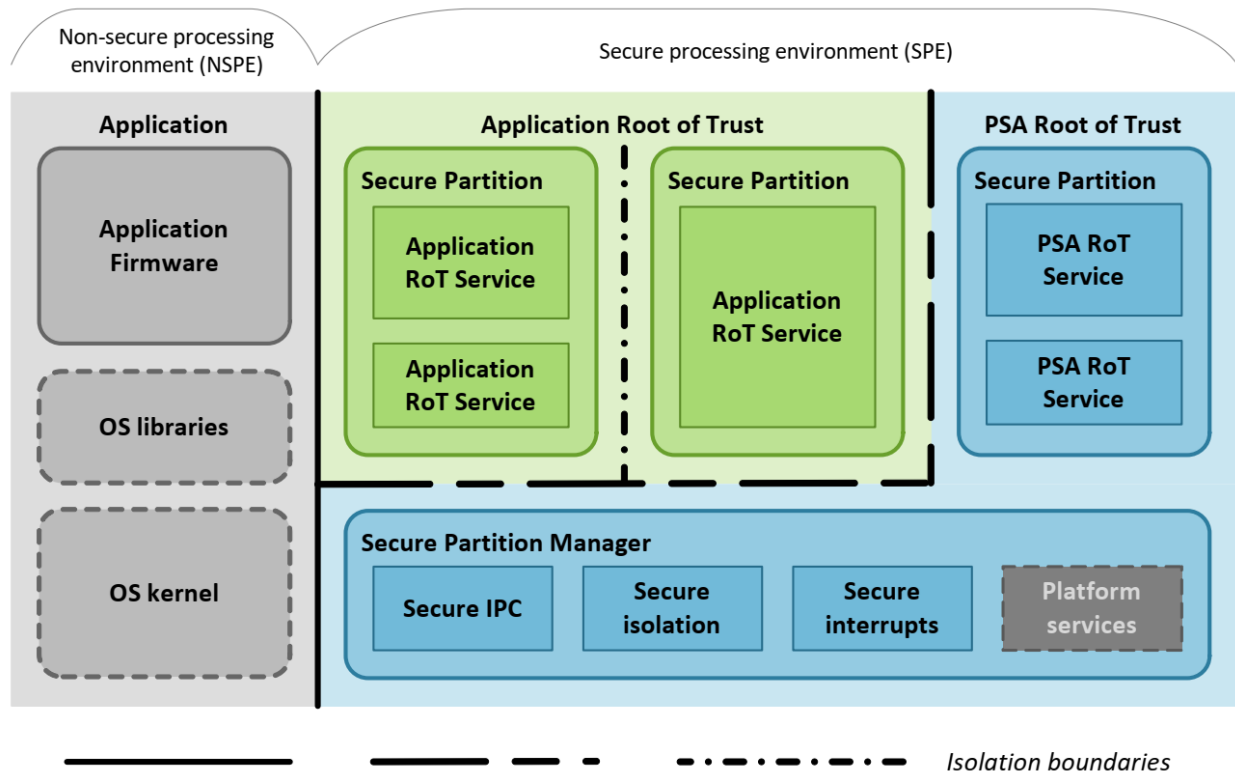
TOE is the entity on which threat modeling is performed. The logic behind this process is to firstly investigate the TOE which could be a system, solution or use case. This first step helps to identify the assets to be protected in TOE.

According to TOE and assets, Trust Boundaries can be determined. The Data Flow Diagram (DFD) across Trust Boundaries is then defined to help identify the threats.

Those threats should be prioritized based on a specific group of principles and metrics. The principles and metrics should also be specified.

Target of Evaluation

A typical TF-M system diagram from a high-level overview is shown below. TF-M is running in the Secure Processing Environment (SPE) and Non-Secure (NS) software is running in Non-Secure Processing Environment (NSPE). For more details, please refer to Platform Security Architecture Firmware Framework for M (FF-M) [?] and FF-M 1.1 Extensions [?].



The TOE in this general model is the SPE, including TF-M and other components running in SPE.

The TOE can vary in different TF-M models, RoT services and usage scenarios. Refer to dedicated threat models for the specific TOE definitions.

Asset identification

In this threat model, assets include the general items listed below:

- Hardware Root of Trust data, e.g.
 - Hardware Unique Key (HUK)
 - Root authentication key
 - Other embedded root keys
- Software RoT data, e.g.
 - Secure Partition Manager (SPM) code and data
 - Secure partition code and data
 - NSPE data stored in SPE

- Data generated in SPE as requested by NSPE
- Availability of entire RoT service
- Secure logs, including event logs

Assets may vary in different use cases and implementations. Additional assets can be defined in an actual usage scenario and a dedicated threat model.

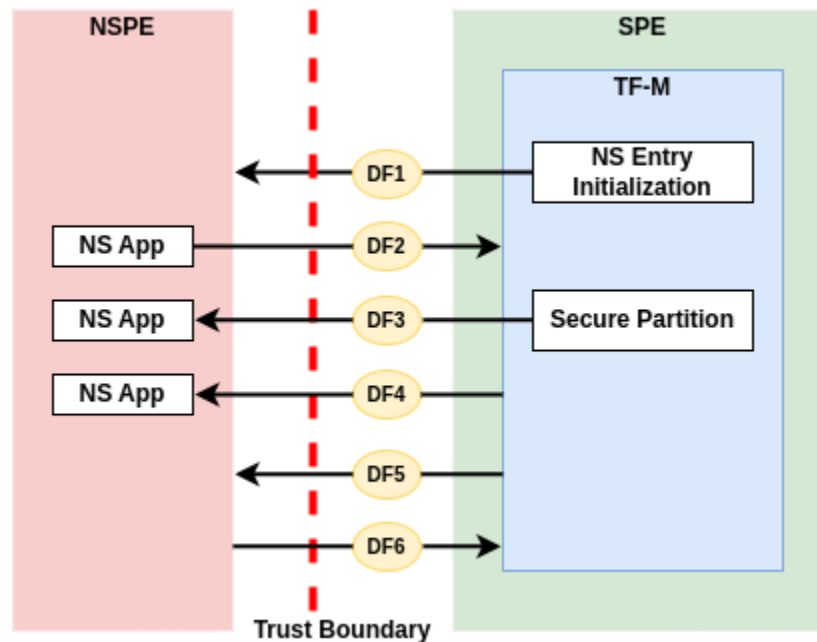
For example, in a network camera use case, the following data can be defined as assets too:

- Certificate for connecting to cloud
- Session keys for encryption/decryption in the communication with cloud
- Keys to encrypt/decrypt the videos and photos

Data Flow Diagram

The Trust Boundary isolates SPE from NSPE, according to the TOE definition in *Target of Evaluation*. The Trust Boundary mapped to block diagram is shown in the figure below. Other modules inside SPE stay in the same TOE as TF-M does.

Valid data flows across the Trust Boundary are also shown in the figure below. This threat model only focuses on the data flows related to TF-M.



More details of data flows are listed below.

Table 2:: TF-M Data Flows between NSPE and SPE

Data flow	Description
DF1	TF-M initializes NS entry and activates NSPE. - On Armv8-M platforms with TrustZone, TF-M will hand over the control to Non-secure state. - On multi-core platforms, Secure core starts NS core booting.
DF2	NSPE requests TF-M RoT services. NSPE requests RoT services via PSA Client APIs defined in [?]. In Armv8-M TrustZone scenarios, SG instruction is executed in a Non-secure Callable region to trigger a transition from Non-secure state to Secure state. On multi-core platforms, non-secure core sends PSA Client calls to secure core via mailbox.
DF3	Secure Partitions fetch input data from NS and write back output data to NS. As required in [?], Secure Partitions should not directly access NSPE memory. Instead, RoT services rely on TF-M SPM to access NSPE memory.
DF4	TF-M returns RoT service results to NSPE after NS request to RoT service is completed. In Armv8-M TrustZone scenarios, it also triggers a transition from Secure state back to Non-secure state. On multi-core platforms, secure core returns the result to non-secure core via mailbox.
DF5	Non-secure interrupts preempt SPE execution in Armv8-M TrustZone scenarios.
DF6	Secure interrupts preempt NSPE execution in Armv8-M TrustZone scenarios.

Note

All the other data flows across the Trust Boundary besides the valid ones mentioned above should be prohibited by default. Proper isolation must be configured to prevent NSPE directly accessing SPE.

Threats irrelevant to data flows in *TF-M Data Flows between NSPE and SPE* may be specified in *Miscellaneous threats*.

Data flows inside SPE (informative)

Since all the SPE components stay in the TOE within the same Trust Boundary in this threat model, the data flows between SPE components are not covered in this threat model. Instead, those data flows and corresponding threats will be identified in the dedicated threat model documents of TF-M RoT services and usage scenarios.

Those data flows inside SPE include following examples:

- Data flows between TF-M and BL2
- Data flows between RoT services and SPM
- Data flows between RoT services and corresponding secure hardware and assets, such as secure storage device,

crypto hardware accelerator and Hardware Unique Key (HUK).

Threat identification

Threat priority

Threat priority is indicated by the score calculated via Common Vulnerability Scoring System (CVSS) Version 3.1 [?]. The higher the threat scores, the greater severity the threat is with and the higher the priority is.

CVSS scores can be mapped to qualitative severity ratings defined in CVSS 3.1 specification [?]. This threat model follows the same mapping between CVSS scores and threat priority rating.

As a generic threat model, this document focuses on *Base Score* which reflects the constant and general severity of a threat according to its intrinsic characteristics.

The *Impacted Component* defined in [?] refers to the assets listed in *Asset identification*.

Threats and mitigation list

This section lists generic threats and corresponding mitigation, based on the the analysis of data flows in *Data Flow Diagram*.

Threats are identified following STRIDE model. Please refer to [?] for more details.

The field CVSS `Score` reflects the threat priority defined in *Threat priority*. The field CVSS `Vector String` contains the textual representation of the CVSS metric values used to score the threat. Refer to [?] for more details of CVSS vector string.

For each threat, a mitigation strategy is determined:

- **Controlled:** a mitigation has been implemented by TF-M.
- **Accepted:** no mitigation is currently implemented as the risk is acceptable, and the mitigation description justifies the acceptance.
- **Suppressed:** the feature can be disabled to remove the threat.
- **Transferred:** the threat cannot be fully mitigated within the scope of TF-M. The threat must be handled by downstream users for their specific platform.
- **Out-of-scope:** some threats are out-of-scope and so do not have a mitigation but are included for completeness.

Note

A generic threat may have different behaviors and therefore require different mitigation, in diverse TF-M models and usage scenarios.

This threat model document focuses on general analysis of the following threats. For the details in a specific configuration and usage scenario, please refer to the dedicated threat model document.

NS entry initialization

This section identifies threats on DF1 defined in *Data Flow Diagram*.

Table 3:: TFM-GENERIC-NS-INIT-T-1

Index	TFM-GENERIC-NS-INIT-T-1
Description	The NS image can be tampered by an attacker
Justification	An attack may tamper the NS image to inject malicious code
Category	Tampering
Mitigation	Controlled: TF-M BL2 uses MCUBoot to validate the integrity and authenticity of the NS image during secure boot, before the Secure firmware jumps to the NS entry or boots up the NS core. Refer to [?] for more details. Platforms may replace the TF-M Chain of Trust (CoT) implementation, in which case this threat is transferred .
CVSS Score	3.5 (Low)
CVSS Vector String	CVSS:3.1/AV:P/AC:L/PR:N/UI:N/S:U/C:L/I:L/A:N

Table 4:: TFM-GENERIC-NS-INIT-T-2

Index	TFM-GENERIC-NS-INIT-T-2
Description	An attacker may replace the current NS image with an older version.
Justification	The attacker downgrades the NS image with an older version which has been deprecated due to known security issues. The older version image can pass the image signature validation and its vulnerabilities can be exploited by attackers.
Category	Tampering
Mitigation	Controlled: TF-M relies on MCUBoot to perform anti-rollback checks. Transferred: TF-M defines a non-volatile counter API to support anti-rollback. Each platform must implement it using specific trusted hardware non-volatile counters. For more details, refer to [?]. The anti-rollback protection implementation can vary on diverse platforms.
CVSS Score	3.5 (Low)
CVSS Vector String	CVSS:3.1/AV:P/AC:L/PR:N/UI:N/S:U/C:L/I:L/A:N

Table 5:: TFM-GENERIC-NS-INIT-T-I-1

Index	TFM-GENERIC-NS-INIT-T-I-1
Description	If SPE doesn't complete isolation configuration before NSPE starts, NSPE can access secure regions which it is disallowed to.
Justification	Secure data can be tampered or disclosed if NSPE is activated and accesses secure regions before isolation configuration is completed by SPE.
Category	Tampering/Information disclosure
Mitigation	Transferred: TF-M defines isolation configuration HALs, which platforms must implement using platform- specific isolation hardware. TF-M provides a reference implementation of the isolation HAL for Armv8-M platforms with TrustZone. On multi-core platform, platform specific initialization must halt NS core until isolation is completed, as defined in [?] (same applies for [?]). TF-M executes isolation configuration at an early stage of secure initialization before starting NS execution.
CVSS Score	9.0 (Critical)
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:N

Table 6:: TFM-GENERIC-NS-INIT-T-I-2

Index	TFM-GENERIC-NS-INIT-T-I-2
Description	If SPE doesn't complete isolation configuration before NSPE starts, NSPE can control devices or peripherals which it is disallowed to.
Justification	On some platforms, devices and peripherals can be configured as Secure state in runtime. If security status configuration of those device and peripherals are not properly completed before NSPE starts, NSPE can control those device and peripherals and may be able to tamper data or access secure data.
Category	Tampering/Information disclosure
Mitigation	Transferred: Platforms must implement the TF-M isolation HALs to complete and enable proper configuration and isolation to protect critical devices and peripherals from being accessed by the NSPE. TF-M executes isolation configuration of devices and peripherals at an early stage of secure initialization, before jumping to NS entry or booting up NS core.
CVSS Score	9.0 (Critical)
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:N

Table 7:: TFM-GENERIC-NS-INIT-I-2

Index	TFM-GENERIC-NS-INIT-I-2
Description	If SPE leaves some SPE information in non-secure memory or shared registers when NSPE starts, NSPE may access those SPE information.
Justification	If NSPE can access those SPE information from shared registers or non-secure memory, secure information may be disclosed.
Category	Information disclosure
Mitigation	Controlled: TF-M clears registers that are not banked between security states before handing over execution to the NSPE on Armv8-M platforms with TrustZone. TF-M does not store SPE information in Non-secure memory. Transferred: Platform-specific code in the SPE must not store SPE information in Non-secure memory. On multi-core platforms, shared registers are implementation defined, such as Inter-Processor Communication registers. Multi-core platforms must not store any data which may disclose secure information in the shared registers.
CVSS Score	4.3 (Medium)
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:L/I:N/A:N

Table 8:: TFM-GENERIC-NS-INIT-D-1

Index	TFM-GENERIC-NS-INIT-D-1
Description	An attacker may block NS to boot up
Justification	An attacker may block NS to boot up, such as by corrupting NS image, to stop the whole system from performing normal functionalities.
Category	Denial of service
Mitigation	Out-of-scope: No SPE information will be disclosed and TF-M won't be directly impacted. It relies on NSPE and platform specific implementation to mitigate this threat. It is out of scope of this threat model.
CVSS Score	4.0 (Medium)
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:L

NSPE requests TF-M secure service

This section identifies threats on DF2 defined in *Data Flow Diagram*.

Table 9:: TFM-GENERIC-REQUEST-SERVICE-S-1

Index	TFM-GENERIC-REQUEST-SERVICE-S-1	
Description	A malicious NS application may pretend as a secure client to access secure data which NSPE must not directly access.	
Justification	[?] defines <code>Client ID</code> to distinguish clients which request RoT services. Secure clients are assigned with positive IDs and non-secure clients are assigned with negative ones. A malicious NS application may provide a positive <code>Client ID</code> to pretend as a secure client to access secure data.	
Category	Spoofing	
Mitigation	Controlled: TF-M checks the <code>Client ID</code> from NSPE. If the NS <code>Client ID</code> is not a valid one, TF-M will report this as a security error.	
CVSS Score	8.4 (High)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:C/C:H/I:H/A:N	

Table 10:: TFM-GENERIC-REQUEST-SERVICE-T-1

Index	TFM-GENERIC-REQUEST-SERVICE-T-1	
Description	An attacker in NSPE may tamper the service request input or output vectors between check and use (Time-Of-Check-to-Time-Of-Use (TOCTOU)).	
Justification	If SPE validates the content in input/output vectors locally in NSPE memory, an attacker in NSPE can have a chance to tamper the content after the validation successfully passes. Then SPE will provide RoT service according to the corrupted parameters and it may cause further security issues.	
Category	Tampering	
Mitigation	Controlled: In TF-M implementation, the validation of NS input/output vectors are only executed after those vectors are copied from NSPE into SPE. It prevents an attack from NSPE to tamper those parameters after validation in TF-M.	
CVSS Score	7.8 (High)	
CVSS Vector String	CVSS:3.1/AV:L/AC:H/PR:N/UI:N/S:C/C:H/I:H/A:N	

Table 11:: TFM-GENERIC-REQUEST-SERVICE-T-2

Index	TFM-GENERIC-REQUEST-SERVICE-T-2	
Description	A malicious NS application may request to tamper data belonging to SPE.	
Justification	A malicious NS application may request SPE RoT services to write malicious value to SPE data. The malicious NS application may try to tamper SPE assets, such as keys, or modify configurations in SPE. The SPE data belongs to components in SPE and must not be accessed by NSPE.	
Category	Tampering	
Mitigation	Controlled: TF-M executes memory access check to all the RoT service requests. If a request doesn't have enough permission to access the target memory region, TF-M will refuse this request and assert a security error.	
CVSS Score	7.1 (High)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:N/I:H/A:N	

Table 12:: TFM-GENERIC-REQUEST-SERVICE-R-1

Index	TFM-GENERIC-REQUEST-SERVICE-R-1	
Description	A NS application may repudiate that it has requested services from a RoT service.	
Justification	A malicious NS application may call a RoT service to access critical data in SPE, which it is disallowed to, via a non-public vulnerability. It may refuse to admit that it has accessed that data.	
Category	Repudiation	
Mitigation	Transferred: This threat can be mitigated with an audit logging secure service that records significant security events, such as access to sensitive data. If this threat is in-scope for a particular integration, then the system integrator must implement the mitigation.	
CVSS Score	0.0 (None)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:N/I:N/A:N	

Table 13:: TFM-GENERIC-REQUEST-SERVICE-I-1

Index	TFM-GENERIC-REQUEST-SERVICE-I-1	
Description	A malicious NS application may request to read data belonging to SPE.	
Justification	A malicious NS application may request SPE RoT services to copy SPE data to NS memory. The SPE data belongs to components in SPE and must not be disclosed to NSPE, such as root keys.	
Category	Information disclosure	
Mitigation	Controlled: TF-M executes memory access checks for all the RoT service requests. If a request doesn't have enough permission to access the target memory region, TF-M will refuse the request and assert a security error.	
CVSS Score	7.1 (High)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:H/I:N/A:N	

Table 14:: TFM-GENERIC-REQUEST-SERVICE-T-I-1

Index	TFM-GENERIC-REQUEST-SERVICE-T-I-1	
Description	A malicious NS application may request to control secure device and peripherals, on which it doesn't have the permission.	
Justification	A malicious NS application may request RoT services to control secure device and peripherals, on which it doesn't have the permission.	
Category	Tampering/Information disclose	
Mitigation	Controlled: TF-M performs client check to validate whether the client has the permission to access the secure device and peripherals.	
CVSS Score	9.0 (Critical)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:H/I:H/A:N	

Table 15:: TFM-GENERIC-REQUEST-SERVICE-D-1

Index	TFM-GENERIC-REQUEST-SERVICE-D-1
Description	A Malicious NS applications may frequently call secure services to block secure service requests from other NS applications.
Justification	TF-M runs on IoT devices with constrained resource. Even though multiple outstanding NS PSA Client calls can be supported in system, the number of NS PSA client calls served by TF-M simultaneously are still limited. Therefore, if a malicious NS application or multiple malicious NS applications continue calling TF-M secure services frequently, it may block other NS applications to request secure service from TF-M.
Category	Denial of service
Mitigation	Out-of-scope: TF-M is unable to manage behavior of NS applications. Assets are not disclosed and neither is TF-M directly impacted in this threat. It relies on NS OS to enhance scheduling policy and prevent a single NS application to occupy entire CPU time. It is beyond the scope of this threat model.
CVSS Score	4.0 (Medium)
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:L

Table 16:: TFM-GENERIC-REQUEST-SERVICE-D-2

Index	TFM-GENERIC-REQUEST-SERVICE-D-2
Description	A malicious NS application may provide invalid NS memory addresses as the addresses of input and output data in RoT service requests.
Justification	SPE may be unable to achieve full knowledge of NS memory mapping. SPE may fail to capture those invalid NS memory addresses during memory access check since those invalid addresses may not be included in isolation configuration. In that case, SPE will access those invalid NS memory addresses later to read or write data. It may trigger a system error to crash the whole system immediately. The malicious NS application may be blocked by NS MPU from directly accessing that invalid NS memory address. But it may manipulate SPE to access that address instead.
Category	Denial of service
Mitigation	Controlled: TF-M executes memory access check to the memory addresses in all the NS requests. On Armv8-M platforms with TrustZone, TF-M invokes TT instructions to execute memory address check. If a NS memory area is not matched in any valid SAU or MPU region, it will be marked as invalid and any access permission is disallowed. Therefore, SPM will reject any NS request containing invalid NS memory addresses and reports it as a security error. On multi-core platforms, TF-M implements a default memory access check. If a NS memory area is not found in any memory region configured for isolation, it will be marked as invalid and therefore SPM will reject the corresponding NS request. It will be reported as a security error. Transferred: multi-core platforms may implement platform specific memory check to replace the default one. It relies on platform specific implementation to capture invalid memory address.
CVSS Score	3.2 (Low)
CVSS Vector String	CVSS:3.1/AV:L/AC:H/PR:N/UI:N/S:C/C:N/I:N/A:L

RoT services read and write NS data

This section identifies threats on DF3 defined in *Data Flow Diagram*.

RoT services can either directly access NS memory or rely on TF-M SPM to obtain NS input data and send response data back to NS memory.

Table 17:: TFM-GENERIC-SECURE-SERVICE-RW-T-1

Index	TFM-GENERIC-SECURE-SERVICE-RW-T-1	
Description	An attacker may tamper NS input data while the RoT service is processing those data.	
Justification	A RoT service may access NS input data multiple times during its data processing. For example, it may validate or authenticate the NS input data before it performs further processing. If the NS input data remains in NSPE memory during the RoT service execution, an attacker may tamper the NS input data in NSPE memory after the validation passes.	
Category	Tampering	
Mitigation	Controlled: If RoT services request SPM to read and write NS data, SPM follows [?] to copy the NS input data into SPE memory region owned by the RoT service, before the RoT service processes the data. Therefore, the NS input data is protected during the RoT service execution from being tampered. Transferred: If RoT services can directly access NS memory and read NS input data multiple times during data processing, then the RoT service implementer must review and confirm the implementation of the RoT service copies NS input data into SPE memory area before it processes the data. TF-M-provided RoT services implement this mitigation.	
CVSS Score	3.2 (Low)	
CVSS Vector String	CVSS:3.1/AV:L/AC:H/PR:N/UI:N/S:C/C:N/I:L/A:N	

Table 18:: TFM-GENERIC-SECURE-SERVICE-RW-T-2

Index	TFM-GENERIC-SECURE-SERVICE-RW-T-2	
Description	A malicious NS application may embed secure memory addresses into a structure in RoT service request input vectors, to tamper secure memory which the NS application must not access.	
Justification	[?] limits the total number of input/output vectors to 4. If a RoT service requires more input/output vectors, it may define a parameter structure which embeds multiple input/output buffers addresses. However, as a potential security risk, a malicious NS application can put secure memory addresses into a valid parameter structure to bypass TF-M validation on those memory addresses. The parameter structure can pass TF-M memory access check since itself is valid. However, if the RoT service parses the structure and directly write malicious data from NSPE to the secure memory addresses in parameter structure, the secure data will be tampered.	
Category	Tampering	
Mitigation	Transferred: The RoT service implementer must not embed memory addresses into a single input/output vector. If more than 4 memory addresses are required in a RoT service request, it is recommended to split this request into two or multiple service calls and therefore each service call requires no more than 4 input/output vectors. If RoT services request SPM to read and write NS data, SPM will validate the target addresses and can detect the invalid addresses to mitigate this threat. If RoT services can directly access NS memory, it is required to review and confirm the implementation of RoT service request doesn't embed memory addresses. TF-M-provided RoT services implement these mitigations.	
CVSS Score	7.1 (High)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:N/I:H/A:N	

Table 19:: TFM-GENERIC-SECURE-SERVICE-RW-I-1

Index	TFM-GENERIC-SECURE-SERVICE-RW-I-1
Description	Similar to <i>TFM-GENERIC-SECURE-SERVICE-RW-T-2</i> , a malicious NS application can embed secure memory addresses in a parameter structure in RoT service request input vectors, to read secure data which the NS application must not access.
Justification	Similar to the description in <i>TFM-GENERIC-SECURE-SERVICE-RW-T-2</i> , the secure memory addresses hidden in the RoT service input/output vector structure may bypass TF-M validation. Without a proper check, the RoT service may copy secure data to NSPE according to the secure memory addresses in structure, secure information can be disclosed.
Category	Information disclosure
Mitigation	Transferred: The RoT service implementer must not embed memory addresses into a single input/output vector. If more than 4 memory addresses are required in a RoT service request, it is recommended to split this request into two or multiple service calls and therefore each service call requires no more than 4 input/output vectors. If RoT services request SPM to read and write NS data, SPM will validate the target addresses and can detect the invalid addresses to mitigate this threat. If RoT services can directly access NS memory, it is required to review and confirm the implementation of RoT service request doesn't embed memory addresses. TF-M-provided RoT services implement these mitigations.
CVSS Score	7.1 (High)
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:H/I:N/A:N

TF-M returns secure service result

This section identifies threats on DF4 defined in *Data Flow Diagram*.

When RoT service completes the request from NSPE, TF-M returns the success or failure error code to NS application.

In Armv8-M TrustZone scenarios, TF-M writes the return code value in the general purpose register and returns to Non-secure state.

On multi-core platforms, TF-M writes the return code to NS mailbox message queue via mailbox.

Table 20:: TFM-GENERIC-RETURN-CODE-I-1

Index	TFM-GENERIC-RETURN-CODE-I-1
Description	SPE may leave secure data in the registers not banked after the SPE completes PSA Client calls and executes BXNS to switch Armv8-M back to Non-secure state.
Justification	If SPE doesn't clean up the secure data in registers not banked before switching into NSPE in Armv8-M core, NSPE can read the SPE context from those registers.
Category	Information disclosure
Mitigation	Controlled: In Armv8-M TrustZone scenarios, TF-M cleans general purpose registers not banked before switching into NSPE to prevent NSPE probing secure context from the registers. When FPU is enabled in TF-M, secure FP context belonging to a secure partition will be saved on this partition's stack and cleaned by hardware during context switching. Also TF-M cleans secure FP context in FP registers before switching into NSPE to prevent NSPE from probing secure FP context.
CVSS Score	4.3 (Medium)
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:L/I:N/A:N

NS interrupts preempts SPE execution

This section identifies threats on DF5 defined in *Data Flow Diagram*.

Table 21:: TFM-GENERIC-NS-INTERRUPT-I-1

Index	TFM-GENERIC-NS-INTERRUPT-I-1	
Description	Shared registers may contain secure data when NS interrupts occur.	
Justification	The secure data in shared registers should be cleaned up before NSPE can access shared registers. Otherwise, secure data leakage may occur.	
Category	Information disclosure	
Mitigation	Controlled: On Armv8-M processors with TrustZone, Armv8-M architecture automatically cleans up the registers not banked before switching to Non-secure state when taking NS interrupts. When FPU is enabled in TF-M, with setting of FPCCR_S.TS = 1 besides secure FP context in FP caller registers, FP context in FP callee registers will also be cleaned by hardware automatically when NS interrupts occur, to prevent NSPE from probing secure FP context in FP registers. Refer to Armv8-M Architecture Reference Manual [?] for details. Transferred: On multi-core platforms, shared registers are implementation defined, such as Inter-Processor Communication registers. Multi-core platforms must not store any data which may disclose secure information in the shared registers.	
CVSS Score	4.3 (Medium)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:L/I:N/A:N	

Table 22:: TFM-GENERIC-NS-INTERRUPT-D-1

Index	TFM-GENERIC-NS-INTERRUPT-D-1	
Description	An attacker may trigger spurious NS interrupts frequently to block SPE execution.	
Justification	On Armv8-M processors with TrustZone, an attacker may inject a malicious NS application or hijack a NS hardware to frequently trigger spurious NS interrupts to keep preempting SPE and block SPE to perform normal secure execution.	
Category	Denial of service	
Mitigation	Out-of-scope: Availability of the whole system is out of scope of TF-M. Assets protected by TF-M won't be leaked. TF-M won't be directly impacted.	
CVSS Score	4.0 (Medium)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:L	

Secure interrupts preempts NSPE execution

This section identifies threats on DF6 defined in *Data Flow Diagram*.

Table 23:: TFM-GENERIC-S-INTERRUPT-I-1

Index	TFM-GENERIC-S-INTERRUPT-I-1	
Description	Shared registers may contain secure data when Armv8-M core switches back to Non-secure state on Secure interrupt return.	
Justification	Armv8-M architecture doesn't automatically clean up shared registers while returning to Non-secure state during Secure interrupt return. If SPE leaves critical data in the Armv8-M registers not banked, NSPE can read secure context from those registers and secure data leakage may occur.	
Category	Information disclosure	
Mitigation	Controlled: TF-M saves NSPE context in general purpose registers R4~R11 into secure stack during secure interrupt entry. After secure interrupt handling completes, TF-M unstacks NSPE context from secure stack to overwrite secure context in R4~R11 before secure interrupt return. Armv8-M architecture will automatically unstack NSPE context from non-secure stack to overwrite other registers not banked, such as R0~R3 and R12, during secure interrupt return, before NSPE software can access those registers. When FPU is enabled in TF-M, with setting of FPCCR_S.TS = 1 and FPCCR_S.CLRONRET = 1, besides secure FP context in FP caller registers, FP context in callee registers will also be cleaned by hardware automatically during S exception return, to prevent NSPE from probing secure FP context in FP registers. Refer to Armv8-M Architecture Reference Manual [?] for details.	
CVSS Score	4.3 (Medium)	
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:N/UI:N/S:C/C:L/I:N/A:N	

Miscellaneous threats

This section collects threats irrelevant to the valid TF-M data flows shown above.

Table 24:: TFM-GENERIC-STACK-SEAL

Index	TFM-GENERIC-STACK_SEAL	
Description	Armv8-M processor Secure software Stack Sealing vulnerability.	
Justification	On Armv8-M based processors with TrustZone, if Secure software does not properly manage the Secure stacks when the stacks are created, or when performing non-standard transitioning between states or modes, for example, creating a fake exception return stack frame to de-privilege an interrupt, it is possible for Non-secure world software to manipulate the Secure Stacks, and potentially influence Secure control flow. Refer to [?] for details.	
Category	Elevation of privilege	
Mitigation	Controlled: TF-M has implemented a common mitigation against the stack seal vulnerability. Refer to [?] for details on analysis and mitigation in TF-M.	
CVSS Score	5.3 (Medium)	
CVSS Vector String	CVSS:3.1/AV:L/AC:H/PR:L/UI:N/S:C/C:L/I:L/A:L	

Table 25:: TFM-GENERIC-SVC-CALL-SP-FETCH

Index	TFM-GENERIC-SVC-CALL-SP-FETCH
Description	Invoking Secure functions from handler mode may cause TF-M IPC model to behave unexpectedly.
Justification	On Armv8-M based processors with TrustZone, if NSPE calls a secure function via Secure Gateway (SG) from non-secure Handler mode, TF-M selects secure process stack by mistake for SVC handling. It will most likely trigger a crash in secure world or reset the whole system, with a very low likelihood of overwriting some memory contents.
Category	Denial of service/Tampering
Mitigation	Controlled: TF-M has enhanced implementation to mitigate this vulnerability. Refer to [?] for details on analysis and mitigation in TF-M.
CVSS Score	4.5 (Medium)
CVSS Vector String	CVSS:3.1/AV:L/AC:H/PR:N/UI:N/S:C/C:N/I:L/A:L

Table 26:: VLLDM instruction security vulnerability

Index	TFM-GENERIC-FP-VLLDM
Description	Secure data in FP registers may be disclosed to NSPE when VLLDM instruction is abandoned due to an exception mid-way.
Justification	Refer to [?] for details.
Category	Tampering/Information disclosure
Mitigation	Controlled: In the current TF-M implementation, when FPU is enabled in SPE, TF-M configures NSACR to disable NSPE to access FPU. Therefore, secure data in FP registers is protected from NSPE. Refer to [?], for details on analysis and mitigation.
CVSS Score	3.4 (Low)
CVSS Vector String	CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:U/C:L/I:L/A:N

Version control

Table 27:: Version control

Version	Description	TF-M version
v0.1	Initial draft	TF-M v1.1
v1.0	First version	TF-M v1.2.0
v1.1	Update version	TF-M v1.5.0
v1.2	Update details to align FP support in NSPE.	TF-M v1.5.0
v1.3	Update for validity of dual-cpu model Armv8-M	TF-M v2.1.0
v1.4	Clarify mitigation strategies of threats	TF-M v2.1.1
v1.5	Updated terminology to replace ‘dual-cpu’ with ‘multi-core’ for broader applicability to hybrid and multi-core platforms. Also fixed typos.	TF-M v2.2.1

References

SPDX-License-Identifier: BSD-3-Clause

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3.2 Security Advisories

3.2.1 Advisory TFMV-1

Title	NS world may cause the CPU to perform an unexpected return operation due to unsealed stacks.
CVE ID	CVE-2020-16273
Date	16 October 2020
Versions Affected	All versions up to and including TF-M v1.1
Configurations	All
Impact	Most likely a crash in secure world with a very low likelihood of hijacking secure world execution flow via a separate vulnerability.
Fix Version	commit 92e46a3abd0e328fac29ccd1cf161cd482397567
Credit	Matvey Mukha

Background

When the Non-Secure world returns to Secure after a callback (FNC_RETURN) or Non-Secure interrupt handling (EXC_RETURN), the hardware pops the secure return address and RET_PSR from the respective secure stack. The hardware performs a set of integrity checks at this point and will raise a fault if the checks fail. There is a potential vulnerability if the NS attempts a wrong FNC_RETURN or EXC_RETURN and causes the PE to pop from the unexpected stack. Please refer to [ARMv8-M Secure stack sealing advisory notice](#) for more details on the vulnerability.

To prevent such an attack, the architecture expects the secure software to *seal* the stacks. The [ARMv8-M ARM](#) states that the stack should be sealed with the special value, 0xFEFE5EDA5 (informative IGJGL).

Both the MSP_S and the PSP_S stacks need to be sealed to mitigate stack underflow attacks and this is not done currently in TF-M.

Impact

This section analyses the impact of this vulnerability on the upstream TF-M implementation.

All requests coming in from the Non-Secure world uses ARM_LIB_STACK as the PSP_S and then switches to MSP_S as part of SPM scheduling. The MSP_S is fully unwound when the scheduled partition is executing. All partitions run using the PSP_S and this stack can be separate for each partition or be a common stack depending on whether TF-M is in library mode or IPC model. When the partition execution using PSP_S switches to non-secure world due to a non-secure interrupt or a non-secure callback invocation, the non-secure world on return back to secure can use a fake EXC_RETURN or FNC_RETURN operation to trigger an MSP_S stack underflow, and the CPU could fetch the return PC and PSR from the memory just above MSP_S stack (stack grows from higher to lower address).

The memory above MSP_S is the ARM_LIB_STACK (as described by `tfm_common.s.ct/ld`), which because of over-laying or zero initialization is most likely to be initialized memory for most platforms. This is the stack used to run

the initialization code of TF-M and usually there will be some unused free space in the stack. Any underflow of MSP_S will be into this stack and hence is likely to be a deterministic crash given that this memory is initialized. In theory, a separate vulnerability that allows an attacker to control the memory above MSP_S in concert with this vulnerability could enable an attacker to hijack the secure world execution flow but the likelihood of that is very low.

Through analysing TF-M specific initialization and execution flow, we have not found any scenario in TF-M in which Non-secure world can fake a return operation back to secure world and cause underflow of PSP_S.

As described in the white paper, de-privileged interrupt handling is also vulnerable to this problem. The Library mode of TF-M uses de-privileged interrupt handling and does not allow non-secure interrupt to preempt the secure interrupt handling. But if the de-privileged handler makes a Non-Secure callback then there is a chance that Non-Secure world could return back to Secure world via a fake FNC_RETURN. Under certain conditions, this can force the CPU to use the 2 words on top of stack as return PC and PSR. Sealing the top of MSP_S before switching to PSP_S as part of de-privileged interrupt handling mitigates this vulnerability.

The interrupt handling in IPC model uses PSA signal to signal the partition and does not use de-privileged interrupt handling mechanism. The PSA signal gets handled by the partition when it is scheduled to run by the SPM. Hence during interrupt handling in IPC model, there is no additional threat caused by this vulnerability, compared to regular partition execution.

Conclusion

Overall, from analysis of upstream TF-M implementation, the severity of this vulnerability is low, given that the values popped out from secure stack (either via underflow of MSP_S or from top of MSP_S stack in de-privileged interrupt handling) cannot be influenced by the Non Secure world. To mitigate the risk completely, the fix in TF-M is to follow the recommendation in [ARMv8-M ARM](#) which is to seal the base of both handler mode stack (MSP_S) and thread stacks (PSP_S) and, in case of library mode, also seal the top of MSP_S before handing control over to de-privileged interrupt handler.

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3.2.2 Advisory TFMV-2

Title	Invoking Secure functions from handler mode may cause TF-M IPC model to behave unexpectedly.
CVE ID	CVE-2021-27562
Date	Mar 5, 2021
Versions Affected	Affected All versions up to and including TF-M v1.2
Configurations	IPC Model on Armv8-M
Impact	Most likely a crash in secure world or reset whole system, with a very low likelihood of overwriting some memory contents.
Fix Version	commit e212ea1637a66255b44d0e7c19ebe9786ab56ccb
Credit	Øyvind Rønningstad, Senior SW Engineer from Nordic Semiconductor

Background

When a non-secure caller calls the secure function, the execution switches the security state via Secure Gateway (SG), then arrived at the TF-M secure entry if the SG is successful. After SG, TF-M invokes SVC to SPM at first because SPM handles requests from SPE and NSPE in a unified SVC handler. The TF-M SVC handler code relies on the 'SPSEL' bit in 'EXC_RETURN' to get the caller stack pointer:

```

; Armv8m mainline as the example
mrs      r2, msp          ; r2 = msp;
tst      lr, #4           ; EXC_RETURN.SPSEL == ?
ite      eq               ; condition preparation
moveq    r0, r2           ; if (EXC_RETURN.SPSEL == 0) r0 = msp;
movne    r0, psp          ; if (EXC_RETURN.SPSEL == 1) r0 = psp;
mov      r1, lr           ; r1 = EXC_RETURN;
bl       tfm_core_svc_handler ; r0 = sp(context), r1 = EXC_RETURN, r2 = msp

```

If NSPE calls the secure function in ‘Handler mode’, the TF-M secure entry runs in ‘Handler mode’ before invoking SVC because PE mode does not change during the transition from non-secure state to secure state via the successful SG. Main stack pointer (MSP) is always used in ‘Handler mode’, which is irrelevant to the value of SPSEL. However, the code above selects secure process stack pointer (PSP_S, S suffix here indicates the security state) for further SVC handling based on EXC_RETURN.SPSEL, hence the SVC handler accesses the wrong stack for handling the request in the NSPE caller in ‘Handler mode’ case. To prevent this vulnerability, TF-M secure SVC handler should fetch the right stack pointer register by checking both the PE mode and SPSEL bit. The handler should also prevent callers in non-secure *Handler mode* from calling TF-M SVC functionalities. The following section analysis impact of the IPC model. Library model is not vulnerable to this attack because it checks the PE mode in the TF-M secure entry.

Impact

When the vulnerability is happening, PSP_S is the incorrect stack pointer fetched, the impact is decided by the content PSP_S is pointing to. The SVC handler gets the SVC number by referencing the memory at the 2 bytes subtracted position of member ‘Return Address’ in the PSP_S pointing context, and uses the caller saved registers in the context as parameters for the subsequent functions indicated by the SVC number. The PSP_S may point to the bottom (higher address) of secure thread stack if there is no ongoing secure function call, or it points to a preempted context caused by non-secure preempting secure execution. When PSP_S is pointing to the stack bottom when this issue happens, the caller context is pointing to the underflowed stack of which the top 2 words are stack seal, and the rest of the context is unpredictable. These underflowed content are ZERO initialized for most platforms, hence when fetching the SVC Number at memory address ‘Return Address - 2’, a ‘MemFault’ or ‘HardFault’ would happen. Even if a valid SVC number is returned, the stack seal values are part of the parameters for the subsequent functions that have parameters, which cannot pass the validation for the parameters. When PSP_S is pointing to a preempted context, the preempted place can be the TF-M secure entry area or internal thread space. If the preemption happens when the execution is in the TF-M secure entry area, the PSP_S will point to an NS preemption stack frame and hence will fail the context size validation through this entry into TF-M SVC Handler. In the other scenario that TF-M internal thread is preempted, there is a very remote chance that the exception stack frame will contain a context with a valid SVC Number with valid parameters. As a summary, the impacts of the vulnerability depend on whether an SVC number could be fetched from the incorrect stack and what operation the fetched SVC number corresponds to:

- A memory access fault when fetching the SVC number from memory. This fault halts the whole system. A reset can recover the system back to normal.
- The SVC number corresponds to the initialization process in TF-M. SPM initialization is called for the second time. This re-initialization process will fail and halt the whole system. A reset can recover the system back to normal.
- The SVC number corresponds to the boot data fetching function. If the unpredictable data in stack triggers boot data fetching and passes the parameter validation under a very rare case, boot data is copied into an unpredictable memory address. One note here, the TF-M default boot data has no sensitive information.
- The SVC number corresponds to the log output function. If the unpredictable data in stack triggers log output under a very rare case, secure data at unpredictable address might be printed out through the log interface.
- The SVC number corresponds to other valid numbers other than above. The system resets because of parameter validation fail.

Conclusion

Overall, from the analysis of upstream TF-M implementation, the severity of this vulnerability is Medium, given that a software crash or system reset happen most of the time. The probability for causing secure data leakage via log output or tampering of secure memory with boot data is extremely low as the TF-M internal thread exception stack frame contents have to pass all the validations performed by SPM.

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3.2.3 Advisory TFMV-3

Title	abort() function may not take effect in TF-M Crypto multi-part MAC/hashing/cipher operations.
CVE ID	CVE-2021-32032
Public Disclosure Date	May 10, 2021
Versions Affected	Affected all versions up to and including TF-M v1.3.0
Configurations	All
Impact	It can cause memory leakage in TF-M Crypto service, eventually making TF-M Crypto service unavailable and impacting other services relied on it.
Fix Version	commit 7e2e523a1c4e9ac7b9cc4fd551831f7639ed5ff9
Credit	Chongqing Lei, Southeast University Zhen Ling, Associate Professor, Southeast University Xinwen Fu, Professor, University of Massachusetts Lowell

Background

PSA multi-part crypto operation sequence

PSA Crypto API specification defines a common sequence for all multi-part crypto operations. The sequence can be simplified to the following steps:

- `setup()` sets up the multi-part operation.
- `update()` adds data/configurations into the multi-part operation.
- `finish()` completes the multi-part operation.

PSA Crypto API specification requests that the corresponding `abort()` function shall be called when `update()` or `finish()` function fails. The `abort()` function aborts the ongoing multi-part operation and cleans up the operation context.

TF-M multi-part crypto operation functions eventually call the underlying crypto library (Mbed TLS by default) to perform those steps, including `abort()` step.

PSA multi-part crypto operation objects

PSA Crypto API specification defines an operation object for each type of multi-part crypto operations. For example, `psa_mac_operation_t` for multi-part MAC operations and `psa_hash_operation_t` for multi-part hashing operations.

TF-M Crypto service relies on the underlying crypto library (Mbed TLS by default) to implement those objects. The structures of those objects are crypto library specific and hidden to TF-M. The underlying crypto library usually stores and manages the context of ongoing multi-part crypto operations in the corresponding PSA operation object. For example, Mbed TLS stores multi-part hashing operation context in its `psa_hash_operation_t` implementation.

The context is cleaned up in crypto library `abort()` function when the client calls `abort()` to handle a previous error. The clean-up execution can include zeroing the memory area and freeing allocated memory.

TF-M multi-part crypto operation objects

TF-M Crypto service defines a dedicated operation structure `tfm_crypto_operation_s` to wrap PSA multi-part crypto operation object and maintains its own status, as shown in the code block below.

```
struct tfm_crypto_operation_s {
    ...
    union {
        psa_cipher_operation_t cipher;    /*!< Cipher operation context */
        psa_mac_operation_t mac;          /*!< MAC operation context */
        psa_hash_operation_t hash;       /*!< Hash operation context */
        psa_key_derivation_operation_t key_deriv; /*!< Key derivation operation context */
    } operation;
};
```

TF-M Crypto service assigns a `tfm_crypto_operation_s` object for each multi-part crypto operation sequence during `setup()` step. The `tfm_crypto_operation_s` object content will be cleaned after the sequence completes or fails.

Impact

During multi-part hashing/MAC/cipher operations, if the underlying crypto library function returns an error code, TF-M `update()` and `finish()` functions will immediately clean up the structure `tfm_crypto_operation_s` content and exit.

When `tfm_crypto_operation_s` content is cleaned in TF-M `update()` and `finish()` functions, the content in PSA multi-part crypto operation object inside `tfm_crypto_operation_s` is also cleaned. If the underlying crypto library stores operation context in the PSA operation object, the operation context is lost before clients call `abort()` to handle the error.

Therefore, the underlying crypto library `abort()` function can be unable to perform normal abort operation if it cannot fetch the context or its content. In other words, the underlying crypto library `abort()` may not work normally or take effect.

In theory when the case analyzed above occurs:

- If the underlying crypto library dynamically allocates some memory regions during multi-part operation and stores those memory region pointers in the PSA multi-part operation object, the underlying crypto library will be unable to locate and free those allocated memory regions in `abort()`. It will cause memory leakage in TF-M Crypto service. It may further make TF-M Crypto service unavailable and affect other services relying on TF-M Crypto service.
- The underlying crypto library `abort()` may still consider the field values in the context as valid. `abort()` may perform unexpected behaviors or access invalid memory regions. It may trigger further faults and block TF-M Crypto service or even the whole system.

Note

The actual consequences depend on the implementation of the multi-part operations in the underlying crypto library.

Impacted PSA Crypto API functions

The following PSA multi-part crypto operation functions are impacted:

- Multi-part hashing operations
 - `psa_hash_update()`
 - `psa_hash_finish()`
 - `psa_hash_verify()`
 - `psa_hash_clone()`
- Multi-part MAC operations
 - `psa_mac_update()`
 - `psa_mac_sign_finish()`
 - `psa_mac_verify_finish()`
- Multi-part cipher operations
 - `psa_cipher_generate_iv()`
 - `psa_cipher_set_iv()`
 - `psa_cipher_update()`
 - `psa_cipher_finish()`

Justifications on unaffected multi-part operations

TF-M multi-part AEAD operations and multi-part key derivation operations are not impacted by this issue.

TF-M Crypto service has not implemented multi-part AEAD operations. TF-M multi-part AEAD functions directly return an error of unsupported operations.

In TF-M key derivation implementation, the `psa_key_derivation_operation_t` object is only cleaned in the `abort()` function after the underlying crypto library completes abort.

Mitigation

The clean-up operation shall be removed from error handling routines in the following TF-M Crypto functions:

- Multi-part hashing operations
 - `tfm_crypto_hash_update()`
 - `tfm_crypto_hash_finish()`
 - `tfm_crypto_hash_verify()`
 - `tfm_crypto_hash_clone()`
- Multi-part MAC operations
 - `tfm_crypto_mac_update()`
 - `tfm_crypto_mac_sign_finish()`

- `tfm_crypto_mac_verify_finish()`
- Multi-part cipher operations
 - `tfm_crypto_cipher_generate_iv()`
 - `tfm_crypto_cipher_set_iv()`
 - `tfm_crypto_cipher_update()`
 - `tfm_crypto_cipher_finish()`

Note

This mitigation assumes that client follows the sequence specified in PSA Crypto API specification to call `abort()` when an error occurs during multi-part crypto operations.

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3.2.4 Advisory TFMV-4

Title	NSPE may access secure keys stored in TF-M Crypto service in Profile Small with Crypto key ID encoding disabled.		
CVE ID	CVE-2021-40327		
Public Disclosure Date	Disclo-	sure Date	22nd Nov, 2021
Versions Affected	Af-	ected	TF-M v1.4.0
Configurations	Profile Small		
Impact	In Profile Small, secure keys stored in Crypto service can be leaked to NSPE if NSPE acquires secure key IDs.		
Fix Version	Commit 42e77b561fcfe19819ff1e63cb7c0b672ee8ba41 and v1.4.1		
Credit	N/A		

Background

TF-M Profile Small disabled Crypto key ID encoding with key owner client ID in TF-M v1.4.0 release.

When the Crypto key is stored into TF-M Crypto service, the key ID is not encoded with the client ID of key owner in Profile Small in TF-M v1.4.0. Therefore, TF-M Crypto service is unable to distinguish or validate owners of keys in Profile Small. NSPE can access the keys belonging to SPE in Profile Small in some scenarios.

Details

In TF-M v1.4.0, TF-M Crypto service by default relies on two mechanisms to validate key owners in key management.

- TF-M Crypto service maintains a key handle array. When a key is stored in Crypto service, the key ID and the key owner client ID are stored in the array. When a caller requests to access a key, TF-M Crypto service validates the request by comparing the caller client ID with the stored key client ID.
- Mbed TLS stores a special structure encoded by key owner client ID and the key ID. When a caller requests to access a key, Mbed TLS validates the request by comparing the caller client ID with the key client ID stored in that structure.

Secure clients are not isolated from each other in Profile Small and it doesn't require to validate key owner client ID between secure clients. Therefore, in TF-M v1.4.0, Profile Small disabled both mechanisms above to optimize the key storage size. The key directly or indirectly stored via `psa_import_key()` is not encoded with key owner client ID.

However, it also disables the validation of NS client ID when a NS client accesses keys stored in TF-M Crypto. NS clients can call `psa_export_key()` to access secure clients' keys stored via `psa_import_key()`, if NS clients acquire the key ID of secure clients.

Impact

Only TF-M Profile Small is impacted. All the other configurations or Profiles are not affected.

Analysis of RoT services in Profile Small

TF-M Profile Small enables Internal Trusted Storage (ITS), Crypto and Initial Attestation by default. The following analysis focuses on the impact on RoT services in Profile Small.

- ITS service doesn't create or store its own key in Crypto service. It is not impacted directly.
- Crypto service key derivation may be impacted.
 - `psa_key_derivation_output_key()` eventually stores the derived key in Crypto service. The stored derived keys can be accessed by a NS client if the NS client acquires the derived key ID value.
 - Platform specific implementation may store Hardware Unique Key (HUK) into Crypto service for key derivation from HUK via `psa_import_key()`.
 - * Platform driver may import HUK as a temporary key into Crypto service during derivation and close the temporary key when derivation completes.

If a NS client preempts the derivation and calls PSA Cryptography API to access temporary HUK data stored in Crypto service, the access will be captured by TF-M re-entry detection and rejected by TF-M SPE.

 - * Platform driver may permanently store HUK via Crypto service for derivation and the key is still managed by Crypto service when NSPE is running.

NS client can access HUK data via PSA Cryptography API if it acquires the key ID of stored HUK.
- Symmetric key algorithm based Initial Attestation temporarily stores symmetric Initial Attestation Key (IAK) in Crypto service during Initial Attestation Token generation. It imports symmetric IAK into Crypto service during generation and removes it from Crypto service when generation completes.

If a NS client preempts the generation and calls PSA Cryptography API to access the temporary IAK data stored in Crypto service, the access will be captured by TF-M re-entry detection and rejected by TF-M SPE.

Therefore, Initial Attestation is not impacted directly.

Impact on Profile Small default implementation

Default Profile Small RoT services don't initially call Crypto key derivation or store any secure key into Crypto service.

According to the analysis of RoT services above, device HUK can be accessed by NS clients and leaked to NSPE, in Profile Small default implementation, when all the following conditions are met.

- Platform specific implementation stores HUK in Crypto service, initially or during a derivation requested by NS client.
- HUK is still stored in Crypto service when NSPE is running.
- An NS client acquires the key ID of HUK in Crypto service and accesses HUK key via PSA Cryptography API.

Other vulnerabilities are not found yet so far.

Impact on vendor RoT services

If a vendor RoT service is integrated in Profile Small, its keys stored via `psa_import_key()` or derived from `psa_key_derivation_output_key()` can be accessed by NS client and leaked to NSPE when both following conditions are met.

- The secure key is stored in Crypto service when NSPE is running.
- An NS client acquires the key ID and accesses the key via PSA Cryptography API.

How NS client can acquire secure key ID is related to key management implementation of the underlying crypto library in TF-M Crypto service. With default Mbed TLS, NS hackers can import a NS key at first to obtain the rough base value of Mbed TLS key slots and then try a smaller subset of key ID values by brute-force.

Mitigation

This issue has been fixed by enforcing Mbed TLS key ID encoding with key owner client ID to be enabled.

This patch intended to optimize TF-M Crypto service key handle array and coincidentally fixed the issue.

v1.4.1 fixed this issue as a patch release.

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3.2.5 Advisory TFMV-5

Title	psa_fwu_write() may cause buffer overflow in SPE.
CVE ID	CVE-2021-43619
Public Disclosure Date	Feb 11, 2022
Versions Affected	From 3e7129f552c4cfd2973b9c17f3565df1ba45b334 to 921d0eaa85595e9b992de866b9da8b0d7dfe08eb
Configurations	IPC model with Firmware Update partition enabled
Impact	In IPC model, the caller of <code>psa_fwu_write()</code> from SPE or NSPE can overwrite the stack memory outside of the local buffer in Firmware Update partition.
Fix Version	commit 78f75304269baf4fc349f1a2b687675e71d2955e
Credit	Mark Horvath, Staff Software Engineer from Arm Ltd.

Background

In Firmware Update partition, the `psa_fwu_write()` service is declared as:

```
/**
 * \brief Writes an image to its staging area.
 *
 * Writes the image data 'block' with length 'block_size' to its staging area.
 *
 * \param[in] image_id      The identifier of the image
 * \param[in] block_offset  The offset of the block being passed into block,
 *                          in bytes
 * \param[in] block         A buffer containing a block of image data. This
```

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```

*                               might be a complete image or a subset.
* \param[in] block_size        Size of block. The size must not be greater than
*                               PSA_FWU_MAX_BLOCK_SIZE.
*
*/
psa_status_t psa_fwu_write(psa_image_id_t image_id,
                           size_t block_offset,
                           const void *block,
                           size_t block_size);

```

In IPC model, this service calls the `tfm_fwu_write_ipc()` API to write the input data into the device. In this API, the `block_size` bytes input data (in `block` argument) is read into a 1024 bytes local buffer via the `psa_read()` API. If the input argument `block_size` is greater than 1024, then the memory space starting from the address of the local buffer with `block_size` bytes would be overwritten by the input data in `block` argument.

Impact

In IPC model, the caller of `psa_fwu_write()` from SPE or NSPE can overwrite the memory space in RAM. The overwritten memory space ranges from the address of the local buffer which locates at the stack of Firmware Update partition to the end of the RAM. The overwritten memory may include the data of SPM and device drivers, as well as part of the stack of Firmware Update partition.

Mitigation

Add check against the input buffer length in the `tfm_fwu_write_ipc()` API before reading the input data into the local buffer. See commit [78f75304269baf4fc349f1a2b687675e71d2955e](#).

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3.2.6 Advisory TFMV-6

Title	Partial tag comparison when using Chacha20-Poly1305 on the PSA driver API interface in CryptoCell enabled platforms
CVE ID	CVE-2023-40271
Public Disclosure Date	04/09/2023
Versions Affected	TF-M v1.6.0, TF-M v1.6.1, TF-M v1.7.0, TF-M v1.8.0
Configurations	CC312 enabled platforms, where the legacy driver API is disabled (CC312_LEGACY_DRIVER_API_ENABLED=OFF) and the single part AEAD APIs are implemented through a dedicated function and not by leveraging the multipart functions (CC3XX_CONFIG_ENABLE_AEAD_ONE_SHOT_USE_MULTIPART not set)
Impact	It might allow for unauthenticated payloads to be deemed as authentic by comparing only the first 4 bytes of the authentication tag instead of the full length of 16 bytes
Fix Version	2e82124af29b58abcc98608196a63b3c5daf603b , TF-M v1.8.1
Credit	Nordic Semiconductor

Background

AEAD algorithms

Authenticated Encryption with Associated Data (AEAD) is a common ciphering method where the data to be encrypted is also authenticated as part of the process by creating an authentication tag. When the encrypted data is then decrypted, the authentication tag is verified and if it does not match the expected value, then the entire operation fails. In this way, the operation allows for Authenticity in addition to the confidentiality granted by the encryption process. Using PSA Crypto APIs, it's possible to use several of such algorithms, such as AES in Galois-Counter Mode (GCM), AES in Counter with CBC-Mac Mode (CCM) or Chacha20-Poly1305, which is a combination of the Chacha20 cipher with the Poly1305 authenticator¹.

In particular for Chacha20-Poly1305 the corresponding macro defining the algorithm in PSA Cryptographic API specification is `PSA_ALG_CHACHA20_POLY1305`.

Single part vs multipart API functions

PSA Crypto API specification² allows the usage of AEAD algorithms through several possible APIs, that can be grouped generally in the single part, or integrated, operation type, and in the multipart operation type. The main difference is that for single part operations, the whole encryption and tag production (and on the other hand, the whole decryption and tag authentication) happen with a single API call. This type of approach is simpler but at the same time less flexible than the multipart approach where an operation context must be allocated by the application and the encryption/decryption processes require the call to different APIs to setup the operation context, provide inputs to the process and calculate the output. For example, the integrated APIs defined by the PSA Crypto API spec are called `psa_aead_encrypt()` and `psa_aead_decrypt()`.

It is possible, to reduce the code size of an implementation, to implement the single part APIs by calling the underlying multipart functions, effectively encapsulating the multipart flow in the single part APIs.

PSA Unified Driver API for Cryptoprocessors

The PSA Unified Driver API for Cryptoprocessors spec³ allows a PSA compliant implementation to redirect some of the operations to the underlying hardware accelerated platform. For example, CryptoCell is a cryptographic accelerator available in some of the TF-M supported platforms that can accelerate crypto operations, such as Chacha20-Poly1305. The driver code associated to it has the responsibility of driving the hardware resources with inputs and collect the outputs. When decrypting, the driver must compare the reconstructed authentication tag with the expected value, and return failure in case of mismatch during verification. For Chacha20-Poly1305, the tag size is 16 bytes.

For CryptoCell enabled platforms, the software component implementing the PSA Unified Driver interface is located into the `psa_driver_api` directory in `lib/ext/cryptoCELL-312-runtime/codesafe/src/` in TF-M's codebase. This components can be configured to have single part APIs implemented though the corresponding multipart functions by setting the following define at build time: `CC3XX_CONFIG_ENABLE_AEAD_ONE_SHOT_USE_MULTIPART`.

By default, CryptoCell enabled platforms don't build the PSA Unified Driver API interface layer but rely on the legacy interface. To enable the PSA Driver interface the following TF-M build option during CMake config must be set to OFF: `CC312_LEGACY_DRIVER_API_ENABLED=OFF`.

Impact

When the PSA Driver API interface is used and is not configured to rely on the corresponding multipart functions, when performing the verification of the authentication tag at the end of the authenticated decryption process, the buffer containing the tag is only partially verified (the first 4 bytes only instead of the full 16 bytes). This allows for the possibility of unauthenticated data to be recognized as authentic. An attacker could theoretically construct a malicious payload to actively exploit such partial verification.

¹ Chacha20 and Poly1305 for IETF Protocols: <https://datatracker.ietf.org/doc/html/rfc7539>

² PSA Cryptographic API v1.1: <https://arm-software.github.io/psa-api/crypto/1.1/index.html>

³ PSA Unified Driver interface: <https://github.com/Mbed-TLS/TF-PSA-Crypto/blob/development/docs/proposed/psa-driver-interface.md>

Impacted PSA Crypto API functions

The following PSA single part crypto operation function is impacted:

- `psa_aead_decrypt`

Mitigation

The verification of the authentication tag must happen on the full 16 bytes of instead of just the first 4 bytes. This means that loop that currently performs such verification in the `cc3xx_decrypt_chacha20_poly1305()` function must be changed from this:

```
/* Check tag in "constant-time" */
for (diff = 0, i = 0; i < sizeof(tag_length); i++)
    diff |= tag[i] ^ local_tag_buffer[i];
```

to this:

```
/* Check tag in "constant-time" */
for (diff = 0, i = 0; i < tag_length; i++)
    diff |= tag[i] ^ local_tag_buffer[i];
```

References

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3.2.7 Advisory TFMV-7

Title	ARoT can access PRoT data via debug logging functionality
CVE ID	CVE-2023-51712
Public Disclosure Date	The issue was publicly reported on 2023.12.04
Versions Affected	All version up to TF-M v2.0.0 inclusive
Configurations	IPC mode with <code>TFM_SP_LOG_RAW_ENABLED=1</code>
Impact	A malicious ARoT partition can expose any part of memory via stdio interface if <code>TFM_SP_LOG_RAW_ENABLED</code> is set
Fix Version	TBD
Credit	Roman Mazurak, Infineon

Background

TF-M log subsystem if enabled by `TFM_SP_LOG_RAW_ENABLED` config option, uses a SVC call to print logging messages on the stdio output interface. Since the SVC handler has the highest privilege level and full memory access, this communication channel can be exploited to expose any memory content to stdout device, usually UART. The logging subsystem is available to the secure side only but in isolation level 2 and higher PSA Root of Trust partitions (PRoT) shall be protected from an access from Application Root of Trust (ARoT) partitions. Although a direct call of `tfm_hal_output_sp_log()` from ARoT partition will be blocked by MPU raising the `MemoryManagement()` exception, a malicious ARoT partition can create an alternative SVC call to output any memory data like this:

```
static int tfm_output_unpriv_string(const unsigned char *str, size_t len)
{
```

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```
__ASM volatile("SVC %0          \n"
               "BX LR          \n"
               : : "I" (2));
}
```

Impact

In IPC mode with PSA isolation level 2 and higher and TFM_SP_LOG_RAW_ENABLED option enabled an ARoT partition can expose to the stdout device any memory data using TF-M logging subsystem via SVC call.

Mitigation

Ensure that data sent for logging belongs to the current partition. For that purpose tfm_hal_memory_check(curr_partition->boundary, data, size, TFM_HAL_ACCESS_READABLE) is added to the logging function of the SVC handler. If the check fails then tfm_core_panic() is invoked and system halts.

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3.2.8 Advisory TFMV-8

Title	Unchecked user-supplied pointer via mailbox messages may cause write of arbitrary address.
CVE ID	CVE-2024-45746
Public Disclosure Date	October 02, 2024
Versions Affected	All version from TF-M v1.6.0 up to TF-M v2.1.0 inclusive
Configurations	Platforms with standard mailbox dispatcher tfm_spe_mailbox.
Impact	The mailbox message could contain arbitrary pointers which, in case of psa_call failure, would lead to write to a user-specified address in memory.
Fix Version	5ae0a02e847335f4e35ae6aa0b68b80280794776, TF-M v2.1.1
Credit	Infineon Technologies AG, in collaboration with: Tobias Scharnowski, Simon Wörner and Johannes Willbold from fuzzware.io.

Background

The psa_call message through the mailbox contains input/output vectors along with their respective lengths. This message is provided by a NSPE client. SPE takes the message and pass it to the mailbox dispatcher (tfm_spe_mailbox), which handles the message by performing a copy of the i/o vectors into local arrays. When either the client_id translation or the psa_call fails, the dispatcher replies immediately to the client. At that moment, the outvec is written back for its given length, which may not have been sanitized beforehand, resulting in arbitrary access of memory if the provided length goes beyond the legit vector size.

Impact

When the dispatcher in tfm_spe_mailbox is used, a user through mailbox could write into arbitrary address by first placing the malicious data into the local vectors with a bad message, then subsequently sending a psa_call with an invalid vector length. If both calls fail, the reply routine in tfm_spe_mailbox could take the injected data and write it into a desired location specified by the invalid length. Note that the above sequence would require sending the two messages through two different mailbox slots.

Mitigation

Ensure that the outvec is written back only when the psa operation is successful. Any errors ahead of replying must be taken as a hint to avoid such write-back since they may be due to wrong supplied user-data in the vectors (pointers, length etc). To achieve the above, proper sanitization of input data must also be performed and related errors propagated to the reply subroutine.

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3.2.9 Advisory TFMV-9

Title	FWU does not check the length of the TLV's payload
CVE ID	CVE-2025-53022
Public Disclosure Date	Jul 21, 2025
Versions Affected	All version from TF-M v1.3.0 up to TF-M v2.2.0 inclusive
Configurations	Any with Firmware Update (FWU) partition enabled
Impact	A Type-Length-Value (TLV) payload can be larger than the image dependency resulting in out-of-bounds write
Fix Version	6d074b68d3eb3e34038c1b074a68d92a8fea3f82
Credits	Bartek Piekarski Distinguished Engineer, Product Security, Arm Ltd.

Background

In Firmware Update partition, the `fwu_bootloader_install_image()` function checks the authenticity and integrity of the image candidates. For each image candidate, its dependencies are loaded and when criteria are met, the candidate image is staged. During the dependencies check, the length of the image dependency fetched from the TLV is not verified, and this could lead to overflow in read and write from memory in subsequent steps.

Impact

An incorrect and unchecked TLV data can result in accessing the reading and writing memory for the candidate images that goes beyond the bounds, possibly causing in a redirection of program execution.

Mitigation

Add a length check in the `fwu_bootloader_install_image()` immediately after the next TLV is found by `bootutil_tlv_iter_next()`, during the dependencies fetch. See commit [6d074b68d3eb3e34038c1b074a68d92a8fea3f82](#).

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ID	Title
TFMV-1	NS world may cause the CPU to perform an unexpected return operation due to unsealed stacks.
TFMV-2	Invoking Secure functions from handler mode may cause TF-M IPC model to behave unexpectedly.
TFMV-3	<code>abort()</code> function may not take effect in TF-M Crypto multi-part MAC/ hashing/ cipher operations.
TFMV-4	NSPE may access secure keys stored in TF-M Crypto service in Profile Small with Crypto key ID encoding disabled.
TFMV-5	<code>psa_fwu_write()</code> may cause buffer overflow in SPE.
TFMV-6	Partial tag comparison when using Chacha20-Poly1305 on the PSA driver API interface in CryptoCell enabled platforms
TFMV-7	ARoT can access PRoT data via debug logging functionality
TFMV-8	Unchecked user-supplied pointer via mailbox messages may cause write of arbitrary address
TFMV-9	FWU does not check the length of the TLV's payload

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3.3 Security Recommendations

Organization

Arm Limited

Contact

tf-m@lists.trustedfirmware.org

Security recommendations are listed here.

Advice and guidance provided in this document is intended to improve security overall and prevent vulnerabilities.

3.3.1 Security Recommendation no.1

Title	Compiler-induced constant-time violations
Date	April, 2026
Versions Affected	None. General guidance.
Configurations	General guidance for Clang 18 LLVM compiler.
Impact	Some optimizations for modern versions of Clang may defeat the constant-time behaviour, causing a possible timing side-channel attack.

Background

Modern versions of Clang introduce timing side-channels into padding validation code on some platforms when LLVM's select-optimize feature is enabled.

Some compiler optimizations can defeat the constant-time protection, if the compiler recognizes the functional behavior of the code and emits binary code that is functionally equivalent, but not constant-time.

For detailed explanation see: <https://www.cve.org/CVERecord?id=CVE-2025-66442>

Impact

If TF-PSA-Crypto or Mbed TLS is built with Clang 18 with the LLVM select-optimize feature enabled, compiling for 64-bit RISC-V, some features may be vulnerable to a timing oracle attack.

Mitigation

Force the compiler options with `--disable-select-optimize=true`.

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3.4 Security Disclosures

Trusted Firmware-M (TF-M) disclose all security vulnerabilities, or are advised about, that are relevant to TF-M. TF-M encourage responsible disclosure of vulnerabilities and try the best to inform users about all possible issues.

The TF-M vulnerabilities are disclosed as Security Advisories, all of which are listed in the [Security Advisories](#) section.

3.5 Found a Security Issue?

Although TF-M try to keep secure, it can only do so with the help of the community of developers and security researchers.

Warning

If any security vulnerability was found, please **do not** report it in the [issue tracker](#) or on the [mailing list](#). Instead, please follow the [Security incident process](#).

One of the goals of this process is to ensure providers of products that use TF-M have a chance to consider the implications of the vulnerability and its remedy before it is made public. As such, please follow the disclosure plan outlined in the [Security Incident Process](#). TF-M do the best to respond and fix any issues quickly.

Afterwards, write-up all the findings about the TF-M source code is highly encouraged.

3.6 Attribution

TF-M values researchers and community members who report vulnerabilities and TF-M policy is to credit the contributor's name in the published security advisory.

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ROADMAP

TF-M has been under active development since it was launched in Q1'18. It is being designed to include

1. Secure boot ensuring integrity of runtime images and responsible for firmware upgrade.
2. Runtime firmware consisting of TF-M Core is responsible for secure isolation, execution and communication aspects. and a set of Secure Services providing services to the Non-Secure and Secure Applications. The secures services currently supported are Secure Storage, Cryptography, Firmware Update, Attestation and Platform Services

If you are interested in collaborating on any of the roadmap features or other features, please mail TF-M mailing list

4.1 Supported Features

- PSA Firmware Framework v1.0, 1.1 Extension including IPC and SFN modes.
- PSA Level1, 2 and 3 Isolation.
- Secure Boot (mcuboot upstream) including generic fault injection mitigations
- PSA Protected Storage, Internal Trusted Storage v1.0 and Encrypted ITS
- PSA Cryptov1.0 (uses Mbed TLS v3.4.0)
- PSA Initial Attestation Service v1.0
- PSA Firmware Update v1.0
- PSA ADAC Specification Implementation
- Base Config, kconfig based configuration
- Profile Small, Medium, ARoT-less Medium, Large
- Secure Partition Interrupt Handling, Pre-emption of SPE execution
- Dual CPU
- Open Continuous Integration (CI) System
- Boot and Runtime Crypto Hardware Integration
- Fault Injection Handling library to mitigate against physical attacks
- Threat Model
- Arm v8.1-M Privileged Execute Never (PXN) attribute and Thread reentrancy disabled (TRD)
- FPU, MVE Support
- CC-312 PSA Cryptoprocessor Driver Interface

- Secure Storage - Key Diversification Enhancements
- Build System - Separate Secure and Non-Secure builds
- PSA Crypto layer for mcuboot/BL2
- Support LLVM Embedded Toolchain for Arm
- MISRA testing/documentation
- Switch to using upstream t_cose
- Remote Test Infrastructure

4.2 CQ4'25

- Supporting multiple clients (Hybrid Platforms) i.e. TF-M supporting multiple on core and off core clients on heterogeneous (e.g. Cortex-A + Cortex-M platforms)
- TF-M v2.2.2 and v2.1.4 releases
- Integrate TF-PSACrypto1.0

4.3 Future

- Integrate TF-PSACrypto1.x-LTS
- TF-M v2.3.0 (LTS)
- Fuzz Testing
- Implement support for multiple clients (Hybrid Platforms) contd.
- Build System Enhancements - Simplify build scripts
- TF-M Performance - Further Benchmarking and Optimization
- Scheduler - Multiple Secure Context Implementation
- PSA FWU Service Enhancements
- PSA ADAC Spec - Enhancements and Testing
- Arm v8.1-M Unprivileged Debug
- [Secure Storage] Extended PSA APIs
- [Audit Logs] Secure Storage, Policy Manager
- PSA FF Lifecycle API

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GLOSSARY OF TERMS AND ABBREVIATIONS

AAPCS

ARM Architecture Procedure Call Standard: The AAPCS defines how subroutines can be separately written, separately compiled, and separately assembled to work together. It describes a contract between a calling routine and a called routine

Application RoT

[PSA term](#). The security domain in which additional security services are implemented. Also referred as ARoT.

HAL

Hardware Abstraction Layer: Interface to abstract hardware-oriented operations and provides a set of APIs to the upper layers.

ITS

Internal Trusted Storage

One of PSA services provided by TF-M.

MPC

Memory Protection Controller: Bus slave-side security controller for memory regions.

MPU

Memory Protection Unit: Hardware component providing privilege control.

NSPE

Non Secure Processing Environment: [PSA term](#). In TF-M this means non secure domain typically running an OS using services provided by TF-M.

PPC

Peripheral Protection Controller: Bus slave-side security controller for peripheral access.

PS

Protected Storage

One of PSA services provided by TF-M.

PSA

[PSA term](#). Platform Security Architecture.

PSA RoT

[PSA term](#). This defines the most trusted security domain within a PSA system. Also referred as PRoT.

PSA-FF

[PSA term](#). Platform Security Architecture Firmware Framework.

PSA-FF-M

[PSA term](#). Platform Security Architecture Firmware Framework for M.

RoT

Root of Trust: [PSA term](#). This is the minimal set of software, hardware and data that is implicitly trusted in the

platform — there is no software or hardware at a deeper level that can verify that the Root of Trust is authentic and unmodified.

RoT Service

[PSA term](#). A set of related security operations that are implemented in a Secure Partition.

S/NS

Secure/Non-secure: The separation provided by TrustZone hardware components in the system.

SAU

Secure Attribution Unit: Hardware component providing isolation between Secure, Non-secure Callable and Non-secure addresses.

SFN

Secure Function: The function entry to a secure service. Multiple SFN per SS are permitted.

SP

Secure Partition

A logical container for secure services.

SPE

Secure Processing Environment: [PSA term](#). In TF-M this means the secure domain protected by TF-M.

SPM

Secure Partition Manager

The TF-M component responsible for enumeration, management and isolation of multiple Secure Partitions within the TEE.

SPRT

Secure Partition Runtime: The TF-M component responsible for Secure Partition runtime functionalities.

SPRTL

Secure Partition Runtime Library: A library contains the SPRT code and data.

SS

Secure Service: A component within the TEE that is atomic from a security/trust point of view, i.e. which is viewed as a single entity from a TF-M point of view.

SVC

SuperVisor Call: ARMv7M assembly instruction to call a privileged handler function

TBSA-M

Trusted Base System Architecture for M. TBSA term. See [Trusted Base System Architecture for M](#)

TFM

TF-M

Trusted Firmware-M or Trusted Firmware for M-class. ARM TF-M provides a reference implementation of secure world software for ARMv8-M.

Reference

[Firmware Framework for M \(FF-M\)](#)

[Trusted Base System Architecture for M](#)

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TF-M PLATFORMS

6.1 Infineon platforms

IMPROVEMENT: Add PSC3M6 docs

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BUILD INSTRUCTIONS

Warning

The build process was changed a lot in Q3 2023 and included into the release v2.0. For building instructions for early versions please refer to the documentation of respective versions.

As you know from the introduction TF-M implements *SPE* with a set of secure services. TF-M application as *NSPE* client uses those services through isolation boundary via *PSA-FF-M* API. Both *SPE* and *NSPE* are separate binaries and built independently. *SPE* and *NSPE* binaries are combined and signed making the final image for downloading onto targets when building *NSPE*.

Note

This document describes the process of building a single *SPE* alone. Refer to Building Tests on how to build TF-M regression tests and PSA Arch tests to verify TF-M.

TF-M uses **CMake v3.21** or higher. Before starting please make sure you have all required software installed and configured as explained in the TF-M getting started.

Contents

- *Building TF-M (SPE)*
 - *Getting the source code*
 - *Configuring*
 - *Building binaries*
 - *Dependency management*
- *Building Application (NSPE)*
 - *SPE artifacts structure*
 - *NSPE toolchains*
 - *Basic SPE integration*

The additional building materials you can find in the following links:TF-M source folder

7.1 Documentation generation

Two documents are available for generation:

- Reference Manual (HTML, PDF)
- User Guide (HTML, PDF)

The documentation build is independent from building the binary artifacts.

7.1.1 Tools and building environment

These tools are used to generate TF-M documentation:

- Doxygen v1.8.0 or later
- Graphviz dot v2.38.0 or later
- PlantUML v1.2018.11 or later
- Java runtime environment v1.8 or later (for running PlantUML)
- (Optional) `uv <https://docs.astral.sh/uv/getting-started/installation/#standalone-installer>`

Additionally, for PDFs format:

- LaTeX
- PdfLaTeX

There are two ways of building TF-M reference manual:

1. As a custom target of TF-M CMake build system
2. Directly, using the command line tools

To prepare your building environment execute the following steps:

Linux

Install the required tools:

```
sudo apt-get install -y doxygen graphviz default-jre
mkdir ~/plantuml
curl -L http://sourceforge.net/projects/plantuml/files/plantuml.jar/download --output ~/
↳plantuml/plantuml.jar

# For PDF generation
sudo apt-get install -y doxygen-latex librsvg2-bin

# Setup python venv for the project
python3 -m venv .venv

# NOTE: If your system python install version is <3.10 you can use `uv <https://docs.astral.sh/uv/getting-started/installation/#standalone-installer>` to setup your .venv
uv venv --python 3.12

source .venv/bin/activate
pip install ".[docs]"
```

(continues on next page)

(continued from previous page)

```
# NOTE: If you've used `uv` to setup your `.venv`, prepend the `pip` commands with `uv`
uv pip install ".[docs]"
```

Set the environment variables:

```
export PLANTUML_JAR_PATH=~/.plantuml/plantuml.jar
```

Windows

Download and install the following tools:

- Doxygen 1.8.8
- Graphviz 2.38
- **The Java runtime is part of the Arm DS installation or can be downloaded from [here](#)**
- PlantUML
- MikTeX - for PDF generation only

Set the environment variables, assuming that:

- doxygen, dot, and MikTeX binaries are available on the PATH.
- Java JVM is used from Arm DS installation.

```
set PLANTUML_JAR_PATH=<plantuml_Path>\plantuml.jar
set PATH=$PATH;<ARM_DS_PATH>\sw\java\bin

# Install the required Python modules

# Setup python venv for the project
python3 -m venv .venv

# NOTE: If your system python install version is <3.10 you can use `uv <https://docs.
↪astral.sh/uv/getting-started/installation/#standalone-installer>` to setup your .venv
uv venv --python 3.12

source .venv/bin/activate
pip install ".[docs]"
# If uv is installed
uv pip install ".[docs]"

# NOTE: If you've used `uv` to setup your `.venv`, prepend the `pip` commands with `uv`
uv pip sync pylock.toml
```

7.1.2 Build TF-M Reference Manual

The Reference Manual will be generated in the `build_docs/reference_manual`.

Note

Please make sure to follow *Tools and building environment* before proceeding

Linux

```
cd <TF-M base folder>
source .venv/bin/activate
cmake -S docs -B build_docs
cmake --build build_docs -- tfm_docs_refman_html tfm_docs_refman_pdf
```

Windows

```
cd <TF-M base folder>
source .venv/bin/activate
pip install ".[docs]"
# If uv is installed
uv pip install ".[docs]"
cmake -S docs -B build_docs -G"Unix Makefiles"
cmake --build build_docs -- tfm_docs_refman_html tfm_docs_refman_pdf
```

7.1.3 Build TF-M User Guide

The User Manual will be available under the directory build_docs/user_guide.

Linux

```
cd <TF-M base folder>
source .venv/bin/activate
cmake -S docs -B build_docs
cmake --build build_docs -- tfm_docs_userguide_html tfm_docs_userguide_pdf
```

Windows

```
cd <TF-M base folder>
source .venv/bin/activate
cmake -S docs -B build_docs -G"Unix Makefiles"
cmake --build build_docs -- tfm_docs_userguide_html tfm_docs_userguide_pdf
```

7.1.4 Direct build using a command line tools

The direct build will build both user_guide and reference_manual.

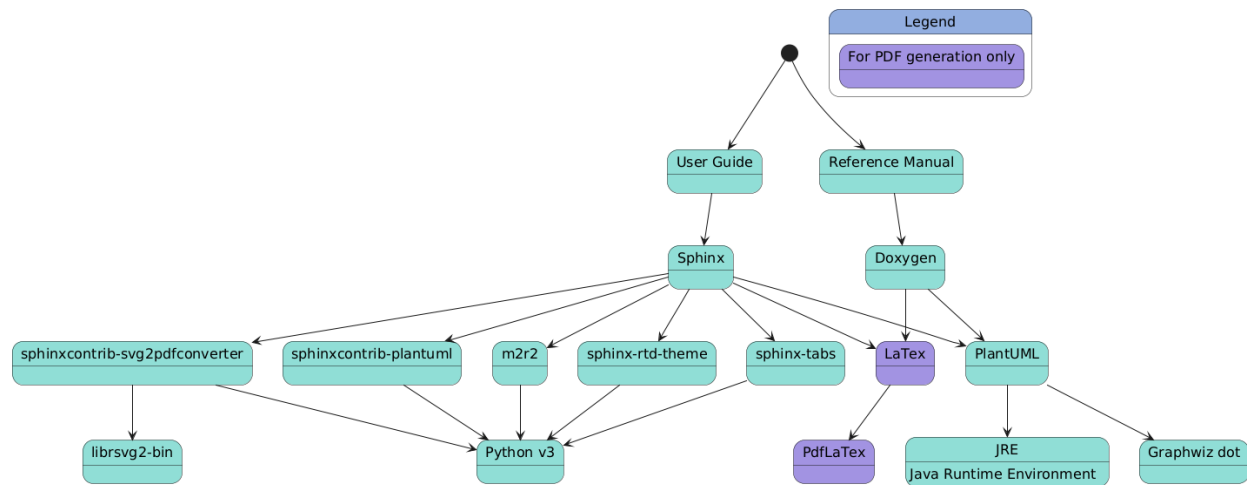
Linux

```
# Build the documentation from build_docs directory
cd <TF-M base folder>
source .venv/bin/activate
mkdir build_docs
cp docs/conf.py build_docs/conf.py
cd build_docs
sphinx-build ./ user_guide
```

Windows

```
# Command line tools is yet not available for Windows
```

7.1.5 Dependencies



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7.2 Additional build instructions for the IAR toolchain

Follow the instructions in *software requirements*, but replace the `-DTFM_TOOLCHAIN_FILE` setting with `toolchain_IARARM.cmake`.

7.2.1 Notes for building with IARARM

IAR Embedded Workbench for ARM (EWARM) versions 9.50.2 or later are required.

Currently the MUSCA_B1 target is not supported with IARARM, due to lack of testing.

CMake needs to be version 3.27.7 or newer.

For all configurations and build options some of the QCBOR tests fail due to the tests not handling double float NaN:s according to the Arm Runtime ABI. This should be sorted out in the future.

Example: building TF-M for AN521 platform using IAR:

```
cd <TF-M base folder>
cmake -S . -B cmake_build -DTFM_PLATFORM=arm/mps2/an521 -DTFM_TOOLCHAIN_FILE=toolchain_
↪ IARARM.cmake
cmake --build cmake_build -- install
```

Alternately using traditional cmake syntax

```
cd <TF-M base folder>
mkdir cmake_build
cd cmake_build
cmake .. -DTFM_PLATFORM=arm/mps2/an521 -DTFM_TOOLCHAIN_FILE=../toolchain_IARARM.cmake
make install
```

Regression Tests for the AN521 target platform

```
cd <TF-M base folder>
cmake -S . -B cmake_build -DTFM_PLATFORM=arm/mps2/an521 -DTFM_TOOLCHAIN_FILE=toolchain_
↳ IARARM.cmake -DTEST_S=ON -DTEST_NS=ON
cmake --build cmake_build -- install
```

Alternately using traditional cmake syntax

```
cd <TF-M base folder>
mkdir cmake_build
cd cmake_build
cmake .. -DTFM_PLATFORM=arm/mps2/an521 -DTFM_TOOLCHAIN_FILE=../toolchain_IARARM.cmake -
↳ DTEST_S=ON -DTEST_NS=ON
make install
```

Note

For PSA API tests, you may need to append `-DTOOLCHAIN=INHERIT` to the build command.

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7.3 Building TF-M (SPE)

This build generates the SPE binary and artifacts, necessary for *Building Application (NSPE)*.

7.3.1 Getting the source code

```
cd <base folder>
git clone https://git.trustedfirmware.org/TF-M/trusted-firmware-m.git
```

In this documentation, the cloned `trusted-firmware-m` repository will be referenced as `<TF-M source dir>`. Additionally, TF-M depends on several external projects, handled by CMake automatically but you can alter that behaviour using *Dependency management*.

Note

- For building with Armclang compiler, please follow the note in TF-M getting started.
- For building with the IAR toolchain, please see the notes in *IAR software requirements*
- Please use “/” instead of “\” for paths when running CMAKE commands under Windows Command Prompt.

7.3.2 Configuring

TF-M has many config options for configuring and fine-tuning. Please check the *Configuration* section for the details. The **base** (default) configuration contains only essential components such as SPM and HW platform support hence the only mandatory argument to TF-M build is a platform name, provided via CMake command-line option `-DTFM_PLATFORM=<platform name>`, it can be:

- A relative path under `<TF-M source dir>/platform/ext/target`, for example `arm/mps2/an521`.
- An absolute path of target platform, mainly used for out-of-tree platform build.
- A basename of the target platform folder, for example `an521`.

Essential Directories

There are 3 essential directories used by CMake for building TF-M:

- Source code directory `<TF-M source dir>`
- Build directory `<Build Dir>` - the location of all intermediate files required to produce a build target.
- Install directory `<Artifact Dir>` - the location of the build output files.

Note::

It's recommended to use absolute paths for all directories. Relative paths may not fully work.

Toolchains

TF-M supports 4 toolchains for cross-compiling and building the project binaries:

- GNU - **default**
- ArmClang
- IAR
- Arm Toolchain for Embedded (ATfE)

Each toolchain has a configuration file for the compiler and linker. They are located at the root directory of TF-M. Use `TFM_TOOLCHAIN_FILE` option to provide the absolute path to the preferred toolchain file, or relative path to working directory. The default **toolchain_GNUARM.cmake** is selected by `config_base.cmake` file if the option is omitted.

Build type

By default, a `MinSizeRel` configuration is built. Alternate build types can be specified with the `CMAKE_BUILD_TYPE` variable. The possible types are:

- Debug
- RelWithDebInfo
- Release
- MinSizeRel - **default**

Debug symbols are added by default to all builds, but can be removed from *Release* and *MinSizeRel* builds by setting `TFM_DEBUG_SYMBOLS` to `OFF`.

RelWithDebInfo, *Release* and *MinSizeRel* all have different optimizations turned on and hence will produce smaller, faster code than *Debug*. *MinSizeRel* will produce the smallest code and hence is often a good idea on RAM or flash-constrained systems.

Output files

In a successful build, a set of files will be created in the `<Artifact Dir>`. By default, it is `<Build Dir>\api_ns` subfolder but you can redirect the output to any location using `CMAKE_INSTALL_PREFIX` option. It can be an absolute path or relative to your current directory. For the contents of the artifact directory please refer to *SPE artifacts structure*.

Other build parameters

The full list of default options is in `config/config_base.cmake` and explained in *Build configuration*. Several important options are listed below.

Parameter	Description	Default value
BL2	Build level 2 secure bootloader.	ON
PROJECT_CONFIG_HEADER_FILE	User defined header file for TF-M config	
TFM_ISOLATION_LEVEL	Set TF-M isolation level.	1
TFM_PROFILE	See <i>TF-M Profiles</i> .	

Project Config Header File

CMake variable `PROJECT_CONFIG_HEADER_FILE` can be set by a user the full path to a configuration header file, which is used to fine-tune component options. The detailed reference for the project config header file is in *The Header File Config System*.

7.3.3 Building binaries

The command below shows a general template for building TF-M as a typical CMake project:

```
cmake -S <TF-M source dir> -B <Build Dir> -DTFM_PLATFORM=<platform>
cmake --build <Build Dir> -- install
```

Note

It is recommended to clean up the build directory before re-build if the config header file is updated. CMake is unable to automatically recognize the dependency when the header file is defined as a macro.

Building default configuration for an521

```
cd <TF-M source dir>
cmake -S . -B build -DTFM_PLATFORM=arm/mps2/an521
cmake --build build -- install
```

The command above is intended to do:

- take TF-M sources in the current `.` folder
- build SPE in the build folder
- for **an521** platform
- using GNU toolchain *by default*. Use `-DTFM_TOOLCHAIN_FILE=<toolchain file>` for alternatives as described in *Toolchains*
- install output files in `build/api_ns` folder *by default*. You can specify a different directory using `-DCMAKE_INSTALL_PREFIX=<Artifact dir>` as described in *Output files*

Note

It is recommended to build each different build configuration in a separate build directory.

CMake can generate code for many native build systems. TF-M is tested with Unix Makefiles (default) and Ninja. The `-G` option can specify alternative generators. For example for building with Ninja in the Debug *Build type* using ArmClang Toolchains you can use the following:

```
cd <TF-M source dir>
cmake -S . -B build -DTFM_PLATFORM=arm/mps2/an521 -GNinja -DTFM_TOOLCHAIN_FILE=toolchain_
  ARMCLANG.cmake -DCMAKE_BUILD_TYPE=Debug
cmake --build build -- install
```

7.3.4 Dependency management

The TF-M build system will fetch all dependencies by default with appropriate versions and store them inside the build tree. In this case, the build tree location is `<build_dir>/lib/ext`.

If you have local copies already and wish to avoid having the libraries downloaded every time the build directory is deleted, then the following variables can be set to the paths to the root directories of the local repos. This will disable the automatic downloading for that dependencies and speed up development iterations or allow usage of a dependency version different from the current one. Additionally, these path variables can be set in `localrepos.cmake` file which will be included in a build if it exists. This file is ignored in TF-M git settings.

The following table lists the commonly used repos. For others, you can refer to `lib/ext`.

Dependency	Cmake variable	Git repo URL
TF-PSA-Crypto	TF_PSA_CRYPTO_PATH	https://github.com/Mbed-TLS/TF-PSA-Crypto
MCUboot	MCUBOOT_PATH	https://github.com/mcu-tools/mcuboot
QCBOR	QCBOR_PATH	https://github.com/laurencelundblade/QCBOR.git

The recommended versions of the dependencies are listed in `config/config_base.cmake`.

Note

- Some repositories might need patches to allow building it as a part of TF-M. While these patches are being upstreamed they are stored in a dependency folder under `lib/ext/`. In order to use local repositories those patches shall be applied to original source. An alternative is to copy out the auto-downloaded repos under the `<build_dir>/lib/ext`. They have been applied with patches and can be used directly.

Example: building TF-M with local TF-PSA-Crypto repo

Preparing a local repository consists of 2 steps: cloning and patching. This is only required to be done once. For dependencies without `.patch` files in their `lib/ext` directory the only required step is cloning the repo and checking out the correct branch.

```
cd <TF-PSA-Crypto base folder>
git clone https://github.com/Mbed-TLS/TF-PSA-Crypto
cd TF-PSA-Crypto
git checkout <TF_PSA_CRYPTO_VERSION> from <TF-M source dir>/config/config_base.cmake>
git apply <TF-M source dir>/lib/ext/tf-psa-crypto/*.patch
```

Note

<TF-PSA-Crypto base folder> does not need to have any fixed position related to the TF-M repo so alternative method to get prepared dependency repos is to let TF-M download it once and then copy them out of the build/lib/ext folder.

Now build TF-M binaries

```
cd <TF-M source dir>
cmake -S . -B build -DTFM_PLATFORM=arm/mps2/an521 -DTF_PSA_CRYPTOPATH=<TF-PSA-Crypto_
base folder>
cmake --build build -- install
```

7.4 Building Application (NSPE)

As a result of *Building TF-M (SPE)* you will get a set of *Output files* in <Artifact Dir> required for building TF-M application. Essentially, SPE exports a binary and a set of C source files for PSA interface and platform. Please note that NSPE and SPE are independent projects and can be built using different toolchains and toolchain options.

7.4.1 SPE artifacts structure

SPE components prepared and installed for NSPE usage in <Artifact Dir> will have the following structure:

```
<Artifact Dir>
├── bin
├── cmake
├── config
├── image_signing
├── interface
├── platform
└── CMakeLists.txt
```

With certain configurations, additional folders may also be installed. These folders have the following content:

- **bin** - binary images of SPE, Bootloader(optional) and combined.
- **cmake** - CMake scripts like SPE configuration and *NSPE toolchains*.
- **config** - Configuration files
- **image_signing** - binary image signing tool and keys.
- **interface** - PSA interface exposed by SPE.
- **platform** - source code for a selected hardware platform.
- **CMakeLists.txt** - CMake script for the artifacts integration in NSPE.

The content of <Artifact Dir> is an exported directory for integration with CMake projects.

Note

Attempting to change any file in <Artifact Dir> may cause incompatibility issues. Instead, please change the corresponding file in the <TF-M source dir>.

7.4.2 NSPE toolchains

SPE prepares and exports CMake toolchain files for building NSPE in all supported *Toolchains* in `<Artifact Dir>/cmake` folder. Toolchain used to build NSPE can be different from what is used to build SPE.

7.4.3 Basic SPE integration

Refer to the [example](#) of TF-M applications in **tf-m-extras** repository.

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CONFIGURATION

8.1 Build configuration

All configuration options are provided by cmake variables, and their default values, with docstrings, can be found in `config/config_base.cmake`.

Configuration is provided in multiple stages. Each stage will not override any config that has already been set at any of the prior stages.

1. Command-line variable settings are applied.
2. If the `TFM_EXTRA_CONFIG_PATH` variable has been set, that file is loaded.
3. If `TEST_PSA_TEST` is set, then PSA API test related config is applied from `config/tests/config_test_psa_api.cmake`.
4. If it exists, `CMAKE_BUILD_TYPE` specific config is applied from `config/build_type/<build_type>.cmake`.
5. Target specific config from `platform/ext/target/<target_platform>/config.cmake` is applied.
6. If `CRYPTO_HW_ACCELERATOR` is set, then a config specific to the accelerator type is applied if it exists.
7. If it exists, TFM Profile specific config is applied from `config/profile/<tfm_profile>.cmake`.
8. `config/config_default.cmake` is loaded.
9. If `TEST_S` or `TEST_NS` or other single test suite config like `TEST_NS_ATTESTATION` (see `test_configuration`) is set, then config from `${TFM_TEST_REPO_PATH}/test/config/set_config.cmake` and `${TFM_TEST_REPO_PATH}/test/config/default_ns_test_config.cmake` or `${TFM_TEST_REPO_PATH}/test/config/default_s_test_config.cmake` or `${TFM_TEST_REPO_PATH}/test/config/default_test_config.cmake` is applied.

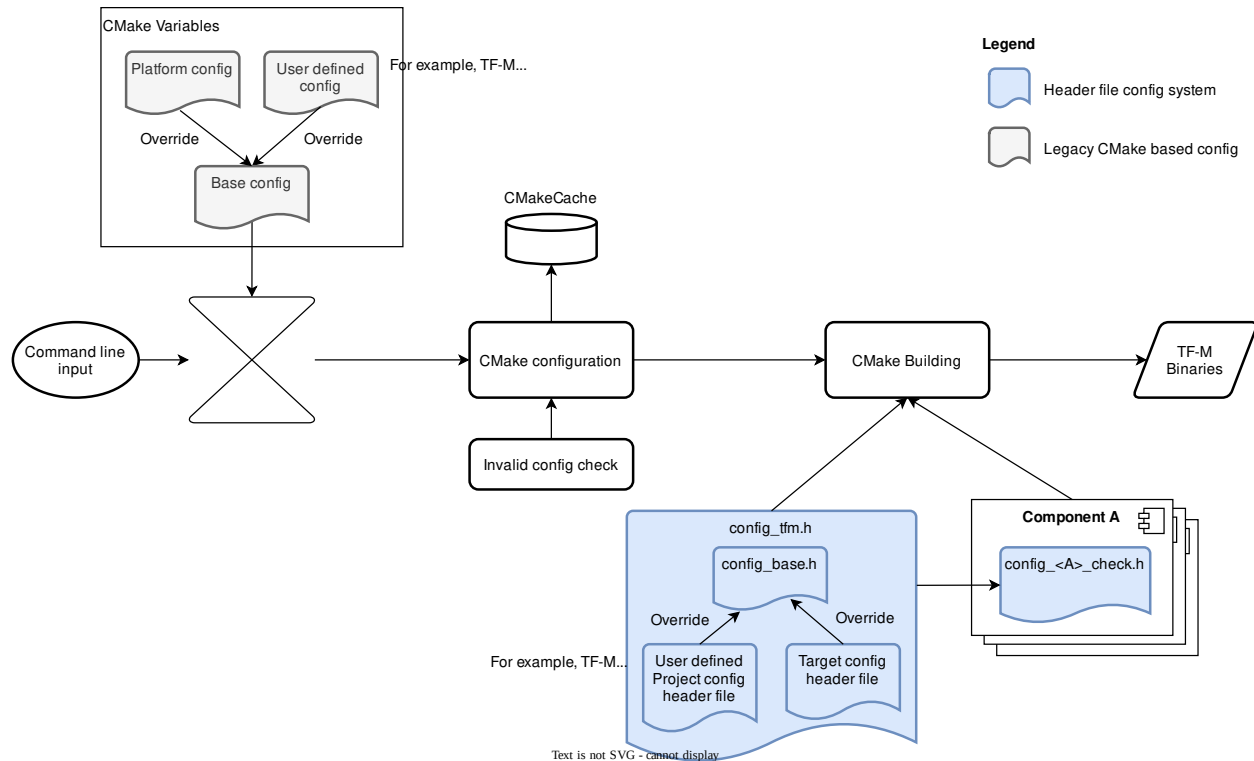
Warning

This means that command-line settings are not applied when they conflict with required platform settings. If it is required to override platform settings (this is not usually a good idea) then `TFM_EXTRA_CONFIG_PATH` should be used.

8.2 The Header File Config System

The header file configurations system is used to fine-tune component options.

The following diagram shows how the system works.



Source files shall include `config_tfm.h` when necessary to fetch Component option settings. It is expected that all Component options are included in `config_tfm.h` to explicitly set values for each option.

The `config_tfm.h` includes base configuration `config_base.h`. Refer to *Base Configuration* for details of the base configurations.

The `config_tfm.h` includes a customized project config file provided via compile definition `PROJECT_CONFIG_HEADER_FILE`. Customized Component options in the project config file overrides those configured in `config_base.h`. The project config header file can be

- Generated by the TF-M Kconfig system `<kconfig_system>`
- One of the header files of Profiles `<tf-m_profiles>`, set via the `TFM_PROFILE` build option.
- Manually customized profile based on pre-set profiles.

Users set CMake variable `PROJECT_CONFIG_HEADER_FILE` with the full path of the configuration header file.

A platform can adjust or place restriction on config options by providing a `config_tfm_target.h` under the root folder of their platforms. If the build system finds the file, it sets the `TARGET_CONFIG_HEADER_FILE` compile definition. Platform specific option settings in `TARGET_CONFIG_HEADER_FILE` overrides those configured in `config_base.h`.

```
#ifdef TARGET_CONFIG_HEADER_FILE
#include TARGET_CONFIG_HEADER_FILE
#endif

#ifdef PROJECT_CONFIG_HEADER_FILE
#include PROJECT_CONFIG_HEADER_FILE
#endif

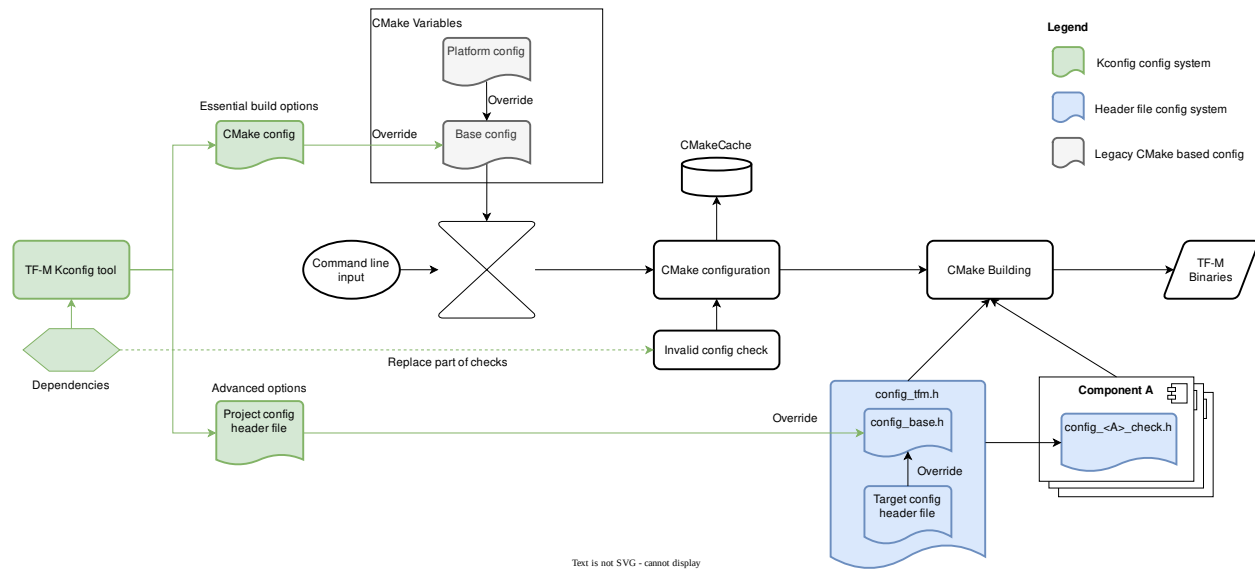
#include "config_base.h"
```

Each component can implement a `config_<comp_name>_check.h` to validate component specific config dependencies. `config_<comp_name>_check.h` shall include `config_tfm.h`

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8.3 The Kconfig System

The Kconfig system is an alternative tool to the CMake config system for users to change config options of TF-M.



It handles dependencies and validations automatically when you change configurations so that the generated configuration options are always valid.

To use the Kconfig system, enable `USE_KCONFIG_TOOL` in command line. And enable `MENUCONFIG` to launch configuration GUI.

The Kconfig system consists of *The Kconfig tool* and the *The Kconfig files*.

8.3.1 The Kconfig tool

The Kconfig tool is a python script based on **Kconfiglib** to generate the following config files:

- **CMake config file**
Contains CMake cache variables of building options.
- **Header file**
Contains component options in the header file system. Component options are gathered together in a separate menu `TF-M component configs` in *The Kconfig files*.
- **The .config and .config.old files**
The `.config` file which contains all the above configurations in the Kconfig format. It will be created after the first execution of the script. It is only used to allow users to make adjustments basing on the previous settings. The Kconfig tool will load it if it exists and `.config.old` will be created to save the previous configurations.

The tool supports loading multiple pre-set configuration files merging into a single one. The first loaded options are overridden by later ones if the config files contain duplicated options. And dependencies between config options are

taken care of. It then launches a configuration GUI for users to change any config options if the `MENUCONFIG` is enabled in build command line.

Integration with TF-M build system

TF-M build system includes `kconfig.cmake` to integrate this tool. It prepares the parameters for the script and invokes it to load multiple configuration files basing on your build setup, including but not limited to

- Build type bound configurations, decided by `CMAKE_BUILD_TYPE`
- Profile configurations, decided by `TFM_PROFILE`

8.3.2 Customizing config options

By default, the Kconfig system only merges configuration files and generated the final config files. To customize the config options, there are several approaches.

Menuconfig

Menuconfig is the recommended approach to adjust the values of the config options because it has a graphic interface for you to easily change the options without worrying about dependencies.

To launch the menuconfig, you need to enable `MENUCONFIG` in addition to enabling `USE_KCONFIG_TOOL`.

```
cmake -S . -B cmake_build -DTFM_PLATFORM=arm/mps2/an521 \  
      -DUSE_KCONFIG_TOOL=ON \  
      -DMENUCONFIG=ON
```

Note

Although the Kconfiglib provides three [menuconfig interfaces](#), only GUI menuconfig can be launched by CMake for the time being.

Command line options

The support of passing configurations via command line is kept for the Kconfig system.

```
cmake -S . -B cmake_build -DTFM_PLATFORM=arm/mps2/an521 \  
      -DUSE_KCONFIG_TOOL=ON \  
      -DTFM_ISOLATION_LEVEL=2
```

Kconfig file

You can also put the frequently used config options into a Kconfig file. When you need to apply the config options in that file, pass it via command line option `-DKCONFIG_CONFIG_FILE`

```
cmake -S . -B cmake_build -DTFM_PLATFORM=arm/mps2/an521 \  
      -DTFM_ISOLATION_LEVEL=2 \  
      -DUSE_KCONFIG_TOOL=ON \  
      -DKCONFIG_CONFIG_FILE=my_config.conf
```

Note

The command line set options override the ones in the config file. And you can always launch menuconfig to do the final adjustments.

8.3.3 The Kconfig files

The Kconfig files are the files written by the [Kconfig language](#) to describe config options. They also uses some Kconfiglib extensions such as optional source `osource` and relative source `rsource` so they can only work with the Kconfiglib.

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8.4 TF-M Profiles

The capabilities and resources may dramatically vary on different IoT devices. Some IoT devices may have very limited memory resource. The program on those devices should keep small memory footprint and basic functionalities. On the other hand, some devices may consist of more memory and extended storage, to support stronger software capabilities.

Diverse IoT use cases also require different levels of security and requirements on device resource. For example, use cases require different cipher capabilities. Selecting cipher suites can be sensitive to memory footprint on devices with constrained resource.

Trusted Firmware-M (TF-M) defines several general profiles, such as Profile Small, Profile Medium, Profile Medium ARoT-less and Profile Large, to provide different levels of security to fit diverse device capabilities and use cases applied on the top of the base configuration.

Each profile specifies a predefined list of features, targeting typical use cases with specific hardware constraints. Profiles can serve as reference designs, based on which developers can continue further development and configurations, according to use case.

TF-M Profiles align with Platform Security Architecture specifications and certification guidelines. It can help vendors to simplify security configuring for PSA certification.

Please check the table below to compare differences while details are discussed in the links below.

8.4.1 Trusted Firmware-M Profile Small Design

Introduction

As one of the TF-M Profiles, TF-M Profile Small (Profile S) consists of lightweight TF-M framework and basic Secure Services to keep smallest memory footprint, supporting fundamental security features on devices with ultra constrained resource.

This profile enables connecting with Edge Gateways and IoT Cloud Services supporting secure connection based solely on symmetric cryptography.

This document summarizes and discusses the features specified in TF-M Profile Small.

Overall design

TF-M Profile Small defines the following features:

- Lightweight framework

- Secure Function (SFN) model²
 - Level 1 isolation
 - Buffer sharing allowed
 - Single secure context
- Crypto
 - Symmetric cipher only
 - Cipher suite for symmetric-key algorithms based protocols, such as cipher suites defined in TLS pre-shared key (TLS-PSK)¹.
 - * Advanced Encryption Standard (AES) as symmetric crypto algorithm
 - * SHA256 as Hash function
 - * HMAC as Message Authentication Code algorithm
 - Only enable multi-part functions in hash, symmetric ciphers, Message Authentication Code (MAC) and Authenticated Encryption with Associated Data (AEAD) operations.
- Internal Trusted Storage (ITS)
 - No encryption
 - No rollback protection
 - Decrease internal transient buffer size
- Initial Attestation
 - Based on symmetric key algorithms
- Lightweight boot
 - Single image boot
 - Anti-rollback protection is enabled

Protected Storage, firmware update and other Secure Services provided by TF-M are disabled by default.

Design details

More details of TF-M Profile Small design are discussed in following sections.

Lightweight framework

TF-M framework model

SFN model is selected by default in Profile Small implementation. SFN model is defined in FF-M 1.1 extensions². It is a more simple implementation of TF-M framework and may also reduce memory footprint, compared with Inter-Process Communication (IPC) model³.

Level 1 isolation

PSA Security Model⁴ defines 3 levels of isolation.

² Arm Firmware Framework for M 1.1 Extensions

¹ Pre-Shared Key Ciphersuites for Transport Layer Security (TLS)

³ Arm Platform Security Architecture Firmware Framework 1.0

⁴ Platform Security Model 1.1

- Level 1 isolation isolates Secure Processing Environment (SPE) from Non-secure Processing Environment (NSPE).
- PSA Root of Trust (PSA RoT) and Application Root of Trust (ARoT) are isolated from each other in level 2 isolation.
- Individual secure partitions are isolated from each other even within a particular security domain (PSA RoT, ARoT), in level 3 isolation.

Profile Small dedicated use cases with simple service model may not require level 2 or level 3 isolation. Devices which Profile Small aims at may be unable to implement stricter isolation, limited by hardware capabilities.

Level 1 isolation reduces requirements enforced by hardware isolation and cost of software for management.

Note

Security note

If a device or a use case enforces level 2 or level 3 isolation, it is suggested to apply other configurations, other than TF-M Profile Small.

Crypto service

TF-M Profile Small only requires symmetric crypto since symmetric algorithms require shorter keys and less computational burden, compared with asymmetric crypto.

By default, TF-M Profile Small requires the same capabilities as defined in TLS-PSK, to support symmetric key algorithms based protocols.

Note

Implementation note

Please note that TF-M Profile Small doesn't require that TLS-PSK is mandatory in applications. Instead, Profile Small only requires the same capabilities as defined in TLS-PSK, such as one symmetric cipher algorithm and one hash function.

TF-M Profile Small selects TLS-PSK cipher suite `TLS_PSK_WITH_AES_128_CCM`⁵ as reference, which requires:

- AES-128-CCM (AES CCM mode with 128-bit key) as symmetric crypto algorithm
- SHA256 as Hash function
- HMAC as Message Authentication Code algorithm

`TLS_PSK_WITH_AES_128_CCM` is selected since it requires small key length and less hardware capabilities, while keeping enough level of security.

Note

Implementation note

Developers can replace default algorithms with others or implement more algorithms.

Proper symmetric key algorithms and cipher suites should be selected according to device capabilities, the use case and the requirement of peers in connection.

⁵ AES-CCM Cipher Suites for Transport Layer Security (TLS)

Refer to *Crypto service configuration* for implementation details of configuring algorithms and cipher suites.

Note

Security note

It is recommended not to use MD5 or SHA-1 for message digests as they are subject to collision attacks⁶⁷.

By default, Profile Small only enables multi-part functions defined in PSA Cryptography API¹³ in hash, symmetric ciphers, MAC and AEAD operations. Disabling single-part functions optimizes the code size of TF-M crypto service. Multi-part operations allows the message data to be processed in fragments instead of all at once. In static memory allocation, single-part operation may require to allocate a large memory space to support long message with unknown length. Therefore single-part operations can help users optimize memory footprint, especially while dealing with streaming data on IoT devices.

It may slightly increase the code size in applications to replace single-part implementation with multi-part implementation. Although the code size increment can be qualified, if users are concerned about the code size increment, they can enable single-part operations by toggling Profile Small default configuration.

It may increase latency and overall time cost to implement cryptography functionality with single-part operations, compared to with multi-part ones. Users can enable single-part operations if the usage scenario requires single-part operations to meet its performance metrics.

Secure Storage

TF-M Profile Small assumes that extremely constrained devices only contain basic on-chip storage, without external or removable storage. As a result, TF-M Profile Small includes ITS service and disables Protected Storage service.

Encryption and rollback protection

Neither encryption nor rollback protection is enabled in current ITS implementation.

It is expected that ITS relies solely on the physical inaccessibility property of on-chip storage, together with PSA isolation, without requiring additional cryptographic protection.

Internal transient buffer

ITS implements a internal transient buffer⁸ to hold the data read from/written to storage, especially for flash, to solve the alignment and security issues.

The internal transient buffer is aligned to the flash device's program unit. Copying data to it from the caller can align all write requests to the flash device's program unit. The internal transient buffer can help protect Flash access from some attacks, such as TOCTOU attack.

Although removing this internal buffer can save some memory consumption, typically 512 bytes, it may bring alignment or security issues. Therefore, to achieve a better trade-off between memory footprint and security, TF-M Profile Small optimizes the internal buffer size to 32 bytes by default.

As discussed in *Crypto service*, TF-M Profile Small requires AES-128 and SHA-256, which use 128-bit key and 256-bit key respectively. Besides, either long public/private keys or PKI-based certificates should be very rare as asymmetric

⁶ Updated Security Considerations for the MD5 Message-Digest and the HMAC-MD5 Algorithms

⁷ Transitioning the Use of Cryptographic Algorithms and Key Lengths

¹³ PSA Cryptography API 1.0

⁸ ITS integration guide

crypto is not supported in Profile Small. Therefore, a 32-byte internal buffer should cover the assets in TF-M Profile Small use cases.

The buffer size can be adjusted according to use case and device Flash attributes. Refer to *Internal Trusted Storage configurations* for more details.

Initial Attestation

Profile Small requires an Initial Attestation secure service based on symmetric key algorithms. Refer to PSA Attestation API document⁹ for details of Initial Attestation based on symmetric key algorithms.

It can heavily increase memory footprint to support Initial Attestation based on asymmetric key algorithms, due to asymmetric ciphers and related PKI modules.

Note

Implementation note

As pointed out by PSA Attestation API document⁹, the use cases of Initial Attestation based on symmetric key algorithms can be limited due to the associated infrastructure costs for key management and operational complexities. It may also restrict the ability to interoperate with scenarios that involve third parties.

If asymmetric key algorithms based Initial Attestation is required in use scenarios, it is recommended to select other TF-M Profiles which support asymmetric key algorithms.

Note

Implementation note

It is recommended to utilize the same MAC algorithm supported in Crypto service to complete the signing in COSE_Mac0, to minimize memory footprint.

Lightweight boot

If MCUBoot provided by TF-M is enabled, single image boot¹⁰ is selected by default in Profile Small. In case of single image boot, secure and non-secure images are handled as a single blob and signed together during image generation.

However, secure and non-secure images must be updated together in single image boot. It may decrease the flexibility of image update and cost longer update process. Since the image sizes should usually be small with limited functionalities in Profile Small dedicated use case, the cost may still be reasonable.

BL2 implementation can be device specific. Devices may implement diverse boot processes with different features and configurations. However, anti-rollback protection is required as a mandatory feature of boot loader. Boot loader should be able to prevent unauthorized rollback, to protect devices from being downgraded to earlier versions with known vulnerabilities.

Implementation

Overview

The basic idea is to add dedicated profile CMake configuration files under folder `config/profile` for TF-M Profile Small default configuration.

⁹ PSA Attestation API 1.0 (ARM IHI 0085)

¹⁰ Secure boot

The top-level Profile Small config file collects all the necessary configuration flags and set them to default values, to explicitly enable the features required in Profile Small and disable the unnecessary ones, during TF-M build.

A platform/use case can provide a configuration extension file to overwrite Profile Small default setting and append other configurations. This configuration extension file can be added via parameter `TFM_EXTRA_CONFIG_PATH` in build command line.

The behavior of the Profile Small build flow (particularly the order of configuration loading and overriding) can be found at *Build configuration*

The details of configurations will be covered in each module in *Implementation details*.

Implementation details

This section discusses the details of Profile Small implementation.

Top-level configuration files

The firmware framework configurations in `config/profile/profile_small` are shown below.

Table 28:: TFM options in Profile Small top-level CMake config file

Configs	Default value	Descriptions
<code>TFM_ISOLATION_LEVEL</code>	1	Select level 2 isolation
<code>TFM_PARTITION_INTERNAL</code>	ON	Enable ITS SP
<code>ITS_BUF_SIZE</code>	32	ITS internal transient buffer size
<code>TFM_PARTITION_CRYPTO</code>	ON	Enable Crypto service
<code>TFM_TF_PSA_CRYPTO_CONF</code>	<code>\${CMAKE_SOURCE_DIR}/lib/ext/tf-psa-crypto/tfpsacrypto_config/crypto_config_profile_small.h</code>	TF-PSA-Crypto config file path
<code>CRYPTO_ASYM_SIGN_MODUL</code>	OFF	Enable asymmetric signature
<code>CRYPTO_ASYM_ENCRYPT_MC</code>	OFF	Enable asymmetric encryption
<code>TFM_PARTITION_INITIAL_</code>	ON	Enable Initial Attestation service
<code>SYMMETRIC_INITIAL_ATT</code>	ON	Enable symmetric attestation
<code>TFM_PARTITION_PROTECT</code>	OFF	Enable PS service
<code>TFM_PARTITION_PLATFORM</code>	OFF	Enable TF-M Platform SP

Note

Implementation note

The following sections focus on the feature selection via configuration setting. Dedicated optimization on memory footprint is not covered in this document.

Device configuration extension

To change default configurations and add platform specific configurations, a platform can add a platform configuration file at `platform/ext<TFM_PLATFORM>/config.cmake`

TF-M framework setting

The top-level Profile Small CMake config file selects SFN model and level 1 isolation.

In SFN model, `-DPSA_FRAMEWORK_HAS_MM_IOVEC` is enabled by default. It reduces memory footprint by avoiding the transient copy from input vectors and copy to output vectors.

Crypto service configuration

Crypto Secure Partition

TF-M Profile Small enables Crypto Secure Partition (SP) in its top-level CMake config file. Crypto SP modules not supported in TF-M Profile Small are disabled. The disabled modules/features are shown below.

- Disable asymmetric cipher
- Disable single-part operations in Hash, MAC, AEAD and symmetric ciphers via selecting `CRYPTO_SINGLE_PART_FUNCS_DISABLED`

Other modules and configurations¹¹ are kept as default values.

Additional configuration flags with more fine granularity can be added to control building of specific crypto algorithms and corresponding test cases.

TF-PSA-Crypto configurations

TF-M Profile Small adds a dedicated TF-PSA-Crypto config file `crypto_config_profile_small.h` under `/lib/ext/tf-psa-crypto/tfpsacrypto_config` folder, instead of the common one `crypto_config_default.h`.

Major cryptographic configurations are set as listed below:

- Enable SHA256
- Enable generic message digest wrappers
- Enable AES
- Enable CCM mode for symmetric ciphers
- Disable other modes for symmetric ciphers
- Disable asymmetric ciphers
- Disable HMAC-based key derivation function (HKDF)

Other configurations can be selected to optimize the memory footprint of Crypto module.

A device/use case can append an extra config header to the Profile Small default TF-PSA-Crypto config file. This can be done by setting the `TFM_TF_PSA_CRYPTO_PLATFORM_EXTRA_CONFIG_PATH` cmake variable in the platform config file `platform/ext<TFM_PLATFORM>/config.cmake`. This cmake variable is a wrapper around the `TF_PSA_CRYPTO_USER_CONFIG_FILE` options, but is preferred as it keeps all configuration in cmake.

¹¹ *Crypto design*

Internal Trusted Storage configurations

ITS service is enabled in top-level Profile Small CMake config file.

The internal transient buffer size `ITS_BUF_SIZE`[?] is set to 32 bytes by default. A platform/use case can overwrite the buffer size in its specific configuration extension according to its actual requirement of assets and Flash attributes.

Profile Small CMake config file won't touch the configurations of device specific Flash hardware attributes[?].

Initial Attestation secure service

TF-M Profile Small provides a reference implementation of symmetric key algorithms based Initial Attestation, using HMAC SHA-256 as MAC algorithm in `COSE_Mac0` structure. The implementation follows PSA Attestation API document[?].

Profile Small top-level config file enables Initial Attestation secure service and selects symmetric key algorithms based Initial Attestation by default.

- Set `TFM_PARTITION_INITIAL_ATTESTATION` to ON
- Set `SYMMETRIC_INITIAL_ATTESTATION` to ON

Symmetric and asymmetric key algorithms based Initial Attestation can share the same generations of token claims, except Instance ID claim.

Profile Small may implement the procedure or rely on a 3rd-party tool to construct and sign `COSE_Mac0` structure.

Details of symmetric key algorithms based Initial Attestation design will be covered in a dedicated document.

Disabled secure services

Protected Storage and Platform Service are disabled by default in Profile Small top-level CMake config file.

Test configuration

Some cryptography tests are disabled due to the reduced cryptographic configuration profile. Some of them are shown in the table below.

Table 29:: TFM options in Profile Small top-level CMake config file

Configs	Default value	Descriptions
<code>TFM_CRYPT0_TEST_ALG_CBC</code>	OFF	Test CBC cryptography mode
<code>TFM_CRYPT0_TEST_ALG_CCM</code>	ON	Test CCM cryptography mode
<code>TFM_CRYPT0_TEST_ALG_CFB</code>	OFF	Test CFB cryptography mode
<code>TFM_CRYPT0_TEST_ALG_ECB</code>	OFF	Test ECB cryptography mode
<code>TFM_CRYPT0_TEST_ALG_CTR</code>	OFF	Test CTR cryptography mode
<code>TFM_CRYPT0_TEST_ALG_OFB</code>	OFF	Test OFB cryptography mode
<code>TFM_CRYPT0_TEST_ALG_GCM</code>	OFF	Test GCM cryptography mode
<code>TFM_CRYPT0_TEST_ALG_SHA_384</code>	OFF	Test SHA-384 cryptography algorithm
<code>TFM_CRYPT0_TEST_ALG_SHA_512</code>	OFF	Test SHA-512 cryptography algorithm
<code>TFM_CRYPT0_TEST_HKDF</code>	OFF	Test HKDF key derivation algorithm
<code>TFM_CRYPT0_TEST_ECDH</code>	OFF	Test ECDH key agreement algorithm
<code>TFM_CRYPT0_TEST_CHACHA20</code>	OFF	Test ChaCha20 stream cipher
<code>TFM_CRYPT0_TEST_CHACHA20_POLY1305</code>	OFF	Test ChaCha20-Poly1305 AEAD algorithm
<code>TFM_CRYPT0_TEST_SINGLE_PART_I</code>	OFF	Test single-part operations in hash, MAC, AEAD and symmetric ciphers

BL2 setting

Profile Small enables MCUBoot provided by TF-M by default. A platform can overwrite this configuration by disabling MCUBoot in its configuration extension file `platform/ext<TFM_PLATFORM>/config.cmake`.

If MCUBoot provided by TF-M is enabled, single image boot is selected in TF-M Profile Small top-level CMake config file.

If a device implements its own boot loader, the configurations are implementation defined.

Table 30:: BL2 options in Profile Small top-level CMake config file

Configs	Default value	Descriptions
BL2	ON	Enable MCUBoot bootloader
MCUBOOT_IMAGE_NUMBER	1	Combine S and NS images

Platform support

Building Profile Small

To build Profile Small, argument `TFM_PROFILE` in build command line should be set to `profile_small`.

Take AN521 as an example.

The following commands build Profile Small without test cases on **AN521** with build type **MinSizeRel**, built by **Arm-clang**. SFN model is selected by default.

```
cd <TFM root dir>
mkdir build && cd build
cmake -DTFM_PLATFORM=arm/mps2/an521 \
      -DTFM_TOOLCHAIN_FILE=../toolchain_ARMCLANG.cmake \
      -DTFM_PROFILE=profile_small \
      -DCMAKE_BUILD_TYPE=MinSizeRel \
      ../
cmake --build ./ -- install
```

The following commands build Profile Small with regression test cases on **AN521** with build type **MinSizeRel**, built by **Armclang**. SFN model is selected by default.

```
cd <TFM root dir>
mkdir build && cd build
cmake -DTFM_PLATFORM=arm/mps2/an521 \
      -DTFM_TOOLCHAIN_FILE=../toolchain_ARMCLANG.cmake \
      -DTFM_PROFILE=profile_small \
      -DCMAKE_BUILD_TYPE=MinSizeRel \
      -DTEST_NS=ON \
      ../
cmake --build ./ -- install
```

Note

- For devices with more constrained memory and flash requirements, it is possible to build with either only `TEST_S` enabled or only `TEST_NS` enabled. This will decrease the size of the test images. Note that both test suites must still be run to ensure correct operation.

More details of building instructions and parameters can be found TF-M build instruction guide¹².

Reference

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8.4.2 Trusted Firmware-M Profile Medium-ARoT-less

Introduction

TF-M Profile Medium-ARoT-less is a reference implementation to align with security requirements defined in PSA Certified ARoT-less Level 2 protection profile (PSA Certified ARoT-less)¹.

TF-M Profile Medium-ARoT-less is defined based on TF-M Profile Medium², which aligns with PSA Certified Level 2 Protection Profile³.

Overall design

TF-M Profile Medium-ARoT-less defines the following feature set:

- Firmware Framework
 - Secure Function (SFN) model⁴
 - Isolation level 1⁵
- Internal Trusted Storage (ITS)
- Crypto
 - Support both symmetric cryptography and asymmetric cryptography
 - Asymmetric key based cipher suite suggested in TLS/DTLS profiles for IoT⁶ and CoAP⁷, including
 - * Authenticated Encryption with Associated Data (AEAD) algorithm
 - * Asymmetric key algorithm based signature and verification
 - * Public-key cryptography based key exchange
 - * Hash function
 - * HMAC for default Pseudorandom Function (PRF)
 - Asymmetric digital signature and verification for Initial Attestation Token (IAT)
- Initial Attestation
 - Asymmetric key algorithm based Initial Attestation
- Lightweight boot
 - Anti-rollback protection

¹² *TF-M build instruction*

¹ *SESIP Profile for PSA Certified ARoT-less Level 2*

² *Trusted Firmware-M Profile Medium Design*

³ *SESIP Profile for PSA Certified Level 2*

⁴ *Arm Firmware Framework for M 1.1 Extensions*

⁵ *Arm Platform Security Architecture Firmware Framework 1.0*

⁶ *Transport Layer Security (TLS) / Datagram Transport Layer Security (DTLS) Profiles for the Internet of Things*

⁷ *The Constrained Application Protocol (CoAP)*

- Multiple image boot
- Firmware Update (FWU) RoT Service

Design details

Most of design in Profile Medium-ARoT-less is identical to that in Profile Medium. Refer to Profile Medium document⁷ for details. Only the differences between Profile Medium-ARoT-less and Profile Medium are specified below.

Firmware framework

PSA Certified ARoT-less⁷ is only applicable to devices that don't support Application RoT (ARoT) services. The platform only consists of PSA RoT domain(s) in SPE making it unnecessary to implement the isolation between ARoT and PSA RoT. Therefore, this profile selects isolation level 1 to simplify implementation and optimize memory footprint and performance.

Since only isolation level 1 is required, this profile enables SFN model rather than IPC mode for further simplification.

Protected Storage

Protected Storage (PS) is implemented as an Application RoT service in TF-M by default. Therefore, PS is disabled by default in this profile.

FWU RoT Service

As PSA Certified ARoT-less requests, FWU RoT Service is enabled by default to support secure update of platform if the platform supports FWU.

Note

Implementation note

The entire secure update sequence involves multiple agents and components, including bootloader, TF-M FWU RoT Service, image update application(s), remote server(s), etc.

The secure update sequence is implementation-defined. Here is a reference of TF-M integration with FreeRTOS OTA⁸.

Implementation

The basic idea is to add dedicated profile CMake configuration files `config/profile/profile_medium_arotless` for this profile default configuration.

This top-level config file collects all the fundamental configuration flags and set them to default values, to explicitly configured the features required in this profile, during TF-M build.

A platform/use case overwrite the default settings to configure this profile.

The behavior of the build flow (particularly the order of configuration loading and overriding) can be found at *Build configuration*.

The default configurations in `config/profile/profile_medium_arotless` are shown below.

⁸ Secure OTA Updates for Cortex-M Devices with FreeRTOS

Table 31:: Config flags in Profile Medium-ARoT-less top-level CMake config file

Configs	Default value	Descriptions
TFM_ISOLATION_LEVEL	1	Select level 1 isolation
CONFIG_TFM_SPM_BACKE	SFN	Select SFN model
TFM_PARTITION_INTERN	ON	Enable ITS SP
ITS_BUF_SIZE	32	ITS internal transient buffer size
TFM_PARTITION_CRYPTO	ON	Enable Crypto service
CRYPTO_ASYM_ENCRYPT_	OFF	Enable Crypto asymmetric encryption operations
TFM_TF_PSA_CRYPTO_CO	\${CMAKE_SOURCE_DIR}/lib/ext/tf-psa-crypto/tfpsacrypto_config/crypto_config_profile_medium.h	TF-PSA-Crypto config file path
TFM_PARTITION_INITIA	ON	Enable Initial Attestation service
TFM_PARTITION_FIRMWA	ON	Enable Firmware Update service
TFM_PARTITION_PROTEC	OFF	Disable PS service
TFM_PARTITION_PLATFO	ON	Enable TF-M Platform SP

Note

Where a configuration is the same as the default in `config/config_default.cmake`, it might be omitted from the profile configuration file.

Note**Implementation note**

If the platform doesn't support secure update functionalities, FWU service will be disabled automatically. A warning will be thrown out during build.

Regression test configuration

FWU regression tests and SFN regression tests are selected by default when regression tests are enabled. Other implementations are the same as those in Profile Medium.

Platform support

To enable Profile Medium-ARoT-less on a platform, the platform specific CMake file should be added into the platform support list in top-level Profile Medium-ARoT-less CMake config file.

Building Profile Medium-ARoT-less

To build Profile Medium-ARoT-less, argument `TFM_PROFILE` in build command line should be set to `profile_medium_arotless`.

Take Musca-B1 as an example. The following commands build Profile Medium-ARoT-less without test cases on **Musca-B1** with build type **MinSizeRel**, built by **Armclang**.

```
cd <TFM root dir>
mkdir build && cd build
cmake -DTFM_PLATFORM=musca-b1 \
      -DTFM_TOOLCHAIN_FILE=../toolchain_ARMCLANG.cmake \
      -DTFM_PROFILE=profile_medium_arotless \
      -DCMAKE_BUILD_TYPE=MinSizeRel \
      ../
cmake --build ./ -- install
```

The following commands build Profile Medium-ARoT-less with regression test cases on **Musca-B1** with build type **MinSizeRel**, built by **Armclang**.

```
cd <TFM root dir>
mkdir build && cd build
cmake -DTFM_PLATFORM=musca-b1 \
      -DTFM_TOOLCHAIN_FILE=../toolchain_ARMCLANG.cmake \
      -DTFM_PROFILE=profile_medium_arotless \
      -DCMAKE_BUILD_TYPE=MinSizeRel \
      -DTEST_S=ON -DTEST_NS=ON \
      ../
cmake --build ./ -- install
```

Note

- For devices with more constrained memory and flash requirements, it is possible to build with either only `TEST_S` enabled or only `TEST_NS` enabled. This will decrease the size of the test images. Note that both test suites must still be run to ensure correct operation.

More details of building instructions and parameters can be found TF-M build instruction guide⁹.

Reference

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⁹ TF-M build instruction

8.4.3 Trusted Firmware-M Profile Medium Design

Introduction

Compared with Profile Small, Profile Medium aims to securely connect devices to Cloud services with asymmetric cipher support. Profile Medium target devices need more resources for more cipher algorithms and higher isolation levels.

Overall design

TF-M Profile Medium defines the following feature set:

- Firmware Framework
 - Inter-Process Communication (IPC) model [?]
 - Isolation level 2 [?]
- Internal Trusted Storage (ITS)
- Crypto
 - Support both symmetric cryptography and asymmetric cryptography
 - Asymmetric key based cipher suite suggested in TLS/DTLS profiles for IoT [?] and CoAP [?], including
 - * Authenticated Encryption with Associated Data (AEAD) algorithm
 - * Asymmetric key algorithm based signature and verification
 - * Public-key cryptography based key exchange
 - * Hash function
 - * HMAC for default Pseudorandom Function (PRF)
 - Asymmetric digital signature and verification for Initial Attestation Token (IAT)
- Initial Attestation
 - Asymmetric key algorithm based Initial Attestation
- Lightweight boot
 - Anti-rollback protection
 - Multiple image boot
- Protected Storage (PS) if off-chip storage device is integrated
 - Data confidentiality
 - Data integrity
 - Rollback protection

Design details

More details of TF-M Profile Medium design are described in following sections.

Firmware framework

Profile Medium with IPC model and isolation level 2 aims to support usage scenarios which require more complicated secure service model and additional protection to PSA RoT.

Level 2 isolation

Profile Medium selects isolation level 2 by default. In addition to isolation level 1, the PSA Root of Trust (PSA RoT) is also protected from access by the Application Root of Trust (App RoT) in level 2 isolation.

IPC model

Profile Medium enables IPC model by default. IPC model can achieve a more flexible framework and higher levels of isolation, but may require more memory footprint and bring in longer latency, compared to SFN model.

TF-M IPC model implementation follows the PSA Firmware Framework for M (PSA-FF-M) [?].

Crypto service

Compared to Profile Small, Profile Medium includes asymmetric cryptography to support direct connection to Cloud services via common protocols, such as TLS/DTLS 1.2.

As suggested in CoAP [?] and [?], TF-M Profile Medium by default selects TLS_ECDHE_ECDSA_WITH_AES_128_CCM as reference, which requires:

- ECDHE_ECDSA as key exchange algorithm.
- AES-128-CCM (AES CCM mode with 128-bit key) as AEAD algorithm. Platforms can implement AES-128-CCM with truncated authentication tag to achieve less network bandwidth [?].
- SHA256 as Hash function.
- HMAC as Message Authentication Code algorithm.

Applications can also support TLS PSK [?] cipher suites, such as TLS_PSK_WITH_AES_128_CCM [?].

Note

Implementation note

Developers can replace default algorithms with others or implement more algorithms according to actual usage scenarios and device capabilities.

If a Crypto hardware accelerator is integrated, the cipher suites and algorithms also depend on those accelerator features.

More details of cipher suite are described below.

Digital signature and verification

ECDSA is selected by default in Profile Medium. ECDSA requires much shorter keys compared with RSA at the same security level. Therefore, ECDSA can cost less storage area for assets and less network bandwidth to setup a TLS connection. ECDSA is also preferred for forward compatibility of future TLS versions.

As requested in [?], ECC curve secp256r1 should be supported. More ECC curves can be added based on the requirements in production.

If usage scenarios require RSA algorithm for backward compatibility and legacy applications, platforms can add RSA support or replace ECDSA with RSA. The cipher suite should be switched accordingly.

AEAD algorithm

If Protected Storage (PS) is implemented, it is recommended to select the same AEAD algorithm for PS service as the one used by TLS/DTLS cipher suite.

Internal Trusted Storage

The configuration of ITS is the same as those in Profile Small [?].

Lightweight boot

BL2 implementation can be device specific. Devices may implement diverse boot processes with different features and configurations. However, the boot loader must support anti-rollback protection. Boot loader must be able to prevent unauthorized rollback, to protect devices from being downgraded to earlier versions with known vulnerabilities.

MCUBoot in TF-M is configured as multiple image boot by default in Profile Medium. In multiple image boot, secure and non-secure images can be signed independently with different keys and they can be updated separately. It can support multiple vendors scenarios, in which non-secure and secure images are generated and updated by different vendors. Multiple image boot may require more storage area compared with single image boot.

Protected Storage

PS service is required if an off-chip storage device is integrated and used on the platform.

TF-M PS service relies on an AEAD algorithm to ensure data confidentiality and integrity. It is recommended to select the same AEAD algorithm as the one used for TLS/DTLS cipher suite.

Anti-rollback protection in PS relies on non-volatile counter(s) provided by TF-M Platform Secure Partition (SP).

Implementation

Overview

The basic idea is to add dedicated profile CMake configuration files under folder `config/profile` for TF-M Profile Medium default configuration, the same as Profile Small does.

The top-level Profile Medium config file collects all the necessary configuration flags and set them to default values, to explicitly enable the features required in Profile Medium and disable the unnecessary ones, during TF-M build.

A platform/use case can provide a configuration extension file to overwrite Profile Medium default setting and append other configurations. This configuration extension file can be added via parameter `TFM_EXTRA_CONFIG_PATH` in build command line.

The behaviour of the Profile Medium build flow (particularly the order of configuration loading and overriding) can be found at *Build configuration*

The details of configurations will be covered in each module in *Implementation details*.

Implementation details

This section discusses the details of Profile Medium implementation.

Top-level configuration files

The firmware framework configurations in `config/profile/profile_medium` are shown below.

Table 32:: Config flags in Profile Medium top-level CMake config file

Configs	Default value	Descriptions
TFM_ISOLATION_LEVEL	2	Select level 2 isolation
TFM_PARTITION_INTERN	ON	Enable ITS SP
ITS_BUF_SIZE	32	ITS internal transient buffer size
TFM_PARTITION_CRYPTO	ON	Enable Crypto service
CRYPTO_ASYM_ENCRYPT_	OFF	Enable Crypto asymmetric encryption operations
TFM_TF_PSA_CRYPTOCO	\${CMAKE_SOURCE_DIR}/lib/ext/tf-psa-crypto/tfpsacrypto_config/crypto_config_profile_medium.h	TF-PSA-Crypto config file path
TFM_PARTITION_INITIA	ON	Enable Initial Attestation service
TFM_PARTITION_PROTEC	ON	Enable PS service
TFM_PARTITION_PLATFO	ON	Enable TF-M Platform SP

Note

Where a configuration is the same as the default in `config/config_base.cmake`, it is omitted from the profile configuration file.

Test configuration

Standard regression test configuration applies. This means that enabling regression testing via

`-DTEST_S=ON -DTEST_NS=ON`

Will enable testing for all enabled partitions. See above for details of enabled partitions. Because Profile Medium enables IPC model, the IPC tests are also enabled.

Some cryptography tests are disabled due to the reduced cryptographic configuration profile.

¹ PS service is enabled by default. Platforms without off-chip storage devices can turn off `TFM_PARTITION_PROTECTED_STORAGE` to disable PS service. See *Protected Storage Secure Partition* for details.

Table 33:: TFM options in Profile Medium top-level CMake config file

Configs	Default value	Descriptions
TFM_CRYPTO_TEST_ALG_CBC	OFF	Disable CBC mode test
TFM_CRYPTO_TEST_ALG_CCM	ON	Enable CCM mode test
TFM_CRYPTO_TEST_ALG_CFB	OFF	Disable CFB mode test
TFM_CRYPTO_TEST_ALG_ECB	OFF	Disable ECB mode test
TFM_CRYPTO_TEST_ALG_CTR	OFF	Disable CTR mode test
TFM_CRYPTO_TEST_ALG_OFB	OFF	Disable OFB mode test
TFM_CRYPTO_TEST_ALG_GCM	OFF	Disable GCM mode test
TFM_CRYPTO_TEST_ALG_SHA_384	OFF	Disable SHA-384 algorithm test
TFM_CRYPTO_TEST_ALG_SHA_512	OFF	Disable SHA-512 algorithm test
TFM_CRYPTO_TEST_HKDF	OFF	Disable HKDF algorithm test
TFM_CRYPTO_TEST_ECDH	ON	Enable ECDH key agreement test
TFM_CRYPTO_TEST_CHACHA20	OFF	Disable ChaCha20 stream cipher test
TFM_CRYPTO_TEST_CHACHA20_POLY1305	OFF	Disable ChaCha20-Poly1305 AEAD algorithm test
TFM_CRYPTO_TEST_SINGLE_PART_I	OFF	Test single-part operations in hash, MAC, AEAD and symmetric ciphers

Device configuration extension

To change default configurations and add platform specific configurations, a platform can add a platform configuration file at `platform/ext<TFM_PLATFORM>/config.cmake`

Crypto service configurations

Crypto Secure Partition

TF-M Profile Medium enables Crypto SP in top-level CMake config file. The following PSA Crypto operations are enabled by default.

- Hash operations
- Message authentication codes
- Symmetric ciphers
- AEAD operations
- Asymmetric key algorithm based signature and verification
- Key derivation
- Key management

TF-PSA-Crypto configurations

TF-M Profile Medium adds a dedicated TF-PSA-Crypto config file `crypto_config_profile_medium.h` under `/lib/ext/tf-psa-crypto/tfpsacrypto_config` folder, instead of the common one `crypto_config_default.h` [?].

Major cryptographic configurations are set as listed below:

- Enable SHA256
- Enable generic message digest wrappers
- Enable AES

- Enable CCM mode for symmetric ciphers
- Disable other modes for symmetric ciphers
- Enable ECDH
- Enable ECDSA
- Select ECC curve secp256r1
- Other configurations required by selected option above

Other configurations can be selected to optimize the memory footprint of Crypto module.

A device/use case can append an extra config header to the Profile Medium default TF-PSA-Crypto config file. This can be done by setting the `TFM_TF_PSA_CRYPT_PLATFORM_EXTRA_CONFIG_PATH` cmake variable in the platform config file `platform/ext<TFM_PLATFORM>/config.cmake`. This cmake variable is a wrapper around the `TF_PSA_CRYPT_USER_CONFIG_FILE` options, but is preferred as it keeps all configuration in cmake.

Internal Trusted Storage configurations

ITS service is enabled in top-level Profile Medium CMake config file by default.

The internal transient buffer size `ITS_BUF_SIZE` [?] is set to 32 bytes by default. A platform/use case can overwrite the buffer size in its specific configuration extension according to its actual requirement of assets and Flash attributes.

Profile Medium CMake config file won't touch the configurations of device specific Flash hardware attributes [?].

Protected Storage Secure Partition

Data confidentiality, integrity and anti-rollback protection are enabled by default in PS.

If PS is selected, AES-CCM is used as AEAD algorithm by default. It requires to enable PS implementation to select diverse AEAD algorithm.

If platforms don't integrate any off-chip storage device, platforms can disable PS in platform specific configuration extension file via `platform/ext<TFM_PLATFORM>/config.cmake`.

BL2 setting

Profile Medium enables MCUBoot provided by TF-M by default. A platform can overwrite this configuration by disabling MCUBoot in its configuration extension file `platform/ext<TFM_PLATFORM>/config.cmake`.

If MCUBoot provided by TF-M is enabled, multiple image boot is selected by default in TF-M Profile Medium top-level CMake config file.

If a device implements its own boot loader, the configurations are implementation defined.

Platform support

To enable Profile Medium on a platform, the platform specific CMake file should be added into the platform support list in top-level Profile Medium CMake config file.

Building Profile Medium

To build Profile Medium, argument `TFM_PROFILE` in build command line should be set to `profile_medium`.

Take AN521 as an example:

The following commands build Profile Medium without test cases on **AN521** with build type **MinSizeRel**, built by **Armclang**.

```
cd <TFM root dir>
mkdir build && cd build
cmake -DTFM_PLATFORM=arm/mps2/an521 \
      -DTFM_TOOLCHAIN_FILE=../toolchain_ARMCLANG.cmake \
      -DTFM_PROFILE=profile_medium \
      -DCMAKE_BUILD_TYPE=MinSizeRel \
      ../
cmake --build ./ -- install
```

The following commands build Profile Medium with regression test cases on **AN521** with build type **MinSizeRel**, built by **Armclang**.

```
cd <TFM root dir>
mkdir build && cd build
cmake -DTFM_PLATFORM=arm/mps2/an521 \
      -DTFM_TOOLCHAIN_FILE=../toolchain_ARMCLANG.cmake \
      -DTFM_PROFILE=profile_medium \
      -DCMAKE_BUILD_TYPE=MinSizeRel \
      -DTEST_S=ON -DTEST_NS=ON \
      ../
cmake --build ./ -- install
```

Note

- For devices with more constrained memory and flash requirements, it is possible to build with either only TEST_S enabled or only TEST_NS enabled. This will decrease the size of the test images. Note that both test suites must still be run to ensure correct operation.

More details of building instructions and parameters can be found TF-M build instruction guide [?].

Reference

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8.4.4 Trusted Firmware-M Profile Large Design

Introduction

As one of TF-M Profiles, Profile Large protects less resource-constrained Arm Cortex-M devices.

Compared to Profile Small¹ and Profile Medium², Profile Large aims to enable more secure features to support higher level of security required in more complex usage scenarios.

- Isolation level 3 enables additional isolation between *Application RoT* (App RoT) services.
- More crypto algorithms and cipher suites are selected to securely connect devices to remote services offered by various major Cloud Service Providers (CSP)
- Basic software countermeasures against physical attacks can be enabled.

¹ *Trusted Firmware-M Profile Small Design*

² *Trusted Firmware-M Profile Medium Design*

Profile Large can be aligned as a reference implementation with the requirements defined in PSA Certified Level 3 Lightweight Protection Profile³.

Overall design

TF-M Profile Large defines the following feature set:

- Firmware Framework
 - Inter-Process Communication (IPC) model⁴
 - Isolation level 3²
- Internal Trusted Storage (ITS)
- Crypto
 - Support both symmetric ciphers and asymmetric ciphers
 - Asymmetric key based cipher suites defined in TLS 1.2⁵ to support direct secure connection to major CSPs, including
 - * Authenticated Encryption with Associated Data (AEAD) algorithm
 - * Asymmetric key algorithm based signature and verification
 - * Public-key cryptography based key exchange
 - * Hash function
 - * HMAC for default Pseudorandom Function (PRF)
 - Asymmetric digital signature and verification for Initial Attestation Token (IAT)
 - Asymmetric algorithms for firmware image signature verification
 - Key derivation
- Initial Attestation
 - Asymmetric key algorithm based Initial Attestation
- **Secure boot**
 - Anti-rollback protection
 - Multiple image boot
- Protected Storage (PS) if off-chip storage device is integrated
 - Data confidentiality
 - Data integrity
 - Rollback protection
- Software countermeasures against physical attacks

Design details

More details of TF-M Profile Large design are described in following sections.

³ PSA Certified Level 3 Lightweight Protection Profile

⁴ Arm Platform Security Architecture Firmware Framework 1.0

⁵ The Transport Layer Security (TLS) Protocol Version 1.2

Firmware framework

Profile Large selects IPC model and isolation level 3 by default.

Isolation level 3 supports additional isolation between App RoT services, compared to isolation level 2. It can protect *RoT* services from each other when their vendors don't trust each other.

Crypto service

Profile Large supports direct connection to Cloud services via common protocols, such as TLS 1.2.

In some usage scenarios, PSA RoT can be managed by device manufacturer or other vendors and is out of control of application developers. Profile Large selects alternative crypto algorithms for each crypto function to support multiple common cipher suites required by various major CSPs. Therefore, application developers can support services for diverse CSPs on same devices with Profile Large, without relying on PSA RoT upgrades of crypto.

Devices meeting Profile Large should be in a position to offer at least two alternatives to every cryptographic primitive for symmetric, asymmetric and hash, and be able to use them for encryption, AEAD, signature and verification.

It will cost more resource in Profile Large to support more crypto algorithms and cipher suites, compared to Profile Medium⁶.

Boot loader

BL2 implementation can be device specific. Devices may implement diverse boot processes with different features and configurations. However, the boot loader must support anti-rollback protection. Boot loader must be able to prevent unauthorized rollback, to protect devices from being downgraded to earlier versions with known vulnerabilities.

MCUBoot in TF-M is configured as multiple image boot by default in Profile Large. In multiple image boot, secure and non-secure images can be signed independently with different keys and they can be updated separately. It can support multiple vendors scenarios, in which non-secure and secure images are generated and updated by different vendors. Multiple image boot may cost larger memory footprint compared with single image boot.

Boot loader can implement software countermeasures to mitigate physical attacks.

Protected Storage

PS service is required if an off-chip storage device is integrated and used on the platform.

Anti-rollback protection in PS relies on non-volatile counter(s) provided by TF-M Platform *Secure Partition* (SP).

Software countermeasures against physical attacks

TF-M Profile Large enables TF-M Fault Injection Hardening (FIH) library Profile Medium by default. It enables the following countermeasure techniques:

- Control flow monitor
- Failure loop hardening
- Complex constants
- Redundant variables and condition checks

Refer to TF-M physical attack mitigation design document⁶ for FIH library details.

⁶ *Physical attack mitigation in Trusted Firmware-M*

Note**TF-M FIH library is still under development.**

TF-M FIH library hardens TF-M critical execution steps to make physical attacks more difficult, together with device hardware countermeasures. It is not guaranteed that TF-M FIH library is able to mitigate all kinds of physical attacks.

Note**Implementation note**

TF-M FIH library doesn't cover platform specific critical configurations. Platforms shall implement software countermeasures against physical attacks to protect platform specific implementation.

Implementation

Overview

The basic idea is to add dedicated profile CMake configuration files under folder `config/profile` for TF-M Profile Large default configuration, the same as other TF-M Profiles do.

The top-level Profile Large config file collects all the necessary configuration flags and set them to default values, to explicitly enable the features required in Profile Large and disable the unnecessary ones, during TF-M build.

A platform/use case can provide a configuration extension file to overwrite Profile Large default setting and append other configurations. This configuration extension file can be added via parameter `TFM_EXTRA_CONFIG_PATH` in build command line.

The behaviour of the Profile Large build flow (particularly the order of configuration loading and overriding) can be found at *Build configuration*

The details of configurations will be covered in each module in *Implementation details*.

Implementation details

This section discusses the details of Profile Large implementation.

Top-level configuration files

The firmware framework configurations in `config/profile/profile_large` are shown below.

Table 34:: Config flags in Profile Large top-level CMake config file

Configs	Descriptions	Default value
TFM_ISOLATION_LEVEL	Select level 3 isolation	3
TFM_PARTITION_INTERNAL	Enable ITS SP	ON
TFM_PARTITION_CRYPTO	Enable Crypto service	ON
TFM_TF_PSA_CRYPTO_CONF	TF-PSA-Crypto config file path	\${CMAKE_SOURCE_DIR}/lib/ext/tf-psa-crypto/tfpsacrypto_config/crypto_config_profile_large.h
TFM_PARTITION_INITIAL	Enable Initial Attestation service	ON
TFM_PARTITION_PROTECTE[?]	Enable PS service	ON
TFM_PARTITION_PLATFORM	Enable TF-M Platform SP	ON

Crypto service configurations

Crypto Secure Partition

TF-M Profile Large enables Crypto SP in top-level CMake config file and selects all the Crypto modules.

TF-PSA-Crypto configurations

TF-M Profile Large adds a dedicated TF-PSA-Crypto config file `crypto_config_profile_large.h` under `/lib/ext/tf-psa-crypto/tfpsacrypto_config` folder, instead of the common one `crypto_config_default.h`⁷.

Major cryptographic configurations are set as listed below:

- Enable SHA256, SHA384 and SHA512
- Enable generic message digest wrappers
- Enable AES
- Enable CCM mode, GCM mode, CTR mode, CFB mode and CBC mode for symmetric ciphers
- Disable other modes for symmetric ciphers
- Enable ECDH
- Enable ECDSA
- Enable RSA
- Select ECC curve `secp256r1` and `secp384r1`
- Enable HMAC-based key derivation function
- Other configurations required by selected option above

A device/use case can append an extra config header to the Profile Large default TF-PSA-Crypto config file to override the default settings. This can be done by setting the `TFM_TF_PSA_CRYPTO_PLATFORM_EXTRA_CONFIG_PATH` cmake variable in the platform config file `platform/ext<TFM_PLATFORM>/config.cmake`. This cmake variable is a wrapper around the `TF_PSA_CRYPTO_USER_CONFIG_FILE` options, but is preferred as it keeps all configuration in cmake.

⁷ *Crypto design*

Internal Trusted Storage configurations

ITS service is enabled in top-level Profile Large CMake config file by default.

The internal transient buffer size `ITS_BUF_SIZE`⁸ is set to 64 bytes by default. A platform/use case can overwrite the buffer size in its specific configuration extension according to its actual requirement of assets and Flash attributes.

Profile Large CMake config file won't touch the configurations of device specific Flash hardware attributes.

Protected Storage Secure Partition

Data confidentiality, integrity and anti-rollback protection are enabled by default in PS.

If PS is selected, AES-CCM is used as AEAD algorithm by default. If platform hardware crypto accelerator supports the AEAD algorithm, the AEAD operations can be executed in hardware crypto accelerator.

If platforms don't integrate any off-chip storage device, platforms can disable PS in platform specific configuration extension file via `platform/ext<TFM_PLATFORM>/config.cmake`.

BL2 setting

Profile Large enables MCUBoot provided by TF-M by default. A platform can overwrite this configuration by disabling MCUBoot in its configuration extension file `platform/ext<TFM_PLATFORM>/config.cmake`.

If MCUBoot provided by TF-M is enabled, multiple image boot is selected by default.

If a device implements its own boot loader, the configurations are implementation defined.

Software countermeasure against physical attacks

Profile Large selects TF-M FIH library Profile Medium by specifying `-DTFM_FIH_PROFILE=MEDIUM` in top-level CMake config file.

System integrators shall implement software countermeasures in platform specific implementations.

Device configuration extension

To change default configurations and add platform specific configurations, a platform can add a platform configuration file at `platform/ext<TFM_PLATFORM>/config.cmake`

Test configuration

Some cryptography tests are disabled due to the reduced cryptographic configuration profile. Profile Large specific test configurations are also specified in Profile Large top-level CMake config file `config/profile/profile_large_test.cmake`.

⁸ ITS integration guide

Table 35:: Profile Large crypto test configuration

Configs	Default value	Descriptions
TFM_CRYPTO_TEST_ALG_CBC	ON	Test CBC cryptography mode
TFM_CRYPTO_TEST_ALG_CCM	ON	Test CCM cryptography mode
TFM_CRYPTO_TEST_ALG_CFB	OFF	Test CFB cryptography mode
TFM_CRYPTO_TEST_ALG_ECB	OFF	Test ECB cryptography mode
TFM_CRYPTO_TEST_ALG_CTR	OFF	Test CTR cryptography mode
TFM_CRYPTO_TEST_ALG_OFB	OFF	Test OFB cryptography mode
TFM_CRYPTO_TEST_ALG_GCM	ON	Test GCM cryptography mode
TFM_CRYPTO_TEST_ALG_SHA_384	OFF	Test SHA-384 cryptography algorithm
TFM_CRYPTO_TEST_ALG_SHA_512	ON	Test SHA-512 cryptography algorithm
TFM_CRYPTO_TEST_HKDF	ON	Test HMAC-based key derivation function
TFM_CRYPTO_TEST_ECDH	ON	Test ECDH key agreement algorithm
TFM_CRYPTO_TEST_CHACHA20	OFF	Test ChaCha20 stream cipher
TFM_CRYPTO_TEST_CHACHA20_POLY1305	OFF	Test ChaCha20-Poly1305 AEAD algorithm
TFM_CRYPTO_TEST_SINGLE_PART_I	OFF	Test single-part operations in hash, MAC, AEAD and symmetric ciphers

Platform support

To enable Profile Large on a platform, the platform specific CMake file should be added into the platform support list in top-level Profile Large CMake config file.

Building Profile Large

To build Profile Large, argument TFM_PROFILE in build command line should be set to `profile_large`.

Take AN521 as an example:

The following commands build Profile Large without test cases on **AN521** with build type **MinSizeRel**, built by **Armclang**.

```
cd <TFM root dir>
mkdir build && cd build
cmake -DTFM_PLATFORM=arm/mps2/an521 \
      -DTFM_TOOLCHAIN_FILE=../toolchain_ARMCLANG.cmake \
      -DTFM_PROFILE=profile_large \
      -DCMAKE_BUILD_TYPE=MinSizeRel \
      ../
cmake --build ./ -- install
```

The following commands build Profile Large with regression test cases on **AN521** with build type **MinSizeRel**, built by **Armclang**.

```
cd <TFM root dir>
mkdir build && cd build
cmake -DTFM_PLATFORM=arm/mps2/an521 \
      -DTFM_TOOLCHAIN_FILE=../toolchain_ARMCLANG.cmake \
      -DTFM_PROFILE=profile_large \
      -DCMAKE_BUILD_TYPE=MinSizeRel \
      -DTEST_S=ON -DTEST_NS=ON \
      ../
```

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```
../
cmake --build ./ -- install
```

More details of building instructions and parameters can be found TF-M build instruction guide⁹.

Reference

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Option	Base	Small	ARoT-less	Medium	Large
TFM_ISOLATION_LEVEL	1	1	1	2	3
CONFIG_TFM_SPM_BACKEND	SFN	SFN	SFN	IPC	IPC
TFM_PARTITION_CRYPTO	OFF	ON	ON	ON	ON
TFM_PARTITION_INTERNAL_TRUSTED_STORAGE	OFF	ON	ON	ON	ON
TFM_PARTITION_PLATFORM	OFF	OFF	ON	ON	ON
TFM_PARTITION_PROTECTED_STORAGE	OFF	OFF	OFF	ON	ON
TFM_PARTITION_INITIAL_ATTESTATION	OFF	ON	ON	ON	ON
SYMMETRIC_INITIAL_ATTESTATION	OFF	ON	OFF	OFF	OFF
TFM_PARTITION_FIRMWARE_UPDATE	OFF	OFF	ON	OFF	OFF
PS_CRYPT_O_AEAD_ALG	GCM	-	-	CCM	CCM
PSA_FRAMEWORK_HAS_MM_IOVEC	OFF	ON	OFF	OFF	OFF
MCUBOOT_IMAGE_NUMBER ¹	2	1	2	2	2
<i>Advanced options, defined in the corresponded header (.h) file</i>					
CRYPTO_ENGINE_BUF_SIZE	0x2080	0x400	0x2080	0x2080	0x2380
CRYPTO_ASYNC_SIGN_MODULE_ENABLED	ON	OFF	ON	ON	ON
CRYPTO_ASYNC_ENCRYPT_MODULE_ENABLE	ON	OFF	OFF	OFF	ON
CRYPTO_SINGLE_PART_FUNCS_DISABLED	OFF	ON	OFF	OFF	OFF
CRYPTO_CONC_OPER_NUM	8	4	8	8	8
CONFIG_TFM_CONN_HANDLE_MAX_NUM	8	3	8	8	8
ITS_BUF_SIZE ²	512	32	32	32	512

1. *MCUBOOT_IMAGE_NUMBER* value is taken from MCUBoot default configuration, except profile Small.

2. Many platforms redefine *ITS_BUF_SIZE* value.

Each profile has predefined configuration for the cryptographic library, located in `/lib/ext/tf-psa-crypto/tfpsacrypto_config/`

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TF-M is highly configurable project with many configuration options to meet a user's needs. A user can select the desired set of services and fine-tune them to their requirements. There are two types of configuration options

Build configuration

Specifies which file or component to include into compilation and build. These are options, usually used by a build system to enable/disable modules, specify location of external dependency or other selection, global

⁹ TF-M build instruction

to a project. These option set shall be considered while adopting TF-M to other build systems. In the *Base Configuration* table these options have *Build* type.

Component configuration

To adjust a particular parameter to a desired value. Those options are local to a component or externally referenced when components are coupled. Options are in C header file. The *The Header File Config System* has more details about it. In the *Base Configuration* table these options have *Component* type.

Note

Originally, TF-M used CMake variables for both building and component tuning purposes. It was convenient to have a single system for both building and component's configurations. To simplify and improve configurability and better support build systems other than a CMake, TF-M introduced a *The Header File Config System* and moved component options into a dedicated config headers.

8.5 How to configure

TF-M Project provides a base build, defined in `/config/config_base.cmake` and `/config/config_base.h`. Starting from the base, users can enable required services and features using several independent methods to configure TF-M.

Use *TF-M Profiles*.

There are 4 sets of predefined configurations for a elected use cases, called profiles. A user can select a profile by providing `-DTFM_PROFILE=<profile file name>`. Each profiles represented by a pair of configuration files for Building (CMake) options and Component options (.h file)

Use a custom profile.

Another method is to take an existing TF-M profile and adjust the desired options manually editing CMake and config header files. This is for users familiar with TF-M.

Use *The Kconfig System*.

This method is recommended for beginners. Starting from the *base configuration* a user can enable necessary services and options. KConfig ensures that all selected options are consistent and valid. This is new in v1.7.0 and it covers only SPM and PSA services. As an output KConfig produces a pair of configuration files, similar to a profile.

Note

In contrast, before TF-M v1.7.0, the default build includes all possible features. With growing functionality, such rich default build became impractical by not fitting into every platform and confusing of big memory requirements.

8.6 Priorities

A project configuration performed in multiple steps with priorities. The list below explains the process but for the details specific to *Build configuration* or *The Header File Config System* please check the corresponded document.

1. The base configuration with default values is used as a starting point
2. A profile options applied on top of the base
3. A platform can check the selected configuration and apply restrictions
4. Finally, command line options can modify the composed set

Note

To ensure a clear intention and conscious choice, all options must be provided explicitly via a project configuration file. Default values on step 1 will generate warnings which are expected to break a build.

8.7 Base Configuration

The base configuration is the ground for configuring TF-M, provided defaults are defined in `/config/config_base.cmake` and `/config/config_base.h`. The base build includes SPM and platform code only.

This table lists the config option categorizations of the SPM and Secure Partitions.

8.7.1 Crypto

Options	Type	Base Value
TFM_PARTITION_CRYPT0	Build	OFF
CRYPTO_TFM_BUILTIN_KEYS_DRIVER	Build	ON
CRYPTO_NV_SEED	Component	ON
CRYPTO_ENGINE_BUF_SIZE	Component	0x2080
CRYPTO_IOVEC_BUFFER_SIZE	Component	5120
CRYPTO_STACK_SIZE	Component	0x1B00
CRYPTO_CONC_OPER_NUM	Component	8
CRYPTO_RNG_MODULE_ENABLED	Component	1
CRYPTO_KEY_MODULE_ENABLED	Component	1
CRYPTO_AEAD_MODULE_ENABLED	Component	1
CRYPTO_MAC_MODULE_ENABLED	Component	1
CRYPTO_HASH_MODULE_ENABLED	Component	1
CRYPTO_CIPHER_MODULE_ENABLED	Component	1
CRYPTO_ASYNC_SIGN_MODULE_ENABLED	Component	1
CRYPTO_ASYNC_ENCRYPT_MODULE_ENABLED	Component	1
CRYPTO_KEY_DERIVATION_MODULE_ENABLED	Component	1
CRYPTO_SINGLE_PART_FUNCS_ENABLED	Component	1

8.7.2 Initial Attestation

Options	Type	Base Value
TFM_PARTITION_INITIAL_ATTESTATION	Build	OFF
SYMMETRIC_INITIAL_ATTESTATION	Build	OFF
ATTEST_KEY_BITS	Build	256
ATTEST_TOKEN_PROFILE	Component	"PSA_IOT_1"
ATTEST_INCLUDE_OPTIONAL_CLAIMS	Component	1
ATTEST_INCLUDE_COSE_KEY_ID	Component	0
ATTEST_STACK_SIZE	Component	0x800

8.7.3 Internal Trusted Storage

Options	Type	Base Value
TFM_PARTITION_INTERNAL_TRUSTED_STORAGE	Build	OFF
ITS_CREATE_FLASH_LAYOUT	Component	1
ITS_RAM_FS	Component	0
ITS_VALIDATE_METADATA_FROM_FLASH	Component	1
ITS_MAX_ASSET_SIZE	Component	512
ITS_NUM_ASSETS	Component	10
ITS_BUF_SIZE	Component	ITS_MAX_ASSET_SIZE
ITS_STACK_SIZE	Component	0x720

8.7.4 Protected Storage

Options	Type	Base Value
TFM_PARTITION_PROTECTED_STORAGE	Build	OFF
PS_ENCRYPTION	Build	ON
PS_SUPPORT_FORMAT_TRANSITION	Build	OFF
PS_CRYPTO_AEAD_ALG	Build	PSA_ALG_GCM
PS_AES_KEY_USAGE_LIMIT	Build	0
PS_CREATE_FLASH_LAYOUT	Component	1
PS_RAM_FS	Component	0
PS_VALIDATE_METADATA_FROM_FLASH	Component	1
PS_MAX_ASSET_SIZE	Component	2048
PS_NUM_ASSETS	Component	10
PS_ROLLBACK_PROTECTION	Component	1
PS_STACK_SIZE	Component	0x700

8.7.5 Firmware Update

Options	Type	Base Value
PLATFORM_HAS_FIRMWARE_UPDATE_SUPPORT	Build	OFF
TFM_PARTITION_FIRMWARE_UPDATE	Build	OFF
TFM_CONFIG_FWU_MAX_WRITE_SIZE	Build	1024
TFM_CONFIG_FWU_MAX_MANIFEST_SIZE	Build	0
FWU_DEVICE_CONFIG_FILE	Build	""
FWU_SUPPORT_TRIAL_STATE	Build	Depends on MCU-BOOT_UPGRADE_STRATEGY
TFM_FWU_BOOTLOADER_LIB	Build	"mcuboot"
FWU_DEVICE_IMPL_INFO_DEF_FILE	Build	""
TFM_FWU_BUF_SIZE	Component	PSA_FWU_MAX_BLOCK_SIZE
FWU_STACK_SIZE	Component	0x600

8.7.6 Platform Secure Partition

Options	Type	Base Value
TFM_PARTITION_PLATFORM	Build	OFF
PLATFORM_SERVICE_INPUT_BUFFER_SIZE	Component	64
PLATFORM_SERVICE_OUTPUT_BUFFER_SIZE	Component	64
PLATFORM_SP_STACK_SIZE	Component	0x500
PLATFORM_NV_COUNTER_MODULE_DISABLED	Component	0

8.7.7 NS Agent Mailbox Secure Partition

Options	Type	Base Value
NS_AGENT_MAILBOX_STACK_SIZE	Component	0x800
MAILBOX_IS_UNCACHED_S	Component	1
MAILBOX_IS_UNCACHED_NS	Component	1

8.7.8 NS Agent TrustZone Secure Partition

Options	Type	Base Value
TFM_TZ_REENTRANCY_CHECK	Component	OFF

8.7.9 Secure Partition Manager

Options	Type	Base Values
TFM_ISOLATION_LEVEL	Build	1
PSA_FRAMEWORK_HAS_MM_IOVEC	Build	OFF
CONFIG_TFM_SPM_BACKEND	Build	“SFN”
TFM_SPM_LOG_LEVEL	Build	1
CONFIG_TFM_AUTO_BOOT_NS_CORE	Component	1
CONFIG_TFM_STACK_WATERMARKS	Build	OFF
CONFIG_TFM_CONN_HANDLE_MAX_NUM	Component	8
CONFIG_TFM_DOORBELL_API	Component	0
CONFIG_TFM_SCHEDULE_WHEN_NS_INTERRUPTED	Component	0
CONFIG_TFM_HYBRID_PLAT_SCHED_TYPE	Component	0
CONFIG_TFM_PARTITION_META_DYNAMIC_ISOLATION	Component	0

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INTEGRATION GUIDE

The purpose of this document is to provide a guide on how to integrate TF-M with other hardware platforms and operating systems.

9.1 Source Structure

TF-M is designed to provide a user-friendly source structure to ease integration and service development. This document introduces the source structure and its design intention of TF-M. Additionally, the folders below are important for TF-M integration and described in a separate documents:

9.1.1 Details for the platform folder

Interfacing with TF-M

`platform/ext/target/tfm_peripherals_def.h`

This file should enumerate the hardware peripherals that are available for TF-M on the platform. The name of the peripheral used by a service is set in its manifest file. The platform have to provide a macro for each of the provided peripherals, that is substituted to a pointer to type `struct platform_data_t`. The memory that the pointer points to is allocated by the platform code. The pointer gets stored in the partitions database in SPM, and it is provided to the SPM HAL functions.

Peripherals currently used by the services in TF-M

- `TFM_PERIPHERAL_UART1- UART1`

Note

If a platform doesn't support a peripheral, that is used by a service, then the service cannot be used on the given platform. Using a peripheral in TF-M that is not supported by the platform results in compile error.

`platform/include/tfm_spm_hal.h`

This file contains the declarations of functions that a platform implementation has to provide for TF-M's SPM. For details see the comments in the file.

`platform/include/tfm_platform_system.h`

This file contains the declarations of functions that a platform implementation has to provide for TF-M's Platform Service. For details see `docs/user_guides/services/tfm_platform_integration_guide.rst`

System Startup

Before calling `main()`, platform startup code should initialise all system clocks, perform runtime initialisation and other steps such as setting the vector table. Configuration of the following features is optional as TF-M architecture code will check and initialise them:

- The Security Extension.
- The Floating-point Extension.

Debug authentication settings

A platform may provide the option to configure debug authentication. TF-M core calls the HAL function `enum tfm_hal_status_t tfm_hal_platform_init(void)` in which debug authentication is configured based on the following defines:

- `DAUTH_NONE`: Debugging the system is not enabled.
- `DAUTH_NS_ONLY`: Invasive and non invasive debugging of non-secure code is enabled.
- `DAUTH_FULL`: Invasive and non-invasive debugging of non-secure and secure code is enabled.
- `DAUTH_CHIP_DEFAULT`: The debug authentication options are used that are set by the chip vendor.

The desired debug authentication configuration can be selected by setting one of the options above to the `cmake` command with the `-DDEBUG_AUTHENTICATION=<define>` option. The default value of `DEBUG_AUTHENTICATION` is `DAUTH_CHIP_DEFAULT`.

Note

`enum tfm_hal_status_t tfm_hal_platform_init(void)` is called during the TF-M core initialisation phase, before initialising secure partition. This means that BL2 runs with the chip default setting.

Sub-folders

include

This folder contains the interfaces that TF-M expects every target to provide. The code in this folder is created as a part of the TF-M project therefore it adheres to the BSD 3.0 license.

ext

This folder contains code that has been imported from other projects so it may have licenses other than the BSD 3.0 used by the TF-M project.

Please see the *Details for the platform/ext folder* for details.

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9.1.2 Details for the platform/ext folder

This folder has code that has been imported from other projects. This means the files in this folder and subfolders have Apache 2.0 license which is different to BSD 3.0 license applied to the parent TF-M project.

Note

This folder is strictly Apache 2.0 with the exception of cmake files. Maintainers should be consulted if this needs to be revisited.

Sub-folders**accelerator**

This folder contains cmake and code files to interact cryptographic accelerators.

In order to use a cryptographic accelerator, a platform must set `CRYPTO_HW_ACCELERATOR_TYPE` in `preload.cmake`. This option maps directly to the subdirectories of the accelerator directory. Currently available accelerators are : the CryptoCell cc312, the STMicroelectronics accelerator `stm`.

A minimal API is exposed to interact with accelerators, the details of this api are in `accelerator/interface/crypto_hw.h`. Implementation of the API is inside the subdirectory of the individual accelerator.

To configure a cryptographic accelerator at build time, two cmake options can be specified.

- **CRYPTO_HW_ACCELERATOR**
 - ON All possible cryptographic operations will be offloaded to the accelerator.
 - OFF The cryptographic accelerator will be ignored and software cryptography will be used.

cmsis

This folder contains core and compiler specific header files imported from the CMSIS_5 project.

common**armclang and gcc**

These contain the linker scripts used to configure the memory regions in TF-M regions.

template

This directory contains platform-independent dummy implementations of the interfaces in `platform/include`. These implementations can be built directly for initial testing of a platform port, or used as a basic template for a real implementation for a particular target. They **must not** be used in production systems.

driver

This folder contains the headers with CMSIS compliant driver definitions that that TF-M project expects a target to provide.

target_cfg.h

This file is expected to define the following macros respectively.

- **TFM_DRIVER_STDIO** - This macro should expand to a structure of type `ARM_DRIVER_USART`. TFM redirects its standard input and output to this instance of USART.
- **NS_DRIVER_STDIO** - This macro should expand to a structure of type `ARM_DRIVER_USART`. Non-Secure application redirects its standard input and output to this instance of USART.

target

This folder contains the files for individual target. For a buildable target, the directory path from the `target` directory to its `CMakeLists.txt` file is the argument that would be given to `-DTFM_PLATFORM=`.

The standard directory structure is as follows:

- **target**
 - **<Vendor name>**
 - * `common`
 - * `<buildable target 1>`
 - * `<buildable target 2>`
 - * `<buildable target 3>`

Each buildable target must contain the cmake files mandated in the section below.

The `common` directory is not required, but can be used to contain code that is used by multiple targets.

There must not be any directories inside the vendor directory that is not either the `common` directory or a buildable platform, to avoid confusion about what directories are a valid `TFM_PLATFORM`.

Buildable target required cmake files

A buildable target must provide 3 mandatory cmake files. These files must all be placed in the root of the buildable target directory.

preload.cmake

This file contains variable definitions that relate to the underlying hardware of the target.

- `TFM_SYSTEM_PROCESSOR`: The processor used by the target. The format is that same as the format used in the `-mcpu=` argument of `GNUARM` or `ARMCLANG`. The special `+modifier` syntax must not be used.
- `TFM_SYSTEM_ARCHITECTURE`: The architecture used by the target. The format is that same as the format used in the `-march=` argument of `GNUARM` or `ARMCLANG`. The special `+modifier` syntax must not be used.
- `TFM_SYSTEM_DSP`: Whether the target has the DSP feature of the given `TFM_SYSTEM_PROCESSOR`
- `CRYPTO_HW_ACCELERATOR_TYPE`: The type of cryptographic accelerator the target has, if it has one. This maps exactly to the subdirectories of `platform/ext/accelerator`
- `CONFIG_TFM_FP_ARCH`: The “FPU architecture” (Armclang) or “FP hardware” (GNU) used by the target for compiling C source files. The value will be transferred to `-mfpu` argument of `GNUARM` or `ARMCLANG`
- `CONFIG_TFM_FP_ARCH_ASM`: The “FPU architecture” (Armclang) used by the target for assembling ASM source files. This value will be transferred to the `--fpu` argument of `ARMCLANG`. This value is not used by `GNUARM`.

For more details on `CONFIG_TFM_FP_ARCH` and `CONFIG_TFM_FP_ARCH_ASM`, please refer to *Floating-Point Support*.

Other than these particular cmake variables, it is permissible for the `preload.cmake` file to contain `add_definitions` statements, in order for set compile definitions that are global for the hardware. This is commonly used to select a particular set of code from a vendor SDK.

It is not permissible to contains code other than the above in a `preload.cmake` file, any general cmake code should be placed in `CMakeLists.txt` and any configuration options should be contained in `config.cmake`

config.cmake

This file collects platform-specific overrides to the configuration options. This should only contain cmake options that are included in `config_base.cmake`. These options should be set as `CACHE` variables, as they are in `config_base.cmake`.

CMakeLists.txt

This file should contain all other required cmake code for the platform. This primarily consists of the following:

- Adding an include directory to the target `platform_region_defs`, which contains the headers `flash_layout.h` and `region_defs.h`
- Adding startup and scatter files to the `tfm_s`, `tfm_ns` and `bl2` targets.
- linking `CMSIS_5_tfm_ns` to the correct version of the CMSIS RTX libraries, as defined in `lib/ext/CMSIS_5/CMakeLists.txt`
- Adding required source files, include directories and compile definitions to the `platform_s`, `platform_ns` and `platform_bl2` targets.

preload_ns.cmake

This optional cmake file is required only if the target runs the NSPE on a core that requires different compiler options than the SPE core. This file has the same format as `preload.cmake`, but instead details the hardware of the NS core that is **not** running the main TF-M secure code.

install.cmake

This optional cmake file is required only if additional files need to be installed for the platform.

Flash layout header file

Target must provide a header file, called `flash_layout.h`, which defines the information explained in the follow subsections. The defines must be named as they are in the subsections.

BL2 bootloader

The BL2 bootloader requires the following definitions:

- `FLASH_BASE_ADDRESS` - Defines the first valid address in the flash.
- `FLASH_AREA_BL2_OFFSET` - Defines the offset from the flash base address where the BL2 - MCUBOOT area starts.
- `FLASH_AREA_BL2_SIZE` - Defines the size of the BL2 area.
- `FLASH_AREA_SCRATCH_OFFSET` - Defines the offset from the flash base address where the scratch area starts, which is used during image swapping.
- `FLASH_AREA_SCRATCH_SIZE` - Defines the size of the scratch area. The minimal size must be as the biggest sector size in the flash.
- `FLASH_DEV_NAME` - Specifies the flash device used by BL2.

The BL2 requires further definitions depending on the number of images, the meaning of these macros are also slightly different:

- Required definitions in case of 1 image (S and NS images are concatenated and handled together as one binary blob):

- FLASH_AREA_0_OFFSET - Defines the offset from the flash base address where the primary image area starts, which hosts the active firmware image.
- FLASH_AREA_0_SIZE - Defines the size of the primary image area.
- FLASH_AREA_2_OFFSET - Defines the offset from the flash base address where the secondary image area starts, which is a placeholder for new firmware images.
- FLASH_AREA_2_SIZE - Defines the size of the secondary image area.
- Required definitions in case of 2 images (S and NS images are handled and updated separately):
 - FLASH_AREA_0_OFFSET - Defines the offset from the flash base address where the primary image areas start, which host the active firmware images. It is also the offset of the primary (active) secure image area.
 - FLASH_AREA_0_SIZE - Defines the size of the primary secure image area.
 - FLASH_AREA_1_OFFSET - Defines the offset from the flash base address where the primary (active) non-secure image area starts.
 - FLASH_AREA_1_SIZE - Defines the size of the primary non-secure image area.
 - FLASH_AREA_2_OFFSET - Defines the offset from the flash base address where the secondary image areas start, which are placeholders for new firmware images. It is also the offset of the secondary secure image area.
 - FLASH_AREA_2_SIZE - Defines the size of the secondary secure image area.
 - FLASH_AREA_3_OFFSET - Defines the offset from the flash base address where the secondary non-secure image area starts.
 - FLASH_AREA_3_SIZE - Defines the size of the secondary non-secure image area.

The table below shows a fraction of the flash layout in case of 2 and 1 updatable images with the related flash areas that hold the firmware images:

Image number: 2		Image number: 1	
Flash area	Content	Flash area	Content
FLASH_AREA_0	Secure image primary slot	FLASH_AREA_0	Secure + Non-secure image primary slot
FLASH_AREA_1	Non-secure image primary slot		
FLASH_AREA_2	Secure image secondary slot	FLASH_AREA_2	Secure + Non-secure image secondary slot
FLASH_AREA_3	Non-secure image secondary slot		
FLASH_AREA_SCRATCH	Scratch area	FLASH_AREA_SCRATCH	Scratch area

- **IMAGE_EXECUTABLE_RAM_START** - Defines the start of the region to which images are allowed to be loaded. Only used if **MCUBOOT_UPGRADE_STRATEGY** is configured to be **RAM_LOAD**.
- **IMAGE_EXECUTABLE_RAM_SIZE** - Defines the size of the region to which images are allowed to be loaded. Only used if **MCUBOOT_UPGRADE_STRATEGY** is configured to be **RAM_LOAD**.

Assemble tool

The `assemble.py` tool is used to concatenate secure and non-secure binary to a single binary blob. It requires the following definitions:

- **SECURE_IMAGE_OFFSET** - Defines the offset from the single binary blob base address, where the secure image starts.
- **SECURE_IMAGE_MAX_SIZE** - Defines the maximum size of the secure image area.
- **NON_SECURE_IMAGE_OFFSET** - Defines the offset from the single binary blob base address, where the non-secure image starts.
- **NON_SECURE_IMAGE_MAX_SIZE** - Defines the maximum size of the non-secure image area.

Image tool

The `imgtool.py` tool is used to handle the tasks related to signing the binary. It requires the following definition:

- **S_IMAGE_LOAD_ADDRESS** - Defines the address to where the Secure (or combined Secure and Non-secure) image is loaded and is executed from. Only used if **MCUBOOT_UPGRADE_STRATEGY** is configured to be **RAM_LOAD**.
- **NS_IMAGE_LOAD_ADDRESS** - Defines the address to where the Non-secure image is loaded and is executed from. Only used if **MCUBOOT_UPGRADE_STRATEGY** is configured to be **RAM_LOAD** and **MCUBOOT_IMAGE_NUMBER** is greater than 1.

Expose target support for HW components

Services may require HW components to be supported by the target to enable some features (e.g. PS service with rollback protection, etc). The following definitions need to be set in the `.cmake` file if the target has the following HW components:

- **TARGET_NV_COUNTERS_ENABLE** - Specifies that the target has non-volatile (NV) counters.

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9.1.3 / (TF-M root)

This table describes the structure under the root folder with part of possible folders.

Folder name	Description
bl1	The 1st stage immutable bootloader
bl2	MCUBoot based 2nd stage bootloader
cmake	Cmake files of the build system
config	Configuration system files
docs	The documentation
interface	RoT service API for client calls
lib	The 3rd party libraries
<i>platform</i>	Platform intermedia files
<i>secure_fw</i>	The secure firmware
tools	Tools in scripts for building

9.1.4 platform

The *platform* folder contains SW ports of all *supported platforms* and the files necessary for *adding a new platform*. Please refer to *platform folder document* for more information.

Folder name	Description
include	HAL and platform public headers
ext	Platform ports and related files

platform/ext

This folder can include imported files licensed differently from TF-M's BSD-3 license. More details are in the dedicated document *Details for the platform/ext folder*.

Folder name	Description
accelerator	Supported Crypto HW accelerators
cmsis	A copy of essential CMSIS headers
common	Common HAL implementation
driver	Driver headers for porting
target/<vendor>	Vendor specific folders with ported platforms

Each *vendor* is assigned one folder for usage under *target/*. A *vendor* contributes platform's ports and manages the structure inside it.

9.1.5 secure_fw

This folder contains components needed by secure firmware and the exported interfaces for application, service development and HAL integration.

Folder name	Description
include	Public headers of <i>secure_fw</i>
partitions	Default services and SPRTL
spm	PSA FF-M SPM implementation
shared	Sources shared out of SPRTL

The shared sources can be referenced by the building system out of SPRTL. Generally, they are runtime and PSA APIs.

secure_fw/include

This folder holds public headers for external references by clients, services and platforms. Avoid putting private headers, not referenced by other modules in this *include* folder.

secure_fw/partitions

This folder contains default services implemented as partitions and runtime utilities used and provided by TF-M.

Folder name	Description
lib/runtime	The SPRTL sources and intermedia files
<partition_x>	Sources of <i>partition_x</i>
<partition_x>/include	RoT Service API headers of <i>partition_x</i>
<partition_y>	Sources of <i>partition_y</i>
<partition_y>/include	RoT Service API headers of <i>partition_y</i>

Here *partition_x* and *partition_y* are examples of RoT services without detailed structure of them.

secure_fw/spm

The SPM is the core component to provide a mechanism for providing secure services complied with PSA FF-M.

Folder name	Description
include	SPM public headers.
core	SPM base functionalities
*ext	Extended SPM functionalities

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9.2 SPM Backends

This document briefly introduces the backends of Secure Partition Manager (SPM) in TF-M and how to select one for building.

9.2.1 IPC and SFN

The Firmware Framework M (FF-M)^{1, 2} provides two different programming models for Secure Partitions.

- IPC Model

The Secure Partition processes signals in any order, and can defer responding to a message while continuing to process other signals.

- SFN Model

The Secure Partition is made up of a collection of callback functions which implement secure services.

¹ FF-M v1.0 Specification

² FF-M v1.1 Extension

Although the programming model is different, they share the same APIs to interact with the SPM. The behaviours of the APIs share the same implementation with slight differences for the two programming models. This is regarded as the **frontend**.

The TF-M runtime implementations behind **frontend** are different. TF-M provides two backends correspondingly.

- IPC backend

In this backend, the SPM and each Secure Partition have their own execution contexts, which is required to support the IPC model Secure Partitions. This also enables the SPM to provide higher isolation levels. This SPM backend acts like a multiple-process system. It can also adopt SFN model Secure Partitions.

- SFN backend

The SFN backend provides more efficient executions because it shares a single-thread execution context with all the Secure Partitions. This SPM backend acts like a single library. Therefore, it can only adopt SFN model Secure Partitions. And it does not support higher isolation levels. On the other hand, it consumes less memory compared to the IPC backend.

The following table summaries the relationships between SPM backends, Secure Partition models and isolation levels.

SPM backend	Supported Partition model	Supported Isolation Level
SFN	SFN Partition	1
IPC	IPC and SFN Partition	1, 2 and 3

9.2.2 Implementation Recommendations

If an implementation doesn't contain any IPC model Secure Partition and only requires isolation level 1, then it is recommended to select the SFN backend to optimize memory consumption and execution performance.

If an implementation contains any IPC model Secure Partition or requires isolation level 2 or 3, then the IPC backend is required.

9.2.3 TF-M Configuration Switches

In the TF-M build system, the `CONFIG_TFM_SPM_BACKEND` configuration is used to select the backend of SPM. The valid values are SFN and IPC.

```
-DCONFIG_TFM_SPM_BACKEND=SFN
```

If `CONFIG_TFM_SPM_BACKEND` is not set, then IPC is the default value.

9.2.4 References

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9.3 Non-secure Client Extension Integration Guide

This document introduces TF-M Non-secure Client Extension (NSCE) and how to integrate it with Non-secure Processing Environment (NSPE) RTOS.

9.3.1 What is NSCE for

Besides the secure services provided via PSA APIs, there are two interactions between TF-M and NSPE RTOS.

- Non-secure context management in TF-M

When a NS task calls a secure service, a context is maintained in TF-M. If TF-M supports multiple secure service calls, the context needs to be loaded and saved with the corresponding NS task when the RTOS kernel (the kernel) does scheduling.

- Non-secure client ID (NSID) management

As per PSA Firmware Framework specification, NSID is required for a secure service call from the NSPE. The NSID can be managed by either SPM (the same NSID must be used for all connections) or the NSPE RTOS. For the latter case, the NSPE RTOS must provide the NSID for each connection.

NSCE is implemented to support the interactions above.

- Provide the interface to NSPE RTOS for managing the context in TF-M.
- Provide a mechanism to NSPE RTOS for providing the NSID for each connection.

9.3.2 NSCE component diagram

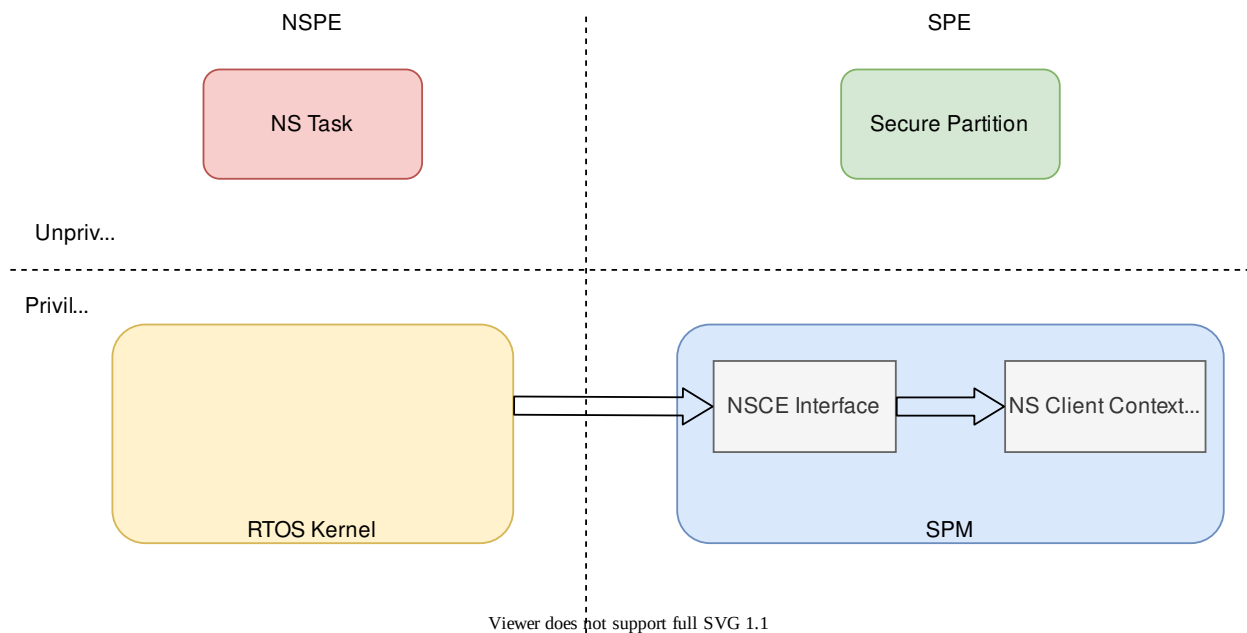


Figure 2:: Non-secure Client Extension Component Diagram

This diagram shows the components of NSCE.

- **NSCE interface:** The NSCE API interface to NSPE RTOS.
- **NS Client Context Manager:** It is the internal component of SPM that is used to manage the NS client context. It is not accessible to NSPE.

9.3.3 Group-based context management

NSCE introduces a group-based context management for NS clients. The purpose is to support diverse use scenarios in the IoT world.

For rich-resource devices, TF-M could assign separate contexts for each connection. For resource-constrained devices, TF-M may support very limited context slots. Multiple NS connections could share the same context provided they have mutual exclusion access to the secure services.

To support this flexibility, a group ID (*gid*) and a thread ID (*tid*) are specified for the connection of the NS client. NSCE allocates only one context for a given group. Threads in the same group share the context.

The *gid* and *tid* are specified by NSPE RTOS via NSCE interface.

9.3.4 NSCE interface

NSCE defines a set of APIs as the interface for NSPE RTOS to manage the context in TF-M. NSID is provided as an input parameter of *tfm_nsce_load_ctx()*.

NSCE APIs are typically called by the kernel and must be called from the non-secure handler mode.

```
uint32_t tfm_nsce_init(uint32_t ctx_requested)
```

This function should be called before any other NSCE APIs below to do non-secure context initialization in NSCE for the calling NS entity. *ctx_requested* is the number of the contexts requested by the NS entity. The number of assigned context will be returned. It may be equal to or smaller than the requested context number based on the current available resource in TF-M. If *ctx_requested* is 0, then the maximum available context number will be assigned and returned. If the initialization is failed, 0 should be returned.

Note

As TF-M only supports one context for now, so the return value is always 1 if no error. Currently, it is safe to skip calling *tfm_nsce_init()*. But, for future compatibility, it is recommended to do so.

```
uint32_t tfm_nsce_acquire_ctx(uint8_t group_id, uint8_t thread_id)
```

This function allocates a context for the NS client connection. The *gid* and *tid* are the input parameters. A token will be returned to the NSPE if TF-M has an available context slot. Otherwise, *TFM_NS_CLIENT_INVALID_TOKEN* is returned. It is the responsibility of NSPE RTOS to assign *gid* and *tid* for each NS client.

```
uint32_t tfm_nsce_release_ctx(uint32_t token)
```

This function tries to release the context which was retrieved by *tfm_nsce_acquire_ctx()*. As the context may be shared with other threads within the same group, the context is only freed and back to the available context pool after all threads in the same group release the context.

```
uint32_t tfm_nsce_load_ctx(uint32_t token, int32_t nsid)
```

This function should be called when NSPE RTOS schedules in a NS client. *token* is returned by *tfm_nsce_acquire_ctx()*. *nsid* is the non-secure client ID used for the following PSA service calls.

The assignment of NSID is managed by the NSPE RTOS. It is not required to use the same NSID for a NS client when calling this function each time. This allows the NS client changing its NSID in the lifecycle. For example, the provisioning task may need to switch NSID to provision the keys for different NS clients created afterwards.

```
uint32_t tfm_nsce_save_ctx(uint32_t token)
```

This function should be called when NSPE RTOS schedules out a NS client. The input parameter *token* is returned by *tfm_nsce_acquire_ctx()*. After the context is saved, no secure service call can be made from NSPE until a context is loaded via *tfm_nsce_load_ctx()*.

9.3.5 NSCE integration guide

Enable NSCE in TF-M

To enable NSCE in TF-M, set the build flag *TFM_NS_MANAGE_NSID* to *ON* (default *OFF*).

Support NSCE in an RTOS

For supporting NSCE in an RTOS, the integrator needs to do the following major work:

- Integrate the NSCE API calls into the NS task lifecycle. For example, creating/scheduling/destroying a NS task.
- Manage the assignment for *gid*, *tid* and NSID.

The typical steps are listed below:

- Before programming, the integrator should plan the group assignment for the NS tasks that need to call secure services. If the number of tasks is less than or equal to TF-M non-secure context slots, then different *gid* can be assigned to each task for taking dedicated context in TF-M. Otherwise, the integrator need to think about grouping the tasks to share the context.
- In the kernel initialization stage, it calls *tfm_nsce_init()* with requested context number to initialize the non-secure context in TF-M. The actual allocated context number will be returned. *0* means initialization failed. The kernel could use different group assignment sets according to the context number allocated to it. TF-M only supports one context for now.
- The kernel calls *tfm_nsce_acquire_ctx()* when creating a new task. This should be done before the new task calls any secure service. A valid token returned should be saved as part of the task context in NSPE RTOS.
- When the kernel schedules in a task with a valid *token* associated, *tfm_nsce_load_ctx()* should be called before resuming the execution of that task. The NSID is specified by the kernel through the *nsid* parameter. The mapping between NSID and task is managed by the kernel. *tfm_nsce_load_ctx()* can be called multiple times without calling *tfm_nsce_save_ctx()* for switching the NSID for the same task (same *tid* and *gid*).
- *tfm_nsce_save_ctx()* should be called if the current task has a valid *token* before being switched to another task. Calling *tfm_nsce_load_ctx()* for another task before saving the current context may result in NS context lost in TF-M for the running task.
- When the task is terminated, destroyed or crashed, the kernel should call *tfm_nsce_release_ctx* to make sure the associated resource is back to the pool in TF-M.

Integration example

This is the software module diagram of a typical RTOS/NSCE integration example.

- Built-in Secure Context Manager: An RTOS may have an existing built-in secure context manager with a group of secure context management APIs defined. Let's take RTX which uses [Armv8-M TrustZone APIs](#) as the example.

Note

RTOS may define the NS task context in the secure side as “secure context”. It is the same thing as the “non-secure context” (context for a secure service call from NS client) from TF-M’s point of view.

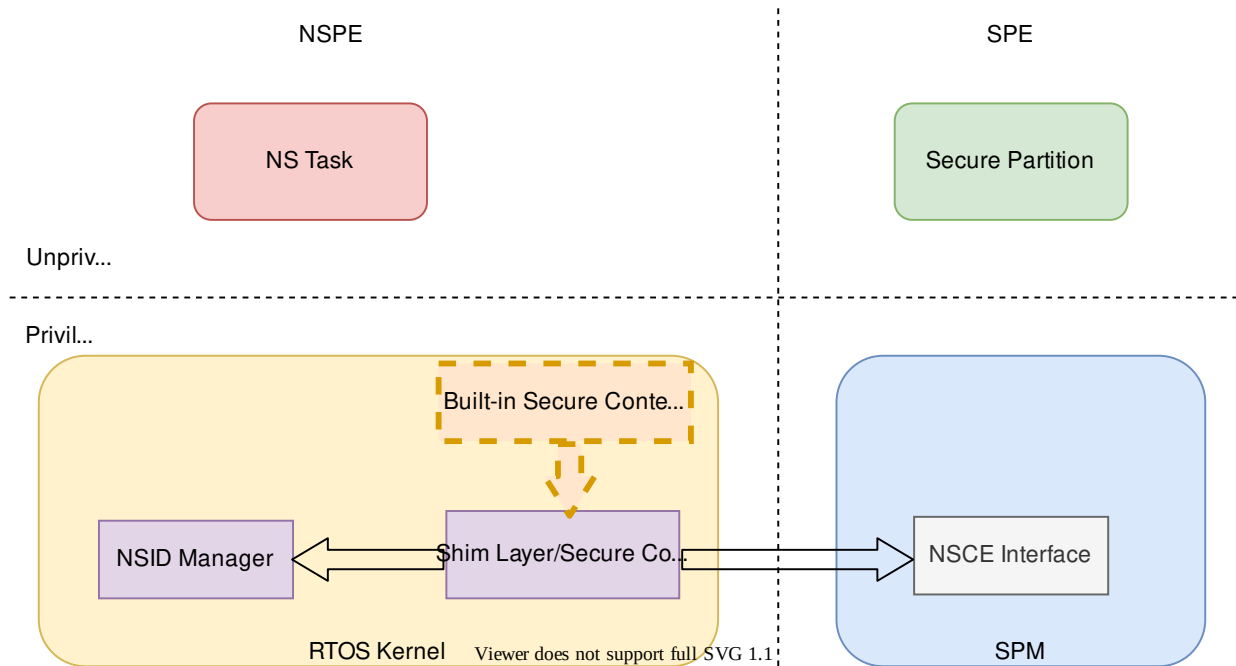


Figure 3:: RTOS/NSCE integration example

- **Shim Layer/Secure Context Manager:** If the RTOS has a “Built-in Secure Context Manager”, then a shim layer should be provided to translate the built-in API calls into the NSCE API calls. A reference shim layer for RTX TrustZone APIs is [here](#).

If the RTOS has no existing one, then a “Secure Context Manager” should be implemented based on the NSCE APIs. You can refer to the shim layer example above for the implementation. The timing of calling NSCE APIs in the kernel is introduced in the *Support NSCE in an RTOS* section above.

- **NSID Manager:** If NSPE RTOS manages the NSID, then this module is used by the secure context manager or shim layer to manage the NSID assignment for the NS tasks. The assignment is implementation defined. A [task name based NSID manager](#) is provided as a reference.

Integration notes

- **gid:** It is a `uint8_t` value (valid range is 0 - 255). So, maximum 256 groups (NSCE context slots) are supported by the NSCE interface. TF-M only supports single context for now. So, it is recommended to use single group ID at this stage.
- **tid:** It is a `uint8_t` value (valid range is 0 - 255). Thread ID is used to identify a NS client within a given group. *tid* has no special meaning for TF-M. So, usually the kernel only needs to ensure a NS task has a unique *tid* within a group.
- **gid and tid management:** It is the responsibility of NSPE RTOS to manage the assignment of *gid* and *tid*. Based on the explanation above, the *gid* could be assigned as a constant value. And, the *tid* can be increased globally when calling `tfm_nsce_acquire_ctx()` for a new task. Just notice *tid* may overflow.
- **NSID management:** It is the responsibility of NSPE RTOS to manage the assignment of the NSID for each task. It is highly recommended that a NS client uses the same NSID following a reboot or update. The binding of a NS client to a specific NSID will ensure the correct access to the assets. For example, the data saved in the protected storage.
- **Integrate with the existing secure context management APIs of NSPE RTOS:** The NSCE APIs are designed to

be compatible with most known existing secure context management APIs. A shim layer is needed to translate the API calls. See the integration example above.

- `tfm_nsce_acquire_ctx()` must be called before calling `tfm_nsce_load_ctx()`, `tfm_nsce_save_ctx()` or `tfm_nsce_release_ctx()`.
- `tfm_nsce_release_ctx()` can be called without calling `tfm_nsce_save_ctx()` ahead.

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9.4 Generic OS migration from Armv7-M to Armv8-M architecture

The purpose of this document is to list a set of requirements needed for migrating a generic OS kernel running on Armv7-M to the Armv8-M architecture.

9.4.1 List of requirements

- If the same OS codebase is used for both Secure and Non Secure builds, it is suggested to put specific code targeted to the Non Secure build under a compile time switch, e.g. `#if (DOMAIN_NS == 1U)`. The OS build system in this case needs to be amended accordingly to support this new switch.
- If the OS implements stack limit checking, the PSPLIM register needs to be initialized and properly handled during thread context switch operations.
- If the OS manipulates directly the Link Register, the default Link Register value used in Handler mode transitions needs to be differentiated between Secure and Non Secure builds, i.e. `0xFD` and `0xBC`, respectively.
- If the OS manages the non-secure client identification, please check the *Non-secure Client Extension Integration Guide*.

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9.5 Floating-Point Support

TF-M adds several configuration flags to control Floating point (FP)¹ support in TF-M Secure Processing Environment (SPE) and Non Secure Processing Environment (NSPE).

- Support FP in SPE or NSPE.
- Support FP Application Binary Interface (ABI)² types: software, hardware. SPE and NSPE shall use the same FP ABI type.
- Support lazy stacking enable/disable in SPE only, NSPE is not allowed to enable/disable this feature.
- Support GNU Arm Embedded Toolchain³. GNU Arm Embedded Toolchain 10.3- 2021.10 and later version shall be used to mitigate VLLDM instruction security vulnerability⁴.
- Support both IPC⁵ and SFN¹¹ models in TF-M.
- Support Armv8-M mainline.

¹ High-Performance Hardware Support for Floating-Point Operations

² Float Point ABI

³ GNU Arm Embedded Toolchain

⁴ VLLDM instruction Security Vulnerability

⁵ Arm® Platform Security Architecture Firmware Framework 1.0

¹¹ FF-M v1.1 Extension

- Support isolation level 1,2,3.
- Support Arm Compiler for Embedded¹⁰. Arm Compiler for Embedded 6.17 and later version shall be used to mitigate VLLDM instruction security vulnerability⁷.
- Does not support use FPU in First-Level Interrupt Handling (FLIH)⁶ at current stage.

Please refer to Arm AN521 or AN552 platform as a reference implementation when you enable FP support on your platforms.

Note

Alternatively, if you intend to use FP in your own NSPE application but the TF-M SPE services that you enable do not require FP, you can set the CMake configuration `CONFIG_TFM_ENABLE_CP10CP11` to ON and **ignore** any configurations described below.

Note

`CONFIG_TFM_DISABLE_CP10CP11` can be set to ON to disable CP10/CP11 coprocessors in cases when it's expected that FPU must not be used neither by secure nor by non-secure images.

Note

GNU Arm Embedded Toolchain 10.3-2021.10 may have issue that reports `'-mcpu=cortex-m55'` conflicts with `'-march=armv8.1-m.main'` warning⁸. This issue has been fixed in the later version.

9.5.1 FP ABI type for SPE and NSPE

FP design in Armv8.0-M⁹ architecture requires consistent FP ABI types between SPE and NSPE. Furthermore, both sides shall set up CPACR individually when FPU is used. Otherwise, No Coprocessor (NOCP) usage fault will be asserted during FP context switch between security states.

Secure and non-secure libraries are compiled with `COMPILER_CP_FLAG` and linked with `LINKER_CP_OPTION` for different FP ABI types. All those libraries shall be built with `COMPILER_CP_FLAG`.

If FP ABI types mismatch error is generated during build, please check whether the library is compiled with `COMPILER_CP_FLAG`. Example:

```
target_compile_options(lib
PRIVATE
    ${COMPILER_CP_FLAG}
)
```

9.5.2 CMake configurations for FP support

The following CMake configurations configure `COMPILER_CP_FLAG` in TF-M SPE.

- `CONFIG_TFM_ENABLE_FP` is used to enable/disable FPU usage.

¹⁰ Arm Compiler for Embedded

⁶ Secure Interrupt Integration Guide

⁸ GCC Issue on `'-mcpu=cortex-m55'` conflicts with `'-march=armv8.1-m.main'` Warning

⁹ Armv8-M Architecture Reference Manual

CONFIG_TFM_ENABLE_FP	FP support
off (default)	FP disabled
on	FP enabled

Note

CONFIG_TFM_FLOAT_ABI depends on CONFIG_TFM_ENABLE_FP. If CONFIG_TFM_ENABLE_FP is set to on, CONFIG_TFM_FLOAT_ABI is automatically set to hard.

Note

If TF-M SPE is built with CONFIG_TFM_ENABLE_FP=on, then SPE takes care of enabling floating point coprocessors CP10 and CP11 on the NS side. This is done to avoid NOCP usage fault.

- CONFIG_TFM_LAZY_STACKING is used to enable/disable lazy stacking feature. This feature is only valid for FP hardware ABI type. NSPE is not allowed to enable/disable this feature. Let SPE decide the secure/non-secure shared setting of lazy stacking to avoid the possible side-path brought by flexibility.

CONFIG_TFM_LAZY_STACKING	Description
OFF	Disable lazy stacking
ON (default)	Enable lazy stacking

- CONFIG_TFM_FP_ARCH specifies which FP architecture is available on the target, valid for FP hardware ABI type.

FP architecture is processor dependent. For GNUARM compiler, example value are: auto, fpv5-d16, fpv5-sp-d16, etc. For armclang, example value are: none, softvfp, fpv5-d16, fpv5-sp-d16, etc.

This parameter shall be specified by platform in preload.cmake. Please check compiler reference manual and processor hardware manual for more details to set correct FPU configuration for platform.

- CONFIG_TFM_FP_ARCH_ASM specifies the target FPU architecture name shared by Arm Compiler armasm and armlink. It is only used in the --fpu= argument by Arm Compiler and shall be aligned with CONFIG_TFM_FP_ARCH.

FP architecture is processor dependent. For armasm and armlink, example value are: SoftVFP, FpV5_D16, etc.

This parameter shall be specified by platform in preload.cmake. Please check compiler reference manual and processor hardware manual for more details to set correct FPU configuration for platform.

Reference

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9.6 Secure Interrupt Integration Guide

9.6.1 Introduction

This document describes how to enable an interrupt in TF-M. The target audiences are mainly platform integrators and Secure Partition developers.

This document assumes that you have read the PSA Firmware Framework (FF-M) v1.0¹ and the FF-M v1.1 extensions² thus have knowledge on the terminologies such as Secure Partitions and manifests.

9.6.2 Interrupt Handling Model

TF-M supports the two interrupt handling models defined by FF-M:

- First-Level Interrupt Handling (FLIH)

In this model, the interrupt handling is carried out immediately when the interrupt exception happens.

The interrupt handling can optionally set an interrupt signal for the Secure Partition Thread to have further data processing.

- Second-Level Interrupt Handling (SLIH)

In this model, the interrupt handling is deferred after the interrupt exception. The handling occurs in the Secure Partition Thread thus is subject to scheduling.

The FLIH supports handling an interrupt in a bounded time, but very limited APIs are allowed in the FLIH handling because the handling occurs in an special exception context.

The SLIH is deferred and subject to scheduling but all Secure Partition APIs are allowed as the SLIH handling is in the Secure Partition Thread.

Both the FLIH and the SLIH can be used by Secure Partitions which conform to Firmware Framework v1.1.

While the SLIH is the only supported model for Secure Partitions which conform to Firmware Framework v1.0.

Please refer to chapter 6.2 of FF-M v1.1³ for more details on the interrupt handling models.

9.6.3 Enabling an Interrupt

To enable an interrupt, you need to do the following:

- Binding the interrupt to a Secure Partition.
- Granting the Secure Partition access permissions to the device of the interrupt.
- Initializing the interrupt.
- Integrating the interrupt handling function

TF-M has two Test Partitions as good examples for both FLIH³ and SLIH⁴. See also *Enabling the Interrupt Tests* on how to integrate them to platforms.

Binding an Interrupt to a Secure Partition

To bind an interrupt to a Secure Partition, you need to add an item to the `irqs` attribute of the Secure Partition manifest. `irqs` is a list of Interrupt Request (IRQ) assigned to the Secure Partition.

Secure Partitions are not allowed to share IRQs with other Secure Partitions.

Different Firmware Framework versions have different definitions of manifest.

¹ FF-M v1.0 Specification

² FF-M v1.1 Extension

³ https://git.trustedfirmware.org/plugins/gitiles/TF-M/tf-m-tests.git/+HEAD/tests_reg/test/secure_fw/suites/spm/irq/service/tfm_flih_test_service/

⁴ https://git.trustedfirmware.org/plugins/gitiles/TF-M/tf-m-tests.git/+HEAD/tests_reg/test/secure_fw/suites/spm/irq/service/tfm_slih_test_service/

FF-M v1.0

Here is an example manifest of Secure Partitions conform to Firmware Framework version 1.0:

```
{
  "irqs": [
    {
      "source": 5,
      "signal": "DUAL_TIMER_SIGNAL"
    },
    {
      "source": "TIMER_1_SOURCE",
      "signal": "TIMER_1_SIGNAL"
    }
  ]
}
```

- source

Required, Unique.

The `source` is a string that identifies the interrupt source. It can be a valid exception number or a symbolic name defined in platform codes.

- signal

Required, Unique.

The `signal` attribute is a symbolic name used by TF-M to identify which interrupt is asserted. It is also used by the Secure Partition to receive the interrupt signal by calling `psa_wait` for interrupt handling.

It is defined in the Secure Partition header file `<psa_manifest/manifestfilename.h>` generated by TF-M:

```
#define signal VALUE
```

The interrupt handling model is SLIH by default as it is the only supported one for FF-M v1.0.

FF-M v1.1

Here is an example manifest of Secure Partitions conform to Firmware Framework version 1.1:

```
{
  "irqs": [
    {
      "source" : "TIMER_1_SOURCE",
      "name" : "TIMER_1",
      "handling": "FLIH"
    },
    {
      "source" : 5,
      "name" : "DUAL_TIMER",
      "handling": "SLIH"
    }
  ]
}
```

- source

The source is the same as the one in Firmware Framework Version 1.0.

- name

Required, Unique.

The name is used to construct the following two elements:

- the interrupt signal symbol: `{{name}}_SIGNAL`, the equivalent of `signal` in FF-M v1.0
- the FLIH Function for handling FLIH IRQs provided by Secure Partition:

```
psa_flih_result_t {{name}}_flih(void);
```

It is also declared in `<psa_manifest/manifestfilename.h>`.

- handling

Required.

The handling attribute specifies the interrupt handling model and must have one of the following values:

- FLIH - First-Level Interrupt Handling
- SLIH - Second-Level Interrupt Handling

Granting Permissions to Devices for Secure Partitions

A secure partition shall be granted two parts of permissions to access a device. One is the Memory Mapped I/O (MMIO) region of the device. The other is the driver codes to access the device.

The MMIO Regions

You need to declare the MMIO region in the `mmio_regions` attributes in the Secure Partition manifest, to enable the Secure Partition to access it.

An MMIO region can be described as either `numbered_region` or `named_region`. A numbered region consists of a base address and a size. A named region consists of a string `name` to describe the region.

Here is an example of named region:

```
{
  "mmio_regions": [
    {
      "name": "TFM_PERIPHERAL_TIMER0",
      "permission": "READ-WRITE"
    }
  ]
}
```

- name

Required.

The name attribute is a symbolic name defined by platforms. It is a pointer to structure instance that usually includes the base address and size of the region and some other platform specific attributes that are required to set up permissions to the MMIO region.

The structure is defined by platforms and the name must be `struct platform_data_t`.

- permission

Required.

The permission attribute must have one of the following values:

- READ-ONLY
- READ-WRITE

The Device Drivers

To give permissions of devices drivers to Secure Partitions, it is recommended to put the driver codes to the Partition's CMake library:

```
target_sources(some_partition_lib
PRIVATE
    some_driver_code.c
)
```

Initializing the Interrupts

Platforms must define an interrupt initialization function for each Secure interrupt.

The prototype of the function is:

```
enum tfm_hal_status_t {source_symbol}_init(void *p_pt,
                                           const struct irq_load_info_t *p_ildi)
```

The {source_symbol} is:

- irq_{source}, if the source attribute of the IRQ in Partition manifest is a number
- Lowercase of source attribute, if source is a symbolic name

For example if the manifest declares "source": 5, then the function name is irq_5_init. If the manifest declares "source" : "TIMER_1_IRQ" then the function name is timer_1_irq_init.

The function will be called by the Framework automatically during initialization. The function can be put in any source file that belongs to SPM, for example a tfm_interrupts.c added to the tfm_spm CMake target.

The initialization of an interrupt must include:

- setting the priority
- ensuring that the interrupt targets the Secure State.
- saving the interrupt information

Setting Priority

The priority of external interrupts must be in the following range: (0, N / 2), where N is the number of configurable priorities. Smaller values have higher priorities.

For example if the number of configurable priority of your interrupt controller is 16, you must use the priorities in range (0, 8) only, boundaries excluded.

Note that these are not the values set into the interrupt controllers. Different platforms may have different values for those priorities. But if you use the NVIC_SetPriority function provided by CMSIS to set priorities, you can pass the values directly.

Platforms have the flexibilities on the assignment of priorities.

Targeting Interrupts to Secure

In single core systems, platform integrators must ensure that the Secure interrupts target to Secure State by setting the Interrupt Controller.

In multi-core systems, this might be optional.

Saving the Interrupt Information

The initialization function is called during Partition loading with the following information:

- `p_pt` - pointer to Partition runtime struct of the owner Partition
- `p_ildi` - pointer to `irq_load_info_t` struct of the interrupt

Platforms must save the information for the future use. See *Integrating the Interrupt Handling Function* for the usage.

The easiest way is to save them in global variables for each interrupt. TF-M provides a struct for saving the information:

```
struct irq_t {
    void *p_pt;
    const struct irq_load_info_t *p_ildi;
};
```

Integrating the Interrupt Handling Function

TF-M provides an interrupt handling entry for Secure interrupts:

```
void spm_handle_interrupt(void *p_pt, const struct irq_load_info_t *p_ildi)
```

The `p_pt` and `p_ildi` are the information passed to interrupt initialization functions and saved by platforms.

Platforms should call this entry function in the interrupt handlers held in Vector Table with the information saved by the interrupt initialization functions. If the information is saved as global variables, then the interrupt handlers can be put in the same source file that contains the initialization functions.

Here is an example:

```
void TFM_TIMER0_IRQ_Handler(void) /* The handler in Vector Table */
{
    spm_handle_interrupt(p_timer0_pt, p_tfm_timer0_irq_ldinf);
}
```

9.6.4 Enabling the Interrupt Tests

TF-M provides test suites for FLIH and SLIH interrupts respectively. They are disabled by default.

Note

FLIH interrupt test and SLIH interrupt test share the same timer `TFM_TIMER0_IRQ` thus cannot be enabled at the same time.

To enable the tests, please follow steps in the previous sections. In addition, you need to implement the following APIs of timer control:

- `void tfm_plat_test_secure_timer_start(void)`
- `void tfm_plat_test_secure_timer_clear_intr(void)`

- `void tfm_plat_test_secure_timer_stop(void)`

You shall also select the following flags in platform specific `config.cmake` to indicate that FLIH and SLIH interrupt tests are supported respectively.

- `PLATFORM_FLIH_IRQ_TEST_SUPPORT`: platform implements support of FLIH interrupt tests
- `PLATFORM_SLIH_IRQ_TEST_SUPPORT`: platform implements support of SLIH interrupt tests

The following configurations control SLIH and FLIH interrupt tests:

- `TEST_NS_FLIH_IRQ`
- `TEST_NS_SLIH_IRQ`

They can be enabled via build command line or via `TEST_NS`.

9.6.5 Migrating to Firmware Framework v1.1

Please refer to [Migrating Secure Partitions to version 1.1 of FF-M v1.1](#)[?].

9.6.6 References

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9.7 Platform Provisioning

TF-M stores any data that should be provisioned at the factory in OTP memory. The default is that this OTP memory is actually implemented using on-chip flash, the same that is used to implement the ITS service.

If the lifecycle state is in the `TFM_SLC_ASSEMBLY_AND_TEST`¹ state (which is the default for non-provisioned boards), then TF-M will attempt to provision the: - HUK instead of booting. It will read the data from the `assembly_and_test_prov_data` struct, and will then provision it to OTP. The lifecycle state will then transition to `TFM_SLC_PSA_ROT_PROVISIONING`[?].

If the lifecycle state is in the `TFM_SLC_PSA_ROT_PROVISIONING`[?] state, then TF-M will attempt to provision the:

- IAK
- boot seed
- implementation id
- certification reference
- bl2 ROTPKs (of which there are up to 4)
- entropy seed

Once all of these have been loaded from the `psa_rot_prov_data` struct and provisioned to OTP, the LCS will transition to `TFM_SLC_SECURED`[?]. Note that this provisioning step will be run immediately after the `TFM_SLC_ASSEMBLY_AND_TEST`[?] provisioning stage if the lifecycle transition completed successfully.

¹ For the definitions of these lifecycle states, please refer to the Platform Security Model <https://developer.arm.com/documentation/den0128/0100/>

9.7.1 Provisioning development hardware

If `TFM_DUMMY_PROVISIONING` is enabled in the cmake config (as it is by default), a set of dummy keys / data will be provisioned. The dummy IAK matches the IAK tested by the TF-M tests, and the dummy bl2 ROTPKs match the dummy bl2 keys used by default. `TFM_DUMMY_PROVISIONING_MUST_` not be used in production hardware, as the keys are insecure.

9.7.2 Provisioning production hardware

For provisioning of real hardware, firstly `TFM_DUMMY_PROVISIONING` must be disabled. Then it is required to inject the keys into RAM so they populate the `assembly_and_test_prov_data` and `psa_rot_prov_data` structs, at the beginning of the TF-M boot. These structs each require a magic value to be set to be accepted by the provisioning code, which is detailed in `platform/ext/common/provisioning.c`. Two suggestions for how to do this are:

- Attach a debugger, and inject the values into RAM.
- Flash an image that contains the required data. Care must be taken with this approach that the keys are not left in RAM after provisioning, so a different image (without provisioning data embedded) must be flashed afterwards, without erasing the OTP flash area.

Provisioning on CryptoCell-312 enabled platforms

On boards that have a CC312 accelerator, and that have the default flash-backed OTP disabled by setting `PLATFORM_DEFAULT_OTP=OFF` in cmake, the CC312 OTP will be used as a backing for the OTP HAL.

Due to the CC312 requiring a power-cycle to transition LCS, you will be prompted to manually power-cycle the board between provisioning stages.

Boards with real OTP memory cannot be reprovisioned - care should be taken that the data being provisioned is the desired data.

Platform-specific OTP backing

If a platform has a medium that is suitable for storing data with OTP semantics (Where a bit cannot transition from a 1 to a 0), such as physical OTP memory, then it can provide a backing for the OTP HAL by implementing the methods described in `tfm_plat_otp.h`.

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9.8 Branch Protection

The Branch Protection is an optional feature that can help with mitigation of common exploits through undesired branching in code.

The Branch Protection feature leverages the Pointer Authentication and Branch Target Identification extension (PACBTI) which is optional and available in Armv8.1-M architecture.

PACBTI is designed to mitigate Return-Oriented Programming (ROP) and Jump-Oriented Programming (JOP) security exploit attacks.

9.8.1 Pointer authentication

Pointer authentication (PAC) is a technique by which the pointer is “signed”. This signed pointer is generated by the hardware by combining a modifier, a secret key and the pointer itself through a hardware cryptographic mechanism.

The generated signed pointer is produced at the beginning of a subroutine or function. On returning from the subroutine/function another signed pointer is generated and compared with the previous signed pointer. If they match, then the pointer (return address) has not been tampered and there are no side-effects. Otherwise the validating instruction generates a UsageFault exception.

All the above functionality is achieved by introducing specific instructions at the beginning and at the end of the subroutine/function. This addition is controlled by compiler options, when supported.

To enable PAC in your platform, TF-M supports the following build options:

- `BRANCH_PROTECTION_PACRET`
- `BRANCH_PROTECTION_PACRET_LEAF`

One of the above values can be assigned to `CONFIG_TFM_BRANCH_PROTECTION_FEAT`

`BRANCH_PROTECTION_PACRET_LEAF` extends `BRANCH_PROTECTION_PACRET` when PAC is desired also in leaf functions.

9.8.2 Branch target identification

Branch Target Identification (BTI) can mitigate some of the JOP attacks so that indirect branches can only allow target instructions. Such instructions are also called landing pads. In other words, an indirect branch can only “land” where a specific instruction is present, otherwise the branch has been compromised and a UsageFault exception is raised.

The above functionality is achieved by introducing specific instructions at the beginning of executable code.

To enable BTI in your platform, TF-M supports the following build option:

- `BRANCH_PROTECTION_BTI`

One of the above values can be assigned to `CONFIG_TFM_BRANCH_PROTECTION_FEAT`

9.8.3 PAC and BTI

It is possible to have some combinations of the PAC and BTI options above. The allowed values for `CONFIG_TFM_BRANCH_PROTECTION_FEAT` are listed below, but only one option can be used:

- `BRANCH_PROTECTION_DISABLED` - PACBTI feature is disabled
- `BRANCH_PROTECTION_NONE` - PACBTI instructions are not generated
- `BRANCH_PROTECTION_STANDARD` - Enables pointer authentication and branch target identification, leaf functions excluded
- `BRANCH_PROTECTION_PACRET` - Enables pointer authentication only, leaf functions excluded
- `BRANCH_PROTECTION_PACRET_LEAF` - Enables pointer authentication on all functions, including leaf functions
- `BRANCH_PROTECTION_BTI` - Enables branch target identification only

Note that PACBTI is conditional to both architecture support and compiler support.

The build system performs checks to ensure that support is available and terminates the build process if the required conditions are not met.

9.8.4 Limitations

Currently PACBTI is supported only for Arm Compiler. Even though GNUARM supports PACBTI, some features are missing that prevent all the component integration to successfully work together. It may be possible in the future to extend support for other compilers.

For further details on PACBTI features refer to:

[Armv8-M Architecture Reference Manual](#)

[Armv8.1-M Pointer Authentication and Branch Target Identification Extension](#)

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9.9 Adding a new platform

9.9.1 Porting TF-M to a New Hardware

The purpose of this document is to provide a guide on how to integrate TF-M with another hardware platform. This document will give general guidance on how to port a platform on the TF-M build system and which interfaces must exist on the platform for TF-M (S and NS) to run on this new platform.

Prerequisites

Building environment

Make sure you have a working build environment and that you can build TF-M on AN521 following the *Build instructions*.

Toolchains and software requirements

Please follow the Getting started guide.

CMSIS Drivers

The TF-M stack requires at least two CMSIS HAL implementations:

- [USART](#)
- [FLASH](#)

Porting flow

In a nutshell, this should be a 6 iterative steps process:

1. Adding all the mandatory files and expected objects/functions declarations
2. Booting and configuring the core(s)
 - startup(s) code and SystemInit
3. Adding the USART drivers
 - CMSIS HAL
4. Adding the FLASH drivers
 - CMSIS HAL

5. Enabling/Configuring/Disabling features including templated features

- E.G. NV Counters, attestation, crypto keys....

6. Adding the optional platform SVC handling

Some platforms may have their own SVC requests in addition to the TF-M built-in ones.

7. Running the regression tests

- See Running TF-M on Arm platforms as an example

File architecture

The platform selection when building TF-M is set via the CMake variable `TFM_PLATFORM`. This variable holds part of the path to the platform. When using `-DTFM_PLATFORM=arm/mps2/an521` or `-DTFM_PLATFORM=an521` TF-M build system will look for the platform in `<TF-M ROOT>/platform/ext/target/arm/mps2/an521`. Therefore all hardware dependent code for your platform should go to `<TF-M ROOT>/platform/ext/target/`.

platform/ext/target

This folder contains a first level of board vendor (such as ARM, STM, NXP, Cypress ...), each folder will usually contain a second level for each board. This second level is not mandatory.

`platform/ext/target/<vendor>/[<board name>/]`

From now on this will be referred to as the `platform` folder.

platform/ext/common

This folder contains files and folder commons to the platforms, such as the shims to the CMSIS drivers. It also contains the scatter files that can be used for the bl2, tfm_s, tfm_ns partitions.

This folder also contains another folder named `template`. The latter contains example implementations that are used for platforms by default, but which can be altered or replaced if other functionality is required.

name	description
<code>PLAT-FORM_DEFAULT_ATTEST_HAL</code>	Use the default implementation of the attestation HAL (default True)
<code>PLAT-FORM_DEFAULT_NV_COUNTERS</code>	Use the default implementation of the counters in NV (default True)
<code>PLAT-FORM_DEFAULT_CRYPTOKEYS</code>	Use the default implementation of crypto keys (default True)
<code>PLATFORM_DEFAULT_ROTTPK</code>	Use the default implementation of the RoT Public Key (default True)
<code>PLATFORM_DEFAULT_IATK</code>	Use the default implementation of the initial attestation key (default True)
<code>PLAT-FORM_DEFAULT_UART_STDOUT</code>	Use the default implementation of the uart for stdout output (default True)
<code>PLAT-FORM_DEFAULT_NV_SEED</code>	Use the default implementation of the NV seed in the RNG (default True)
<code>PLATFORM_DEFAULT_OTP</code>	Use the default implementation of the OTP (default True)

Platform Folder

Description

Depending on the level of integration you want with TF-M some files or information will be mandatory for the build system to build working firmware.

Please note that platform folder provides source for building both *SPE* and *NSPE* parts. The *SPE* builds directly from the source tree while files necessary for *NSPE* platform support are installed to <Artifact folder> for building TF-M application as described in the *Build instructions*.

Questions to be answered:

- Will the platform use MCUboot as the second stage bootloader?

BL2/MCUboot provides a secure bootloader that enables simple software upgrades.

This optional second stage bootloader is set-up via the bl2 target in the CMakeLists.txt file (see below).

- Will the platform support the Non-Secure world application?

A platform can be designed to only support the secure world, in which case we would refer to it as a secure enclave. TF-M build system allows the developer to strip all Non-Secure world related code out of the final image. Most platforms, and especially the ones intended to be generic or to have a Non-Secure application will require Non-Secure world support. In that case a platform shall instruct build system on the file set for exporting to Non-Secure world.

- How does the non-secure world communicate with the secure world?

TF-M supports running the non-secure world on the same CPU as the secure world, communicating via TrustZone or running the non-secure world on a separate CPU, communicating via a mailbox. The platform is responsible for configuring toolchains with correct CPU and architecture related features for secure and non-secure worlds.

The architecture for secure world is configured in the cpuarch.cmake file (see below).

- How does the FLASH need to be split between worlds?

The flash split is very dependent on the support of BL2 and NS world. When porting a new platform, one shall arrange enough flash size for each of them.

If supporting upgrades (via MCUboot), additional flash area will be required to store the updates before upgrading the whole system.

- How does the RAM need to be split between worlds?

The RAM split is very dependent on the support of the NS world.

If you're not porting the platform for a specific project but are enabling the Non-Secure world, you should ensure that you leave enough RAM available for it to run.

Note

TF-M S world size in RAM and Flash varies greatly with different build options.

TF-M project provides [metrics](#) of the S world size for existing platforms, which may help to get a rough guide to the sizes needed.

Files

CMakeLists.txt :

(MANDATORY)

This is the entry point for the build system to build your platform on the secure side and also export files to build Non-Secure side.

it must:

- Add a folder to the target platform_region_defs. [*PLATFORM_REGION_DEFS*]

This folder will contain two files *flash_layout.h* and *region_defs.h*

- Add scatter files to the bl2 and tfm_s targets. [*SCATTER*]

Please note that TF-M provides a common scatter file for the bl2, tfm_s and tfm_ns targets, which can be used in most cases.

- Add startup files to the bl2 and tfm_s targets. [*STARTUP*]
- Add required sources and includes for the bl2 and tfm_s targets [*SOURCES_INCLUDES*]
- Install all files required for building the platform in the Non-secure application [*INSTALL*]

The installation section expands the common installation script with the platform specific files. The following predefined variables are available to address the respective subfolders of the target <Artifact folder>.

name	description
INSTALL_INTERFACE_INC_DIR	interface/include - interface header files
INSTALL_INTERFACE_SRC_DIR	interface/src - interface source files
INSTALL_INTERFACE_LIB_DIR	interface/lib - interface libraries
INSTALL_IMAGE_SIGNING_DIR	image_signing tools and files
INSTALL_CMAKE_DIR	CMake modules for Non-secure app build
INSTALL_PLATFORM_NS_DIR	NS platform source files

config.cmake:

(MANDATORY)

This file is used to setup default build configurations for TF-M and platform configurations which have fixed values depending on hardware and software supportness. These configurations should be set as normal CMake variables while others are cache variables.

The platform configurations in the below table are required.

name	description
CON- FIG_TFM_USE_TRUSTZONE	Use TrustZone to transition between NSPE and SPE on the same CPU
TFM_MULTI_CORE_TOPOLOGY	NSPE runs on a separate CPU to SPE

The platform configurations in the below table control optional features which rely on platform specific implementation. These features are disabled by default. Platforms shall implement corresponding functionalities and explicitly set the configuration to enable the feature.

name	description
PLATFORM_HAS_ISOLATION_L3_SUPPORT	Whether the platform has isolation level 3 support
PLATFORM_HAS_FIRMWARE_UPDATE_SUPPORT	Whether the platform has firmware update support
PSA_API_TEST_TARGET	The target platform name of PSA API test
PLATFORM_SVC_HANDLERS	Whether the platform has specific SVC handling

For build configurations, please refer to `config_base.cmake`.

[*config_cmake*]

cpuarch.cmake:

(MANDATORY)

This file contains hardware information such as the main processor and architecture of the SPE CPU. On single-core platforms, it should be installed to <Artifact folder> for NSPE build. On multi-core platforms, two `cpuarch.cmake` files should be added.

- a SPE specific `cpuarch.cmake` used in SPE build
- an NSPE one which should be installed to <Artifact folder> with filename `cpuarch.cmake` for NSPE build. See *ns/cpuarch_ns.cmake*.

name	description
TFM_SYSTEM_PROCESSOR	The SPE Processor the platform is using
TFM_SYSTEM_ARCHITECTURE	The architecture of the processor
CONFIG_TFM_FP_ARCH	The Float Point architecture flag for toolchain
CONFIG_TFM_FP_ARCH_ASM	The Float Point architecture flag for assembly code

tests/tfm_tests_config.cmake:

(OPTIONAL)

This file contains platform-specific config options for TF-M regression tests. The `tests` folder should be installed to <Artifact folder>/platform for NSPE build. Here are some examples.

name	description
PLATFORM_SLIH_IRQ_TEST_SUPPORT	Whether the platform has SLIH test support
PLATFORM_FLIH_IRQ_TEST_SUPPORT	Whether the platform has FLIH test support

tests/psa_arch_tests_config.cmake:

(OPTIONAL)

This file contains platform-specific config options for PSA API tests. Here are some examples.

name	description
PSA_API_TEST_TARGET	The target platform name of PSA API test

startup files:

(MANDATORY)

These files (one for BL2, one for S, one for NS) are the expected startup files. The reset handler should call SystemInit and then should end up calling `__START` which should be defined as `_start` if not defined elsewhere.

flash_layout.h:

(MANDATORY)

This file can be anywhere in the platform folder, usually in a sub folder named `partition`. TF-M doesn't provide a template for this file, common practice is to copy it from another platform (e.g. `arm/mps2/an521`) and update the following entries.

Note: all size are in bytes

name	description	Requisiteness
FLASH_S_PARTITION_SIZE	Size of the Secure partition in flash	Yes
FLASH_NS_PARTITION_SIZE	Size of the Non-Secure partition in flash	if <code>tfm_ns</code> is built
FLASH_AREA_IMAGE_SECURE	Size of the flash sector	if <code>bl2</code> is built
FLASH_TOTAL_SIZE	Flash total size	Yes
FLASH_BASE_ADDRESS	Flash base memory address	if <code>bl2</code> is built
FLASH_AREA_BL2_OFFSET	BL2 offset in flash	if <code>bl2</code> is built
FLASH_AREA_BL2_SIZE	BL2 flash size	if <code>bl2</code> is built
FLASH_PS_AREA_SIZE	Allocated size for the protected storage data in flash	Yes
FLASH_ITS_AREA_SIZE	Allocated size for the internal trusted storage data in flash	Yes
SECURE_IMAGE_OFFSET	Offset of the secure image data in flash	if <code>bl2</code> is built
FLASH_DEV_NAME	Name as defined in the CMSIS flash drivers	Yes
TFM_HAL_PS_FLASH_DRIVERS	Name as defined in the CMSIS flash drivers	used by protected storage partition
TFM_HAL_PS_SECTORS_PER_BLOCK	Number of physical erase sectors per logical FS block	used by protected storage partition
TFM_HAL_PS_PROGRAMMABLE_UNIT	Smallest flash programmable unit in bytes	used by protected storage partition
TFM_HAL_ITS_FLASH_DRIVERS	Name as defined in the CMSIS flash drivers	used by internal trusted storage partition
TFM_HAL_ITS_SECTORS_PER_BLOCK	Number of physical erase sectors per logical ITS block	used by internal trusted storage partition
TFM_HAL_ITS_PROGRAMMABLE_UNIT	Smallest flash programmable unit in bytes	used by internal trusted storage partition
TFM_NV_COUNTERS_AREA_SIZE	Allocated size for the NV counters data in flash	if using TF-M templates

region_defs.h:

(MANDATORY)

This file can be anywhere in the platform folder, usually in a sub folder named `partition`. TF-M doesn't provide a template for this file, common practice is to copy it from another platform (e.g. `arm/mps2/an521`)

and update the following entries.

General advice: if you don't know beforehand the size you will want for these elements you will have to make it iterative from an arbitrary value taken from another platform (e.g. arm/mps2/an521)

Note: all size are in bytes

name	description	Requisiteness
BL2_HEAP_SIZE	Size of the Bootloader (MCUboot) heap	if bl2 is built
BL2_MSP_STACK_SIZE	(if bl2 is built) Size of the Bootloader (MCUboot) Main stack	if bl2 is built
S_HEAP_SIZE	Size of the Secure (S) world Heap	yes
S_MSP_STACK_SIZE	Size of the Secure (S) world Main stack	yes
S_PSP_STACK_SIZE	Size of the Secure (S) world Process stack	no for IPC model
NS_HEAP_SIZE	Size of the Non-Secure (NS) world Heap	if tfm_ns is built
NS_STACK_SIZE	Size of the Non-Secure (NS) world stack	if tfm_ns is built
PSA_INITIAL_ATTEST_MAX	Size of the buffer that will store the initial attestation	used by initial attestation partition
TFM_ATTEST_BOOT_RECOF	Size of buffer that can store the encoded list of boot records	used by delegated attestation partition
BL2_HEADER_SIZE	Size of the Header for the Bootloader (MCUboot)	if bl2 is built
BL2_TRAILER_SIZE	Size of the Trailer for the Bootloader (MCUboot)	if bl2 is built
SHARED_SYMBOL_AREA_S	Size of shared common code between bl2 and tfm_s	if bl2 is built and want to reduce image size

(OPTIONAL)

If the TF-M common linker script is used then:

name	description	Requisiteness
S_CODE_START	Start address for the S code	Yes
S_CODE_SIZE	Size of the S code	Yes
S_DATA_START	Start address for the S data	Yes
S_DATA_SIZE	Size of the S data	Yes
S_RAM_CODE_START	Start address for the S code	if no XIP on flash
S_RAM_CODE_SIZE	Size of the S code	if no XIP on flash

CMSIS_Driver/Config/cmsis_driver_config.h:

(location as defined in CMakeLists.txt)

This file should include the CMSIS drivers implementation headers.

CMSIS_Driver/Config/RTE_Device.h:

(location as defined in CMakeLists.txt)

This is the Run-Time Environment file from CMSIS, which is there to allow enabling or disabling drivers prior to building. If your platform is designed as a general use platform, this file should contain all the available CMSIS drivers, and you should provide a recommended configuration. If your platform is designed for a specific use-case then you should reference and enable only the mandatory drivers.

CMSIS_Driver/Driver_Flash.c:

(location as defined in CMakeLists.txt)

TF-M relies on CMSIS Drivers, as such it requires the CMSIS functions to be implemented. As a platform owner you can decide to either implement the drivers in the CMSIS functions or to use the CMSIS functions as a shim to your native drivers.

Refer to the CMSIS [FLASH](#) documentation.

CMSIS_Driver/Driver_USART.c:

(location as defined in CMakeLists.txt)

TF-M relies on CMSIS Drivers, as such it requires the CMSIS functions to be implemented. As a platform owner you can decide to either implement the drivers in the CMSIS functions or to use the CMSIS functions as a shim to your native drivers.

Refer to the CMSIS [USART](#) documentation.

target_cfg.[ch]:

(location as defined in CMakeLists.txt)

It is expected that these files contain all platform specific code related to memory protection (e.g. SAU/PPC/MPC). These functions will not be called by TF-M directly, but are expected to be called from the function `tfm_hal_set_up_static_boundaries()` in `tfm_hal_isolation.c`.

tfm_hal_platform.c:

(location as defined in CMakeLists.txt)

Each platform is expected to implement the following API declared in `platform/include/tfm_hal_platform.h`

```
enum tfm_hal_status_t tfm_hal_platform_init(void);
```

The function will be called before SPM initialization.

tfm_hal_isolation.c:

(location as defined in CMakeLists.txt)

Each platform is expected to implement all the functions declared in `platform/include/tfm_hal_isolation.h`.

A reference implementation for Armv8-M platforms is provided in `platform/ext/common/tfm_hal_isolation_v8m.c`. Platforms using the common TF-M linker scripts and scatter files can use it to implement standard TF-M isolation with Armv8-M MPU regions. Platform-specific MPU regions can be appended by defining `PLATFORM_STATIC_MPU_REGIONS` in the platform's `tfm_peripherals_def.h` header.

These functions will be called from TF-M.

tfm_platform_system.c:

(location as defined in CMakeLists.txt)

Each platform is expected to implement all the functions declared in platform/include/tfm_platform_system.h.

check_config.cmake:

As a platform owner you may want to enforce some configuration or to prevent the use of unsupported configurations.

This file (CMake format) allows you to do so by allowing you to check for invalid configuration values.

This file is optional.

TF-M build system already provides a generic configuration checker that will be called on top of one provided by the platform owner. The generic checker is located in <TF-M ROOT>/config/.

[*check_config.cmake*]

platform_svc_numbers.h

(OPTIONAL)

If your platform has its own SVC handling, then you need to

- create the `platform_svc_numbers.h` which defines the platform SVC numbers.

The bit [7] of the number must be set to 1 to reflect that it is a platform SVC number. The bit [6] indicates whether this SVC should be called from Handler mode or Thread mode. For more details of the bit assignments, please check the `svc_num.h`. TF-M provides two Macros `TFM_SVC_NUM_PLATFORM_THREAD(index)` and `TFM_SVC_NUM_PLATFORM_HANDLER(index)` to easily construct a valid number.

- implement the `platform_svc_handlers` function which handles SVC.
- enable `PLATFORM_SVC_HANDLERS` config option.

ns/CMakeLists.txt

(MANDATORY)

This is CMake script for building the platform support on NSPE side. It's copied to <Artifact folder> in the installation phase and instructs on how to build **platform_ns** target. The default NSPE build script expects this target definition and extends it with files, common for all TF-M platforms.

Note::

This file shall define and use paths of installed directories in <Artifact folder>, instead of paths in TF-M platform folder.

[*NSCMakeLists.txt*]

ns/cpuarch_ns.cmake

(MANDATORY for multi-core platforms)

This file contains the hardware information for the NSPE CPU. It should be installed to <Artifact folder>/platform for NSPE build, renamed to `ns/cpuarch_ns.cmake`.

[ns/cpuarch_ns.cmake]

Functions

There are a few functions that need to be declared and properly initialized for TF-M to work. The function declarations can be found in `platform/include/tfm_platform_system.h` and `platform/include/tfm_hal_*.h`.

tfm_platform_hal_system_reset:

This function will in most cases end up calling the NVIC System Reset.

The platform can uninitialized or store some resources before reset.

```
void tfm_platform_hal_system_reset(void);
```

tfm_platform_hal_ioctl:

A single entry point to platform-specific code across the HAL is provided by the IOCTL service.

```
enum tfm_platform_err_t tfm_platform_hal_ioctl(tfm_platform_ioctl_req_t request, psa_
↳ invec *in_vec, psa_outvec *out_vec);
```

tfm_hal_set_up_static_boundaries:

Sets up the static isolation boundaries which are constant throughout the runtime of the system, including the SPE/NSPE and partition boundaries.

```
enum tfm_hal_status_t tfm_hal_set_up_static_boundaries(uintptr_t *p_spm_boundary);
```

tfm_hal_activate_boundary:

Activates one Secure Partition boundary.

```
enum tfm_hal_status_t tfm_hal_activate_boundary(const struct partition_load_info_t *p_
↳ ldinf, uintptr_t boundary);
```

tfm_hal_memory_check:

Checks if a given range of memory can be accessed with specified access types in boundary. The boundary belongs to a partition which contains all asset info.

A default implementation for Armv8-M platforms with TrustZone is provided in `platform/ext/common/tfm_hal_isolation_v8m.c`. Multi-core topology platforms without TrustZone may use the *Memory Check APIs* to implement this HAL.

```
enum tfm_hal_status_t tfm_hal_memory_check(uintptr_t boundary, uintptr_t base, size_t_
↳ size, uint32_t access_type);
```

tfm_hal_bind_boundary:

Binds partition boundaries with the platform. The platform maintains the platform-specific settings for SPM further usage, such as update partition boundaries or check resource accessibility. The platform needs to manage the settings with internal mechanism, and return a value to SPM. SPM delivers this value

back to platform when necessary. And SPM checks this value to decide if the platform-specific settings need to be updated. Hence multiple partitions can have the same value if they have the same platform-specific settings, depending on isolation level.

```
enum tfm_hal_status_t tfm_hal_bind_boundary(const struct partition_load_info_t *p_ldinf,
↪ uintptr_t *p_boundary);
```

tfm_hal_boundary_need_switch:

Lets the platform decide if a boundary switch is needed.

```
bool tfm_hal_boundary_need_switch(uintptr_t boundary_from, uintptr_t boundary_to);
```

tfm_hal_irq_clear_pending:

This function clears any pending IRQ.

```
void tfm_hal_irq_clear_pending(uint32_t irq_num);
```

tfm_hal_irq_enable:

This function enable an IRQ.

```
void tfm_hal_irq_enable(uint32_t irq_num);
```

tfm_hal_irq_disable:

This function disable an IRQ.

```
void tfm_hal_irq_disable(uint32_t irq_num);
```

platform_svc_handlers

This function is the platform's SVC handler. It should return the result for callers and the SPM will then return it to the caller.

```
int32_t platform_svc_handlers(uint8_t svc_num, uint32_t *svc_args, uint32_t exc_return);
```

Annex

CMake build system snippets examples

CMakeLists.txt: Defining regions for Secure world platform and all linked to it.

```
target_include_directories(platform_region_defs
    INTERFACE
    <folder name under the platform folder - usually named platform>
)
```

CMakeLists.txt: Scatter files for SPE platform and bootloader

```

target_add_scatter_file(bl2
    $<${C_COMPILER_ID:ARMClang}>:${PLATFORM_DIR}/ext/common/armclang/tfm_common_bl2.sct>
    $<${C_COMPILER_ID:GNU}>:${PLATFORM_DIR}/ext/common/gcc/tfm_common_bl2.ld>
    $<${C_COMPILER_ID:IAR}>:${PLATFORM_DIR}/ext/common/iar/tfm_common_bl2.icf>
)
target_add_scatter_file(tfm_s
    $<${C_COMPILER_ID:ARMClang}>:${PLATFORM_DIR}/ext/common/armclang/tfm_isolation_s.sct>
    $<${C_COMPILER_ID:GNU}>:${PLATFORM_DIR}/ext/common/gcc/tfm_isolation_s.ld>
    $<${C_COMPILER_ID:IAR}>:${PLATFORM_DIR}/ext/common/iar/tfm_isolation_s.icf>
)

```

CMakeLists.txt: Startup files for SPE platform and bootloader

```

target_sources(bl2
    PRIVATE
    ${CMAKE_CURRENT_SOURCE_DIR}/platform/ext/target/<folder to platform>/device/source/
    ↪startup_<platform name>.c
)
target_sources(tfm_s
    PRIVATE
    ${CMAKE_CURRENT_SOURCE_DIR}/platform/ext/target/<folder to platform>/device/source/
    ↪startup_<platform name>.c
)

```

CMakeLists.txt: The Secure world platform sources

```

target_include_directories(platform_bl2
    PUBLIC
)
target_include_directories(platform_s
    PUBLIC
)

target_sources(platform_bl2
    PRIVATE
)
target_sources(platform_s
    PRIVATE
)
target_sources(tfm_spm
    PRIVATE
    target_cfg.c
    tfm_hal_isolation.c
    tfm_hal_platform.c
)

```

CMakeLists.txt: installation for the Non-Secure world platform build

```

install(FILES ${PLATFORM_DIR}/ext/common/uart_stdout.c
    native_drivers/arm_uart_drv.c
    native_drivers/timer_cmsdk/timer_cmsdk.c
    cmsis_drivers/Driver_USART.c
    retarget/platform_retarget_dev.c

```

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```

        cmsis_core/an521_ns_init.c
        DESTINATION ${INSTALL_PLATFORM_NS_DIR})

install(DIRECTORY ${PLATFORM_DIR}/ext/common
        DESTINATION ${INSTALL_PLATFORM_NS_DIR}/ext)

```

config.cmake

```

set(CONFIG_TFM_USE_TRUSTZONE      ON)
set(TFM_MULTI_CORE_TOPOLOGY      OFF)
set(BL2                          OFF          CACHE BOOL      "Whether to build BL2
↪")
set(NS                          FALSE        CACHE BOOL      "Whether to build NS_
↪app" FORCE)

```

check_config.cmake

```

function(tfm_invalid_config)
    if (${ARGV})
        string (REPLACE ";" " " ARGV_STRING "${ARGV}")
        string (REPLACE "STREQUAL"      "=" ARGV_STRING "${ARGV_STRING}")
        string (REPLACE "GREATER"       ">" ARGV_STRING "${ARGV_STRING}")
        string (REPLACE "LESS"          "<" ARGV_STRING "${ARGV_STRING}")
        string (REPLACE "VERSION_LESS" "<" ARGV_STRING "${ARGV_STRING}")
        string (REPLACE "EQUAL"         "=" ARGV_STRING "${ARGV_STRING}")
        string (REPLACE "IN_LIST"       "in" ARGV_STRING "${ARGV_STRING}")

        message(FATAL_ERROR "INVALID CONFIG: ${ARGV_STRING}")
    endif()
endfunction()

# Requires armclang >= 6.10.1
tfm_invalid_config((CMAKE_C_COMPILER_ID STREQUAL "ARMClang") AND (CMAKE_C_COMPILER_
↪VERSION VERSION_LESS "6.10.1"))

```

/ns/CMakeLists.txt:

```

add_library(platform_ns)

target_sources(platform_ns
    PRIVATE
        arm_uart_drv.c
        timer_cmsdk.c
        uart_stdout.c
        Driver_USART.c
    PUBLIC
        cmsis_core/startup_an521.c
)

target_include_directories(platform_ns
    PUBLIC
        include
        cmsis

```

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```

    cmsis_core
)

target_compile_definitions(platform_ns
    PUBLIC
    $<$<BOOL:${PLATFORM_DEFAULT_CRYPTO_KEYS}>:PLATFORM_DEFAULT_CRYPTO_KEYS>
)

```

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9.9.2 Platform Documentation

This document describes conventions for platform documentation for easy and smooth integration into TF-M online documentation.

Principles

- Minimise TF-M requirements and maximise flexibility to let platform vendors reuse existing documentation.
- Platforms are grouped under vendors' name. There is space to keep general information not specific to any of the vendors' platforms.
- A Table Of Contents (TOC) structures platforms' documentation. There is no defined template for platform documents.

Rules

- Platforms documentation is gathered in `/docs/platform` folder. This folder is similar to the platforms implementation folder `/platform/ext/target` but does not need to mirror its structure.
- A vendor is represented by a subfolder with `index.rst` as an entry point. This file may contain general materials about a vendor not specific to a particular platform.
- A vendor's `index.rst` shall carry a TOCtree structure with a list of all supported platforms at the top level.
- An explicit platform name is preferred in a vendor's TOCtree instead of use of platform's title by default.

Note

Do not forget to update the summary file after adding a new platform `/docs/platform/platform_introduction.rst` included into TF-M introduction.

Example

Assuming a vendor *ABCD123* wants to add and document a new platform *Secure 1024* in TF-M. Here are the basic steps:

1. Create a folder `/docs/platform/ABCD123/`
2. Create a platform documentation under `/docs/platform/ABCD123/`. A natural choice is to put a new platform under `/docs/platform/ABCD123/Secure_1024` folder but it's not required.
3. Create a file `/docs/platform/ABCD123/index.rst` with a platform entry, presuming it's location in `/docs/platform/ABCD123/Secure_1024/index.rst`:

```
#####  
ABCD123 platforms  
#####  
  
.. toctree::  
Secure 1024 <Secure_1024/index>
```

4. Finally, update `/docs/platform/index.rst` with a new vendor name:

```
#####  
TF-M Platforms  
#####  
  
.. toctree::  
    :maxdepth: 2  
    :glob:  
  
    Arm <arm/index.rst>  
    NXP <nxp/index.rst>  
    Cypress <cypress/index.rst>  
    ...  
    ABCD123 <ABCD123/index.rst>
```

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9.9.3 Platform deprecation and removal

Process for Platform deprecation and removal

A platform may need to be removed from upstream code due to lack of community interest or it may have reached end of life. The below section calls out the process for removing platform support from TF-M.

1. An email to the TF-M mailing list proposing the removal of the platform.
2. The merit of the proposal will be considered by the maintainers for a period of 4 weeks and community can express their opinion on the same during this time window. The platform owner can veto the proposal and in case of multiple platform owners with differing opinion or community having interest in the platform, then the project maintainer can work with the platform owner and use their discretion to decide on the proposal.
3. Once a decision is made to remove the platform, the platform is considered to be in *deprecated* state and the TF-M version after which it will be removed will also be mentioned. Suitable build time or runtime messages needs to be incorporated to the platform to warn about the *deprecation*.
4. The project should strive to keep the *deprecated* platforms building/running till the removal. This relies on platform owners being still actively involved with the project and maintaining the platform.
5. Although this will be the usual process for platform deprecation and eventual removal, the process still leaves room for the platform deprecation to be cancelled after it has been marked as *deprecated* due to evolving project and market requirements. This is left to consensus between project maintainers and platform owner/s.
6. Once a platform has been removed, it can be added back in future and this would follow the same guidelines as for a new platform contribution.

List of Deprecated Platforms

The below list calls out platforms marked for deprecation according to the above process and the platform will be removed soon after the mentioned release.

Deprecated Platform	Removed after release	Comments
arm/mps2/an539	v1.2.0	N.A
arm/mps2/sse-200_aws	v1.3.0	N.A
arm/musca_a	v1.3.0	N.A
arm/mps2/fvp_sse300	v1.4.0	N.A
arm/musca_b1/secure_enclave	v1.6.0	N.A

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9.10 Services

9.10.1 Initial Attestation Service Integration Guide

Introduction

TF-M Initial Attestation Service allows the application to prove the device identity during an authentication process to a verification entity. The initial attestation service can create a token on request, which contains a fix set of device specific data.

TF-M Initial Attestation Service by default enables asymmetric key algorithm based attestation (*asymmetric attestation* for short). Symmetric key algorithm based attestation (*symmetric attestation* for short) can be enabled instead by selecting build option SYMMETRIC_INITIAL_ATTESTATION.

- In asymmetric attestation, device must contain an attestation key pair, which is unique per device. The token is signed with the private part of attestation key pair. The public part of the key pair is known by the verification entity. The public key is used to verify the token authenticity.
- In symmetric attestation, device should contain a symmetric attestation key to generate the authentication tag of token content. The verification entity uses the same symmetric key to verify the token authenticity.

The data items in the token used to verify the device integrity and assess its trustworthiness. Attestation key provisioning is out of scope for the attestation service and is expected to take part during manufacturing of the device.

Claims in the initial attestation token

The initial attestation token is formed of claims. A claim is a data item, which is represented in a key - value structure. The following fixed set of claims are included in the token:

- **Auth challenge:** Input object from caller. Can be a single nonce from server or hash of nonce and attested data. It is intended to provide freshness to report and the caller has responsibility to arrange this. Allowed length: 32, 48, 64 bytes. The claim is modeled to be eventually represented by the EAT standard claim nonce. Until such a time as that standard exists, the claim will be represented by a custom claim. Value is encoded as byte string.
- **Instance ID:** It represents the unique identifier of the instance. In the PSA definition it is:
 - a hash of the public attestation key of the instance in asymmetric attestation.
 - hashes of the symmetric attestation key of the instance in symmetric attestation.

The claim is modeled to be eventually represented by the EAT standard claim UEID of type GUID. Until such a time as that standard exists, the claim will be represented by a custom claim Value is encoded as byte string.

- **Verification service indicator:** Optional, recommended claim. It is used by a Relying Party to locate a validation service for the token. The value is a text string that can be used to locate the service or a URL specifying the address of the service. The claim is modelled to be eventually represented by the EAT standard claim origination. Until such a time as that standard exists, the claim will be represented by a custom claim. Value is encoded as text string.
- **Profile definition:** Optional, recommended claim. It contains the name of a document that describes the ‘profile’ of the token, being a full description of the claims, their usage, verification and token signing. The document name may include versioning. Custom claim with a value encoded as text string.
- **Implementation ID:** Uniquely identifies the underlying immutable PSA RoT. A verification service can use this claim to locate the details of the verification process. Such details include the implementation’s origin and associated certification state. Custom claim with a value encoded as byte string.
- **Client ID:** The partition ID of that secure partition or non-secure thread who called the initial attestation API. Custom claim with a value encoded as a *signed* integer. Negative number represents non-secure caller, positive numbers represents secure callers, zero is invalid.
- **Security lifecycle:** It represents the current lifecycle state of the instance. Custom claim with a value encoded as an integer.
- **Hardware version:** Optional claim. Globally unique number in EAN-13 format identifying the GDSII that went to fabrication, HW and ROM. It can be used to reference the security level of the PSA-ROT via a certification website. Custom claim with a value is encoded as text string.
- **Boot seed:** It represents a random value created at system boot time that will allow differentiation of reports from different system sessions. The size is 32 bytes. Custom claim with a value is encoded as byte string.
- **Software components:** Optional, but required in order to be compliant with the PSA-SM. It represents the software state of the system. The value of the claim is an array of CBOR map entries, with one entry per software component within the device. Each map contains multiple claims that describe evidence about the details of the software component.
- **No software measurements:** Optional, but required if no software component claims are made. In the event that the implementation does not contain any software measurements then it is mandatory to include this claim to indicate this is a deliberate state. Custom claim with a value encoded as an unsigned integer set to 1.

Each software component claim can include the following properties. Any property that is not optional must be included:

- **Measurement type:** Optional claim. It represents the role of the software component. Value is encoded as short(!) text string.
- **Measurement value:** It represents a hash of the invariant software component in memory at start-up time. The value must be a cryptographic hash of 256 bits or stronger. Value is encoded as byte string.
- **Version:** Optional claim. It represents the issued software version. Value is encoded as text string.
- **Signer ID:** Optional claim, but required in order to be compliant with the PSA-SM. It represents the hash of a signing authority public key. Value is encoded as byte string.
- **Measurement description:** Optional claim. It represents the way in which the measurement value of the software component is computed. Value is encoded as text string containing an abbreviated description (name) of the measurement method.

Initial attestation token (IAT) data encoding

The initial attestation token is planned to be aligned with future version of [Entity Attestation Token](#) format. The token is encoded according to the [CBOR](#) format and signed according to [COSE](#) standard.

Code structure

The PSA interface for the Initial Attestation Service is located in `interface/include`. The only header to be included by applications that want to use functions from the PSA API is `psa/initial_attestation.h`.

The TF-M Initial Attestation Service source files are located in `secure_fw/partitions/initial_attestation`.

Service source files

- **CBOR library**

- `lib/ext/qcbor` This library is used to create a proper CBOR token. It can be used on 32-bit and 64-bit machines. It was designed to suite constrained devices with low memory usage and without dynamic memory allocation. Its source code is fetched automatically during the build configuration step from an external repository: [QCBOR library](#).
- `<qcbor_src>/inc/qcbor/qcbor_encode.h`: Public API documentation of CBOR library (encoding).
- `<qcbor_src>/inc/qcbor/qcbor_decode.h`: Public API documentation of CBOR library (decoding).

- **COSE library:**

- `lib/ext/t_cose`: This library is used to sign a CBOR token and create the COSE header and signature around the initial attestation token. Only a subset of the [COSE](#) standard is implemented. The `COSE_Sign`, `COSE_Sign1`, `COSE_Mac0` signature and the `COSE_Encrypt`, `COSE_Encrypt0` encryption schemes are supported. Its source code is fetched automatically during the build configuration step from an external repository: [t_cose library](#).
- `<t_cose_src>/src/t_cose_crypto.h`: Expose an API to bind `t_cose` library with available crypto library in the device.
- `<t_cose_src>/crypto_adapters/t_cose_psa_crypto.c`: Implements the exposed API and ports `t_cose` to the PSA Crypto API.

- **Initial Attestation Service:**

- `attest_core.c`: Implements core functionalities such as implementation of APIs, retrieval of claims and token creation.
- `attest_boot_data.c`: Implements core functionalities for measured boot.
- `attest_token_encode.c`: Implements the token creation functions such as start and finish token creation and adding claims to the token.
- `attest_asymmetric_key.c`: Calculate the Instance ID value based on asymmetric initial attestation key.
- `tfm_attest.c`: Implements the SPM abstraction layer, and bind the attestation service to the SPM implementation in TF-M project.
- `tfm_attest_req_mgr.c`: Includes the initialization entry of attestation service and handles attestation service requests in IPC model.
- `attest_symmetric_key.c`: Calculate the Instance ID value based on symmetric initial attestation key.

Service interface definitions

- **Boot loader interface:** The attestation service might include data in the token about the distinct software components in the device. This data is provided by the boot loader and must be encoded in the TLV format, definition is described below in the boot loader interface paragraph. Possible claims in the boot status are described above in the software components paragraph.
- **Hardware abstraction layer:**
 - Headers are located in `platform/include` folder.
 - `tfm_attest_hal.h`: Expose an API to get the following claims: security lifecycle, verification service indicator, profile definition.
 - `tfm_plat_boot_seed.h`: Expose an API to get the boot seed claim.
 - `tfm_plat_device_id.h`: Expose an API to get the following claims: implementation ID, hardware version.
- **SPM interface:**
 - `attestation.h`: Expose an API to bind attestation service to an SPM implementation.
- **PSA interface:**
 - `psa/initial_attestation.h`: Public API definition of initial attestation service.
- **Crypto interface:**
 - `t_cose_crypto.h`: Expose an API to bind the `t_cose` implementation to any cryptographic library.

PSA interface

The TF-M Initial Attestation Service exposes the following PSA interface:

```
psa_status_t
psa_initial_attest_get_token(const uint8_t *auth_challenge,
                           size_t        challenge_size,
                           uint8_t       *token_buf,
                           size_t        token_buf_size,
                           size_t        *token_size);

psa_status_t
psa_initial_attest_get_token_size(size_t challenge_size,
                                 size_t *token_size);
```

The caller must allocate a large enough buffer, where the token is going to be created by Initial Attestation Service. The size of the created token is highly dependent on the number of software components in the system and the provided attributes of these. The `psa_initial_attest_get_token_size()` function can be called to get the exact size of the created token.

System integrators might need to port these interfaces to a custom secure partition manager implementation (SPM). Implementations in TF-M project can be found here:

- `interface/src/tfm_attest_api.c`: interface implementation.

Secure Partition Manager (SPM) interface

The Initial Attestation Service defines the following interface towards the secure partition manager (SPM). System integrators **must** port this interface according to their SPM implementation.

```
enum psa_attest_err_t
attest_get_boot_data(uint8_t major_type, void *ptr, uint32_t len);

enum psa_attest_err_t
attest_get_caller_client_id(int32_t *caller_id);
```

- `attest_get_boot_data()`: Service can retrieve the relevant data from shared memory area between boot loader and runtime software. It might be the case that only SPM has direct access to the shared memory area, therefore this function can be used to copy the service related data from shared memory to a local memory buffer. In TF-M implementation this function must be called during service initialization phase, because the shared memory region is deliberately overlapping with secure main stack to spare some memory and reuse this area during execution. If boot loader is not available in the system to provide attributes of software components then this function must be implemented in a way that just initialize service's memory buffer to:

```
struct shared_data_tlv_header *tlv_header = (struct shared_data_tlv_header *)ptr;
tlv_header->tlv_magic = 2016;
tlv_header->tlv_tot_len = sizeof(struct shared_data_tlv_header *tlv_header);
```

- `attest_get_caller_client_id()`: Retrieves the ID of the caller thread.
- `tfm_client.h`: Service relies on the following external definitions, which must be present or included in this header file:

```
typedef struct psa_invec {
    const void *base;
    size_t len;
} psa_invec;

typedef struct psa_outvec {
    void *base;
    size_t len;
} psa_outvec;
```

Hardware abstraction layer

The following API definitions are intended to retrieve the platform specific claims. System integrators **must** implement these interface according to their SoC and software design. Detailed definition of the claims are above in the claims in the initial attestation token paragraph.

- `tfm_attest_hal_get_security_lifecycle()`: Get the security lifecycle of the device.
- `tfm_attest_hal_get_verification_service()`: Get the verification service indicator for initial attestation.
- `tfm_attest_hal_get_profile_definition()`: Get the name of the profile definition document for initial attestation.
- `tfm_plat_get_boot_seed()`: Get the boot seed, which is a constant random number during a boot cycle.
- `tfm_plat_get_implementation_id`: Get the implementation ID of the device.
- `tfm_plat_get_cert_ref`: Get the hardware version of the device.

Boot loader interface

It is **recommended** to have a secure boot loader in the boot chain, which is capable of measuring the runtime firmware components (calculates the hash value of firmware images) and provide other attributes of these (version, type, etc). If the used boot loader is not capable of sharing these information with the runtime software then the `BOOT_DATA_AVAILABLE` compiler flag **must** be set to OFF (see *Related compile time options*).

The shared data between boot loader and runtime software is TLV encoded. The definition of TLV structure is described in `bl2/include/tfm_boot_status.h`. The shared data is stored in a well known location in secure internal memory and this is a contract between boot loader and runtime SW.

The structure of shared data must be the following:

- At the beginning there must be a header: `struct shared_data_tlv_header` This contains a magic number and a size field which covers the entire size of the shared data area including this header.

```
struct shared_data_tlv_header {
    uint16_t tlv_magic;
    uint16_t tlv_tot_len;
};
```

- The header is followed by the entries which are composed from an entry header structure: `struct shared_data_tlv_entry` and the data. In the entry header there is a type and a length field. The `tlv_type` field identifies the consumer of the entry in the runtime software and specify the subtype of that data item. The `tlv_len` field covers the length of the data (not including the size of the entry header).

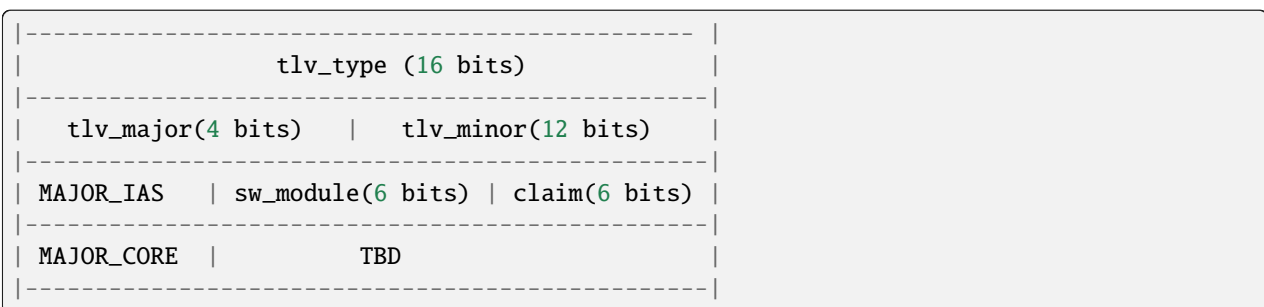
After the entry header structure comes the actual data.

```
struct shared_data_tlv_entry {
    uint16_t tlv_type;
    uint16_t tlv_len;
};
```

- Arbitrary number and size of data entry can be in the shared memory area.

The figure below gives an overview about the `tlv_type` field in the entry header. The `tlv_type` is always composed from a major and a minor number. Major number identifies the addressee in runtime software, which the data entry is sent to. Minor number is used to encode more info about the data entry. The actual definition of minor number could change per major number. In case of boot status data, which is going to be processed by initial attestation service, the minor number is split further to two part: `sw_module` and `claim`. The `sw_module` identifies the SW component in the system which the data item belongs to and the `claim` part identifies the exact type of the data.

`tlv_type` description:



Overall structure of shared data:

Magic number(uint16_t) Shared data total length(uint16_t)	

Major_type(4 bits) Minor_type(12 bits) Length(uint16_t)	

Raw data	

.	
.	
.	

Major_type(4 bits) Minor_type(12 bits) Length(uint16_t)	

Raw data	

Crypto interface

Asymmetric key algorithm based attestation

Device **must** contain an asymmetric key pair. The private part of it is used to sign the initial attestation token. Current implementation supports only the ECDSA P256 signature over SHA256. The public part of the key pair is used to create the key identifier (kid) in the unprotected part of the COSE header. The kid is used by the verification entity to look up the corresponding public key to verify the signature in the token. The *t_cose* part of the initial attestation service implements the signature generation and kid creation. But the actual calculation of token's hash and signature is done by the Crypto service in the device. System integrators might need to re-implement the following functions if they want to use initial attestation service with a different cryptographic library than Crypto service:

- `t_cose_crypto_pub_key_sign()`: Calculates the signature over a hash value.
- `t_cose_crypto_get_ec_pub_key()`: Get the public key to create the key identifier.
- `t_cose_crypto_hash_start()`: Start a multipart hash operation.
- `t_cose_crypto_hash_update()`: Add a message fragment to a multipart hash operation.
- `t_cose_crypto_hash_finish()`: Finish the calculation of the hash of a message.

Interface needed by verification code:

- `t_cose_crypto_pub_key_verify()`: Verify the signature over a hash value.

Key handling

The provisioning of the initial attestation key is out of scope of the service and this document. It is assumed that device maker provisions the unique asymmetric key pair during the manufacturing process. Software integrators **must** make sure that `TFM_BUILTIN_KEY_SLOT_IAK` is available via the Crypto service, which will then be used by the Attestation partition to perform the required signing operations via the PSA crypto interface.

Symmetric key algorithm based attestation

Device **must** contain a symmetric key to generate the authentication tag of the initial attestation token. A key identifier (kid) can be encoded in the unprotected part of the COSE header. It helps verification entity look up the symmetric key to verify the authentication tag in the token.

The *t_cose* part of the initial attestation service implements the authentication tag generation. The authentication tag generation is done by the Crypto service. System integrators might need to re-implement the following functions if

platforms provide a different cryptographic library than Crypto service:

- `t_cose_crypto_hmac_sign_setup()`: Set up a multi-part HMAC calculation operation.
- `t_cose_crypto_hmac_update()`: Add a message fragment to a multi-part HMAC operation.
- `t_cose_crypto_hmac_sign_finish()`: Finish the calculation of the HMAC of a message.

Interface needed by verification code:

- `t_cose_crypto_hmac_verify_setup()`: Set up a multi-part HMAC verification operation.
- `t_cose_crypto_hmac_verify_finish()`: Finish the verification of the HMAC of a message.

It also requires the same hash operations as listed in asymmetric key algorithm based initial attestation above, in attestation test cases.

Key handling

The provisioning of the initial attestation key is out of scope of the service and this document. It is assumed that device maker provisions the symmetric key during the manufacturing process. The following API is defined to retrieve the symmetric attestation key from platform layer. Software integrators **must** port this interface according to their SoC design and make sure that key is available by Crypto service:

- `tfm_plat_get_symmetric_iak()`: Get the symmetric initial attestation key raw data.
- `tfm_plat_get_symmetric_iak_id()`: Get the key identifier of the symmetric initial attestation key. The key identifier can be used as `kid` parameter in COSE header. Optional.

Note

Asymmetric initial attestation and symmetric initial attestation may share the same HAL APIs in future development.

Initial Attestation Service compile time options

There is a defined set of flags that can be used to compile in/out certain service features. The `config_base.cmake` file sets the default values of those flags. The list of flags are:

- `ATTEST_INCLUDE_OPTIONAL_CLAIMS`: Include also the optional claims to the attestation token. Default value: ON in base configure and profile large while OFF in profile small and medium.
- `ATTEST_INCLUDE_COSE_KEY_ID`: COSE key-id is an optional field in the COSE unprotected header. Key-id is calculated and added to the COSE header based on the value of this flag. Default value: OFF.
- `SYMMETRIC_INITIAL_ATTESTATION`: Select symmetric initial attestation. Default value: OFF.
- `ATTEST_STACK_SIZE`- Defines the stack size of the Initial Attestation Partition. This value mainly depends on the build type(debug, release and minisizerel) and compiler. Default value: Depends on the profile.
- `ATTEST_KEY_BITS` Defines the size of the initial attestation key, in bits. Default value: 256.
- `PSA_INITIAL_ATTEST_MAX_TOKEN_SIZE` Defines the maximum possible size of a token. Default value: 0x250.

Related compile time options

- `BOOT_DATA_AVAILABLE`: The boot data is expected to be present in the shared data area between the boot loader and the runtime firmware when set to ON. Otherwise, when it's OFF TF-M does not check the content of the shared data area but instead assumes that the TLV header is present and valid (the magic number is correct) and there are no data entries. Its default value depends on the BL2 flag.

Comparison of asymmetric and symmetric algorithm based token authentication

The symmetric key based authentication requires a more complex infrastructure for key management. Symmetric keys must be kept secret because they are sensitive asset, like the private key in case of asymmetric cryptographic algorithms. The main difference is that private keys are only stored on device, with proper hardware protection against external access, but symmetric keys must be known by both party (device and verifier), so they must also be stored in a central server of a relying party (who verifies the tokens). If keys are revealed then devices can be impersonated. If the database with the symmetric keys becomes compromised then all corresponding devices become untrusted. Since a centralised database of symmetric keys may need to be network connected, this can be considered to be a valuable target for attackers. The advantage of ECDSA based token authentication is that sensitive assets is only stored one place (in the device) and only one unique key per device. So if a device is compromised then only that single device become untrusted. In this case, the database of the relying party contains the corresponding public keys, which are not considered to be a confidential assets, so they can be shared with anybody. This shows the main advantage of asymmetric based authentication, because verification of attestation tokens can be done by a third party, such as cloud service providers (CSP). Thus Device Maker (DM) or Chip Maker (CM) does not need to operate such a service.

	Symmetric	Asymmetric
Authentication mode	HMAC over SHA256	ECDSA P256 over SHA256
Crypto key type in HW	Symmetric key	ECDSA private key (secp256r1)
Secrets are stored	Device and database	Device only
Verification database contains	Same symmetric key	Public keys
COSE authentication tag in the token	COSE_Mac0	COSE_Sign1
Verification entity	CM or DM, who provisioned the symmetric key	Can be anybody: third party provisioning service, cloud service provider, CM, DM

Verification

Regression test

The initial attestation token is verified by the attestation test suite in `test/secure_fw/suites/attestation`. The test suite is responsible for verifying the token signature and parsing the token to verify its encoding and the presence of the mandatory claims. This test suite can be executed on the device. It is part of the regression test suite. The test suite is configurable in the `test/secure_fw/suites/attestation/attest_token_test_values.h` header file. In this file there are two attributes for each claim which are configurable (more details in the header file):

- Requirements of presence: optional or mandatory
- Expected value: Value check can be disabled or expected value can be provided here.

For initial attestation tests, the built-in IAK is used. Initial attestation regression test verifies the IAT generated by initial attestation service with the exported public key.

iat-verifier

There is another possibility to verify the attestation token. This addresses the off-device testing when the token is already retrieved from the device and verification is done on the requester side. There is a Python script for this purpose in the `tf-m-tools` repo called `iat-verifier`. It does the same checking as the attestation test suite. The following steps describe how to simulate an off-device token verification on a host computer. It is described how to retrieve an initial attestation token when TF-M code is executed on FVP and how to use the `iat-verifier` script to check the token. This example assumes that user has license for DS-5 and FVP models:

- Build TF-M with any of the `ConfigRegression*`.cmake build configurations for MPS2 AN521 platform. More info in `tfm_build_instruction`.

- Lunch FVP model in DS-5. More info in Run TF-M examples on Arm platforms.
 - Set a breakpoint in `test/secure_fw/suites/attestation/attest_token_test.c` in `decode_test_internal(..)` after the `token_main_alt(..)` returned, i.e. on line 859. Execute the code in the model until the breakpoint hits second time. At this point the console prints the test case name:
 - For asymmetric initial attestation, the console prints ECDSA signature test of attest token.
 - For symmetric initial attestation, the console prints Symmetric key algorithm based Initial Attestation test.
 - At this point the token resides in the model memory and can be dumped to host computer.
 - The ADDRESS and SIZE attributes of the initial attestation token is stored in the `completed_token` local variable. Their value can be extracted in the (x)=Variables debug window.
 - Apply commands below in the Commands debug window to dump the token in binary format to the host computer:
 - For asymmetric initial attestation `dump memory <PATH>/iat_01.cbor <ADDRESS> +<SIZE>`
 - For symmetric initial attestation `dump memory <PATH>/iat_hmac_02.cbor <ADDRESS> +<SIZE>`
 - Execute commands below on the host computer to verify the token:
 - For asymmetric initial attestation `check_iat -p -K -k platform/ext/common/template/tfm_initial_attestation_key.pem <PATH>/iat_01.cbor`
 - For symmetric initial attestation `check_iat -m mac -p -K -k platform/ext/common/template/tfm_symmetric_iak.key <PATH>/iat_hmac_02.cbor`
 - Documentation of the iat-verifier can be found in the `tf-m-tools-iat-verifier`.
-

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9.10.2 Crypto Service Integration Guide

Introduction

The TF-M Crypto service allows Non Secure world applications and Secure services to use cryptographic functionalities such as symmetric ciphering, message signing and verification, asymmetric encryption and decryption, cryptographic hashes, message authentication codes (MACs), key derivation and agreement, authenticated encryption with associated data (AEAD). It exposes the PSA Crypto APIs¹.

The secure service resides in the Crypto partition as a single entry point and is made of different components:

- An interface layer that exposes the PSA Crypto API to either NS or S entities and is implemented in `interface/src/tfm_crypto_api.c`. The interface is based on the Uniform Secure Service Signature and communicates with the Secure Partition Manager available in TF-M
- An init module `secure_fw/partitions/crypto/crypto_init.c` that implements functionalities requested by TF-M during the initialisation phase, and an API dispatcher that at runtime receives the requests from the interface and dispatches them to the component that processes that particular API request
- A set of components that process cryptographic API requests, each component dispatching to a subset of functionalities, i.e. AEAD, Asymmetric, Ciphering, Hashing, Key derivation, Key management, MACs, and Random Number Generation

¹ PSA Crypto APIs: <https://arm-software.github.io/psa-api/crypto/1.1/>

- An alloc module `secure_fw/partitions/crypto/crypto_alloc.c` that manages the partition secure memory, storing multipart application contexts, input / outputs of the APIs being requested, inaccessible from NS or other secure partitions
- A library abstraction module `secure_fw/partitions/crypto/crypto_library.c` which is used to abstract the details of the cryptographic library used by the service *backend* to provide the actual implementation of the crypto functionalities. The backend library must expose the PSA Crypto APIs, and must provide support to encode *key ownership* into key identifiers. This is not standardized by the PSA Crypto APIs so it must be provided as an extension to the APIs. The backend library also needs to provide the subsystem for key storage and retrieval, and, in case, the interface to the cryptographic accelerator of the underlying platform, using the PSA cryptoprocessor driver interface specification². For this reason, it must provide a mechanism to access platform *builtin* keys, and permanent key slots using the *TF-M Internal Trusted Storage (ITS) service*, if available.

Code structure

The PSA interfaces for the Crypto service are located in `interface/include`. The only header to be included by applications that want to use functions from the PSA API is `psa/crypto.h`. The TF-M Crypto service source files are located in `secure_fw/partitions/crypto`.

PSA interfaces

The TF-M Crypto service exposes the PSA interfaces detailed in the header `psa/crypto.h`. This header, in turn, includes several other headers which are not meant to be included directly by user applications. For a detailed description of the PSA API interface, please refer to the comments in the `psa/crypto.h` header itself.

Service source files

A brief description of what is implemented by each source file is as below:

- `crypto_cipher.c` : Dispatcher for symmetric crypto operations
- `crypto_hash.c` : Dispatcher for hash operations
- `crypto_mac.c` : Dispatcher for MAC operations
- `crypto_aead.c` : dispatcher for AEAD operations
- `crypto_key_derivation.c` : Dispatcher for key derivation and key agreement operations
- `crypto_key_management.c` : Dispatcher for key management operations towards the key slot management system provided by the backend library
- `crypto_rng.c` : Dispatcher for the random number generation requests
- `crypto_asymmetric.c` : Dispatcher for message signature/verification and encryption/decryption using asymmetric crypto
- `crypto_init.c` : Init module for the service. The module stores also the internal buffer used to allocate temporarily the IOVECs needed, which is not required in case of SFN model. The size of this buffer is controlled by the `CRYPTO_IOVEC_BUFFER_SIZE` config define
- `crypto_library.c` : Library abstractions to interface the dispatchers towards the underlying library providing *backend* crypto functions. Currently this only supports the TF-PSA-Crypto library. In particular, the TF-PSA-Crypto library requires to provide a static buffer to be used as heap for its internal allocation. The size of this buffer is controlled by the `CRYPTO_ENGINE_BUF_SIZE` config define
- `crypto_alloc.c` : Takes care of storing multipart operation contexts in a secure memory not visible outside of the crypto service. The `CRYPTO_CONC_OPER_NUM` config define determines how many concurrent contexts are

² PSA cryptoprocessor driver interface: <https://github.com/Mbed-TLS/TF-PSA-Crypto/blob/development/docs/proposed/psa-driver-interface.md>

supported at once. In a multipart operation, the client view of the contexts is much simpler (i.e. just an handle), and the Alloc module keeps track of the association between handles and contexts

- `tfm_crypto_api.c` : This module is contained in `interface/src` and implements the PSA Crypto API client interface exposed to both S/NS clients. This module allows a configuration option `CONFIG_TFM_CRYPTO_API_RENAME` to be set to 1 in case the NS environment or integrators want to rename the API symbols exported by the TF-M Crypto service. The renaming adds a default prefix, `tfm_crypto__` to all functions. The prefix can be changed editing the interface file. This config option is for the NS environment or integration setup only, hence it is not accessible through the TF-M config
- `tfm_mbedcrypto_alt.c` : This module is specific to the Mbed TLS³ library integration and provides some alternative implementation of Mbed TLS APIs that can be used when a optimised profile is chosen. Through the `_ALT` mechanism it is possible to replace at link time default implementations available in Mbed TLS with the ones available in this file

Note

The `_ALT` mechanism will be deprecated in future releases of the Mbed TLS library

Considerations on service configuration

Crypto *backend* library configuration

The TF-M Crypto service relies on a cryptographic library to provide the functionalities specific by the PSA Crypto API spec and the PSA cryptoprocessor driver interface spec. At the moment, the only supported library is TF-PSA-Crypto[?].

The configuration of the backend library is supplied using the `TFM_TF_PSA_CRYPTO_CONFIG_PATH` config option that points to a configuration header following the configuration scheme of TF-PSA-Crypto.

Platforms can specify an extra config file by setting the `TFM_TF_PSA_CRYPTO_PLATFORM_EXTRA_CONFIG_PATH` variable (which is a wrapper around the `TF_PSA_CRYPTO_USER_CONFIG_FILE` option). This is preferred for platform configuration over `TFM_TF_PSA_CRYPTO_CONFIG_PATH` as it does not interfere with config changes due to TFM Profile.

Note

The default entropy source configured for TF-PSA-Crypto is `MBEDTLS_ENTROPY_NV_SEED` with a unique seed. For production devices, an alternative hardware entropy source can be specified using the config option `MBEDTLS_PSA_DRIVER_GET_ENTROPY`

Note

Starting from Mbed TLS 3.3.0, the Python package `jsonschema` must be available when building as it is required by the autogen framework for the driver integrations into the PSA Crypto core and driver wrapper modules

Crypto service build time options

- `CRYPTO_STACK_SIZE` : Defines the stack size of the Crypto Secure Partition. This value might depend on several parameters such as the build type, the compiler being used, the cryptographic functionalities that are enabled at build time

³ TF-PSA-Crypto library: <https://github.com/Mbed-TLS/TF-PSA-Crypto/>

- `CRYPTO_<COMPONENT>_MODULE_ENABLED` : A series of defines, one per each `<COMPONENT>` that processes cryptographic operations, that are used to disable modules at build time. Each define corresponds to a component as described in *the components list*.

Crypto service *builtin* keys integration

A detailed description of how the service interacts with *builtin* keys is available in the `tfm_builtin_key_loader` *design document*.

Note

The crypto service integration with builtin keys relies on implementation details of TF-PSA-Crypto that are not standardized in the spec and might change between releases due to ongoing work⁴

References

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9.10.3 TF-M Internal Trusted Storage Service Integration Guide

Introduction

TF-M Internal Trusted Storage (ITS) service implements PSA Internal Trusted Storage APIs.

The service is backed by hardware isolation of the flash access domain and relies on hardware to isolate the flash area from access by the Non-secure Processing Environment, as well as the Application Root of Trust at higher levels of isolation.

The current ITS service design relies on hardware abstraction provided by TF-M. The ITS service provides a non-hierarchical storage model, as a filesystem, where all the assets are managed by a linearly indexed list of metadata.

The design addresses the following high level requirements as well:

- **Confidentiality** - Resistance to unauthorised accesses through hardware/software attacks. Assumed to be provided by the internal flash device, backed by hardware isolation.
- **Access Authentication** - Mechanism to establish requester's identity (a non-secure entity, secure entity, or a remote server).
- **Integrity** - Resistance to tampering by attackers with physical access is assumed to be provided by the internal flash device itself, while resistance to tampering by Non-secure or App RoT attackers also requires hardware isolation.
- **Reliability** - Resistance to power failure scenarios and incomplete write cycles.
- **Configurability** - High level of configurability to scale up/down memory footprint to cater for a variety of devices with varying requirements.
- **Performance** - Optimized to be used for resource constrained devices with very small silicon footprint, the PPA (power, performance, area) should be optimal.

⁴ Interface for platform keys: <https://github.com/ARM-software/psa-crypto-api/issues/550> (private)

Current ITS Service Limitations

- **Fragmentation** - The current design does not support fragmentation, as an asset is stored in a contiguous space in a block. This means that the maximum asset size can only be up-to a block size. Each block can potentially store multiple assets. A delete operation implicitly moves all the assets towards the top of the block to avoid fragmentation within block. However, this may also result in unutilized space at the end of each block.
- **Non-hierarchical storage model** - The current design uses a non-hierarchical storage model, as a filesystem, where all the assets are managed by a linearly indexed list of metadata. This model locates the metadata in blocks which are always stored in the same flash location. That increases the number of writes in a specific flash location as every change in the storage area requires a metadata update.
- **Protection against physical storage medium failure** - Complete handling of inherent failures of storage mediums (e.g. bad blocks in a NAND based device) is not supported by the current design.
- **Lifecycle management** - Currently, it does not support any subscription based keys and certificates required in a secure lifecycle management. Hence, an asset's validity time-stamp can not be invalidated based on the system time.
- **Provisioning vs user/device data** - In the current design, all assets are treated in the same manner. In an alternative design, it may be required to create separate partitions for provisioning content and user/device generated content. This is to allow safe update of provisioning data during firmware updates without the need to wipe out the user/device generated data.

Code Structure

TF-M Internal Trusted Storage service code is located in `secure_fw/partitions/internal_trusted_storage/` and is divided as follows:

- Core files
- Flash filesystem interfaces
- Flash interfaces

The PSA ITS interfaces for the TF-M ITS service are located in `interface/include/psa`.

PSA Internal Trusted Storage Interfaces

The TF-M ITS service exposes the following mandatory PSA ITS interfaces version 1.0:

```
psa_status_t psa_its_set(psa_storage_uid_t uid, size_t data_length, const void *p_data,
↳ psa_storage_create_flags_t create_flags);
psa_status_t psa_its_get(psa_storage_uid_t uid, size_t data_offset, size_t data_size,
↳ void *p_data, size_t *p_data_length);
psa_status_t psa_its_get_info(psa_storage_uid_t uid, struct psa_storage_info_t *p_info);
psa_status_t psa_its_remove(psa_storage_uid_t uid);
```

These PSA ITS interfaces and TF-M ITS types are defined and documented in `interface/include/psa/storage_common.h`, `interface/include/psa/internal_trusted_storage.h`, and `interface/include/tfm_its_defs.h`

Core Files

- `tfm_its_req_mgr.c` - Contains the ITS request manager implementation which handles all requests which arrive to the service. This layer extracts the arguments from the input and output vectors, and it calls the internal trusted storage layer with the provided parameters.

- `tfm_internal_trusted_storage.c` - Contains the TF-M internal trusted storage API implementations which are the entry points to the ITS service. Constructs a filesystem configuration using information from the ITS HAL, allocates a filesystem context for ITS and makes appropriate FS calls. Also handles requests from the PS partition with a separate FS config and context.
- `its_utils.c` - Contains common and basic functionalities used across the ITS service code.

Flash Filesystem Interface

- `flash_fs/its_flash_fs.h` - Abstracts the flash filesystem operations used by the internal trusted storage service. The purpose of this abstraction is to have the ability to plug-in other filesystems or filesystem proxies (supplciant).
- `flash_fs/its_flash_fs.c` - Contains the `its_flash_fs` implementation for the required interfaces.
- `flash_fs/its_flash_fs_mblock.c` - Contains the metadata block manipulation functions required to implement the `its_flash_fs` interfaces in `flash_fs/its_flash_fs.c`.
- `flash_fs/its_flash_fs_dblock.c` - Contains the data block manipulation functions required to implement the `its_flash_fs` interfaces in `flash_fs/its_flash_fs.c`.

The system integrator **may** replace this implementation with its own flash filesystem implementation or filesystem proxy (supplciant).

Flash Interface

The ITS filesystem flash interface is defined by `struct its_flash_fs_ops_t` in `flash_fs/its_flash_fs.h`.

Implementations of the ITS filesystem flash interface for different types of storage can be found in the ``internal_trusted_storage/flash` directory.

- `flash/its_flash.h` - Helper header that includes the correct ITS flash interface implementation for the target, and abstracts the allocation of different flash device types.
- `flash/its_flash_nand.c` - Implements the ITS flash interface for a NAND flash device, on top of the CMSIS flash interface implemented by the target. This implementation writes entire block updates in one-shot, so the CMSIS flash implementation **must** be able to detect incomplete writes and return an error the next time the block is read.
- `flash/its_flash_nor.c` - Implements the ITS flash interface for a NOR flash device, on top of the CMSIS flash interface implemented by the target.
- `flash/its_flash_ram.c` - Implements the ITS flash interface for an emulated flash device using RAM, on top of the CMSIS flash interface implemented by the target.

The CMSIS flash interface **must** be implemented for each target based on its flash controller.

The ITS flash interface depends on target-specific definitions from `platform/ext/target/<TARGET_NAME>/partition/flash_layout.h`. Please see the *Internal Trusted Storage Service HAL* section for details.

ITS Service Integration Guide

This section describes mandatory (i.e. **must** implement) or optional (i.e. **may** implement) interfaces which the system integrator has to take in to account in order to integrate the internal trusted storage service in a new platform.

Flash Interface

For ITS service operations, a contiguous set of blocks must be earmarked for the internal trusted storage area. The design requires either 2 blocks, or any number of blocks greater than or equal to 4. Total number of blocks can not be 0, 1 or 3. This is a design choice limitation to provide power failure safe update operations.

Maximum Asset Size

An asset is stored in a contiguous space in a logical filesystem block. The maximum size of an asset can be up-to the size of the data block. Typically, each logical block corresponds to one physical flash erase sector (the smallest unit that can be erased), but the `TFM_HAL_ITS_SECTORS_PER_BLOCK` configuration below allows a number of contiguous erase sectors to form one logical block.

Internal Trusted Storage Service HAL

The ITS service requires the platform to implement the ITS HAL, defined in `platform/include/tfm_hal_its.h`.

The following C definitions in the HAL are mandatory, and must be defined by the platform in a header named `flash_layout.h`:

- `TFM_HAL_ITS_FLASH_DRIVER` - Defines the identifier of the CMSIS Flash `ARM_DRIVER_FLASH` object to use for ITS. It must have been allocated by the platform and will be declared extern in the HAL header.
- `TFM_HAL_ITS_PROGRAM_UNIT` - Defines the size of the ITS flash device's physical program unit (the smallest unit of data that can be individually programmed to flash). It must be equal to `TFM_HAL_ITS_FLASH_DRIVER.GetInfo()->program_unit`, but made available at compile time so that filesystem structures can be statically sized. Valid values are powers of two between 1 and the flash sector size, inclusive.

The following C definitions in the HAL may optionally be defined by the platform in the `flash_layout.h` header:

- `TFM_HAL_ITS_FLASH_AREA_ADDR` - Defines the base address of the dedicated flash area for ITS.
- `TFM_HAL_ITS_FLASH_AREA_SIZE` - Defines the size of the dedicated flash area for ITS in bytes.
- `TFM_HAL_ITS_SECTORS_PER_BLOCK` - Defines the number of contiguous physical flash erase sectors that form a logical erase block in the filesystem. The typical value is 1, but it may be increased so that the maximum required asset size will fit in one logical block.

If any of the above definitions are not provided by the platform, then the `tfm_hal_its_fs_info()` HAL API must be implemented instead. This function is documented in `tfm_hal_its.h`.

The sectors reserved to be used for Internal Trusted Storage **must** be contiguous.

Internal Trusted Storage Service Optional Platform Definitions

The following optional platform definitions may be defined in `flash_layout.h`:

- `ITS_RAM_FS_SIZE` - Defines the size of the RAM FS buffer when using the RAM FS emulated flash implementation. The buffer must be at least as large as the area earmarked for the filesystem by the HAL.
- `ITS_FLASH_NAND_BUF_SIZE` - Defines the size of the write buffer when using the NAND flash implementation. The buffer must be at least as large as a logical filesystem block.
- `ITS_MAX_BLOCK_DATA_COPY` - Defines the buffer size used when copying data between blocks, in bytes. If not provided, defaults to 256. Increasing this value will increase the memory footprint of the service.

More information about the `flash_layout.h` content, not ITS related, is available in *Details for the platform/ext folder* along with other platform information.

ITS Service Build Definitions

The ITS service uses a set of C definitions to compile in/out certain features, as well as to configure certain service parameters. When using the TF-M build system, these definitions are controlled by build flags of the same name. The `config/config_base.cmake` file sets the default values of those flags, but they can be overwritten based on platform capabilities by setting them in `platform/ext/target/<TARGET_NAME>/config.cmake`. The list of ITS service build definitions is:

- **ITS_CREATE_FLASH_LAYOUT**- this flag indicates that it is required to create an ITS flash layout. If this flag is set, ITS service will generate an empty and valid ITS flash layout to store assets. It will erase all data located in the assigned ITS memory area before generating the ITS layout. This flag is required to be set if the ITS memory area is located in a non-persistent memory. This flag can be set if the ITS memory area is located in a persistent memory without a valid ITS flash layout in it. That is the case when it is the first time in the device life that the ITS service is executed.
- **ITS_VALIDATE_METADATA_FROM_FLASH**- this flag allows to enable/disable the validation mechanism to check the metadata store in flash every time the flash data is read from flash. This validation is required if the flash is not hardware protected against data corruption.
- **ITS_RAM_FS**- setting this flag to ON enables the use of RAM instead of the persistent storage device to store the FS in the Internal Trusted Storage service. This flag is OFF by default. The ITS regression tests write/erase storage multiple time, so enabling this flag can increase the life of flash memory when testing. If this flag is set to ON, **ITS_RAM_FS_SIZE** must also be provided. This specifies the size of the block of RAM to be used to simulate the flash.

Note

If this flag is disabled when running the regression tests, then it is recommended that the persistent storage area is erased before running the tests to ensure that all tests can run to completion. The type of persistent storage area is platform specific (eFlash, MRAM, etc.) and it is described in corresponding `flash_layout.h`

- **ITS_MAX_ASSET_SIZE** - Defines the maximum asset size to be stored in the ITS area. This size is used to define the temporary buffers used by ITS to read/write the asset content from/to flash. The memory used by the temporary buffers is allocated statically as ITS does not use dynamic memory allocation.
- **ITS_NUM_ASSETS** - Defines the maximum number of assets to be stored in the ITS area. This number is used to dimension statically the filesystem metadata tables in RAM (fast access) and flash (persistent storage). The memory used by the filesystem metadata tables is allocated statically as ITS does not use dynamic memory allocation.
- **ITS_BUF_SIZE**- Defines the size of the partition's internal data transfer buffer. If not provided, then **ITS_MAX_ASSET_SIZE** is used to allow asset data to be copied between the client and the filesystem in one iteration. Reducing the buffer size will decrease the RAM usage of the partition at the expense of latency, as data will be copied in multiple iterations. *Note:* when data is copied in multiple iterations, the atomicity property of the filesystem is lost in the case of an asynchronous power failure.
- **ITS_STACK_SIZE**- Defines the stack size of the Internal Trusted Storage Secure Partition. This value mainly depends on the platform specific flash drivers, the build type (Debug, Release and MinSizeRel) and compiler.

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9.10.4 Platform Service Integration Guide

Introduction

TF-M Platform service is a trusted service which allows secure partitions and non-secure applications to interact with some platform-specific components. There are a number of features which requires some interaction with platform-specific components which are at the same time essential for the security of the system. Therefore, those components need to be handled by a secure partition which is part of the trusted compute base.

These platform-specific components include system reset, power management, Debug, GPIO, etc.

TF-M Platform interfaces

The TF-M interfaces for the Platform service are located in `interface/include/`. The TF-M Platform service source files are located in `secure_fw/partitions/platform`.

TF-M Platform service

The Platform service interfaces and types are defined and documented in `interface/include/tfm_platform_api.h`

- `platform_sp.h/c` : These files define and implement functionalities related to the platform service
- `tfm_platform_api.c` : This file implements `tfm_platform_api.h` functions to be called from the secure partitions. This is the entry point when the secure partitions request an action to the Platform service (e.g system reset).

Platform HAL

The Platform Service relies on a platform-specific implementation to perform some functionalities. Mandatory functionality (e.g. system reset) that are required to be implemented for a platform to be supported by TF-M have their dedicated HAL API functions. Additional platform-specific services can be provided using the IOCTL function call.

For API specification, please check: `platform/include/tfm_platform_system.h`

An implementation is provided in all the supported platforms. Please, check `platform/ext/target/<SPECIFIC_TARGET_FOLDER>/services/src/tfm_platform_system.c` for examples.

The API **must** be implemented by the system integrators for their targets.

IOCTL

A single entry point to platform-specific code across the HAL is provided by the IOCTL service and HAL function:

```
enum tfm_platform_err_t tfm_platform_hal_ioctl(tfm_platform_ioctl_req_t request,
                                              psa_invec *in_vec,
                                              psa_outvec *out_vec);
```

A request type is provided by the client, with additional parameters contained in the optional `in_vec` parameter. An optional output buffer can be passed to the service in `out_vec`. An IOCTL request type not supported on a particular platform should return `TFM_PLATFORM_ERR_NOT_SUPPORTED`

Non-Volatile counters

The Platform Service provides an abstracted service for exposing the NV counters to secure partitions or non-secure callers. The following operations are supported:

- Increment a counter.
- Read a counter value to a preallocated buffer.

```
enum tfm_platform_err_t
tfm_platform_nv_counter_increment(uint32_t counter_id);

enum tfm_platform_err_t
tfm_platform_nv_counter_read(uint32_t counter_id,
                           uint32_t size, uint8_t *val);
```

The range of counters id is defined in : `platform/include/tfm_plat_nv_counters.h`

For Level 2,3 isolation implementations, secure partitions in the Application Root of Trust, should have `TFM_PLATFORM_SERVICE` set as a dependency for access to the NV counter API.

Current Service Limitations

- **system reset** - The system reset functionality is only supported in isolation level 1. Currently the mechanism by which PSA-RoT services should run in privileged mode in level 3 is not in place due to an ongoing work in TF-M Core. So, the `NVIC_SystemReset` call performed by the service is expected to generate a memory fault when it tries to access the `SCB->AIRC` register in level 3 isolation.

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9.10.5 Protected Storage Service Integration Guide

Introduction

TF-M Protected Storage (PS) service implements PSA Protected Storage APIs.

The service is usually backed by hardware isolation of the flash access domain and, in the current version, relies on hardware to isolate the flash area from non-secure access. In absence of hardware isolation, the secrecy and integrity of data is still maintained.

The PS service implements an AES-GCM based AEAD encryption policy, as a reference, to protect data integrity and authenticity.

The PS reuses the non-hierarchical filesystem provided by the TF-M Internal Trusted Storage service to store encrypted, authenticated objects.

The design addresses the following high level requirements as well:

- **Confidentiality** - Resistance to unauthorised accesses through hardware/software attacks.
- **Access Authentication** - Mechanism to establish requester's identity (a non-secure entity, secure entity, or a remote server).
- **Integrity** - Resistant to tampering by either the normal users of a product, package, or system or others with physical access to it. If the content of the protected storage is changed maliciously, the service is able to detect it.
- **Reliability** - Resistant to power failure scenarios and incomplete write cycles.
- **Configurability** - High level configurability to scale up/down memory footprint to cater for a variety of devices with varying security requirements.
- **Performance** - Optimized to be used for resource constrained devices with very small silicon footprint, the PPA (power, performance, area) should be optimal.

Current PS Service Limitations

- **Asset size limitation** - An asset is stored in a contiguous space in a block/sector. Hence, the maximum asset size can be up-to the size of the data block/sector. Detailed information about the maximum asset size can be found in the section *Maximum asset size* below.
- **Fragmentation** - The current design does not support fragmentation, as an asset is stored in a contiguous space in a block. Each block can potentially store multiple assets. A delete operation implicitly moves all the assets towards the top of the block to avoid fragmentation within block. However, this may also result in unutilized space at the end of each block.

- **Non-hierarchical storage model** - The current design uses a non-hierarchical storage model, as a filesystem, where all the assets are managed by a linearly indexed list of metadata. This model locates the metadata in blocks which are always stored in the same flash location. That increases the number of writes in a specific flash location as every change in the storage area requires a metadata update.
- **PSA internal trusted storage API** - In the current design, the service does not use the PSA Internal Trusted Storage API to write the rollback protection values stored in the internal storage.
- **Protection against physical storage medium failure** - Complete handling of inherent failures of storage mediums (e.g. bad blocks in a NAND based device) is not supported by the current design.
- **Key diversification** - In a more robust design, each asset would be encrypted through a different key.
- **Lifecycle management** - Currently, it does not support any subscription based keys and certificates required in a secure lifecycle management. Hence, an asset's validity time-stamp can not be invalidated based on the system time.
- **Provisioning vs user/device data** - In the current design, all assets are treated in the same manner. In an alternative design, it may be required to create separate partitions for provisioning content and user/device generated content. This is to allow safe update of provisioning data during firmware updates without the need to wipe out the user/device generated data.

Code Structure

Protected storage service code is located in `secure_fw/partitions/protected_storage/` and is divided as follows:

- Core files
- Cryptographic interfaces
- Non-volatile (NV) counters interfaces

The PSA PS interfaces for PS service are located in `interface/include/psa`

PSA Protected Storage Interfaces

The PS service exposes the following mandatory PSA PS interfaces, version 1.0:

```
psa_status_t psa_ps_set(psa_storage_uid_t uid, size_t data_length, const void *p_data,
↳ psa_storage_create_flags_t create_flags);
psa_status_t psa_ps_get(psa_storage_uid_t uid, size_t data_offset, size_t data_size,
↳ void *p_data, size_t *p_data_length);
psa_status_t psa_ps_get_info(psa_storage_uid_t uid, struct psa_storage_info_t *p_info);
psa_status_t psa_ps_remove(psa_storage_uid_t uid);
uint32_t psa_ps_get_support(void);
```

For the moment, it does not support the extended version of those APIs.

These PSA PS interfaces and PS TF-M types are defined and documented in `interface/include/psa/protected_storage.h`, `interface/include/psa/storage_common.h` and `interface/include/tfm_ps_defs.h`

Core Files

- `tfm_ps_req_mgr.c` - Contains the PS request manager implementation which handles all requests which arrive to the service. This layer extracts the arguments from the input and output vectors, and it calls the protected storage layer with the provided parameters.

- `tfm_protected_storage.c` - Contains the TF-M protected storage API implementations which are the entry points to the PS service.
- `ps_object_system.c` - Contains the object system implementation to manage all objects in PS area.
- `ps_object_table.c` - Contains the object system table implementation which complements the object system to manage all object in the PS area. The object table has an entry for each object stored in the object system and keeps track of its version and owner.
- `ps_encrypted_object.c` - Contains an implementation to manipulate encrypted objects in the PS object system.
- `ps_utils.c` - Contains common and basic functionalities used across the PS service code.
- `ps_filesystem_interface.c` - Contains the interface to directly use ITS implementation of services.

Flash Filesystem and Flash Interfaces

The non-hierarchical filesystem and flash interfaces reside under Internal Trusted Storage service directory. The way PS service uses them depends on the config option `TFM_PARTITION_INTERNAL_TRUSTED_STORAGE`.

When TF-M Internal trusted storage service is active (i.e. `TFM_PARTITION_INTERNAL_TRUSTED_STORAGE = ON`), all file system and storage interfaces are built as part of ITS service library. Thus, the PS service stores encrypted, authenticated objects by making service calls to the ITS service.

When TF-M Internal trusted storage service is disabled (i.e. `TFM_PARTITION_INTERNAL_TRUSTED_STORAGE = OFF`), all file system and storage interfaces are built as part of PS service library. Thus, the PS service stores encrypted, authenticated objects by making standard function calls to the file system within its own partition code.

The ITS filesystem and flash interfaces and their implementation can be found in `secure_fw/partitions/internal_trusted_storage/flash_fs` and `secure_fw/partitions/internal_trusted_storage/flash` respectively. More information about the filesystem and flash interfaces can be found in the *ITS integration guide*.

The ITS service implementation in `secure_fw/partitions/internal_trusted_storage/tfm_internal_trusted_storage.c`, constructs a filesystem configuration for Protected Storage based on target-specific definitions from the Protected Storage HAL. Please see the *Protected Storage Service HAL* section for details of these.

Cryptographic Interface

- `crypto/ps_crypto_interface.h` - Abstracts the cryptographic operations for the protected storage service.
- `crypto/ps_crypto_interface.c` - Implements the PS service cryptographic operations with calls to the TF-M Crypto service.

Non-volatile (NV) Counters Interface

The current PS service provides rollback protection based on NV counters. PS defines and implements the following NV counters functionalities:

- `nv_counters/ps_nv_counters.h` - Abstracts PS non-volatile counters operations. This API detaches the use of NV counters from the TF-M NV counters implementation, provided by the platform, and provides a mechanism to compile in a different API implementation for test purposes. A PS test suite **may** provide its own custom implementation to be able to test different PS service use cases.
- `nv_counters/ps_nv_counters.c` - Implements the PS NV counters interfaces based on TF-M NV counters implementation provided by the platform.

PS Service Integration Guide

This section describes mandatory (i.e. **must** implement) or optional (i.e. **may** implement) interfaces which the system integrator have to take in to account in order to integrate the protected storage service in a new platform.

Maximum Asset Size

An asset is stored in a contiguous space in a block/sector. The maximum size of an asset can be up-to the size of the data block/sector minus the object header size (PS_OBJECT_HEADER_SIZE) which is defined in `ps_object_defs.h`. The PS_OBJECT_HEADER_SIZE changes based on the PS_ENCRYPTION flag status.

Protected Storage Service HAL

The PS service requires the platform to implement the PS HAL, defined in `platform/include/tfm_hal_ps.h`.

The following C definitions in the HAL are mandatory, and must be defined by the platform in a header named `flash_layout.h`:

- `TFM_HAL_PS_FLASH_DRIVER` - Defines the identifier of the CMSIS Flash `ARM_DRIVER_FLASH` object to use for PS. It must have been allocated by the platform and will be declared extern in the HAL header.
- `TFM_HAL_PS_PROGRAM_UNIT` - Defines the size of the PS flash device's physical program unit (the smallest unit of data that can be individually programmed to flash). It must be equal to `TFM_HAL_PS_FLASH_DRIVER.GetInfo()` -> `program_unit`, but made available at compile time so that filesystem structures can be statically sized. Valid values are powers of two between 1 and the flash sector size, inclusive.

The following C definitions in the HAL may optionally be defined by the platform in the `flash_layout.h` header:

- `TFM_HAL_PS_FLASH_AREA_ADDR` - Defines the base address of the dedicated flash area for PS.
- `TFM_HAL_PS_FLASH_AREA_SIZE` - Defines the size of the dedicated flash area for PS in bytes.
- `TFM_HAL_PS_SECTORS_PER_BLOCK` - Defines the number of contiguous physical flash erase sectors that form a logical erase block in the filesystem. The typical value is 1, but it may be increased so that the maximum required asset size will fit in one logical block.

If any of the above definitions are not provided by the platform, then the `tfm_hal_ps_fs_info()` HAL API must be implemented instead. This function is documented in `tfm_hal_ps.h`.

The sectors reserved to be used for Protected Storage **must** be contiguous sectors starting at `TFM_HAL_PS_FLASH_AREA_ADDR`.

The design requires either 2 blocks, or any number of blocks greater than or equal to 4. Total number of blocks can not be 0, 1 or 3. This is a design choice limitation to provide power failure safe update operations.

Protected Storage Service Optional Platform Definitions

The following optional platform definitions may be defined in `flash_layout.h`:

- `PS_RAM_FS_SIZE` - Defines the size of the RAM FS buffer when using the RAM FS emulated flash implementation. The buffer must be at least as large as the area earmarked for the filesystem by the HAL.
- `PS_FLASH_NAND_BUF_SIZE` - Defines the size of the write buffer when using the NAND flash implementation. The buffer must be at least as large as a logical filesystem block.

More information about the `flash_layout.h` content, not ITS related, is available in *Details for the platform/ext folder* along with other platform information.

TF-M NV Counter Interface

To have a platform independent way to access the NV counters, TF-M defines a platform NV counter interface. For API specification, please check: `platform/include/tfm_plat_nv_counters.h`

The system integrators **may** implement this interface based on the target capabilities and set the `PS_ROLLBACK_PROTECTION` flag to compile in the rollback protection code.

Note

If this flag is enabled, the lifecycle of the PS service depends on the shorter write endurance of the assets storage device and the NV counters storage device.

Secret Platform Unique Key

The encryption policy relies on a secret hardware unique key (HUK) per device. It is system integrator's responsibility to provide an implementation which **must** be a non-mutable target implementation. For API specification, please check: `platform/include/tfm_plat_crypto_keys.h`

A stub implementation is provided in `platform/ext/common/template/crypto_keys.c`

Non-Secure Identity Manager

TF-M core tracks the current client IDs running in the secure or non-secure processing environment. It provides a dedicated API to retrieve the client ID which performs the service request.

Non-secure Client Extension Integration Guide provides further details on how client identification works.

PS service uses that TF-M core API to retrieve the client ID and associate it as the owner of an asset. Only the owner can read, write or delete that asset based on the creation flags.

The *integration guide* provides further details of non-secure implementation requirements for TF-M.

Cryptographic Interface

The reference encryption policy is built on AES-GCM, and it **may** be replaced by a vendor specific implementation.

The PS service abstracts all the cryptographic requirements and specifies the required cryptographic interface in `secure_fw/partitions/protected_storage/crypto/ps_crypto_interface.h`

The PS service cryptographic operations are implemented in `secure_fw/partitions/protected_storage/crypto/ps_crypto_interface.c`, using calls to the TF-M Crypto service.

PS Service Build Definitions

The PS service uses a set of C definitions to compile in/out certain features, as well as to configure certain service parameters. When using the TF-M build system, these definitions are controlled by build flags of the same name. The `config/config_base.cmake` file sets the default values of those flags, but they can be overwritten based on platform capabilities by setting them in `platform/ext/target/<TARGET_NAME>/config.cmake`. The list of PS service build definitions is:

- `PS_ENCRYPTION`- this flag allows to enable/disable encryption option to encrypt the protected storage data.
- `PS_CRYPTO_AEAD_ALG` - this flag indicates the AEAD algorithm to use for authenticated encryption in Protected Storage.

Note

For GCM/CCM it is essential that IV doesn't get repeated. If this flag is set to `PSA_ALG_GCM` or `PSA_ALG_CCM`, `PS_ROLLBACK_PROTECTION` must be enabled to protect against IV rollback.

- `PS_CREATE_FLASH_LAYOUT`- this flag indicates that it is required to create a PS flash layout. If this flag is set, PS service will generate an empty and valid PS flash layout to store assets. It will erase all data located in the assigned PS memory area before generating the PS layout. This flag is required to be set if the PS memory area is located in a non-persistent memory. This flag can be set if the PS memory area is located in a persistent memory without a valid PS flash layout in it. That is the case when it is the first time in the device life that the PS service is executed.
- `PS_VALIDATE_METADATA_FROM_FLASH`- this flag allows to enable/disable the validation mechanism to check the metadata store in flash every time the flash data is read from flash. This validation is required if the flash is not hardware protected against malicious writes. In case the flash is protected against malicious writes (i.e embedded flash, etc), this validation can be disabled in order to reduce the validation overhead.
- `PS_ROLLBACK_PROTECTION`- this flag allows to enable/disable rollback protection in protected storage service. This flag takes effect only if the target has non-volatile counters and `PS_ENCRYPTION` flag is on.
- `PS_AES_KEY_USAGE_LIMIT` - setting this to a value other than zero limits the number of AES blocks that will be encrypted/decrypted using any one key. When the limit is reached for a given object, a new key will be derived and the data will be encrypted with the new key before being stored. Note that setting this limit too low may reduce the maximum asset size because PS will reject objects that are too large to be encrypted and decrypted without hitting this limit.
- `PS_RAM_FS`- setting this flag to ON enables the use of RAM instead of the persistent storage device to store the FS in the Protected Storage service. This flag is OFF by default. The PS regression tests write/erase storage multiple time, so enabling this flag can increase the life of flash memory when testing. If this flag is set to ON, `PS_RAM_FS_SIZE` must also be provided. This specifies the size of the block of RAM to be used to simulate the flash.

Note

If this flag is disabled when running the regression tests, then it is recommended that the persistent storage area is erased before running the tests to ensure that all tests can run to completion. The type of persistent storage area is platform specific (eFlash, MRAM, etc.) and it is described in corresponding `flash_layout.h`

- `PS_MAX_ASSET_SIZE` - Defines the maximum asset size to be stored in the PS area. This size is used to define the temporary buffers used by PS to read/write the asset content from/to flash. The memory used by the temporary buffers is allocated statically as PS does not use dynamic memory allocation.
- `PS_NUM_ASSETS` - Defines the maximum number of assets to be stored in the PS area. This number is used to dimension statically the object table size in RAM (fast access) and flash (persistent storage). The memory used by the object table is allocated statically as PS does not use dynamic memory allocation.
- `PS_TEST_NV_COUNTERS`- this flag enables the virtual implementation of the PS NV counters interface in `test/secure_fw/suites/ps/secure/nv_counters` of the `tf-m-tests` repo, which emulates NV counters in RAM, and disables the hardware implementation of NV counters provided by the secure service. This flag is enabled by default, but has no effect when the secure regression test is disabled. This flag can be overridden to OFF when building the regression tests. In this case, the PS rollback protection test suite will not be built, as it relies on extra functionality provided by the virtual NV counters to simulate different rollback scenarios. The remainder of the PS test suites will run using the hardware NV counters. Please note that running the tests in this configuration will quickly increase the hardware NV counter values, which cannot be decreased again. Overriding this flag from its default value of OFF when not building the regression tests is not currently supported.

- `PS_STACK_SIZE`- Defines the stack size of the Protected Storage Secure Partition. This value mainly depends on the build type(debug, release and minisizerel) and compiler.

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9.10.6 Adding Secure Partition

Terms and abbreviations

This document uses the following terms and abbreviations.

Table 36:: term table

Term	Meaning
FF-M ID	Firmware Framework for M Identifier
IPC	Interprocess communication
IPC model	The secure IPC framework
irqs	Interrupt requests
MMIO	Memory Mapped I/O
PSA	Platform Security Architecture
RoT	Root of Trust
SFN	Secure Function
SFN model	Secure Function model
SID	RoT Service ID
SP	Secure Partition
SPM	Secure Partition Manager
TF-M	Trusted firmware M

Introduction

Secure Partition is an execution environment that provides the following functions to Root of Trust (RoT) Services:

- Access to resources, protection of its own code and data.
- Mechanisms to interact with other components in the system.

Each Secure Partition is a single thread of execution and the smallest unit of isolation.

This document mainly describes how to add a secure partition in TF-M and focuses on the configuration, manifest, implement rules. The actual source-level implementation is not included in this document.

Note

If not otherwise specified, the steps are identical for IPC and SFN model.

The IPC and SFN model conforms to the *PSA Firmware Framework for M (FF-M) v 1.1* changes. Refer to [PSA Firmware Framework specification](#) and [Firmware Framework for M 1.1 Extensions](#) for details.

Process

The main steps to add a secure partition are as follows:

- *Add source folder*
- *Add manifest*
- *Update the Build System*
- *Implement the RoT services*

Add source folder

Add a source folder under `<TF-M base folder>/secure_fw/partitions` for the new secure partition (Let's take `example` as the folder name):

This folder should include those parts:

- Manifest file
- CMake configuration files
- Source code files

Add manifest

Each Secure Partition must have resource requirements declared in a manifest file. The Secure Partition Manager (SPM) uses the manifest file to assemble and allocate resources within the SPE. The manifest includes the following:

- A Secure Partition name.
- A list of implemented RoT Services.
- Access to other RoT Services.
- Memory requirements.
- Scheduling hints.
- Peripheral memory-mapped I/O regions and interrupts.

Note

The current manifest format in TF-M is “yaml” which is different from the requirement of PSA FF.

Note

The users can set LOW, NORMAL or HIGH in `priority` attribute in a Secure Partition manifest.

The manifest tool `tools/tfm_parse_manifest_list.py` calculates the load priority of Secure Partitions based on their `priority` values and their dependencies. SPM determines the loading and initialization order of Secure Partitions based on their load priority values during TF-M initialization.

- A Secure Partition with a higher `priority` is loaded and initialized before Secure Partitions with lower `priority`.
- A Secure Partition is loaded and initialized after its dependencies are.

Here is a manifest reference example for the IPC model:

Note

To use SFN model, the user needs to replace "model": "IPC" to "model": "SFN". The user also needs to remove the attribute "entry_point", and optionally replace it with "entry_init".

```
{
  "psa_framework_version": 1.1,
  "name": "TFM_SP_EXAMPLE",
  "type": "APPLICATION-ROT",
  "priority": "NORMAL",
  "model": "IPC",
  "entry_point": "tfm_example_main",
  "stack_size": "0x0200",
  "services" : [
    {
      "name": "ROT_A",
      "sid": "0x000000E0",
      "non_secure_clients": true,
      "connection_based": true,
      "version": 1,
      "version_policy": "STRICT"
      "mm_iovec": "disable"
    }
  ],
  "mmio_regions": [
    {
      "name": "TFM_PERIPHERAL_A",
      "permission": "READ-WRITE"
    }
  ],
  "irqs": [
    {
      "source": "TFM_A_IRQ",
      "name": "A_IRQ",
      "handling": "SLIH"
    }
  ]
  "dependencies": [
    "TFM_CRYPT0",
    "TFM_INTERNAL_TRUSTED_STORAGE_SERVICE"
  ]
}
```

Update manifest list

The <TF-M base folder>/tools/tfm_manifest_list.yaml is used to collect necessary information of secure partition. The manifest tool tools/tfm_parse_manifest_list.py processes it and generates necessary files while building.

Please refer to the *Manifest List* for the format of manifest lists.

Reference configuration example:

```
{
  "description": "TFM Example Partition",
  "manifest": "secure_fw/partitions/example/tfm_example_partition.yaml",
  "conditional": "@TFM_PARTITION_EXAMPLE@",
  "output_path": "partitions/example",
  "version_major": 0,
  "version_minor": 1,
  "pid": 290,
  "linker_pattern": {
    "library_list": [
      "*tfm_*partition_example*"
    ]
  }
}
```

TF-M also supports out-of-tree Secure Partition build where you can have your own manifest lists. Please refer to *Out-of-tree Secure Partition build* for details.

Secure Partition ID Distribution

Every Secure Partition has an identifier (ID). TF-M will generate a header file that includes definitions of the Secure Partition IDs. The header file is `<TF-M build folder>generated/interface/include/psa_manifest/pid.h`. Each definition uses the name attribute in the manifest as its name and the value is allocated by SPM.

The Partition ID can be set to a fixed value or omitted to be auto allocated.

```
#define name id-value
```

Table 37:: PID table

Secure Partitions	PID Range
TF-M Internal Partitions	0 - 255
PSA and user Partitions	256 - 2999
TF-M test Partitions	3000 - 4999
Firmware Framework test Partitions	5000 - 5999
Reserved	6000 -

The full list of valid PIDs is included in `<TF-M base folder>/tools/parse_tfm_manifest_list.yaml`.

Please refer to `<TF-M base folder>/tools/tfm_manifest_list.yaml`, `<TF-M extras repo>/partitions/*/*_manifest_list.yaml`, `<TF-M test repo>/tests_reg/test/secure_fw/tfm_test_manifest_list.yaml` and `<TF-M test repo>/tests_psa_arch/spe/tfm_psa_ff_test_manifest_list.yaml` for the detailed PID allocations.

About where to add the definition, please refer to the chapter *Update manifest list*.

RoT Service ID (SID) Distribution

An RoT Service is identified by its RoT Service ID (SID). A SID is a 32-bit number that is associated with a symbolic name in the Secure Partition manifest. The bits [31:12] uniquely identify the vendor of the RoT Service. The remaining bits [11:0] can be used at the discretion of the vendor.

Here is the RoT Service ID table used in TF-M.

Table 38:: SID table

Partitions	Vendor ID(20 bits)	Function ID(12 bits)
initial_attestation	0x00000	0x020-0x03F
platform	0x00000	0x040-0x05F
protected_storage	0x00000	0x060-0x06F
internal_trusted_storage	0x00000	0x070-0x07F
crypto	0x00000	0x080-0x09F
firmware_update	0x00000	0x0A0-0x0BF
ns_agent_mailbox	0x00000	0x0C0-0x0DF
tfm_secure_client	0x0000F	0x000-0x01F
tfm_ipc_client	0x0000F	0x060-0x07F
tfm_ipc_service	0x0000F	0x080-0x09F
tfm_slih_test_service	0x0000F	0x0A0-0x0AF
tfm_flih_test_service	0x0000F	0x0B0-0x0BF
tfm_ps_test_service	0x0000F	0x0C0-0x0DF
tfm_secure_client_2	0x0000F	0x0E0-0x0FF
tfm_sfn_test_service_1	0x0000F	0x100-0x11F
tfm_sfn_test_service_2	0x0000F	0x120-0x13F
tfm_attest_test_service	0x0000F	0x140-0x15F

RoT Service Stateless Handle Distribution

A Secure partition may include stateless services. They are distinguished and referenced by stateless handles. In manifest, a `stateless_handle` attribute is set for indexing stateless services. It must be either "auto" or a number in the range [1, 32] in current implementation and may extend. Also the `connection-based` attribute must be set to `false` for stateless services.

The indexes of stateless handles are divided into two ranges for different usages. Indexes [1, 16] are assigned to TF-M Secure Partitions. The rest indexes [17, 32] are reserved for any other Secure Partitions, for example Secure Partitions in `tf-m-tests` and `tf-m-extras`.

The following table summaries the stateless handle allocation for the TF-M Secure Partitions.

Table 39:: Stateless Handle table

Partition name	Stateless Handle
TFM_SP_CRYPT0	1
TFM_SP_PS	2
TFM_SP_ITS	3
TFM_SP_INITIAL_ATTESTATION	4
TFM_SP_FWU	5
TFM_SP_PLATFORM	6
TFM_NS_MAILBOX_AGENT	7

For the indexes of other Secure Partitions, please refer to their manifests or documentations.

stack_size

The `stack_size` is required to indicate the stack memory usage of the Secure Partition. The value of this attribute must be a decimal or hexadecimal value in bytes. It can also be a build configurable with default value defined in `config_base.cmake`. The value of the configuration can be overridden to fit different use cases.

heap_size

This attribute is optional. The default value is 0. It indicates the heap memory usage of the Secure Partition. The allowed values are the same as the `stack_size`.

mmio_regions

This attribute is a list of MMIO region objects which the Secure Partition needs access to. TF-M only supports the `named_region` current. Please refer to PSA FF for more details about it. The user needs to provide a name macro to indicate the variable of the memory region.

TF-M uses the below structure to indicate a peripheral memory.

```
struct platform_data_t {  
    uint32_t periph_start;  
    uint32_t periph_limit;  
    int16_t periph_ppc_bank;  
    int16_t periph_ppc_loc;  
};
```

Note

This structure is not expected by TF-M, it's only that the current implementations are using. Other peripherals that need different information to create isolation need to define a different structure with the same name.

Here is an example for it:

```
struct platform_data_t tfm_peripheral_A;  
#define TFM_PERIPHERAL_A (&tfm_peripheral_A)
```

mm_iovec

Memory-mapped iovecs (MM-IOVEC) provides direct mapping of client input and output vectors into the Secure Partition. When this attribute is set to `enable`, it enables Secure Partitions to use the MM-IOVEC APIs if the framework supports MM-IOVEC.

Using MM-IOVEC provides a memory and runtime optimization for larger buffers, but reduces mitigation for common security vulnerabilities. Please refer to [Firmware Framework for M 1.1 Extensions](#) for more details. Whether to use MM-IOVEC depends on the requirements of memory and runtime optimization and security.

Update the Build System

The following changes to the build system are required for the newly added secure partition.

Add a CMakeLists.txt file

Each Secure Partition must have a corresponding `CMakeLists.txt`, in this case, `<TF-M base folder>/secure_fw/partitions/example/CMakeLists.txt`, which is the compilation configuration for this Secure Partition.

Here is a reference example for `CMakeLists.txt`

The CMake file should include the following contents

- Add library `tfm_app_rot_partition_example` and associated source files.

```
add_library(tfm_app_rot_partition_example STATIC)

target_sources(tfm_app_rot_partition_example
    PRIVATE
        tfm_example_partition.c
)

```

Note

The secure partition must be built as a standalone static library, and the name of the library must follow this pattern, as it affects how the linker script will lay the partition in memory:

- tfm_psa_rot_partition* in case of a PSA RoT partition
- tfm_app_rot_partition* in case of an Application RoT partition

- Add source files generated by the manifest tool.

```
# The intermedia file defines the partition stack.
target_sources(tfm_app_rot_partition_example
    PRIVATE
        ${CMAKE_BINARY_DIR}/generated/example_partition/auto_generated/intermedia_
→tfm_example_partition.c
)

# The load info file includes the static data of the partition.
target_sources(tfm_partitions
    INTERFACE
        ${CMAKE_BINARY_DIR}/generated/example_partition/auto_generated/load_info_
→tfm_example_partition.c
)

```

- Add dependency with manifest tool.

To make sure the above generated files are up-to-date when the Secure Partition library is built, dependencies between the library and the manifest tool target should be set up.

```
add_dependencies(tfm_app_rot_partition_example manifest_tool)

```

- Link tfm_sprt for the PSA API interfaces.

```
target_link_libraries(tfm_app_rot_partition_example
    PRIVATE
        tfm_sprt
)

```

- Link the Secure Partition library to tfm_partitions so that it can be included in the final image.

```
target_link_libraries(tfm_partitions
    INTERFACE
        tfm_app_rot_partition_example
)

```

Finally, the build of this Secure Partition should be added to <TF-M base folder>/secure_fw/partitions/CMakeLists.txt.

```
add_subdirectory(example)
```

Update the Config System

If the Secure Partition has the build config to enable or disable it, the config option should be added to config systems.

CMake Config

The default value of the config option should be added to the <TF-M base folder>/config/config_base.cmake.

```
set(TFM_PARTITION_EXAMPLE OFF CACHE BOOL "Enable the example partition")
```

Kconfig

A menuconfig should be added to <TF-M base folder>/secure_fw/partitions/example/Kconfig.

```
menuconfig TFM_PARTITION_EXAMPLE
    bool "Enable the Example Partition"
    default n
```

And add it to <TF-M base folder>/secure_fw/partitions/Kconfig

```
resource ``example/Kconfig``
```

Note

The Secure Partition building should be skipped if it is not enabled. This should be done by adding the following code at the beginning of its CMakeLists.txt

```
if (NOT TFM_PARTITION_EXAMPLE)
    return()
endif()
```

Implement the RoT services

To implement RoT services, the partition needs a source file which contains the implementations of the services, as well as the partition entry point. The user can create this source file under <TF-M base folder>/secure_fw/partitions/example/tfm_example_partition.c.

As an example, the RoT service with SID **ROT_A** will be implemented.

Entry point for IPC Model Partitions

This function must have a loop that repeatedly waits for input signals and then processes them, following the Secure Partition initialization.

```
#include "psa_manifest/tfm_example.h"
#include "psa/service.h"
```

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```

void tfm_example_main(void)
{
    psa_signal_t signals = 0;

    /* Secure Partition initialization */
    example_init();

    /*
     * Continually wait for one or more of the partition's RoT Service or
     * interrupt signals to be asserted and then handle the asserted
     * signal(s).
     */
    while (1) {
        signals = psa_wait(PSA_WAIT_ANY, PSA_BLOCK);
        if (signals & ROT_A_SIGNAL) {
            rot_A();
        } else {
            /* Should not come here */
            psa_panic();
        }
    }
}

```

Entry init for SFN Model Partitions

In the SFN model, the Secure Partition consists of one optional initialization function, which is declared as the `entry_init` symbol as mentioned in section *Add manifest*. After initialization, the `entry_init` function returns the following values:

- Return `PSA_SUCCESS` if initialization succeeds.
- Return `PSA_SUCCESS` if initialization is partially successful, and you want some SFNs to receive messages. RoT Services that are non-operational must respond to connection requests with `PSA_ERROR_CONNECTION_REFUSED`.
- Return an error status if the initialization failed, and no SFNs within the Secure Partition must be called.

Service implementation for IPC Model

The service is implemented by the `rot_A()` function, which is called upon an incoming signal. This implementation is up to the user, however an example service has been included for reference. The following example sends a message “Hello World” when called.

```

#include "psa_manifest/tfm_example.h"
#include "psa/service.h"

/* Some other type of services. */
#define SOME_ROT_A_SERVICE_TYPE (1)

static void rot_A(void)
{
    const int BUFFER_LEN = 32;
    psa_msg_t msg;
    int i;

```

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```

uint8_t rec_buf[BUFFER_LEN];
uint8_t send_buf[BUFFER_LEN] = "Hello World";

psa_get(ROT_A_SIGNAL, &msg);
switch (msg.type) {
case PSA_IPC_CONNECT:
case PSA_IPC_DISCONNECT:
    /*
     * This service does not require any setup or teardown on connect
     * or disconnect, so just reply with success.
     */
    psa_reply(msg.handle, PSA_SUCCESS);
    break;
default:
    /* Handling services requested by psa_call. */
    if (msg.type == PSA_IPC_CALL) {
        for (i = 0; i < PSA_MAX_IOVEC; i++) {
            if (msg.in_size[i] != 0) {
                psa_read(msg.handle, i, rec_buf, BUFFER_LEN);
            }
            if (msg.out_size[i] != 0) {
                psa_write(msg.handle, i, send_buf, BUFFER_LEN);
            }
        }
        psa_reply(msg.handle, PSA_SUCCESS);
    } else if (msg.type == SOME_ROT_A_SERVICE_TYPE) {
        /* Operations for SOME_ROT_A_SERVICE_TYPE */
    } else {
        /* Invalid type for this Secure Partition. */
        return PSA_ERROR_PROGRAMMER_ERROR;
    }
}
}

```

Service implementation for SFN Model

SFN model consists of a set of Secure Functions (SFN), one for each RoT Service. The connection, disconnection and request messages do not cause a Secure Partition signal to be asserted for SFN Secure Partitions. Instead, the Secure Function (SFN) for the RoT Service is invoked by the framework, with the message details provided as a parameter to the SFN. To add a secure function (SFN) to process messages for each RoT Service, each SFN will have following prototype.

```
psa_status_t <<name>>_sfn(const psa_msg_t *msg);
```

A connection-based example service has been included for reference which sends a message “Hello World” when called.

```

#include "psa_manifest/tfm_example.h"
#include "psa/service.h"

/* Some other type of services. */
#define SOME_ROT_A_SERVICE_TYPE (1)

```

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```

psa_status_t rot_a_sfn(const psa_msg_t *msg)
{
    const int BUFFER_LEN = 32;
    int i;
    uint8_t rec_buf[BUFFER_LEN];
    uint8_t send_buf[BUFFER_LEN] = "Hello World";

    switch (msg->type) {
    case PSA_IPC_CONNECT:
    case PSA_IPC_DISCONNECT:
        /*
         * This service does not require any setup or teardown on connect
         * or disconnect, so just reply with success.
         */
        return PSA_SUCCESS;
    default:
        /* Handling services requested by psa_call. */
        if (msg->type == PSA_IPC_CALL) {
            for (i = 0; i < PSA_MAX_IOVEC; i++) {
                if (msg->in_size[i] != 0) {
                    psa_read(msg->handle, i, rec_buf, BUFFER_LEN);
                }
                if (msg->out_size[i] != 0) {
                    psa_write(msg->handle, i, send_buf, BUFFER_LEN);
                }
            }
            return PSA_SUCCESS;
        } else if (msg->type == SOME_ROT_A_SERVICE_TYPE) {
            /* Operations for SOME_ROT_A_SERVICE_TYPE */
        } else {
            /* Invalid type for this Secure Partition. */
            return PSA_ERROR_PROGRAMMER_ERROR;
        }
    }
}

```

Test suites and test partitions

A regression test suite can be added to verify whether the new secure partition works as expected. Refer to [Adding TF-M Regression Test Suite](#) for the details of adding a regression test suite.

Some regression tests require a dedicated RoT service. The implementations of the RoT service for test are similar to secure partition addition. Refer to [Adding partitions for regression tests](#) to get more information.

Out-of-tree Secure Partition build

TF-M supports out-of-tree Secure Partition build, whose source code folders are maintained outside TF-M repo. Developers can configure `TFM_EXTRA_MANIFEST_LIST_FILES` and `TFM_EXTRA_PARTITION_PATHS` in build command line to include out-of-tree Secure Partitions.

- `TFM_EXTRA_MANIFEST_LIST_FILES`

A list of the absolute path(s) of the manifest list(s) provided by out-of-tree Secure Partition(s). Use semicolons

; to separate multiple manifest lists. Wrap the multiple manifest lists with double quotes.

- **TFM_EXTRA_PARTITION_PATHS**

A list of the absolute directories of the out-of-tree Secure Partition source code folder(s). TF-M build system searches CMakeLists.txt of partitions in the source code folder(s). Use semicolons ; to separate multiple out-of-tree Secure Partition directories. Wrap the multiple directories with double quotes.

A single out-of-tree Secure Partition folder can be organized as the figure below.

```
secure partition folder
├── CMakeLists.txt
├── manifest_list.yaml
├── out_of_tree_partition_manifest.yaml
└── source code
```

In the example above, TFM_EXTRA_MANIFEST_LIST_FILES and TFM_EXTRA_PARTITION_PATHS in the build command can be configured as listed below.

```
-DTFM_EXTRA_MANIFEST_LIST_FILES=<Absolute-path-sp-folder/manifest_list.yaml>
-DTFM_EXTRA_PARTITION_PATHS=<Absolute-path-sp-folder>
```

Multiple out-of-tree Secure Partitions can be organized in diverse structures. For example, multiple Secure Partitions can be maintained under the same directory as shown below.

```
top-level folder
├── Partition 1
│   ├── CMakeLists.txt
│   ├── partition_1_manifest.yaml
│   └── source code
├── Partition 2
│   └── ...
├── Partition 3
│   └── ...
├── manifest_list.yaml
└── Root CMakeLists.txt
```

In the example above, a root CMakeLists.txt includes all the partitions' CMakeLists.txt, for example via add_subdirectory(). The manifest_list.yaml lists all partitions' manifest files. TFM_EXTRA_MANIFEST_LIST_FILES and TFM_EXTRA_PARTITION_PATHS in build command line can be configured as listed below.

```
-DTFM_EXTRA_MANIFEST_LIST_FILES=<Absolute-path-top-level-folder/manifest_list.yaml>
-DTFM_EXTRA_PARTITION_PATHS=<Absolute-path-top-level-folder>
```

Alternatively, out-of-tree Secure Partitions can be separated in different folders.

<pre>partition 1 folder ├── CMakeLists.txt ├── manifest_list.yaml ├── partition_1_manifest.yaml └── source code</pre>	<pre>partition 2 folder ├── CMakeLists.txt ├── manifest_list.yaml ├── partition_2_manifest.yaml └── source code</pre>
---	---

In the example above, each Secure Partition manages its own manifest files and CMakeLists.txt. TFM_EXTRA_MANIFEST_LIST_FILES and TFM_EXTRA_PARTITION_PATHS in build command line can be configured as listed below. Please note those input shall be wrapped with double quotes.

```
-DTFM_EXTRA_MANIFEST_LIST_FILES="<Absolute-path-part-1-folder/manifest\_list.yaml>;  
↔<Absolute-path-part-2-folder/manifest\_list.yaml>"  
-DTFM_EXTRA_PARTITION_PATHS="<Absolute-path-part-1-folder>; <Absolute-path-part-2-folder>"
```

Note

Manifest list paths in TFM_EXTRA_MANIFEST_LIST_FILES do NOT have to be one-to-one mapping to Secure Partition directories in TFM_EXTRA_PARTITION_PATHS. The orders don't matter either.

TFM_EXTRA_MANIFEST_LIST_FILES and TFM_EXTRA_PARTITION_PATHS can be configured in multiple extra sources. It is recommended to use CMake list APPEND method to avoid unexpected override.

Further Notes

- In the IPC model, Use PSA FF proposed memory accessing mechanism. SPM provides APIs and checking between isolation boundaries, a free accessing of memory can cause program panic.
- In the IPC model, the memory checking inside partition runtime is unnecessary. SPM handles the checking while memory accessing APIs are called.
- In the IPC model, the client ID had been included in the message structure and secure partition can get it when calling psa_get() function. The secure partition does not need to call tfm_core_get_caller_client_id() to get the caller client ID anymore.
- In the IPC model, SPM will check the security policy and partition dependence between client and service. So the service does not need to validate the secure caller anymore.

Reference

[PSA Firmware Framework specification](#)

[Firmware Framework for M 1.1 Extensions](#)

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9.10.7 TF-M Manifest Tool User Guide

This document describes the TF-M manifest tool and its usage. The target audiences are mainly platform integrators and Secure Partition developers.

This document assumes that audiences have knowledge about the terminologies defined in PSA Firmware Framework (FF-M) v1.0¹ and the FF-M v1.1 extensions², such as Secure Partition and manifests.

Introduction

The TF-M manifest tool is a python script which is used to parse Secure Partition manifests and generate source header files defined by FF-M³ such as psa_manifest/pid.h and psa_manifest/sid.h. It also generates TF-M specific files for building.

In the TF-M build system, the manifest tool is invoked during building automatically. Arguments are passed through CMake variables and they are customizable. See *Usage in TF-M Build System* for more details.

It can be also used as a standalone tool.

¹ FF-M v1.0 Specification

² FF-M v1.1 Extension

Arguments

The tool takes 5 arguments at most.

```
tfm_parse_manifest_list.py [-h] -o out_dir
                           -m manifest list [manifest list ...]
                           -f file-list [file-list ...]
                           -c config-files [config-files ...]
                           [-q]
```

-o/--outdir

Required.

The directory to hold the output files.

-m/--manifest-lists

Required. At least one item must be provided.

A list of TF-M Secure Partition manifest lists which contain one or more Secure Partition manifests in each. See *Manifest List* for details.

-f/--file-list

Required. At least one item must be provided.

A list of files that describe what files the tool should generate. See *Generated File List* for details.

-c/--config-files

Required. At least one item must be provided.

A list of header files which contain build configurations such as isolation level and Secure Partition enabled status. See *Configuration Header File* for details.

-q/--quiet

Optional.

Reduces log messages, only warnings and errors will be printed.

Manifest List

A manifest list is a YAML³ file that describes a list of Secure Partition manifests. Refer to the TF-M default manifest list⁴ as an example.

Each manifest list must contain a `manifest_list` attribute which collects the descriptions of Secure Partition manifests. Following are the supported attributes of in the manifest lists.

- **description**

Required.

Descriptive information for the tool to refer to the Secure Partition, for example in logs.

- **manifest**

Required.

The manifest file of the Secure Partition. Both absolute path and relative path to the manifest list are supported. Environment variables are supported.

³ YAML

⁴ TF-M manifest list

- `output_path`

Optional.

The directory to hold the Secure Partition specific output files. See *Generated File List* for more details.

Both absolute path and relative path to the directory specified by `-o/--outdir` are supported. Environment variables are supported.

It is set to the directory specified by `-o/--outdir` if omitted.

- `conditional`

Optional.

The configuration to enable or disable this Secure Partition. The value must be defined in one of the *Configuration Header File*. If it is omitted, the Secure Partition is always enabled.

- `pid`

Optional.

The Secure Partition ID (PID).

For non-test purpose Secure Partitions, it is required. See *Adding Secure Partition* for the PID allocations.

For test purpose Secure Partitions, this attribute is optional. The manifest tool assigns one for the Secure Partition. The value is not guaranteed to be the same for different builds.

- `linker_pattern`

Optional.

The information for linker to place the symbols of the Secure Partition. It is only required if you are using the linker scripts provided by TF-M. Each Secure Partition is expected to be built as a library. The name of the library must follow the format of `tfm_<type>_partition_<name>`. The valid value for `<type>` is `[psa_rot, app_rot]` corresponding to the type of the Secure Partitions. The `<name>` is any string to distinguish the Secure Partition from others.

Supported patterns are:

- `library_list`, must be `*tfm_*partition_<name>.*`.
- `object_list`

Any object files containing symbols belonging to the Secure Partition that are not included in the Secure Partitions library.

- `non_ffm_attributes`

Optional.

TF-M defines some special manifest attributes for TF-M dedicated Secure Partitions. Those special attributes are not compliant to FF-M. Using the specific attributes requires explicit registration by adding them to this `non_ffm_attributes`. The purpose is to ensure that developers are aware of the compliance issue. The manifest tool will report errors if unregistered Non-FFM attributes are detected. This attribute is for TF-M specific Secure Partitions and using TF-M-specific attributes is not encouraged.

Generated File List

A generated file list is a YAML file that describes the files to be generated by the manifest tool. Refer to TF-M default generated file list⁵ as an example.

⁵ TF-M generated file list

Each one must contain a `file_list` attribute which collects the files to generate. Each item in the `file_list` must contain the following attributes.

- `template`

This attribute is the file path of a Jinja2⁶ template. The TF-M manifest tool uses Jinja2 template engine for file generations. It can be a relative path to TF-M root directory or an absolute path. Environment variables are supported.

- `output`

The output file of the `template`. Both absolute path and relative path to the directory specified by `-o/--outdir` are supported. Environment variables are supported.

The `tfm_generated_file_list.yaml` is essential to build TF-M.

There are several files that are required for each Secure Partition, so they are not in any generated file lists since one template generates multiple files.

- `psa_manifest/<manifestfilename>.h`

`manifestfilename` is the file name of the manifest. This file contains internal definitions for the Secure Partition implementation, such as RoT Service signals and Secure Functions. Refer to FF-M⁷ for more details. The corresponding template is `manifestfilename.template`

- `intermedia_<manifestfilename>.c`

TF-M specific, which holds the stacks of Secure Partitions. This file must be built with the Secure Partition libraries. The corresponding template is `partition_intermedia.template`.

- `load_info_<manifestfilename>.c`

TF-M specific, which contains the load information of Secure Partitions. This file must be built with the TF-M SPM library. The corresponding template is `partition_load_info.template`.

These files are generated to `output_path` specified by each Secure Partition in the manifest lists.

Configuration Header File

The format of each configuration item must be

```
#define CONFIG_NAME  VALUE
```

The following format is also supported for boolean type configurations.

```
#define CONFIG_NAME
```

The configurations can be divided into two categories.

- Generic configurations:

- `PSA_FRAMEWORK_ISOLATION_LEVEL`

The isolation level, required. Valid values are [1, 2, 3].

- `CONFIG_TFM_SPM_BACKEND`

The backend of SPM, required. Valid values are [SFN, IPC]. See *SPM backends* for details of backends.

- Secure Partition enablement configurations

⁶ Jinja2

Configurations used to enable or disable Secure Partitions. The configuration names must match the values of `conditional` attributes in *Manifest List*. Valid values are `[0, 1]`. It's optional for a Secure Partition which does not have the `conditional` attribute.

The configurations can be split to multiple files corresponding to the multiple manifest lists.

Usage in TF-M Build System

In the TF-M build system, the manifest tool is invoked during building automatically. The arguments can be customized by altering the CMake configurations.

The manifest lists are passed to the manifest tool via the `TFM_MANIFEST_LIST` CMake configuration. The default value is the `tfm_manifest_list.yaml`. It can be overridden or appended with other manifest lists.

Corresponding manifest lists of test Secure Partitions are appended if either TF-M regression or PSA compliance tests are enabled.

The generated file lists are passed via `GENERATED_FILE_LISTS`. It can be also overridden or appended with other file lists.

The `-q` argument is appended if `PARSE_MANIFEST_QUIET_FLAG` is enabled.

Paths in manifest lists and generated file lists can have CMake variables as long as they are absolute paths. The lists then must be processed by the CMake command `configure_file()`⁷ before passing to the manifest tool.

The configuration header file is generated by the build system automatically.

References

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9.10.8 ADAC (Authenticated Debug Access Control)

Introduction

Background

In order to make sure debug capabilities of any system do not become attack vectors, PSA provides reference ADAC system architecture. It specifies functional layers that reside above actual physical link.

Authenticated Debug Access Control (ADAC), also referred to as Secure Debug, is a protocol that provides a way to use strong authentication to restrict device debug access to only authorized entities. Across various life cycle states of target device, it permits appropriate access to finely configured domains.

Requirements

Debug capabilities must achieve several safety goals to be fully resilient. It requires

1. Strong authentication
2. Finely grained hardware and firmware domains
3. Enforcing debug limitations

ADAC protocol is fully flexible to meet the above goals. But depending on the system design and topology, it must be implemented so to meet varying attack surface.

⁷ CMake `configure_file()`

ADAC Components (Host Side)

- **Secure Debug Manager (SDM)**: SDM is the host side component adac protocol implementation. Its API is implemented in below external github repository.

`git@github.com:ARM-software/sdm-api.git.`

An example implementation for SDM library is in below external github repository.

`git@github.com:ARM-software/secure-debug-manager.git.`

ADAC Components (Target Side)

- **Secure Debug Authenticator (SDA)**: SDA is the target side component that implements ADAC protocol. It is implemented in below external repository. It authenticates all input credentials and applies specific debug permissions.

`https://git.trustedfirmware.org/shared/psa-adac.git.`

SDA integration

The above mentioned SDA repository includes required HAL updates for a few supported platforms along with transport layer protocol. Hence, the API to initiate the connection with the host debugger and to perform the authentication process is platform-specific.

For example for musca b1 platform, the API to integrate is:

- `tfm_to_psa_adac_musca_b1_secure_debug()`

Once the device is Secured, opening up the debug ports poses a risk to active confidential production secrets. So on debug request, a system reset is required, and most of the time, SDA code is expected to reside in the boot loader (BL2) or equivalent.

However RSE platform has some implementation constraints due to which SDA is integrated in runtime service. For more details, please refer to the Implementation Constraints section in the ADAC Implementation for RSE

Please follow the below link for further information on SDA implementation.

[psa-adac read me](#)

Reference

[ADAC specification](#)

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9.11 How to build TF-M

Follow the *Build instructions*.

9.12 How to export files for building non-secure applications

Explained in the *Build instructions*.

9.13 How to add a new platform

Porting TF-M to a New Hardware contains guidance on how to add a new platform.

9.14 How to integrate another OS

9.14.1 OS migration to Armv8-M platforms

To work with TF-M on Armv8-M platforms, the OS needs to support the Armv8-M architecture and, in particular, it needs to be able to run in the non-secure world. More information about OS migration to the Armv8-M architecture can be found in the *OS requirements*. Depending upon the system configuration this may require configuring drivers to use appropriate address ranges.

9.14.2 Interface with TF-M

The files needed for the interface with TF-M are exported at the `<install_dir>/interface` path. The NS side is only allowed to call TF-M secure functions (veneers) from the NS Thread mode.

TF-M interface header files are exported in `<install_dir>/interface/include` directory. For example, the Protected Storage (PS) service PSA API is declared in the file `<install_dir>/interface/include/psa/protected_storage.h`.

TF-M also exports a reference implementation of PSA APIs for NS clients in the `<install_dir>/interface/src`.

On Armv8-M TrustZone based platforms, NS OS uses `tfm_ns_interface_dispatch()` to integrate with TF-M implementation of PSA APIs. TF-M provides a reference implementation of this function for RTOS and bare metal use cases. RTOS implementation of `tfm_ns_interface_dispatch()` (provided in `interface/src/os_wrapper/tfm_ns_interface_rtos.c`) uses mutex to provide multithread safety. Mutex wrapper functions defined in `interface/include/os_wrapper/mutex.h` are expected to be provided by NS RTOS. When reference RTOS implementation of dispatch function is used NS application should call `tfm_ns_interface_init()` function before first PSA API call. Bare metal implementation `tfm_ns_interface_dispatch()` (provided in `interface/src/os_wrapper/tfm_ns_interface_bare_metal.c`) does not provide multithread safety and does not require implementation of mutex interface. If needed, instead of using reference implementation, NS application may provide its own implementation of `tfm_ns_interface_dispatch()` function.

TF-M provides a reference implementation of NS mailbox on multi-core platforms, under folder `interface/src/multi_core`. See *Mailbox design* for TF-M multi-core mailbox design.

9.14.3 Interface with non-secure world regression tests

A non-secure application that wants to run the non-secure regression tests needs to call the `tfm_non_secure_client_run_tests()`. This function is exported into the header file `test_framework_integ_test.h` inside the `<build_dir>/install` folder structure in the test specific files, i.e. `<build_dir>/install/export/tfm/test/inc`. The non-secure regression tests are precompiled and delivered as a static library which is available in `<build_dir>/install/export/tfm/test/lib`, so that the non-secure application needs to link against the library to be able to invoke the `tfm_non_secure_client_run_tests()` function. The PS non-secure side regression tests rely on some OS functionality e.g. threads, mutexes etc. These functions comply with CMSIS RTOS2 standard and have been exported as thin wrappers defined in `os_wrapper.h` contained in `<build_dir>/install/export/tfm/test/inc`. OS needs to provide the implementation of these wrappers to be able to run the tests.

9.14.4 NS client Identification

The NS client identification (NSID) is specified by either SPM or NSPE RTOS. If SPM manages the NSID (default option), then the same NSID (-1) will be used for all connections from NS clients. For the case that NSPE RTOS manages the NSID and/or different NSIDs should be used for different NS clients. See *Non-secure Client Extension Integration Guide*.

9.15 Non-secure interrupts

Non-secure interrupts are allowed to preempt Secure thread mode. With the current implementation, a NSPE task can spoof the identity of another NSPE task. This is an issue only when NSPE has provisions for task isolation. Note, that AIRCR.PRIS is still set to restrict the priority range available to NS interrupts to the lower half of available priorities so that it wouldn't be possible for any non-secure interrupt to preempt a higher-priority secure interrupt.

9.16 Secure interrupts and scheduling

To ensure correct operation in the general case, the secure scheduler is not run after handling a secure interrupt that pre-empted the NSPE. On systems with specific constraints, it may be desirable to run the scheduler in this situation, which can be done by setting `CONFIG_TFM_SCHEDULE_WHEN_NS_INTERRUPTED` to 1. This could be done if the NSPE is known to be a simple, single-threaded application or if non-secure interrupts cannot preempt the SPE, for example.

When `CONFIG_TFM_SCHEDULE_WHEN_NS_INTERRUPTED` is set to 0, the SPM traces scheduling events triggered by interrupts into a cookie. This feature prevents the system from missing events if interrupts from both SPE and NSPE occur at the same time.

Note that scheduling also depends on the system topology. If your system has multiple cores running multiple NSPEs, then refer to *Scheduling Hybrid Platforms* for more details.

9.17 Integration with non-Cmake systems

9.17.1 Generated Files

Files that are derived from PSA manifests are generated at build-time by cmake. For integration with systems that do not use cmake, the files must be generated manually.

The `tools/tfm_parse_manifest_list.py` script can be invoked manually. Some arguments will be needed to be provided. Please refer to `tfm_parse_manifest_list.py --help` for more details.

Some variables are used in the template files, these will need to be set in the environment before the script will succeed when the script is not run via cmake.

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DESIGN DOCUMENTS

10.1 Multi CPU

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The following documents outline the design and implementation principles of TF-M for multi-cpu and hybrid platforms, where the Secure Processing Environment (SPE) and Non-secure Processing Environment (NSPE) execute on separate cores.

During boot, the secure core initializes the hardware and peripherals that the NSPE uses, establishing an assured execution path.

The cores communicate using a mailbox-based Inter-Processor Communication (IPC) mechanism.

The framework includes a Remote Procedure Call (RPC) layer and an NS Agent Mailbox to abstract platform dependencies. Hybrid platform scheduling extends the original dual-cpu concepts for systems with a local and multiple remote NSPEs, supporting flexible scheduling policies.

10.1.1 Booting a Multi CPU System

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System Architecture

There are many possibly ways to design a multi cpu system. Some important considerations from a boot perspective are:

- Which core has access to which areas of Flash?
 - It is possible that the secure core has no access to the Flash from which the non-secure core will boot, in which case the non-secure core will presumably have a separate root of trust and perform its own integrity checks on boot.
- How does the non-secure core behave on power-up? Is it held in reset, does it jump to a set address, ...?
- What are the performance characteristics of the two core?

- There could be a great disparity in performance

Note

Concept for multi-cpu are extended in the Hybrid Platform section.

TF-M Multi CPU Booting

In an effort to make the problem manageable, as well as to provide a system with good performance, that is flexible enough to work for a variety of multi cpu systems, the following design decisions have been made:

- TF-M will (for now) only support systems where the secure core has full access to the Flash that the non-secure core will boot from
 - This keeps the boot flow as close as possible to the single core design, with the secure core responsible for maintaining the chain of trust for the entire system, and for upgrade of the entire system
- The secure code will make a platform-specific call immediately after setting up hardware protection to (potentially) start the non-secure core running
 - This is the earliest point at which it is safe to allow the non-secure code to start running, so starting it here ensures system integrity while also giving the non-secure code the maximum amount of time to perform its initialization
 - Note that this is after the bootloader has validated the non-secure image, which is the other key part to maintain security
 - This also means that only `tfm_s` and `tfm_ns` have to change, and not `mcuboot`
- Both the secure and non-secure code will make platform-specific calls to establish a synchronization point. This will be after both sides have done any initialization that is required, including setting up inter-core communications. On a single core system, this would be the point at which the secure code jumps to the non-secure code, and at the very start of the non-secure code.
- After completing initialization on the secure core (at the point where on a single core system, it would jump to the non-secure code), the main thread on the secure core will be allowed to die
 - The scheduler has been started at this point, and an idle thread exists. Any additional work that is only required in the multi cpu case will be interrupt-driven.
 - All work related to the non-secure core will take place from a `ns_agent_mailbox` partition, which will establish communication with the non-secure core and then act on its behalf
- Because both cores may be booting in parallel, executing different initialization code, at different speeds, the design must be resilient if either core attempts to communicate with the other before the latter is ready. For example, the client (non-secure) side of the IPC mechanism must be able to handle the situation where it has to wait for the server (secure) side to finish setting up the IPC mechanism.
 - This relates to the synchronization calls mentioned above. It means that those calls cannot utilise the IPC mechanism, but must instead use some platform-specific mechanism to establish this synchronization. This could be as simple as setting aside a small area of shared memory and having both sides set a “ready” flag, but may well also involve the use of interrupts.
 - This also means that the synchronization call must take place after the IPC mechanism has been set up but before any attempt (by either side) to use it.

API Additions

Three new HAL functions are required:

```
void tfm_hal_boot_ns_cpu(uintptr_t start_addr);
```

- Called on the secure core from ns_agent_mailbox partition when it first runs (after protections have been configured).
- Performs the necessary actions to start the non-secure core running the code at the specified address.

```
void tfm_hal_wait_for_ns_cpu_ready(void);
```

- Called on the secure core from ns_agent_mailbox partition after making the above call.
- Flags that the secure core has completed its initialization, including setting up the IPC mechanism.
- Waits, if necessary, for the non-secure core to flag that it has completed its initialisation

```
void tfm_ns_wait_for_s_cpu_ready(void);
```

- Called on the non-secure core from main() after the multi-cpu-specific initialization (on a single core system, this would be the start of the non-secure code), before the first use of the IPC mechanism.
- Flags that the non-secure side has completed its initialization.
- Waits, if necessary, for the secure core to flag that it has completed its initialization.

For all three, an empty implementation will be provided with a weak symbol so that platforms only have to provide the new functions if they are required.

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10.1.2 Scheduling for a Hybrid Platform Solution

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Warning

This feature is completed but documentation, support and testing are limited.

Please provide any feedback or comments via the TF-M mailing list or submit a patch in review.trustedfirmware.org.

Introduction

The Hybrid Platform solution refers to the more generic support for platforms with one local NSPE client and one or more NSPE remote client. The local client resides in the same cpu as the SPE, while the remote clients reside on separate CPUs.

A Hybrid Platform solution requires applications to run from both local and remote clients. This inevitably brings some scheduling trade-offs and limitations on the table.

The implementation provided and explained in this section aims to give a reference implementation for some use-cases where different systems need to satisfy various scheduling preferences.

Those are broadly:

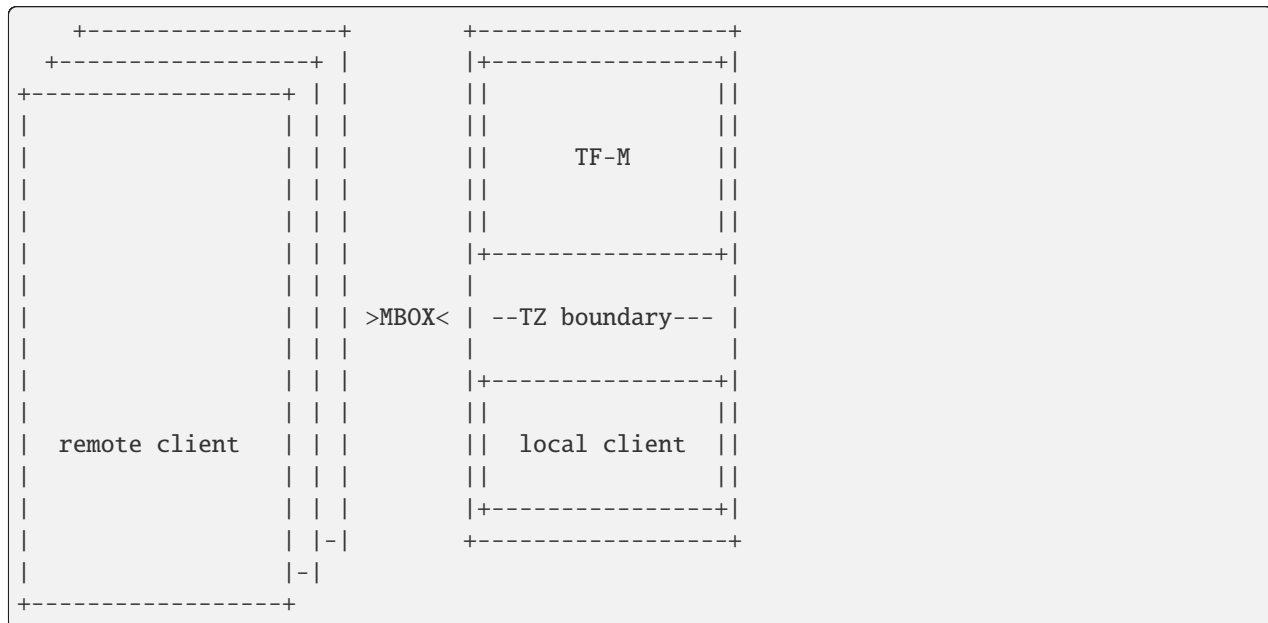
- SPE is in charge of scheduling
- NSPE is in charge of scheduling
- a BALANCED combination of the above

and are explained below.

System Architecture

A Hybrid Platform is a system topology where there is an instance of SPE and one or multiple instances of NSPE running at the same time.

A common layout is:



where *local client* refers to the NSPE running on the same core of SPE, while *remote client* refers to the other instance(s) of NSPE running on other separate cores (irrespective of whether, or not, the same type of core).

Typically, a Hybrid Platform would fall into two categories:

Topology	Description
Heterogeneous topology	M-Class processor running TF-M and secure services. Non-secure applications running in both the Normal World of the M-Class and in one or more other (non M-class) different cores.
Homogeneous topology	M-Class processor running TF-M and secure services. Non-secure applications running in both the Normal World of the M-Class and in one or more other identical M-Class cores.
Remark: When the M-Class core hosting the Secure World does NOT have a Normal World, then the solution is known as “Dual-CPU” System.	

Scheduling Scenarios

There are commonly two classes of scheduling use-cases depending on how the workload importance is distributed:

- Local NSPE can be interrupted anytime when a secure service is requested by the remote clients.
- Local NSPE can NOT be interrupted when a secure service is requested by the remote clients.

Usually, it is known at design stage which types of workloads and applications will run in both local and remote clients, thus it can be decided in advance which class of clients require to be serviced immediately and/or can be preempted. Such knowledge is then used to decide which scheduling option is best for the whole system.

Those options are generalized and provided with three build choices assigned to the build config `CONFIG_TFM_HYBRID_PLAT_SCHED_TYPE`.

CON- FIG_TFM_	Description
TFM_HYE (Default)	No support for Hybrid Platform is provided.
TFM_HYE	NSPE has no control over when the request is processed. SPE preempts NSPE (is scheduled to run) and runs the requested secure service (by the remote client). Once completed, execution then resumes to NSPE.
TFM_HYE	NSPE is in full control over the incoming requests from the remote clients. The incoming requests may be queued but execution does not switch to SPE. The local NSPE makes its own decisions on when it is an appropriate point to yield execution to execute the mailbox requests. To do so, NSPE performs a <code>psa_call</code> to a stateless service in the mailbox partition and let the execution to proceed in SPE. Note that the local NSPE does not have knowledge of any pending messages in the mailbox awaiting processing. It can only attempt to start their processing.
TFM_HYE	When multiple mailbox channels, and their respective interrupts, are available on a platform, then it is possible to schedule independently the different sources. That is, some messages can be configured to be scheduled according to the <code>SCHED_SPE</code> mechanism, while others can be configured to be scheduled according to the <code>SCHED_NSPE</code> (deferred to local NSPE) mechanism. This can be achieved via a special attribute in the manifest: <code>mbox_flags</code> .

The definitions for the options above are available in `secure_fw/spm/include/tfm_hybrid_platform.h`.

Integration examples

Hybrid Platform with SPE scheduling

In the platform configuration settings file `config_tfm_target.h` choose

```
#define CONFIG_TFM_HYBRID_PLAT_SCHED_TYPE    TFM_HYBRID_PLAT_SCHED_SPE
```

Hybrid Platform with NSPE scheduling

If your platform implements a custom set of RPC operations, then add the `process_new_msg` handler:

```
static int32_t platform_process_new_msg(uint32_t *nr_msg)
{
    /* some optional platform tasks */

    /*
     * Note that it's the platform's choice on whether or not to return the
     * number of mailbox messages processed.
     */
}
```

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```

    */
    return platform_handle_req();
}

static struct tfm_rpc_ops_t rpc_ops = {
    .handle_req = platform_handle_req,
    .reply = platform_mailbox_reply,
    .handle_req_irq_src = platform_handle_req_irq_src,
    .process_new_msg = platform_process_new_msg,
};

```

The platform interrupt handler is already expected to call `spm_handle_interrupt()`, so to correctly handle the set/clear of IRQs, the platform shall also call `tfm_multi_core_set_mbox_irq()` right after it as shown below.

```

void mailbox_IRQHandler(void)
{
    /* some optional platform tasks */

    spm_handle_interrupt(p_pt, p_ildi);

#ifdef CONFIG_TFM_HYBRID_PLAT_SCHED_TYPE == TFM_HYBRID_PLAT_SCHED_NSPE
    tfm_multi_core_set_mbox_irq(p_ildi);
#endif
}

```

The counterpart `tfm_multi_core_clear_mbox_irq()` is called instead by the mailbox partition.

In the platform configuration settings file `config_tfm_target.h` choose

```
#define CONFIG_TFM_HYBRID_PLAT_SCHED_TYPE    TFM_HYBRID_PLAT_SCHED_NSPE
```

Then, in the *local client* NSPE, where the scheduling decisions are made, simply call the auxiliary mailbox service to process any pending mailbox messages:

```

psa_status_t status;
uint32_t num_msg;

psa_outvec out_vec[] = {
    {
        .base = &num_msg,
        .len = sizeof(num_msg),
    },
};

status = psa_call(
    NS_AGENT_MBOX_PROCESS_NEW_MSG_HANDLE,
    PSA_IPC_CALL,
    NULL, 0,
    out_vec, IOVEC_LEN(out_vec));
if (status < 0) {
    /* process the error */
} else {
    /* num_msg contains the number of mailbox slots/messages processed */
}

```

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}

Hybrid Platform with BALANCED scheduling

All the above for TFM_HYBRID_PLAT_SCHED_NSPE plus the following changes, where required, for the mailbox manifest.

```
"services": [
{
  "<attribute>" : "<value>"
  ...
},
],
"irqs": [
{
  "<attribute>" : "<value>"
  ...
  "mbbox_flags": MBOX_IRQ_FLAGS_<DEFAULT|DEFER>_SCHEDULE,
},
...
],
```

Currently, the mailbox flags `mbbox_flags` supported are:

- `MBOX_IRQ_FLAGS_DEFAULT_SCHEDULE`: the requests via mailbox transport are managed accordingly to the default hybrid platform scheduling options (this is always the default option)
- `MBOX_IRQ_FLAGS_DEFER_SCHEDULE`: the requests via mailbox transport are deferred until the local NSPE calls the mailbox partition auxiliary service `NS_AGENT_MBOX_PROCESS_NEW_MSG`

The `mbbox_flags` is optional, it can be omitted when this feature is not required.

In the platform configuration settings file `config_tfm_target.h` choose

```
#define CONFIG_TFM_HYBRID_PLAT_SCHED_TYPE    TFM_HYBRID_PLAT_SCHED_BALANCED
```

Hybrid Platform and common binary image

For homogeneous hybrid platform topology, the same binary image is likely to be deployed across cores. In that case, both mailbox and TrustZone agents will be in use. Therefore, the API for PSA calls would be, from each client's perspective, virtually the same. In that event, the APIs provided by both agents need to be available simultaneously, and thus, a component that coordinates the redirection is required.

To achieve this, a `TFM_HYBRID_PLATFORM_API_BROKER` is supplied. This component simply redirects the standard PSA calls from the client to the correct interface. The redirection choice is made at initialization time, where the local or remote NSPE sets the respective interface API.

For such hybrid platforms, the common build flags configuration would be as follows:

Config Option	Set to
CONFIG_TFM_USE_TRUSTZONE	ON
TFM_HYBRID_PLATFORM_API_BROKER	ON
TFM_MULTI_CORE_TOPOLOGY	ON
TFM_MULTI_CORE_NS_OS	user's requirements
TFM_MULTI_CORE_NS_OS_MAILBOX_THREAD	user's requirements
TFM_MULTI_CORE_TEST	user's requirements

The NSPE is expected to have a mechanism to detect its execution target, that is being remote or local. This is *IMPLEMENTATION DEFINED*.

This mechanism should drive the main application to initialize the appropriate interface, where the API broker is configured for the correct execution environment.

See an example below.

```
#define EXEC_TARGET_LOCAL true
#define EXEC_TARGET_REMOTE false
bool plat_is_this_client_local(void)
{
    /*
     * An implementation-defined mechanism to detect the execution target.
     * For example, a RO memory location could hold such information.
     */

    return <EXEC_TARGET_LOCAL | EXEC_TARGET_REMOTE>;
}
```

Then, the main implementation simply calls the mechanism logic function and calls either the interface for TrustZone or Mailbox. Subsequently, the interface sets the execution environment within the API broker.

```
/*
 * The corresponding init function sets the execution environment via the API
 * broker.
 */
if (plat_is_this_client_remote()) {
    tfm_ns_multi_core_boot();
} else {
    tfm_ns_interface_init();
}
```

Hybrid Platform and NSPE reentrancy

With scheduling set to TFM_HYBRID_PLAT_SCHED_SPE, the non-secure state can be interrupted, allowing execution to switch to the secure side until it completes. In systems where NSPE has no knowledge on when - or even if - there is a secure service in action, a non-secure application can interrupt and take over execution. This is the case, for example, when a standard RTOS runs its scheduler (by pre-empting the secure side) and later a new task runs. That task can attempt to request services from the secure state, which could lead to an intentional system panic. To safely deal with such scenario, NSPE has a way to detect if reentrancy is allowed. To do so, enable TFM_TZ_REENTRANCY_CHECK in the build and include a validation check at run-time. The following diagram illustrates an example for a NSPE with an RTOS and a possible scenario of a secure service pre-empted by a new task.

The sequence diagram above demonstrates the denial of re-entry until the secure service completes; non-secure applications must check `tfm_ns_check_safe_entry()` and yield otherwise.

```
psa_status_t ns_status = PSA_ERROR_CONNECTION_REFUSED;
enum tfm_reenter_check_err_t reenter_status;
os_status_t os_status;
```

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```
reenter_status = tfm_ns_check_safe_entry();
if (reenter_status == TFM_REENTRANCY_CHECK_SUCCESS) {
    ns_status = psa_call(
        SERVICE_HANDLE,
        SERVICE_ID,
        in_vec, IOVEC_LEN(in_vec),
        out_vec, IOVEC_LEN(in_vec));
    if (ns_status != PSA_SUCCESS) {
        /* handle error */
    }

    /*
     * Release re-entry guard.
     * Must be called on the same call path after a successful secure entry.
     */
    tfm_ns_check_exit();
} else {
    os_status = osThreadYield(); /* also possible osDelay(delayTime); */
    if (os_status != osOK) {
        /* handle error */
    }
}
```

The RTOS yielding mechanism directly influences task scheduling and the continuation of preempted secure services. There may be better functions than the one shown above that allow to achieve the sought scheduling sequence.

Limitations

Currently Hybrid Platform is supported only for the IPC model. Testing is limited.

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10.1.3 NS Agent Mailbox Design

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Background

The SPE communicates with a remote NSPE through the ‘NS Agent Mailbox’. A physical mailbox mechanism is required for this partition to work.

The NS Agent Mailbox operates like a partition. The agent has an endless loop and waits for signals, calls the FF-M API based on the parsed communication messages.

Because it acts as an agent, you must consider several key design details:

- The SPM treats the FF-M Client API caller's ID as the client ID. However, because the NS Agent Mailbox may represent multiple non-secure clients, it must inform SPM which client it represents. The standard FF-M Client API lacks support for this requirement.
- The FF-M Client API blocks the caller until a reply is received. However, the NS Agent Mailbox must respond to non-secure interrupts promptly. Blocking can delay the handling of non-secure communication messages.

Note

All the concepts for dual-cpu can be extended to multi-cpu without loss of generality. Also see the Hybrid Platform section for further details.

Design

The following figure shows the overall design of the relevant components. NS Agents are implementation-defined components that provide FF-M-compliant Client APIs to non-secure clients. From the client's perspective, these APIs behave according to the FF-M specification. However, NS Agents require customization and additional handling in the SPM because they carry extra requirements compared to standard secure partitions.

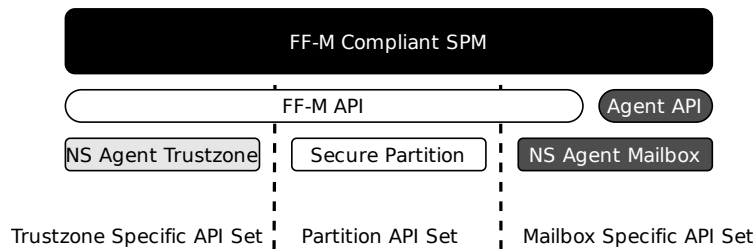


Figure 4:: High-level agents and secure partitions types and the respective callable API sets into SPM

Note

NS Agent TrustZone is mentioned for the comparison purpose. The implementation details are out of scope for this document.

To align with FF-M compliance, the NS Agent Mailbox is designed to:

- operate like a standard Secure Partition under the IPC model, with one single thread and the ability to call standard FF-M APIs
- include a manifest file that describes its attributes and resources
- be assigned a positive partition ID

Services rely on the *client_id* for policy enforcement. When the NS Agent Mailbox calls a Client API, the SPM must know which *client_id* the agent is representing. Since standard APIs treat the caller as the service client, a specific API is needed to support identifying the represented non-secure client.

The SPM sets the non-secure *client_id* into the message just before it sends the message. Until then, it uses the agent's ID.

This specific *Agent API* is non-blocking, unlike the standard FF-M Client APIs. It improves communication efficiency between NS clients and NS Agent Mailbox. This mechanism also requires additional signals and APIs for message acknowledgments.

Note

A standard Secure Partition cannot use the Agent API. If attempted, an error is returned.

Mailbox architecture

The mailbox consists of two components: one in the NSPE and one in the SPE.

The NSPE mailbox handles mailbox functions in the non-secure domain, while the SPE mailbox performs the same functions within TF-M.

On dual-core systems, the NSPE mailbox implements PSA Client APIs by sending client requests and receiving results. Implementations may differ across NSPE operating systems and use cases.

TF-M provides a reference implementation of the NSPE mailbox.

The NSPE mailbox stores data objects in non-secure memory that hold PSA Client call parameters, return values, and other related data. The NSPE mailbox uses platform-specific inter-process communication (IPC) to handle notifications between the NSPE and the SPE.

The SPE mailbox in TF-M receives requests from the NSPE mailbox, parses them, and invokes the TF-M Remote Procedure Call (RPC) layer. The RPC layer delivers the requests to the TF-M core and Secure Partition Manager (SPM). After the PSA Client call completes, the TF-M core and SPM use the RPC layer to return results to the NSPE through the mailbox. The SPE mailbox manages its objects in secure memory. Both NSPE and SPE mailboxes use platform-specific inter-process communication (IPC) to exchange notifications.

The figure below shows the architecture.

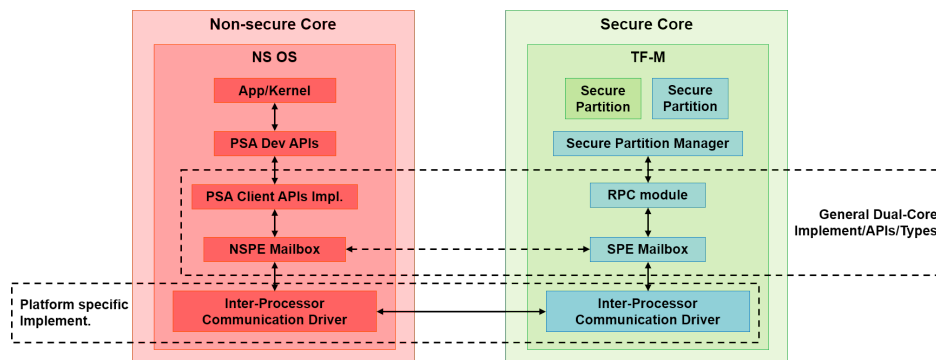


Figure 5:: High-level architecture for SPE and NSPE with mailbox

Mailbox communication for PSA Client calls

NSPE Mailbox queue

The NSPE mailbox maintains a queue in non-secure memory. The queue contains one or more slots, and its size must match that of the SPE mailbox queue. Each slot includes a message and a reply structure. Each slot may also contain fields that identify the non-secure task initiating the PSA Client request. The mailbox message stores the parameters for the PSA Client request inside the slot. The mailbox reply structure receives the PSA Client result from the SPE.

Mailbox messages

In the PSA Client API implementation, the NSPE mailbox selects an empty slot in the queue and writes the parameters into the mailbox message in that slot. The SPE mailbox parses those parameters from the mailbox message. The mailbox message can include additional fields to transfer more information from NSPE to SPE for processing in TF-M.

Mailbox replies

A mailbox reply structure in non-secure memory receives the PSA Client result from the SPE mailbox.

SPE Mailbox queue

The SPE mailbox maintains a queue to store mailbox objects. The number of slots in this queue must match the NSPE mailbox queue. After the SPE is notified of a pending PSA Client request, the mailbox assigns an empty slot, copies the corresponding parameters from non-secure memory, and parses them.

Each slot in SPE mailbox queue can contain the following fields:

- An optional field to hold mailbox message content copied from non-secure memory.
- Index of NSPE mailbox queue slot containing the mailbox message.
- A handle to the mailbox message. Optional. Identify the owner slot of PSA Client result when multiple mailbox messages are under processing.

Additional fields can be included in the slot structure to support mailbox processing in the SPE.

Mailbox notifications between NSPE and SPE

The NSPE mailbox sends a notification to the SPE to signal a pending PSA Client request. In turn, the SPE mailbox notifies the NSPE when the PSA Client result is available. These notifications use platform-specific Inter-Processor Communication (IPC).

To support *full-duplex* communication between NSPE and SPE, it is recommended to assign a dedicated IPC channel to each notification routine. If both routines share the same IPC channel, proper synchronization must be implemented to prevent conflicts.

In the SPE, the IPC interrupt handler processes notifications from the NSPE and triggers the agent mailbox handling. The communication prototype design¹ defines the behavior of the IPC interrupt handler.

The NSPE can use either an interrupt handler or polling to detect IPC notifications from the SPE.

PSA API and Agent API

The PSA API has been expanded to include:

- `agent_psa_connect()`
- `agent_psa_close()`
- `agent_psa_call()`

To support the behavior changes in these APIs, new signals such as `ASYNC_MSG_REPLY` and additional types are introduced. The RPC layer uses the Agent API to forward calls to the targeted service.

When non-secure clients call `psa_call()`, a mailbox message containing its parameters is delivered to the NS Agent Mailbox. The agent extracts the parameters and calls `agent_psa_call()`.

The `agent_psa_call()` interface takes four arguments, unlike the standard `psa_call()` which takes six. Some parameters are combined to align the two interfaces.

¹ Communication prototype between NSPE and SPE in Dual-core systems

A composition type is used to pack the vectors:

```
struct client_params_t {
    int32_t      ns_client_id_stateless;
    const psa_invec *p_invecs;
    psa_outvec    *p_outvecs;
};
```

ns_client_id_stateless identifies the non-secure client ID when accessing a stateless service. This value is ignored for connection-based services.

A macro, *PARAM_PACK*, is introduced to pack vector types and message lengths into a 32-bit *control* variable:

```
uint32_t control = PARAM_PACK(params->psa_call_params.type,
                               params->psa_call_params.in_len,
                               params->psa_call_params.out_len);
```

The agent must indicate whether memory sources for input and output vectors are secure or non-secure. This helps the SPM validate the vectors properly.

To cover the above mentioned scenarios, guidelines are listed for input and output vector processing:

- The agent needs to tell SPM where vectors and descriptors come from, to assist SPM perform proper memory checking. The source information is encoded in the parameter *control*.
- When SPE platforms have the capability to identify the memory sources, platforms can decide whether to skip the indication or not, in the HAL.

Guidelines for vector processing:

- The agent must inform SPM of the memory origin for each vector and descriptor to enable proper memory checks. This information is encoded in the *control* parameter.
- If SPE can determine memory sources independently, it may skip the indication in the HAL.

The *agent_psa_connect()* and *agent_psa_close()* functions include an extra *ns_client_id* parameter to identify the non-secure client. This value must be negative. If the value is zero or positive, SPM ignores it.

This mechanism allows agents to call APIs asynchronously on behalf of the services they represent.

To support the Agent API, the following acknowledgment signal is defined:

```
#define ASYNC_MSG_REPLY          (0x00000004u)
```

This signal enables the agent and SPM to manage partition signals, retrieve messages, and handle replies.

Code Example

See the reference implementation of `secure_fw/partitions/ns_agent_mailbox/ns_agent_mailbox.c`

Custom manifest attributes

Three custom manifest attributes are added for this agent:

Name	Description
<code>ns_agent</code>	Indicates if this partition is an agent (true/false)
<code>client_id_base</code>	The minimum client ID value (<0)
<code>client_id_limit</code>	The maximum client ID value (<0)

`client_id_base` and `client_id_limit` are negative numbers. This means that `client_id_base` $\leq$ `client_id_limit`, but the absolute value of `client_id_base` must be greater than or equal to that of `client_id_limit`. SPM checks for overlapping IDs when initializes the secure partitions and agents.

Non-secure callers must provide a negative client ID (<0) when using the PSA API through their communication layer. All NS Agents follow a uniform ID mapping:

NS client ID	Transformed client ID
-1	<code>client_id_limit</code>
-2	<code>client_id_limit - 1</code>
...	
• $(\text{abs}(\text{client_id_limit}) - \text{abs}(\text{client_id_base}) + 1)$	<code>client_id_base</code>

The functions `tfm_multi_core_hal_client_id_translate` and `tfm_multi_core_register_client_id_range` perform this translation.

Any other IDs provided by the NSPE will result in `PSA_ERROR_INVALID_ARGUMENT` error.

See the reference implementation of `secure_fw/partitions/ns_agent_mailbox/ns_agent_mailbox.yaml`

Runtime and IDLE processing

At runtime, the NS Agent Mailbox should not be blocked by Agent-specific APIs. It may only be blocked when:

- calling standard PSA Client APIs
- calling `psa_wait()`

Only one component should enter IDLE mode. This depends on system topology:

- If a TrustZone-based NSPE exists, that NSPE should enter IDLE, regardless of how many NS Agent Mailbox are in the system.
- If only mailbox-based NSPEs exist, one NS Agent Mailbox can enter IDLE.

The implemented solution:

- SPRTL provides an API for entering IDLE.
- Only partitions with a specific flag can call this API.
- The manifest tooling counts the flagged partitions and returns an error if more than one exists.

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10.1.4 Memory Access Check of Trusted Firmware-M in Multi-Core Topology

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Introduction

On Armv8-M platforms with TrustZone, the platform's TF-M isolation HAL implementation usually relies on the [Armv8-M Security Extension](#) (CMSE) intrinsic `cmse_check_address_range()` to check memory access permissions. However, on multi-core platforms, the secure core may be an Armv6-M or Armv7-M core that does not implement CMSE. Even if CMSE is implemented on a multi-core platform, additional checks on system-level security and memory access management units may be required since CMSE intrinsics and TT instructions are only aware of MPU/SAU/IDAU inside the secure core.

As a result, TF-M provides a common platform memory access check API and an implementation that works without CMSE support. Multi-core topology platforms may use the memory check API `tfm_has_access_to_region()`, which checks whether an access has proper permission to read or write the target memory region, as a substitute for `cmse_check_address_range()` when implementing the TF-M isolation HAL.

The memory check API is defined in `platform/ext/common/mem_check_v6m_v7m.h` and the common implementation is provided in `platform/ext/common/mem_check_v6m_v7m.c`. The HAL defined in `platform/ext/common/mem_check_v6m_v7m_hal.h` must be implemented by each platform that uses the common implementation. This document discusses the design of the memory access check API and implementation.

Overall Design

Memory Access Check Policy

The policies vary in diverse Isolation Levels.

When TF-M works in Isolation Level 1, the access check in multi-core topology checks

- Memory region is valid according to system settings
- Non-secure client call request should not access secure memory.
- Secure services should not directly access non-secure memory. According to PSA Firmware Framework, Secure services should call Secure Partition APIs to ask TF-M SPM to fetch non-secure input/output buffer for them.
- Whether read/write attribute match between access and memory region

When TF-M works in Isolation Level 2, the access check in multi-core topology checks:

- Memory region is valid according to system settings
- Non-secure client call request should not access secure memory.
- Secure services should not directly access non-secure memory. According to PSA Firmware Framework, Secure services should call Secure Partition APIs to ask TF-M SPM to fetch non-secure input/outputs buffer for them.
- Whether read/write attribute match between access and memory region
- Unprivileged secure access should not access privileged secure memory region

The check policy in Isolation Level 3 will be defined according to TF-M future implementation.

The above policies will be adjusted according to TF-M implementation and PSA specs.

General Check Process in TF-M Core

The TF-M SPM calls the `tfm_hal_memory_check()` HAL API in both TrustZone-based and multi-core topology platforms to keep a uniform interface between SPM and platform.

Multi-core platform specific `tfm_hal_memory_check()` implementations can invoke `tfm_has_access_to_region()` to implement the entire memory access check routine.

During the check process, `tfm_has_access_to_region()` compares the access permission with memory region attributes and determines whether the access is allowed to access the region according to policy described in *Memory Access Check Policy* above.

`tfm_has_access_to_region()` invokes 3 HAL APIs to retrieve attributes of target memory region.

- `tfm_hal_get_mem_security_attr()` retrieves the security attributes of the target memory region.
- `tfm_hal_get_secure_access_attr()` retrieves secure access attributes of the target memory region.
- `tfm_hal_get_ns_access_attr()` retrieves non-secure access attributes of the target memory region.

All three functions are implemented by multi-core platform support. The definitions are specified in the section *HAL APIs* below.

The pseudo code of `tfm_has_access_to_region()` is shown below.

```

1  int32_t tfm_has_access_to_region(const void *p, size_t s, uint8_t flags)
2  {
3      struct security_attr_info_t security_attr;
4      struct mem_attr_info_t mem_attr;
5
6      /* Validate input parameters */
7      if (Validation failed) {
8          return error;
9      }
10
11     /* The memory access check should be executed inside TF-M PSA RoT */
12     if (Not in privileged level) {
13         abort;
14     }
15
16     /* Set initial value */
17     security_attr_init(&security_attr);
18
19     /* Retrieve security attributes of memory region */
20     tfm_hal_get_mem_security_attr(p, s, &security_attr);
21
22     /* Compare according to current Isolation Level */
23     if (Input flags mismatch security attributes) {
24         return error;
25     }
26
27     /* Set initial value */
28     mem_attr_init(&mem_attr);
29
30     if (The target memory region is in secure memory space) {
31         /* Retrieve access attributes of secure memory region */
32         tfm_hal_get_secure_access_attr(p, s, &mem_attr);
33
34         if (Not in Isolation Level 1) {
35             /* Secure memory protection unit(s) must be enabled in Isolation Level 2 and
36             ↪ 3 */
37             if (Protection unit not enabled) {
38                 abort;
39             }
40         }
41     }
42 }
```

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```

39     }
40   } else {
41     /* Retrieve access attributes of non-secure memory region. */
42     tfm_hal_get_ns_access_attr(p, s, &mem_attr);
43   }
44
45   /* Compare according to current Isolation Level and non-secure/secure access. */
46   if (Input flags match memory attributes) {
47     return success;
48   }
49
50   return error;
51 }

```

Note

It cannot be guaranteed that TF-M provides a comprehensive memory access check on non-secure memory for NSPE client call. If non-secure memory protection or isolation is required in a multi-core system, NSPE software should implement and execute the check functionalities in NSPE, rather than relying on TF-M access check.

For example, all the access from NSPE client calls to non-secure memory are classified as unprivileged in current TF-M implementation. Multi-core access check may skip the privileged/unprivileged permission check for non-secure access.

If a multi-core system enforces the privileged/unprivileged isolation and protection of non-secure area, NSPE software should execute the corresponding check functionalities before submitting the NSPE client call request to SPE.

Data Types and APIs**Data Types****Access Permission Flags**

The following flags are defined to indicate the access permission attributes. Each flag is mapped to the corresponding CMSE macro. Please refer to [ARMv8-M Security Extensions: Requirements on Development Tools](#) for details of each CMSE macro.

MEM_CHECK_MPU_READWRITE

Mapped to CMSE macro CMSE_MPU_READWRITE to indicate that the access requires both read and write permission to the target memory region.

```
#define MEM_CHECK_MPU_READWRITE    (1 << 0x0)
```

MEM_CHECK_MPU_UNPRIV

Mapped to CMSE macro CMSE_MPU_UNPRIV to indicate that it is an unprivileged access.

```
#define MEM_CHECK_MPU_UNPRIV      (1 << 0x2)
```

MEM_CHECK_MPU_READ

Mapped to CMSE macro CMSE_MPU_READ. It indicates that it is a read-only access to target memory region.

```
#define MEM_CHECK_MPU_READ      (1 << 0x3)
```

MEM_CHECK_NONSECURE

Mapped to CSME macro CMSE_NONSECURE to indicate that it is a access from non-secure client call request. If this flag is unset, it indicates the access is required from SPE.

```
#define MEM_CHECK_AU_NONSECURE  (1 << 0x1)
#define MEM_CHECK_MPU_NONSECURE (1 << 0x4)
#define MEM_CHECK_NONSECURE     (MEM_CHECK_AU_NONSECURE | \
                                   MEM_CHECK_MPU_NONSECURE)
```

Security Attributes Information

The structure `security_attr_info_t` contains the security attributes information of the target memory region. `tfm_hal_get_mem_security_attr()` implementation should fill the structure fields according to the platform specific secure isolation setting.

```
struct security_attr_info_t {
    bool is_valid;
    bool is_secure;
};
```

`is_valid` indicates whether the target memory region is valid according to platform resource assignment and security isolation configurations.

`is_secure` indicates the target memory region is secure or non-secure. The value is only valid when `is_valid` is true.

Memory Attributes Information

The structure `mem_attr_info_t` contains the memory access attributes information of the target memory region. `tfm_hal_get_secure_access_attr()` and `tfm_hal_get_ns_access_attr()` implementations should fill the structure fields according to the memory protection settings.

```
struct mem_attr_info_t {
    bool is_mpu_enabled;
    bool is_valid;
    bool is_xn;
    bool is_priv_rd_allow;
    bool is_priv_wr_allow;
    bool is_unpriv_rd_allow;
    bool is_unpriv_wr_allow;
};
```

`is_mpu_enabled` indicates whether the MPU and other management unit are enabled and work normally.

`is_valid` indicates whether the target memory region is valid according to platform resource assignment and memory protection configurations.

`is_xn` indicates whether the target memory region is Execute Never. This field is only valid when `is_valid` is `true`.

`is_priv_rd_allow` and `is_priv_wr_allow` indicates whether the target memory region allows privileged read/write. Both the fields are valid only when `is_valid` is `true`.

`is_unpriv_rd_allow` and `is_unpriv_wr_allow` indicates whether the target memory region allows unprivileged read/write. Both the fields are valid only when `is_valid` is `true`.

HAL APIs

`tfm_hal_get_mem_security_attr()`

`tfm_hal_get_mem_security_attr()` retrieves the current active security configuration information and fills the `security_attr_info_t`.

```
void tfm_hal_get_mem_security_attr(const void *p, size_t s,
                                   struct security_attr_info_t *p_attr);
```

Parameters	
<code>p</code>	Base address of the target memory region
<code>s</code>	Size of the target memory region
<code>p_attr</code>	Pointer to the <code>security_attr_info_t</code> to be filled
Return	
<code>void</code>	None

<code>p</code>	Base address of the target memory region
<code>s</code>	Size of the target memory region
<code>p_attr</code>	Pointer to the <code>security_attr_info_t</code> to be filled
Return	
<code>void</code>	None

The implementation should be decoupled from TF-M current isolation level or access check policy.

All the fields in `security_attr_info_t` shall be explicitly set in `tfm_hal_get_mem_security_attr()`.

If the target memory region crosses boundaries of different security regions or levels in security isolation configuration, `tfm_hal_get_mem_security_attr()` should determine whether the memory region violates current security isolation. It is recommended to mark the target memory region as invalid in such case, even if the adjoining regions or levels have the same security configuration.

If the target memory region is not explicitly specified in memory security configuration, `tfm_hal_get_mem_security_attr()` can return the following values according to actual use case:

- Either set `is_valid = false`
- Or set `is_valid = true` and set `is_secure` according to platform specific policy.

`tfm_hal_get_secure_access_attr()`

`tfm_hal_get_secure_access_attr()` retrieves the secure memory protection configuration information and fills the `mem_attr_info_t`.

```
void tfm_hal_get_secure_access_attr(const void *p, size_t s,
                                    struct mem_attr_info_t *p_attr);
```

Parameters	
p	Base address of the target memory region
s	Size of the target memory region
p_attr	Pointer to the mem_attr_info_t to be filled
Return	
void	None

The implementation should be decoupled from TF-M current isolation level or access check policy.

All the fields in mem_attr_info_t shall be explicitly set in tfm_hal_get_secure_access_attr(), according to current active memory protection configuration. It is recommended to retrieve the attributes from secure MPU and other hardware memory protection unit(s). The implementation can also be simplified by checking static system-level memory layout.

If the target memory region is not specified in current active secure memory protection configuration, tfm_hal_get_secure_access_attr() can select the following values according to actual use case.

- Either directly set is_valid to false
- Or set is_valid to true and set other fields according to other memory assignment information, such as static system-level memory layout.

If secure memory protection unit(s) is *disabled* and the target memory region is a valid area according to platform resource assignment, tfm_hal_get_secure_access_attr() must set is_mpu_enabled to false and set other fields according to current system-level memory layout.

tfm_hal_get_ns_access_attr()

tfm_hal_get_ns_access_attr() retrieves the non-secure memory protection configuration information and fills the mem_attr_info_t.

```
void tfm_hal_get_ns_access_attr(const void *p, size_t s,
                               struct mem_attr_info_t *p_attr);
```

Parameters	
p	Base address of the target memory region
s	Size of the target memory region
p_attr	Pointer to the mem_attr_info_t to be filled
Return	
void	None

The implementation should be decoupled from TF-M current isolation level or access check policy.

Since non-secure core runs asynchronously, the non-secure MPU setting may be modified by NSPE OS and therefore the attributes of the target memory region can be unavailable during tfm_hal_get_ns_access_attr() execution in TF-M. When the target memory region is not specified in non-secure MPU, tfm_hal_get_ns_access_attr() can set the fields according to other memory setting information, such as static system-level memory layout.

If non-secure memory protection unit(s) is *disabled* and the target memory region is a valid area according to platform resource assignment, tfm_hal_get_ns_access_attr() can set the following fields in mem_attr_info_t to default values:

- is_mpu_enabled = false
- is_valid = true

- `is_xn = true`
- `is_priv_rd_allow = true`
- `is_unpriv_rd_allow = true`

`is_priv_wr_allow` and `is_unpriv_wr_allow` can be set according to current system-level memory layout, such as whether it is in code section or data section.

General retrieval functions

TF-M implements 3 general retrieval functions to retrieve memory region security attributes or memory protection configurations, based on static system-level memory layout. Platform specific HAL functions can invoke those 3 general functions to simplify implementations.

- `tfm_get_mem_region_security_attr()` retrieves general security attributes from static system-level memory layout.
- `tfm_get_secure_mem_region_attr()` retrieves general secure memory protection configurations from static system-level memory layout.
- `tfm_get_ns_mem_region_attr()` retrieves general non-secure memory protection configurations from static system-level memory layout.

If a multi-core platform's memory layout may vary in runtime, it shall not rely on these 3 functions to retrieve static configurations. These 3 functions run through memory layout table to check against each memory section one by one, with pure software implementation. It might cost more time compared to hardware-based memory access check.

`tfm_get_mem_region_security_attr()`

`tfm_get_mem_region_security_attr()` retrieves security attributes of target memory region according to the static system-level memory layout and fills the `security_attr_info_t`.

```
void tfm_get_mem_region_security_attr(const void *p, size_t s,
                                     struct security_attr_info_t *p_attr);
```

Parameters	
<code>p</code>	Base address of the target memory region
<code>s</code>	Size of the target memory region
<code>p_attr</code>	Pointer to the <code>security_attr_info_t</code> to be filled
Return	
<code>void</code>	None

`tfm_get_secure_mem_region_attr()`

`tfm_get_secure_mem_region_attr()` retrieves general secure memory protection configuration information of the target memory region according to the static system-level memory layout and fills the `mem_attr_info_t`.

```
void tfm_get_secure_mem_region_attr(const void *p, size_t s,
                                    struct mem_attr_info_t *p_attr);
```


Parameters	
p	Base address of the target memory region
s	Size of the target memory region
p_attr	Pointer to the mem_attr_info_t to be filled
Return	
void	None

`tfm_get_ns_mem_region_attr()`

`tfm_get_ns_mem_region_attr()` retrieves general non-secure memory protection configuration information of the target memory region according to the static system-level memory layout and fills the `mem_attr_info_t`.

```
void tfm_get_ns_mem_region_attr(const void *p, size_t s,
                               struct mem_attr_info_t *p_attr);
```

Parameters	
p	Base address of the target memory region
s	Size of the target memory region
p_attr	Pointer to the mem_attr_info_t to be filled
Return	
void	None

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10.1.5 Communication Between NSPE And SPE In Dual Core System

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Introduction

This document describes the communication design between the Non-secure Processing Environment (NSPE) and Secure Processing Environment (SPE) in TF-M on a dual-core system.

The dual-core system has the following characteristics:

- NSPE and SPE are properly isolated and protected following PSA specification.
- An Arm M-profile core hosts SPE and acts as the Secure core.
- Another core hosts NSPE and runs non-secure applications.
- An Inter-Processor Communication hardware module is available in the system for communication between NSPE core and SPE core.
- TF-M runs on the Secure core with platform-specific drivers support.

Note

All the concepts for dual-cpu can be extended to multi-cpu without loss of generality. Also see the Hybrid Platform section for further details.

Scope

This design document focuses on the dual-core communication design inside TF-M. Some changes to TF-M core and Secure Partition Manager (SPM) are listed to support the dual-core communication. This document only discusses the implementation in TF-M Inter-Process Communication (IPC) model. The TF-M non-secure interface library depends on the mailbox and NS real-time operating system (RTOS) implementations. These components are not described in detail.

This document defines the requirements for mailbox functionality. For detailed mailbox design, see the dedicated mailbox document¹.

Organization of the document

- *Overall workflow in dual-core communication* provides an overall workflow of dual-core communication between NSPE and SPE.
- *Mailbox communication Requirements* lists general requirements of mailbox, from the perspective of dual-core communication.
- *PSA client call handling flow in TF-M* discusses about the detailed sequence and key modules of handling PSA client call in TF-M.

Overall workflow in dual-core communication

The overall workflow in dual-core scenario can be described as follows

1. A non-secure application calls TF-M non-secure interface library to request Secure service. The TF-M non-secure interface library translates the Secure service into PSA Client calls.
2. TF-M non-secure interface library notifies TF-M of the PSA client call request, via mailbox. The HAL must define generic mailbox APIs to support various IPC implementations.
3. Inter-Processor Communication interrupt handler and mailbox handling in TF-M deal with the inbound mailbox event(s) and deliver the PSA client call request to TF-M SPM.
4. TF-M SPM processes the PSA client call request and dispatches it to the appropriate secure partition or handler.
5. After the PSA Client call is completed, the return value is sent back to NSPE via mailbox.
6. TF-M non-secure interface library fetches return value from mailbox.
7. The return value is returned to non-secure application.

The interfaces between NSPE app and TF-M NSPE interface library are unchanged so the underlying platform specific details are transparent to NSPE application.

Step 3-5 are covered in *PSA client call handling flow in TF-M* in detail.

Note

¹ Mailbox Design in TF-M on Dual-core System

This document is partially complete since the introduction of the Hybrid Platform Scheduling support, which extends the operation of the mailbox communication between local and remote NSPEs with SPE described here. For more details see *Scheduling for a Hybrid Platform Solution*.

Mailbox communication Requirements

The communication between NSPE and SPE relies on mailbox communication implementation. The mailbox functionalities are eventually implemented by platform specific Inter-Processor Communication drivers. This section lists some general requirements on mailbox communication between NSPE and SPE.

Data exchanged between NPSE and SPE

A mailbox message contains the information and parameters of a PSA client call. After SPE is notified by a mailbox event, SPE fetches the parameters from NSPE for PSA Client call processing. The mailbox design document² defines the structure of the mailbox message.

The information and parameters of PSA Client call in the mailbox message include

- PSA Client API
- Parameters required in PSA Client call. The parameters can include the following, according to PSA client call type:
 - Service ID (SID)
 - Handle
 - Request type
 - Input vectors and the lengths
 - Output vectors and the lengths
 - Requested version of secure service
- NSPE Client ID. Optional. The NSPE Client ID is required when NSPE RTOS enforces non-secure task isolation.

The mailbox implementation may define additional members in mailbox message to accomplish mailbox communication between NSPE and SPE.

When the PSA Client call is completed in TF-M, the return result, such as PSA_SUCCESS or a handle, shall be returned from SPE to NSPE via mailbox.

Synchronization between NSPE and SPE

The system must protect shared mailbox objects and variables from concurrent access by NSPE and SPE.

Synchronization mechanisms must prevent race conditions when both cores access the mailbox.

Support of multiple concurrent calls (informative)

If the support of multiple ongoing NS PSA client calls in TF-M is required in dual-core systems, an optional queue can be maintained in TF-M core to store multiple mailbox objects received from NSPE. To identify NS PSA client calls, additional fields can be added in TF-M SPM objects to store the NS PSA Client request identification.

Note that when just a single outstanding PSA client call is allowed, multiple NSPE OS threads can run concurrently and call PSA client functions. The first PSA client call will be processed first, and any other OS threads will be blocked from submitting PSA client calls until the first is completed.

PSA client call handling flow in TF-M

This section describes how TF-M processes PSA client call requests:

1. Platform specific Inter-Processor Communication interrupt handler is triggered after the mailbox event is asserted by NSPE. The interrupt handler calls `spm_handle_interrupt()`.
2. SPM sends a `MAILBOX_INTERRUPT_SIGNAL` to `ns_agent_mailbox` partition.
3. `ns_agent_mailbox` partition deals with the mailbox message(s) which contain(s) the PSA client call information and parameters. Then the PSA client call request is dispatched to dedicated PSA client call handler in TF-M SPM.
4. After the PSA client call completes, the return value is transmitted to NSPE via mailbox.

Key modules and components:

- *Inter-Processor Communication interrupt handler* discusses the Inter-Processor Communication interrupt handler
- *TF-M Remote Procedure Call (RPC) layer* introduces TF-M Remote Procedure Call layer to support dual-core communication.
- *ns_agent_mailbox partition* describes the mailbox agent partition.
- *Return value response routine in TF-M* proposes the routine to send the return value to NSPE.

Inter-Processor Communication interrupt handler

Platform specific driver must implement the Inter-Processor Communication interrupt handler to deal with the Inter-Processor Communication interrupt asserted by NSPE. The platform specific interrupt handler shall complete the interrupt operations, such as interrupt EOI or acknowledge.

The interrupt handler calls `spm_handle_interrupt()` to notify SPM of the interrupt.

The platform's `tfm_peripherals_def.h` file defines a macro `MAILBOX_IRQ` that identifies the interrupt being used. The platform must also provide a function `mailbox_irq_init()` that initialises the interrupt as described in².

Platform specific driver puts Inter-Processor Communication interrupt into a proper exception priority, according to system and application requirements. The priority setting must guarantee that:

- TF-M may respond to a PSA client call request in time according to system and application requirements.
- Other exceptions, which are more latency sensitive or require higher priorities, are not blocked by Inter-Processor Communication interrupt ISR.

The exception priority setting is IMPLEMENTATION DEFINED.

TF-M Remote Procedure Call (RPC) layer

The RPC layer sits between TF-M SPM and mailbox implementation. The purpose of RPC layer is to decouple mailbox implementation and TF-M SPM and enhance the generality of entire dual-core communication.

The RPC layer provides a set of APIs to TF-M SPM to handle and reply PSA client call from NSPE in dual-core scenario. It hides the details of specific mailbox implementation from TF-M SPM. It avoids modifying TF-M SPM to fit mailbox development and changes. It can keep a unified PSA client call process in TF-M SPM in both single Armv8-M scenario and dual-core scenario.

The RPC layer defines a set callback functions for mailbox implementation to hook its specific mailbox operations. When TF-M SPM invokes RPC APIs to deal with NSPE PSA client call, RPC layer eventually calls the callbacks to execute mailbox operations. RPC layer also defines a set of PSA client call handler APIs for mailbox implementation.

² *Secure Interrupt Integration Guide*

RPC specific client call handlers parse the PSA client call parameters and invoke common TF-M PSA client call handlers.

ns_agent_mailbox partition

A dedicated agent acts as a partition and interacts with the NSPE through the mailbox.

First, the agent calls `tfm_hal_boot_ns_cpu()` and `tfm_hal_wait_for_ns_cpu_ready()` to prepare the non-secure core. Then it initialises the SPE mailbox and enable the IPC interrupt.

Once these tasks are complete, it enters an infinite loop waiting for a `MAILBOX_INTERRUPT_SIGNAL` signal indicating that a mailbox message has arrived.

Mailbox handling is performed in the context of the `ns_agent_mailbox` partition, which makes any necessary calls to other partitions on behalf of the non-secure code.

`ns_agent_mailbox` calls RPC API `tfm_rpc_client_call_handler()` to check and handle PSA client call requests from NSPE.

`tfm_rpc_client_call_handler()` invokes request handling callback function to execute specific mailbox message handling operations. More details in document².

The handling process in mailbox operation consists of the following steps:

1. SPE mailbox fetches the PSA client call parameters from NSPE mailbox. Proper protection and synchronization must be implemented in mailbox to guarantee that the operations are not interfered with by NSPE mailbox operations or Inter-Processor Communication interrupt handler. If a queue is maintained inside TF-M core, SPE mailbox can fetch multiple PSA client calls together into the queue, to save the time of synchronization between two cores.
2. SPE mailbox parses the PSA client call parameters copied from NSPE, including the PSA client call type.
3. The PSA client call request is dispatched to the dedicated TF-M RPC PSA client call handler. The PSA client call request is processed in the corresponding handler based on PSA call types.

The dual-core scenario and single Armv8-M scenario in TF-M IPC implementation shares the same PSA client call routines inside TF-M SPM. The current handler definitions can be adjusted to be more generic for dual-core scenario and single Armv8-M implementation.

If multiple requests are pending, the mailbox processes them one by one.

Return value response routine in TF-M

Diverse PSA client calls can be implemented with different return value replying routines.

Replying routine for `psa_framework_version()` and `psa_version()`

For `psa_framework_version()` and `psa_version()`, handlers return values directly during mailbox processing. A compile flag can enable this routine for dual-core builds.

For `psa_connect()`, `psa_call()`, and `psa_close()`, the target secure partition completes the client call and invokes `psa_reply()`. The SVC handler in TF-M SPM then calls `tfm_rpc_client_call_reply()` to return the result via mailbox.

Mailbox reply functions must not trigger context switches inside the SVC handler.

If errors occur, the RPC handlers return the error without invoking the secure partition. The mailbox transmits the error code to NSPE.

Build options

The feature is enabled by `TFM_MULTI_CORE_TOPOLOGY`.

Reference

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10.2 Secure Services

10.2.1 Secure Partition Manager

This document describes the Secure Partition Manager (*SPM*) implementation design in Trusted Firmware-M (*TF-M*).

Note

- The FF-M in this document refers to the accumulated result of two specifications: [FF-M v1.1 Update on FF-M v1.0](#).
- The words marked as *interpreted* are defined terms. Find the terms in referenced documents if it is not described in this document.

Introduction

The service access process of FF-M:

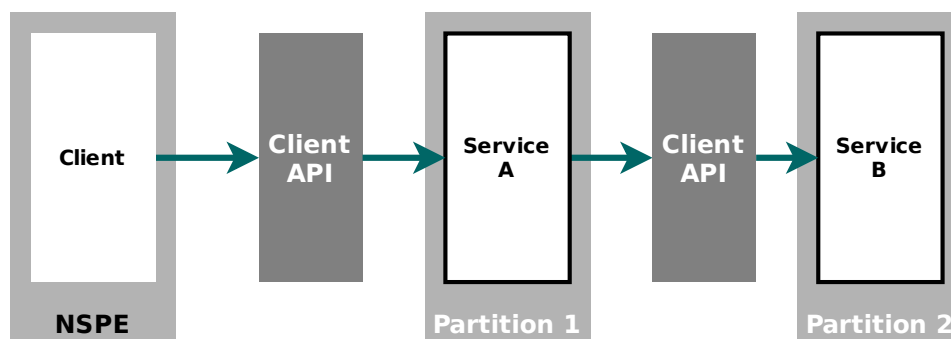


Figure 6:: FF-M service access process

Secure services (aka *Service*) is the component providing secure functionalities in *SPE*, and *Client* is the user of the *Service*. A service acts as a client when it is accessing its depending services.

Services are grouped into *Secure Partition* (aka *partition*). A partition:

- Contains services with the same purpose.

- Provides implementation required isolation boundaries.
- Is a software development unit.

Each service exposes its *Service ID (SID)* and *Handle* for client access usage. Clients access services by *SID* or *Handle* through FF-M *Client API*. Partitions use FF-M *Secure Partition API* when it needs to operate on client data or reply to a client.

SPM is the centre of an FF-M compliant implementation, which sets up and maintains a firmware framework that:

- Implements *Client API* and *Secure Partition API*.
- Manages partition runtime to follow FF-M.
- Involves necessary implementation-defined items to support the implementation.

SPM interfaces consist of these two categories:

- FF-M defined API.
- Extended API to support the implementation.

Both API categories are compliant with FF-M concepts and guidelines. The core concept of TF-M SPM surrounds the FF-M defined service management and access process. Besides this, another important implementation part is partition runtime management.

Partition runtime model

One partition must work under as *ONE* of the runtime models: *Inter-process communication (IPC)* model or *Secure Function (SFN)* model.

A partition that runs under the *IPC* model looks like a classic *process*. There is *ONE* thread inside the partition keeps waiting for signals. SPM converts the service accessing info from the *Client API* call into messages and assert a signal to the partition. The partition calls corresponded service function indicated by the signal and its bound message, and reply service returned result to the client. The advantages of this model:

- It provides better isolation by limiting the interfaces on data interactive. Data are preferred to be processed in a local buffer.
- It provides a mechanism for handling multiple service access. There is no memory mapping mechanism in the MCU system, hence it is hard to provide multiple function call contexts when serving multiple-threaded clients if the service access is implemented in a function-call based mechanism. This model converts multiple service accesses into messages, let the partition handles the service access in messages one by one.

The *Secure Function (SFN)* model partition is close to a *library*. Each service is provided as a function entry inside the partition. SPM launches the target service function after the service is found. The whole procedure (from client to service function) is a function call. This model:

- Saves the workloads spent on *IPC* scheduling.

Meanwhile, it relaxes the data interactive mechanism, for example, allow direct memory access (MMIOVEC). And it is hard to enable multiple-threaded clients service access because of multiple call context-maintenance difficulties.

An implementation contains only *SFN* partitions fits better in the resource-constrained devices, it is called an *SFN model implementation*. And it is an *IPC model implementation* when *IPC* partitions exist in the system.

Note

IPC model implementation can handle access to the services in the *SFN* partition.

Components and isolation levels

There are *THREE* isolation levels defined in *FF-M*. These levels can fulfil different security requirements by defining different isolation boundaries.

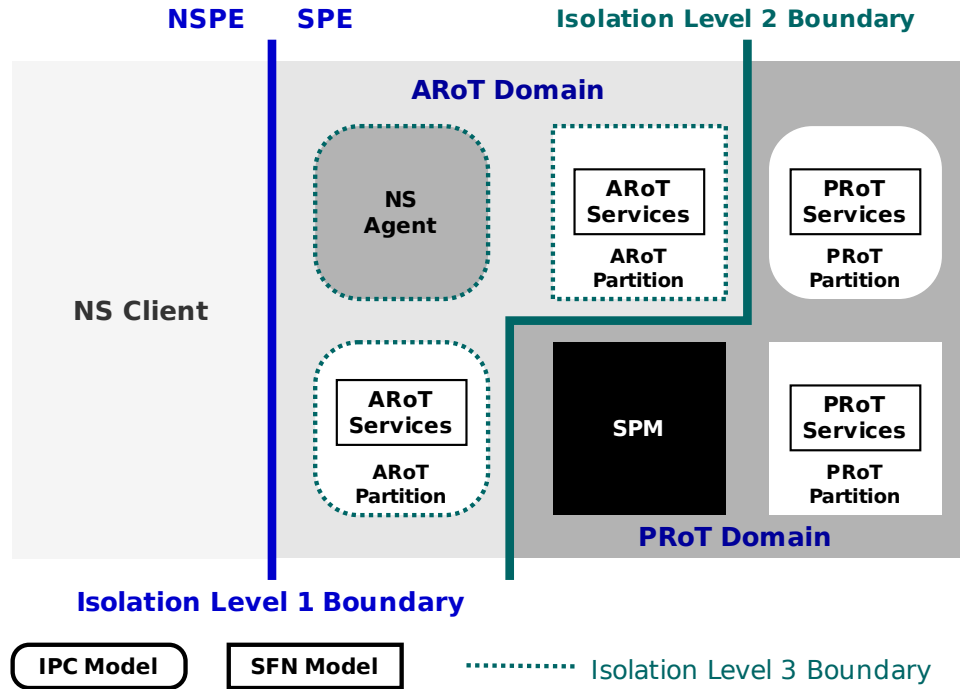


Figure 7:: Components and isolation boundaries

Note

Concept *ARoT*, *PRoT*, *domain*, and boundaries are in the *FF-M* specification.

Not like an *SPE* client that can call *Client API* to access the secure services in one step, an *NSPE* client needs to cross the secure boundaries first before calling *Client API*. The component *NS Agent* in Figure ?? represents *NSPE* clients after they crossed the secure boundaries. This could help *SPM* handles the request in a unified way instead of care about the special boundaries.

Note

NS Agent is a necessary implementation-defined component out of *FF-M* specification. *NS Agent* has a dedicated stack because secure and non-secure can not share the stack. It also has dedicated execution bodies. For example, RPC-based *NS Agent* has a while loop that keeps waiting for messages; and Trustzone-based *NS Agent* has veneer code to take over *NSPE* secure call. This makes *NS Agent* to be a component more like a *process*. Hence in the simplest implementation (*SFN model implementation* mentioned above), *NS Agent* is the only process in the system, the scheduling logic can be extremely simplified since no other process execution needs to be scheduled. But the scheduling interface is still necessary to *SPM*, this could help *SPM* treat both *SFN* and *IPC* model implementation in a unified way.

Check *NS Agent* for details.

Implementation principle

The principles for TF-M SPM implementation:

Important

- SPM can treat these components as the client: NS Agent, SFN Partition, and IPC partition.
- These components can provide services: SFN Partition, IPC partition, and built-in services. A built-in service is built up with SPM together.
- All partition services must be accessed by *Client API*.
- Partitions interact with client data by *Secure Partition API*.
- Built-in services are strongly recommended to be accessed by *Client API*. Customized interfaces are restricted.
- Built-in services can call SPM internal interfaces directly.

Runtime management

The runtime execution runs among the components, there are **4** runtime states:

- *Initializing* state, to set up the SPM runtime environment after system powers up
- *IDLE* state, when SPM runtime environment is set up and partitions are ready for service access.
- *Serving* state, when partition is under initializing or service access handling.
- *Background* state, such as the arrival of secure interrupt or unexpected faults. *Background* state returns to the state it preempts. *Background* state can be nested.

The state transition diagram:

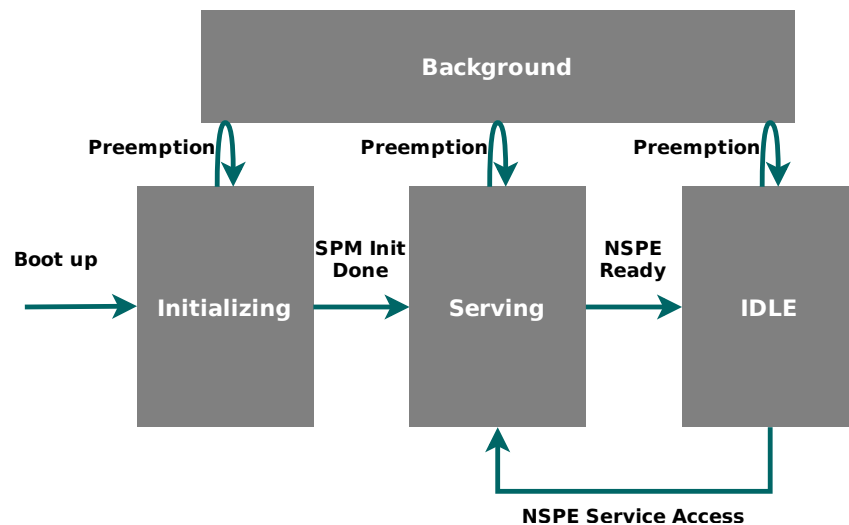


Figure 8:: SPE runtime execution states

Initializing

The goal of TF-M initializing is to perform necessary initialization and move to the *Serving* state. This state starts with platform-specific power on sequence, then *SPM* takes over the execution to perform these operations:

1. A preparation initialization process before *SPM* runtime initialization.
2. *SPM* runtime initialization.
3. A post initialization happens after the *SPM* runtime initialization and before the first partition gets launched.

Note

These procedures and their sub-routines are recommended to be applied with execution measurement mechanism to mitigate the *Hardware Fault Injection* attack.

Preparation initialization

The purpose of this preparation initialization is to provide a chance for performing those security required but generic platform power-on skipped operations, such as:

- Restrict *SPM* execution, for example, set up memory overflow settings for *SPM* runtime memory, or set code out of *SPM* as un-executable, even though *SPM* is a privileged component in general.

Note

The logging-related peripheral can be set up **AT THIS STEP**, if logging is enabled and it needs peripheral support. There is no standalone initializing HAL API proposed for logging, so here is an ideal place for initializing them.

This procedure is abstracted into one *HAL*, and a few example procedures are implemented as its sub-routines for reference:

- Architecture extensions initialization, Check chapter *Architecture security settings* for detailed information.
- Isolation and lifecycle initialization.

The load isolation boundaries need to be set up here, such as SPE/NSPE boundary, and ARoT/PRoT boundary if isolation level 2 is applied.

The lifecycle is initiated by a secure bootloader usually. And in this stage of *SPM* initializing, *SPM* double-checks the lifecycle set up status (following a specific lifecycle management guidelines). Note that the hardware debugger settings can be part of lifecycle settings.

Important

Double-check debugger settings when performing a product release.

SPM runtime initialization

This procedure initializes necessary runtime operations such as memory allocator, loading partitions and partition-specific initialization (binding partitions with platform resources).

The general processes:

1. Initialize runtime functionalities, such as memory allocator.

2. Load partitions by repeating below steps:

- Find a partition load information.
- Allocate runtime objects for this partition.
- Link the runtime objects with load information.
- Init partition contexts (Thread and call context).
- Init partition isolation boundaries (MMIO e.g.).
- Init partition interrupts.

After no more partitions need to be loaded, the SPM runtime is set up but partitions' initialization routines have not run yet - the partition runtime context is initialized for the routine call.

The partition initialization routine is a special service that serves SPM only, because:

- SPM needs to call the initialization routine, just like it calls into the service routine.
- The partition initialization routine can access its depending services. Putting initialization routine in the same runtime environment as common service routines can avoid special operations.

Hence a *Partition initialization client* needs to be created to initialize the SFN partitions, because:

- *SPM runtime initialization* happen inside a special runtime environment compare to the partition runtime execution, then an environment switching is needed.
- IPC partitions are initialized by the scheduler and dependencies are handled by signals and messages asynchronously, hence IPC partitions can process the dependencies by their own.

The *Partition initialization client* is created differently based on the implementation runtime model:

- A SFN client is created under the SFN model implementation.
- A IPC client is created under the IPC model implementation. This client thread has the highest priority.

As the other partitions, the client is created with context standby, and it is executed after the *Post initialization* stage.

Post initialization

Platform code can change specific partition settings in this procedure before partitions start. A few SPM API is callable at this stage, such as set a signal into a specific partition, or customized peripheral settings.

Serving

Two execution categories work under this state:

- *Partition initialization routine execution.*
- *Secure service access.*

This state indicates the serving is ongoing. It is mainly the service routine execution, plus a few SPM executions when SPM API gets called.

Important

The service access process introduced in this chapter (Such as *Secure service access*) is abstracted from the FF-M specification. Reference the FF-M specification for the details of each step.

Partition initialization routine execution

The partition initialization routines get called. One partition may access its depending services during initializing, then this procedure is a *Secure service access*.

The initialization routine gets called initially by *Partition initialization client*, also can be called by Client API before service access, if the target partition is not initialized but a service access request is raised by one client.

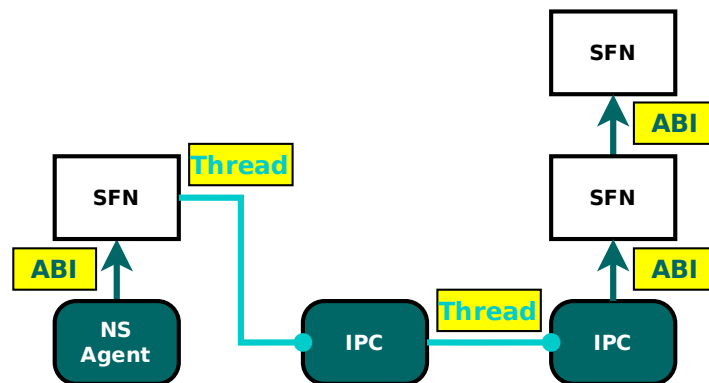
Secure service access

The process of service access:

1. A *client* calls an FF-M Client API.
2. *SPM* validates inputs and looks up for the targeted service.
3. *SPM* constructs the request to be delivered under a proper runtime mechanism.
4. The target service gets executed. It can perform internal executions or access depending services to prepare the response. It also can wait for specific signals.
5. The target service calls FF-M Secure Partition API to request a reply to the client.
6. SPM delivers the response to the client, and the API called by the client returns.

The mechanism of how SPM interact with the target partition depends on the partition runtime model.

- Access to a service in an SFN partition is a function call, which does not switch the current process indicator.
- Access to a service in an IPC partition leads to scheduling, which switches the current process indicator.
- When the execution roams between components because of a function call or scheduling, the isolation boundaries NEED to be switched if there are boundaries between components.



No matter what kind of partition a client is trying to access, the SPM API is called firstly as it is the interface for service access. There are two ABI types when calling SPM API: Cross-boundary or No-cross-boundary.

Calling SPM API

SPM is placed in the PRoT domain. It MAY have isolation boundaries under particular isolation levels. For example:

- There are boundaries between ARoT components and SPM under isolation level 2 and 3.

The API SPM provided needs to support the function call (no boundary switching) and cross-boundary call. A direct call reaches the API entrance directly, while a cross-boundary call needs a mechanism (Supervisor call e.g.) to cross the boundary first before reaching the API entrance.

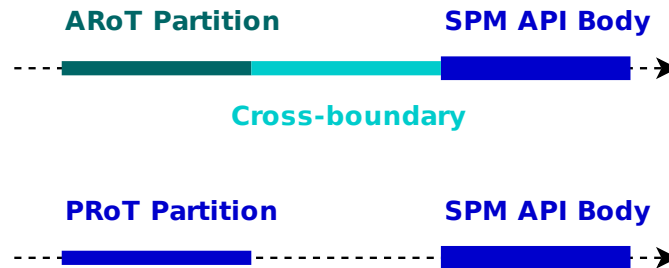


Figure 9:: SPM call types

SPM internal execution flow

SPM internal execution flow as shown in diagram:

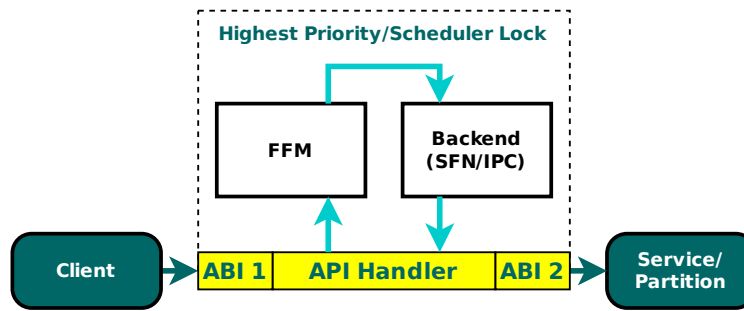


Figure 10:: SPM API runtime

The process:

- PSA API gets called by one of the ABI mentioned in the last chapter as *ABI 1* in the diagram.
- The unified API Handler calls FF-M and backend subroutines in sequence.
- The *FF-M* subroutine performs *FF-M* defined operations.
- The backend operations perform target partition runtime model decided operations. For example, enqueue message into the target partition under the IPC runtime model, or prepare to call context with the message as the parameters under the SFN runtime model.
- API Handler triggers different ABI based on the result of the backends.

The API handler:

- Can process the *PROGRAMMER_ERROR* in a unified place.
- Can see the prepared caller and callee context, with exited SPM context. It is an ideal place for subsequent operations such as context switching.

An example code:

```
void abi(void *p)
{
    status = spm_api(p);
    /*
     * Now both the caller and callee contexts are
```

(continues on next page)

(continued from previous page)

```

    * managed by spm_api.
    */
    if (status == ACTION1) {
        /*
         * Check if extra operations are required
         * instead of a direct return.
         */
        exit_action1();
    }
}

```

The explanation about *Scheduler Lock*:

Some FF-M API runs as a generic thread to prevent long time exclusive execution. When a preemption happens, a new partition thread can call SPM API again, makes SPM API nested. It needs extra memory in SPM to be allocated to store the preempted context. Lock the scheduler while SPM API is executing can ensure SPM API complete execution after preemption is handled. There can be multiple ways to lock the scheduler:

- Set a scheduler lock.
- Set SPM API thread priority as the highest.

Backend service messaging

A message to service is created after the target service is found and the target partition runtime model is known. The preparation before ABI triggers the final accessing:

- The message is pushed into partition memory under a specific ABI mechanism if the target partition model is *SFN* and there are boundaries between SPM and the target partition. After this, requests a specific call type to the SPM ABI module.
- The target service routine is called with the message parameter if there are no boundaries between SPM and the target partition and the partition runtime is *SFN*.
- The message is queued into the partition message list if the target partition runtime model is *IPC*.
- IPC partition replies to the client by *psa_reply*, which is another SPM API call procedure.
- SFN partition return triggers an implied *psa_reply*, which is also another SPM API call procedure.

Note

The backends also handle the isolation boundary switching.

Sessions and contexts

FF-M API allows multiple sessions for a service if the service is classic connection-based. The service can maintain multiple local session data and use *rhandle* in the message body to identify which client this session is bound with.

But this does not mean when an ongoing service accessing is preempted, another service access request can get a chance for new access. This is because of the limited context storage - supporting multiple contexts in a common service costs much memory, and runtime operations (allocation and re-location). Limited the context content in the stack only can mitigate the effort, but this requirement requires too much for the service development.

The implementation-decisions are:

- IPC partitions handles messages one by one, the client gets blocked before the service replying to the client.

- The client is blocked when accessing services are handling a service request in an SFN partition.

ABI type summary

The interface type is decided by the runtime model of the target component. Hence PSA API has two types of ABI: *Cross-boundary ABI* and *Function call ABI*. After SPM operations, one more component runtime type shows up: The IPC partition, hence *schedule* is the mechanism when accessing services inside an IPC partition.

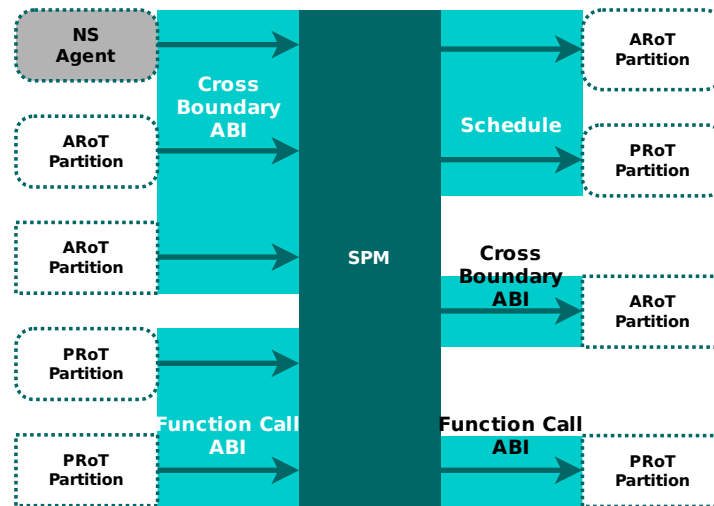


Figure 11:: ABI types

Note

The API that does not switch context returns directly, which is not covered in the above diagram.

IDLE state

The *IDLE state* can be represented by the *NS Agent* action:

- Launching NSPE software (Trustzone case, e.g.), or send a signal to NSPE software (RPC case, e.g.).

It is because *NS Agent* is the last component being initialized in the system. Its execution indicates other partitions' initialization has accomplished.

Background state

Background execution can happen at any time when the arrival of interrupts or execution faults. An ongoing background execution indicates the state is a *Background state*. The characteristics:

- The background state has a higher execution priority than other states - other states stall when the background state is executing.
- Background execution can be nested. For example, an interrupt handler can preempt an ongoing interrupt execution.
- Particular partition code can be involved in the background state, for example, the *First Level Interrupt Handler (FLIH)* of one partition.
- Background state MUST return to the state it preempts.

Note

Interrupt handling is a common background state example. Check Interrupt design document for details.

Practical implementation items

This chapter describes the practical implementation contents.

Important

Arm M-profile architecture is the default hardware architecture when describing architecture-specific items.

The general M-profile programming is not involved in this document. The following chapters introduce the mandatory settings for security requirements.

Architecture security settings

When an *Armv8m Security Extension* (Aka *Trustzone-M*) is available in the system, these settings are required to be set:

- The MSPLIM needs to be set correctly to prevent stack overflow.
- The exception handler priority needs to be decided.
- Boost the secure handler mode priority to prevent NSPE from preempting SPE handler mode execution(*AIRCR.PRIS*).
- Disable NSPE hardware faults when a secure fault is happening. Trap in the secure fault with the highest priority can be a valid option.
- Push seals on the stack top when a stack is allocated (*TFMV-I*). Also check *Stack seal* chapter for details.

Besides *Armv8m Security Extension*, these settings need to care when *Floatpoint Extension* is enabled for partition usage:

- *FPCCR.TS*, *FPCCR.CLRONRET* and *FPCCR.CLRONRETS* need to be set when booting.
- *CPACR.CP10* and *CPACR.CP11* need to be set when booting.

Important

Floatpoint usage is prohibited in SPM and background execution.

Stack seal

When Trustzone-M is applied, the architecture specification recommends sealing the secure stack by:

- Push two *SEAL* values (*0xFE5EDA5*) at the stack bottom, when a stack is allocated.
- Push two *SEAL* values on the stack pointer which is going to be switched out.

Check architecture specification and vulnerability *TFMV-I* for details.

Trustzone-M reentrant

The Trustzone-M has characteristics that:

- SPE keeps the last assigned stack pointer value when execution leaves SPE.
- SPE execution can be preempted by NSPE which causes an execution left.

It is possible that NSPE preemption caused a second thread calls into SPE and re-uses the secure stack contains the first thread's context, which obviously causes information leakage and runtime state inconsistent.

Armv8.1-M provides the hardware setting *CCR_S.TRD* to prevent the reentrant. On an Armv8.0-M architecture, extra software logic needs to be added at the veneer entry:

- Check if the local stack points to a *SEAL* when veneer code get executed.

```
/* This is a theoretical code that is not in a real project. */
veneer() {
    content = get_sp_value();
    if (context != SEAL) /* Error if reentrant detected */
        error();
}
```

SPM Runtime ABI

This chapter describes the runtime implementation of SPM.

Scheduling

The scheduling logic is put inside the PendSV mode. PendSV mode's priority is set as one level higher than the default thread mode priority. If *Trustzone-M* is present, the priority is set as the lowest just above NS exception priority to prevent a preemption in secure exceptions.

PendSV is an ideal place for scheduling logic, because:

- An interrupt triggered scheduling during PendSV execution lead to another PendSV execution before exception return to the thread mode, which can find the latest run-able thread.

Function call ABI

In the diagram Figure ??, the ABI can have two basic types: cross-boundary and direct call (No-cross-boundary).

When applying *SVC* as the cross-boundary mechanism, the implementation can be straight like:

- The SVC handler calls SPM internal routines, and eventually back to the handler before an exit.

Under the IPC model implementation, to reuse *ABI 2* in *No-cross-boundary*, a software ABI needs to be provided.

While under the SFN model plus isolation level 1, both *ABI 1* and *ABI 2* can be a direct function call.

NS Agent

The *NS Agent* (*NSA*) forwards NSPE service access request to SPM. It is a special *partition* that:

- It does not provide FF-M aligned secure services.
- It runs with the second-lowest priority under *IPC model implementation* (The IDLE thread has the lowest priority).
- It has isolation boundaries and an individual stack.

- It requires specific services and mechanisms compared to common partitions.

There are two known types for NS Agent:

- Trustzone-M based.
- Remote Procedure Call (RPC) based.

This process is put inside the ARoT domain, to prevent assign unnecessary PRoT permissions to the NSPE request parsing logic.

Trustzone-M specific

The functionalities of a Trustzone-M specific NSA is:

- Launch NSPE when booting.
- Wait in the veneer code, and get executed when NSPE accesses services.

As there may be multiple NSPE threads calling into SPE, and SPM wants to identify them, special mechanisms can be proposed to provide the identification. Check specific NS ID client ID or context related documents for details.

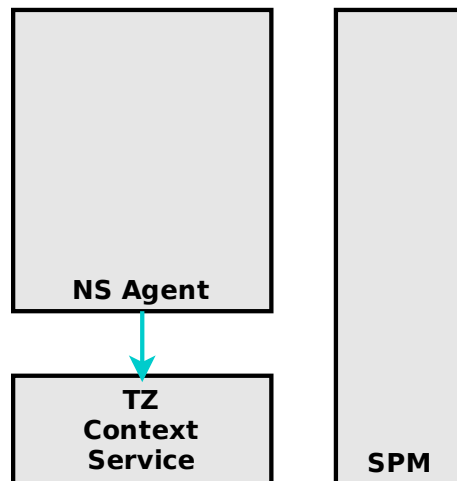


Figure 12:: TZ NSA and specific service

RPC specific

Compared to Trustzone-M NSA, RPC NSA looks closer to a generic partition:

- It has a message loop, keep waiting for RPC events.
- It converts received RPC events into FF-M API call to target services.

And compared to generic partitions, the differences are:

- It parses RPC messages to know which NSPE thread is accessing services. Hence it needs special interfaces to help SPM to identify the NSPE clients.
- It needs to check NSPE client memory and map to local before calling SPM API.
- It cannot be blocked during API calls, which affects handling the RPC requests.

Partition

A partition is a set of services in the same scope. Services are generally implemented as functions, and the partition exposes the services in different ways based on the partition model: *IPC* or *SFN*.

A partition build generates these outputs:

- A partition load information, used by SPM.
- A partition program containing service interface and logic, typically a library.
- An optional service API set for easier client usage, by encapsulating the low-level *FF-M* Client API. These API needs to be integrated into client space.

Partition loading

SPM needs to set up runtime objects to manage partitions by parsing the load information of partitions. In general, the partition load information is stored in a const memory area can be random read directly, hence SPM can direct link runtime objects to the load information without a copy operation. This is called a *Static Load* mechanism.

Each partition has different numbers of dependencies and services, this makes the load information size of each partition different, it would be hard to put such variable size elements in an array. The solution here is putting these elements in a dedicated section, for SPM enumerating while loading. Each partition can define variable size load information type based on the common load info type.

The common load information:

```
struct partition_load_info_t {
    uint32_t    psa_ff_ver;      /* Encode the version with magic */
    int32_t     pid;             /* Partition ID */
    uint32_t    flags;           /* ARoT/PRoT, SFN/IPC, priority */
    uintptr_t   entry;           /* Entry point */
    size_t      stack_size;      /* Stack size */
    size_t      heap_size;       /* Heap size */
    uint32_t    ndeps;           /* Dependency number */
    uint32_t    nservices;       /* Service number */
    uint32_t    nassets;         /* Asset numbers */
    uint32_t    nirqs;           /* Number of IRQ owned by Partition */
    int32_t     client_id_base;   /* The min translated client ID */
    int32_t     client_id_limit; /* The max translated client ID */
    uint16_t    load_order;      /* loading order of this partition */
};
```

And the example for a specific partition load info:

```
struct partition_example_load_info_t {
    struct partition_load_info_t ldi; /* Common info info */
    uint32_t    deps[10]; /* Dependencies */
    /*          ... other infos ... */
};
```

Peripheral binding

A partition can declare multiple peripherals (Interrupts are part of peripherals). The peripherals binding process:

- The tooling references symbols in a fixed pattern in the partition load information.
- The HAL implementation needs to provide the symbols being referenced.

- SPM calls HAL API to bind the partition info with devices when the partition gets loaded.
- The platform HAL acknowledges the binding if validation pass on SPM given load information.

Integration and development

These modules are expected to be object/library level modularised, each module should be generated into object/library at build time:

Table 40:: Object level modularization

Name	Description
SPM	All SPM related modules such as SPM, system, and so on.
Platform	Platform sources are switchable.
Services and Secure Partition	These items should be standalone.
Service Runtime Library	This is a shared runtime library.

HAL

The HAL here mainly refers to the SPM HAL. The SPM HAL implementation is running with the same privilege level and hardware mode with SPM. The implementation is object level modularized with SPM.

Check the *HAL* design document for details.

Configurations

The same TF-M code base is flexible to address different implementation requirements, from the simplest device with isolation level 1 to the most complicated device with isolation level 3 and optional isolation rules.

These configurations are set by switches, during the build time, as runtime support costs extra resources. The common configurations are named *profile*. There are several profiles defined.

History

Table 41:: Revision

Date	Description
2021 Apr-Sep	Updated to cover the implementation for <i>FF-M v1.1</i> features.
2018	Created as 'TF-M Inter-Process Communication' which is deprecated as this document covers whole SPM content.

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10.2.2 Secure Partition Runtime Library

Organization

Arm Limited

Contact

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Background

Trusted Firmware - M (TF-M) uses a toolchain provided runtime library and supervisor calls to easily implement the PSA Firmware Framework (PSA FF) API. This working model works well under isolation level 1 since there are no data isolation requirements. While TF-M is evolving, this model is not suitable because:

- The high-level isolation requires isolating data but some toolchain library interfaces have their own global data which cannot be shared between the Secure Partitions.
- The toolchain libraries are designed without taking security as a core design principle.

A TF-M specific runtime library is needed for the following reasons:

- Easier evaluation or certification by security standards.
- Source code transparency.
- Sharing code to save ROM and RAM space for TF-M.

PSA FF specification also describes the requirements of C runtime API for Secure Partitions.

This runtime library is named the `Secure Partition Runtime Library`, and the abbreviation is `SPRTL`.

Design Principle

The following requirements are mandatory for `SPRTL` implementation:

Important

- **CODE ONLY** - No read-write data should be introduced into runtime library implementation.
- **Thread safe** - All functions are designed with thread-safe consideration. These APIs access caller stack and caller provided memory only.
- **Isolation** - Runtime API code is set as executable and read-only in higher isolation levels.
- **Security first** - `SPRTL` is designed for security and it may come with some performance loss.

API Categories

Several known types of functions are included in `SPRTL`:

- C runtime API.
- RoT Service API.
- PSA Client and Service API.
- [Future expansion, to be detailed later] other secure API.

Security Implementation Requirements

If `malloc/realloc/free` are provided, they must obey additional requirements compared to the C standard: newly allocated memory must be initialized to ZERO, and freed memory must be wiped immediately in case the block contains sensitive data.

The comparison API ('`memcmp`' e.g.), they should not return immediately when the fault case is detected. The implementation should execute in linear time based on input to avoid execution timing side channel attack.

The pointer validation needs to be considered. In general, at least the 'non-NULL' checking is mandatory. A detection for invalid pointer leads to a `psa_panic()`.

The following section describes the first 3 API types and the implementation requirements.

C Runtime API

PSA FF describes a small set of the C standard library. Part of toolchain library API can be used as default if these APIs meet the *Design Principle* and *Security Implementation Requirements*. The toolchain ‘header’ and ‘types’ can be reused to simplify the implementation.

These APIs can take the toolchain provided version, or separately implemented in case there are extra requirements:

Note

- ‘memcpy()/memmove()/memset()’
- String API

These APIs are proposed to be implemented with the security consideration mentioned in *Security Implementation Requirements*:

Note

- ‘memcmp()’
- Other comparison API if referenced (‘strcmp’ e.g.).

The following functions are optional, but if present, they must conform to additional *Security Implementation Requirements*:

Note

- ‘malloc()/free()/realloc()’
- ‘assert()/printf()’

The following APIs are coupled with toolchain library much so applying toolchain library implementation is recommended:

Note

- Division and modulo - arithmetic operations.
- Other low level or compiler specific functions (such as ‘va_list’).

Besides the APIs mentioned above, the following runtime APIs are required for runtime APIs with private runtime context (‘malloc’ e.g.):

Note

- ‘__sprintmain()’ - partition entry runtime wrapper.

RoT Service API

The description of RoT Service API in PSA FF:

Note

Arm recommends that the RoT Service developer also defines an RoT Service API and implementation to encapsulate the use of the IPC protocol, and improve the usability of the service for client firmware.

Part of the RoT Service API have proposed specifications, such as the PSA Cryptography API, PSA Storage API, and PSA Attestation API. It is suggested that the service developer create documents of their RoT Service API and make them publicly available.

The RoT Service API has a large amount and it is the main part of SPRTL. This chapter describes the general implementation of the RoT Service API and the reason for putting them into SPRTL.

In general, a client uses the PSA Client API to access a secure service. For example:

```
/* Example, not a real implementation */
caller_status_t psa_example_service(void)
{
    ...
    handle = psa_connect(SERVICE_SID, SERVICE_VERSION);
    if (INVALID_HANDLE(handle)) {
        return INVALID_RETURN;
    }

    status = psa_call(handle, type, invecs, inlen, outvecs, outlen);

    psa_close(handle);

    return TO_CALLER_STATUS(status);
}
```

This example encapsulates the PSA Client API, and can be provided as a simpler and more generic API for clients to call. It is not possible to statically link this API to each Secure Partition because of the limited storage space. The ideal solution is to put it inside SPRTL and share it to all Secure Partitions. This would simplify the caller logic into this:

```
if (psa_example_service() != STATUS_SUCCESS) {
    /* do something */
}
```

This is the simplest case of encapsulating PSA Client API. If a RoT Service API is connect heavy, then, the encapsulation can be changed to include a connection handle inside a context data structure. This context data structure type is defined in RoT Service headers and the instance is allocated by API caller since API implementation does not have private data.

Note

- Even the RoT Service APIs are provided in SPRTL for all clients, the SPM performs the access check eventually and decides if the access to service can be processed.
- For those RoT Service APIs only get called by a specific client, they can be implemented inside the caller client, instead of putting it into SPRTL.

PSA Client and Service API

Most of the PSA APIs can be called directly with supervisor calls. The only special function is `psa_call`, because it has 6 parameters. This makes the supervisor call handler complex because it has to extract the parameters from the stack. The definition of `psa_call` is the following:

```
psa_status_t psa_call(psa_handle_t handle, int32_t type,
                     const psa_invec *in_vec, size_t in_len,
                     psa_outvec *out_vec, size_t out_len);
```

The parameters need to be packed to avoid passing parameters on the stack, and the supervisor call needs to unpack the parameters back to 6 for subsequent processing.

Privileged Access Supporting

Due to specified API (`printf`, e.g.) need to access privileged resources, TF-M Core needs to provide interface for the resources accessing. The permission checking must happen in Core while caller is calling these interface.

Secure Partition Local Storage

There are APIs that need to reference specific partition private data ('malloc' references local heap, e.g.), and the APIs reference the data by mechanisms other than function parameters. The mechanism in TF-M is called 'Secure Partition Local Storage'.

A straight way for accessing the local storage is to put the local storage pointer in a known position in the stack, but there is a bit of difficulty in particular scenarios.

Note

- The partition's stack is not fixed-size aligned, using stack address aligning method can not work.
- It requires privileged permission to *access* special registers such as *PSPLIMIT*. And Armv6-M and Armv7-M don't have *PSPLIMIT*.

Another common method is to put the pointer in one shared global variable, and the scheduler maintains the value of this variable to point to the running partition's local storage in runtime. It does not fully align with SPRTL design prerequisites listed above, hence extra settings are required to guarantee the isolation boundaries are not broken.

Important

- This variable is put inside a dedicated shared region and it can not hold information not belonging to the owner.

And this mechanism has disadvantages:

- It needs extra maintenance effort from the scheduler and extra resources for containing the variable.

TF-M chooses this common way as the default option for local storage and can be expanded to support more methods.

Tooling Support on Partition Entry

PSA FF requires each Secure Partition to have an entry point. For example:

```
/* The entry point function must not return. */
void entry_point(void);
```


Each partition has its own dedicated local_storage for heap tracking and other runtime state. The local_storage is designed to be saved at the read-write data area of a partition with a specific name. A generic entry point needs to be available to get partition local_storage and do initialization before calling into the actual partition entry. This generic entry point is defined as ‘__sprtmain’:

```
void __sprtmain(void)
{
    /* Get current SP private data from local storage */
    struct p_sp_local_storage_t *m =
        (struct p_sp_local_storage_t *)tfm_sprt_local_storage;

    /* Potential heap init - check later chapter */
    if (m->heap_size) {
        m->heap_instance = tfm_sprt_heap_init(m->heap_sa, m->heap_sz);
    }

    /* Call thread entry 'entry_point' */
    m->thread_entry();

    /* Back to tell Core end this thread */
    SVC(THREAD_EXIT);
}
```

Since SPM is not aware of the ‘__sprtmain’ in SPRTL, it just calls into the entry point listed in partition runtime data structure. And the partition writer may be not aware of running of ‘__sprtmain’ as the generic wrapper entry, tooling support needs to happen to support this magic. Here is an example of partition manifest:

```
{
    "name": "TFM_SP_SERVICE",
    "type": "PSA-ROT",
    "priority": "NORMAL",
    "entry_point": "tfm_service_entry",
    "stack_size": "0x1800",
    "heap_size": "0x1000",
    ...
}
```

Tooling would do manipulation to tell SPM the partition entry as ‘__sprtmain’, and TF-M SPM would maintain the local storage at run time. Finally, the partition entry point gets called and run, tooling helps on the decoupling of SPM and SPRTL implementation. The pseudo code of a tooling result:

```
struct partition_t sp1 {
    .name = "TFM_SP_SERVICE",
    .type = PSA_ROT,
    .priority = NORMAL,
    .id = 0x000000100,
    .entry_point = __sprtmain, /* Tell SPM entry is '__sprtmain' */
    .local_storage = { /* struct sprt_local_storage_t */
        .heap_sa = sp1_heap_buf,
        .heap_sz = sizeof(sp1_heap_buf),
        .thread_entry = sp1_entry, /* Actual Partition Entry */
        .heap_instance = NULL,
    },
}
```

Implementation

The SPRTL C Runtime sources are put under: '\$TFM_ROOT/secure_fw/partitions/lib/runtime/'

The output of this folder is a static library named as 'libtfm_sprt.a'. The code of 'libtfm_sprt.a' is put into a dedicated section so that a hardware protected region can be applied to contain it.

The RoT Service API are put under service interface folder. These APIs are marked with the same section attribute where 'libtfm_sprt.a' is put.

The Formatting API - 'printf' and variants

The 'printf' and its variants need special parameters passing mechanism. To implement these APIs, the toolchain provided builtin macro 'va_list', 'va_start' and 'va_end' cannot be avoided. This is because of some scenarios such as when 'stack canaries' are enabled, only the compiler knows the format of the 'canary' in order to extract the parameters correctly.

To provide a simple implementation, the following requirements are defined for 'printf':

- Format keyword 'xXduscp' needs to be supported.
- Take '%' as escape flag, '%%' shows a '%' in the formatted string.
- To save heap usage, 32 bytes buffer in the stack for collecting formatted string.
- Flush string outputting due to: a) buffer full b) function ends.

The interface for flushing can be a logging device.

Function needs implied inputs

Take 'malloc' as an example. There is only one parameter for 'malloc' in the prototype. Heap management code is put in the SPRTL for sharing with caller partitions. The heap instance belongs to each partition, which means this instance needs to be passed into the heap management code as a parameter. For allocation API in heap management, it needs two parameters - 'size' and 'instance', while for 'malloc' caller it needs a 'malloc' with one parameter 'size' only. As mentioned in the upper chapter, this instance can be retrieved from the Secure Partition Local Storage. The implementation can be:

```
void *malloc(size_t sz)
{
    struct p_sp_local_storage_t *m =
        (struct p_sp_local_storage_t *)tfm_sprt_local_storage;

    return tfm_sprt_alloc(m->heap_instance, sz);
}
```

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10.2.3 TF-M Inter-Process Communication

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Terminology

IPC - Inter-Process Communication

For more terminology please check *Reference* document.

Design Overview

Components for implementing IPC:

- SPM – for partition information and isolation actions
- Core – for exception handling
- Memory pool
- Message manager
- Thread
- Synchronization objects
- PSA API

Implementation Details

Listed modules are all internal modules except PSA API. Prototypes and definitions are not listed for internal modules in this document. For PSA API definitions, check them in PSA Firmware Framework specification in the reference chapter.

SPM and Core

SPM manages Secure Partition information. Enhancements need to be done in SPM data structure for Secure Partition for IPC due to:

- IPC model requires each Secure Partition has its own stack area.
- Multiple services are holding in same Secure Partition and each service has its own information like message queue, SID and priority.
- Changed information related manifest items need to be changed, too.

Modifications in Core:

- More SVC calls need to be added into list since PSA API are implemented as SVC calls in TF-M.
- New PendSV handler for thread scheduling.
- Arch-related context stacking and switching.

Memory Pool

Handles of connection and messages for Secure Partition needs to be allocated dynamically. A memory pool is provided in the system to handle dynamic allocation. Each memory pool item contains below information:

- A list iterator to chain all of memory pool items.
- An information member to record information like size and types.
- The memory item body for caller usage.

A memory area needs to be provided in SPM for the memory pool. It could be an array of memory areas defined in the linker script. Two chains are available to manage the items: free chain and used chain. And an LRU (Last recent used) mechanism is applied for fast seeking while item allocating and destroying.

Message Manager

Message Manager handles message creating, pushing, retrieving and destroy. A message contains below information:

- Message sender and destination
- Message status
- IO vectors for service
- 'psa_msg_t' for service

A checking needs to be performed in SPM before creating a message to detect if a message with the same sender and destination is ongoing. This avoids repeat messages are available in the queue.

Thread

Each Secure Partition has a thread as execution environment. Secure Partition is defined statically in TF-M manifest, which indicates that a number of threads are statically defined. Threads are chained in SPM and sorted with its priority, and there is an extra indicator point to first running thread with the highest priority. This helps fast seeking of running threads while the scheduler is switching threads.

Thread context contains below information:

- Priority
- Status
- Stack pointer
- Stack pointer limitation
- Entry
- Parameter
- Entry return value
- Context
- List iterator

Thread API provides below functions:

- Thread creating and destroying
- Thread status retrieving and changing
- Current thread retrieving
- Thread context switching

PendSV exception in TF-M core is the place thread context APIs been called. Before thread switching taking place, isolation status needs to be changed based on Secure Partition change and current isolation level – a thread is a member of partition which means thread switching caused a partition switching.

Synchronization API

A first synchronization object is an event. This could be applied into event waiting in the partition, and message response handling in IPC. The event object contains below members:

- Owner thread who is waiting for this event
- Event status (Ready or Not-Ready)
- List iterator for synchronization objects management

Event API Limitation: could be waited by one thread only.

PSA API

This chapter describes the PSA API in an implementation manner.

- API type: could be Client API and Service Partition API
- Block-able: Block-able API may block caller thread; Non-Block API does not block caller thread.
- Description: The functionality description and important comments.

```
uint32_t psa_framework_version(void);
uint32_t psa_version(uint32_t sid);
```

- Client API
- Non-Block API
- These 2 functions are finally handled in SPM and return the framework version or version to the caller.

```
psa_handle_t psa_connect(uint32_t sid, uint32_t version);
psa_status_t psa_call(psa_handle_t handle, int32_t type,
                     const psa_invec *in_vec, size_t in_len,
                     psa_outvec *out_vec, size_t out_len);
void psa_close(psa_handle_t handle);
```

- Client API
- Block-able API
- These 3 APIs are implemented in the same manner and just different parameters. SPM converts each call into a corresponding message with a parameter in the message body and pushes the message into service queue to wait for the response. Scheduler switches to a specified thread (partition) and makes Secure Partition to have chance retrieving and process message. After a message response is returned to the caller, the waiting caller gets to go and get the result.

```
psa_signal_t psa_wait(psa_signal_t signal_mask, uint32_t timeout);
```

- Secure Partition API
- Block-able API
- This API blocks caller partition if there is no expected event for it. This function is implemented based on event API.

```
void psa_set_rhandle(psa_handle_t msg_handle, void *rhandle);
psa_status_t psa_get(psa_signal_t signal, psa_msg_t *msg);
size_t psa_read(psa_handle_t msg_handle, uint32_t invec_idx,
               void *buffer, size_t num_bytes);
size_t psa_skip(psa_handle_t msg_handle, uint32_t invec_idx,
               size_t num_bytes);
void psa_write(psa_handle_t msg_handle, uint32_t outvec_idx,
               const void *buffer, size_t num_bytes);
void psa_reply(psa_handle_t msg_handle, psa_status_t status);
void psa_clear(void);
void psa_eoi(psa_signal_t irq_signal);
```

- Secure Partition API

- Non-Block
- These APIs do not take the initiative to change caller status. They process data and return the processed data back to the caller.

```
void psa_notify(int32_t partition_id);
```

- Secure Partition API
- Non-Block
- This API sets DOORBELL bit in destination partition's event. This API does not take the initiative to change caller status.

```
void psa_panic(void);
```

- Secure Partition API
- Block-able API
- This function will terminate execution within the calling Secure Partition and will not return.

Reference

[PSA Firmware Framework specification URL](#)

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10.2.4 Stateless Root of Trust Services Reference

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Introduction

This document describes the implementation for the FF-M v1.1 feature - 'Stateless RoT Service', and the related references when developing RoT services.

It is recommended to refer to the FF-M v1.0 specification¹ and FF-M v1.1 extension² for background and rationale details.

Implementation Details

This chapter describes the implementation-defined items, including stateless handle value definition, tooling update, and programming API changes.

¹ FF-M v1.0 Specification

² FF-M v1.1 Extension

Stateless Handle Value Definition

The index, stateless indicator, and service version information are encoded into a handle by the manifest tooling, and then generated to header file `sid.h`.

Table 42:: Bit Fields of Stateless Handle

Bits	Field Description
bit 31	reserved
bit 30	stateless handle indicator bit, always 1
bit 29 - bit 16	reserved
bit 15 - bit 8	service version requested by client - for client version check
bit 7 - bit 5	reserved
bit 4 - bit 0	the handle index, [0, 31]

Since connection-based services and stateless services share the same PSA API `psa_call()`, an indicator bit is set in the handle indicate the type of the handle. If it is set, the handle is stateless, and definition is as described in the table above. Maximum connection-based handle is 0x3FFFFFFF, thus the indicator bit is always 0.

The index is used for organizing stateless services in manifest tool and locating a stateless service in SPM logic. A range of index [0, 31] is the initial implementation. Future expansion is possible.

Tooling Support

TF-M provides a tool (`tools/tfm_parse_manifest_list.py`) to generate source header files required by partition and services. For example, the generated `sid.h` contains service ID and version. The tooling is extended to generate stateless handle from partition manifests automatically.

The `stateless_handle` attribute in manifest is only supported by partitions with firmware framework version 1.1.

- If `stateless_handle` in manifest is set to an integer, the index is `stateless_handle - 1`.
- If it is `auto` or not set, the first empty index in range [0, 31] is assigned.
- Other settings - tooling reports an error.

Finally, the tooling encodes the handle according to definitions in *Stateless Handle Value Definition* section, and writes them to `sid.h` header file.

Changes in Programming API

This chapter describes the changes in programming API for stateless services. The following APIs' behaviours and message data structure members are updated to support the stateless service.

`psa_connect()`

According to FF-M v1.1, client calling `psa_connect()` with the SID of a stateless RoT Service is a `PROGRAMMER_ERROR`.

`psa_close()`

According to FF-M v1.1, client passing a stateless handle to call this API is a `PROGRAMMER_ERROR`.

psa_call()

```
psa_status_t psa_call(psa_handle_t handle, int32_t type,
                     const psa_invec *in_vec, size_t in_len,
                     psa_outvec *out_vec, size_t out_len)
```

API parameters and behaviour change:

1. The `handle` parameter must be a stateless handle defined in `psa_manifest/sid.h` when requesting a stateless service.
2. This API validates stateless handle, decodes index and service version information from it. SPM uses the index to know which stateless service is requested.
3. This API performs some operations as `psa_connect()` does, such as the authorization check, service and client version check, and handle space allocation.

Service behaviour change during a “psa_call”:

Service does not accept connection and disconnection messages. After a “psa_call” request is serviced, it calls `psa_reply()`, frees the connection handle to handle pool.

psa_set_rhandle()

According to FF-M v1.1, stateless service calling this API on a message is a `PROGRAMMER_ERROR` and it will never return.

psa_msg_t type

The `rhandle` member of a `psa_msg_t` type received by a stateless RoT Service is always `NULL`.

Application Recommendation

There are particular services that do not need sessions. The access to the service is a one-shot connection. These services provide standalone operations that do not require any non-volatile state of resources to be maintained by the RoT service itself or do not expose any kind of context or session to the caller. Such services are recommended to be implemented as stateless, to provide quick access and to avoid extra overheads.

In this case, `rhandle` access would be prohibited as it is used for saving state or non-volatile resources while stateless services do not need them.

Update Feasibility of Existing Services

TF-M native services are used widely. They only need standalone operations and do not need to keep state between sessions. For example, the service in Crypto partition does not do anything during `psa_connect()` or `psa_close()` process. Same for services in other partitions, thus all of them can be implemented as stateless.

Analysis for them:

Table 43:: TF-M Partition Services Update Possibility

Partition	Number of Services	Can be Stateless
ITS	4	All
PS	5	All
Crypto	1	All
FWU	6	All
Platform	4	All
Initial Attestation	2	All

Other services are not analyzed here.

Grouping Services

Stateless service table is stored statically, and TF-M supports 32 stateless services currently.

Similar stateless services in a partition could be grouped, and assign one SID for the group. The `type` parameter in `psa_call()` could be extended to identify the service in group. In this case, it is recommended to use consecutive values for `type`.

It is recommended that each Secure Partition declares one stateless service and uses the `type` parameter to distinguish different stateless services. Therefore, more stateless services can be supported.

Migrating to Stateless Services

Please refer to Chapter 4 “Stateless Root of Trust services”, Appendix B.3.2 “Using a stateless RoT Service”, and Appendix D “Implementing session-less RoT Services” in FF-M v1.1 document for details on which kind of service can be stateless and how to implement a stateless service.

Reference

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10.2.5 Uniform Secure Service Signature

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Declaring secure service interface

The following alternative secure service signature is proposed as an amendment to existing implementation.

Individual signatures - current method

A `<service_name>_veneers.c` file is created in the `secure_fw/ns_callable` directory, that specifies the signature for each veneer function, and calls the secure function from the veneers. The respective `interface/include/<service_name>_veneers.h` file with the veneer declarations have to be created and maintained manually. Note that at present TF-M framework limits the range of valid return values a secure service can provide, reserving a range for framework error codes.

Uniform signatures - proposal

The proposal is to use a uniform signature for all the secure functions of the secure service. There are multiple advantages of this method:

- TF-M Core can do a sanity check on the access rights of the veneer parameters, and there is no need for the secure services to make these checks individually. Please note that in the present implementation sanity check is only fully supported for level 1 isolation.

- The veneer declarations and implementations for the secure functions can be generated automatically from a template (using the secure function list in the secure service's manifest)

The signature for such secure services would look like this:

```
psa_status_t secure_function_name(struct psa_invec *in_vec, size_t in_len,
                                struct psa_outvec *out_vec, size_t out_len);
```

where

Return value:

psa_status_t is a status code whose values are described in PSA Firmware Framework (as in version 1.0-beta-0 chapter 4.3.3).

Note:

The return value limitations imposed by TF-M framework for proprietary secure service veneers would not apply to secure services using the uniform signature. This is analogous to how PSA Firmware Framework handles values returned by `psa_reply()` function.

Arguments:

```
/**
 * A read-only input memory region provided to a RoT Service.
 */
typedef struct psa_invec {
    const void *base;    /*!< the start address of the memory buffer */
    size_t len;          /*!< the size in bytes */
} psa_invec;

/**
 * A writable output memory region provided to a RoT Service.
 */
typedef struct psa_outvec {
    void *base;          /*!< the start address of the memory buffer */
    size_t len;          /*!< the size in bytes */
} psa_outvec;

/**
 * in_len: the number of input parameters, i.e. psa_invecs
 * out_len: the number of output parameters, i.e. psa_outvecs
 */
```

The number of vectors that can be passed to a secure service is constrained:

```
in_len + out_len <= PSA_MAX_IOVEC
```

The veneer function declarations and implementations are generated in the `interface/include/tfm_veneers.h` and `secure_fw\ns_callable\tfm_veneers.c` files respectively. The veneer functions are created with the name `tfm_<secure_function_name>_veneer`

Services that implement the uniform signature do not need to manually fill the template veneer function to call `TFM_CORE_SFN_REQUEST` macro.

Compatibility

Note that the proposal is for the two types of services (those with proprietary signatures and those with uniform signatures) to co-exist, with the intention of eventually phasing out proprietary signatures in favour of the more robust, uniform signature.

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10.2.6 TF-M Crypto Service design

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Abstract

This document describes the design of the TF-M Cryptographic Secure Service (in short, TF-M Crypto service).

Introduction

The TF-M Crypto service provides an implementation of the PSA Certified Crypto APIs in a PSA RoT secure partition in TF-M. It is based on the TF-PSA-Crypto project, which provides a reference implementation of the PSA Crypto APIs as a C software library. For more details on the PSA Crypto APIs refer to¹, while for the TF-PSA-Crypto reference software refer to² and³

¹ PSA Certified Crypto API specifications: <https://arm-software.github.io/psa-api/crypto/>

² Using PSA - Getting started in Mbed TLS: https://mbed-tls.readthedocs.io/en/latest/getting_started/psa/

³ TF-PSA-Crypto repository which holds the reference implementation as a C software library: <https://github.com/Mbed-TLS/TF-PSA-Crypto>

The service can be requested by other services running in the SPE, or by applications running in the NSPE, and its aim is to provide cryptographic primitives in a secure and efficient way, either via software or by routing the calls to any underlying crypto hardware accelerator or secure element that the platform might provide.

Components

The TF-M Crypto service is implemented by a number of different firmware components residing in the Crypto partition, which are listed below:

Table 44:: Components table

Component name	Description	Location
Client API interface	This module exports the PSA Crypto API to be callable from the users, which are called also <i>clients</i> . They could be either other secure partitions or Non Secure world based callers.	interface/src/tfm_crypto_api.c
TF-PSA-Crypto	The TF-PSA-Crypto <code>libtfpsacrypto.a</code> library is used in the service as a cryptographic <i>backend</i> library which provides the APIs to implement a PSA Crypto <i>core</i> , SW based crypto primitives and wrappers for HW crypto accelerators and Secure Elements. It exposes those through the PSA Crypto APIs	Needed as dependency specified by the <code>TF_PSA_CRYPTOPATH</code> CMake configuration option
Init module	This module handles the initialisation of the service objects during TF-M boot and provides the infrastructure to service requests when TF-M is built for IPC or SFN model. The dispatching mechanism of IPC requests is based on a look up table of function pointers. This design allows for better scalability and support of a higher number of Secure functions with minimal overhead and duplication of code. This module is in charge of providing an ID of the caller of each API in the backend, allowing to enforce key ownership policies.	secure_fw/partitions/crypto/crypto_init.c
Alloc module	This module handles the allocation of contexts for multipart operations in the Secure world. This is required because the caller view of contexts, i.e. <i>clients</i> , does not contain any sensible information but just a number handle which is then used by the service itself to match the context to the actual context information which will be stored securely in the TF-M crypto partition private memory. This is enabled by setting <code>MBEDTLS_PSA_CRYPTOPARTITION</code> option on the caller side. Note that setting this option on the client side is a hard requirement in order for the clients to work correctly	secure_fw/partitions/crypto/crypto_alloc.c
Service modules	These modules (AEAD, Asymmetric, Cipher, Hash, Key Derivation, Key Management, MAC, Random Number Generation) represent a thin layer which is in charge of servicing the calls from the clients. They provide parameter sanitization and context retrieval for multipart operations, and dispatching to the corresponding backend library function exposed by the underlying library.	secure_fw/partitions/crypto/crypto_aead.c secure_fw/partitions/crypto/crypto_asymmetric.c secure_fw/partitions/crypto/crypto_cipher.c secure_fw/partitions/crypto/crypto_hash.c secure_fw/partitions/crypto/crypto_key_derivation.c secure_fw/partitions/crypto/crypto_key_management.c secure_fw/partitions/crypto/crypto_mac.c secure_fw/partitions/crypto/crypto_rng.c
Backend library	This module contains several APIs to abstract the interface towards the backend library, which must provide the PSA Crypto core layer, key management, SW based crypto and possibly interfaces for HW crypto accelerators and Secure Elements	secure_fw/partitions/crypto/crypto_library.*

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strac-
tion

Man- The manifest file is a description of the service components.

secure_fw/partitions/
crypto/tfm_crypto.yaml

The interaction between the different components is described by the block diagram in Figure ??:

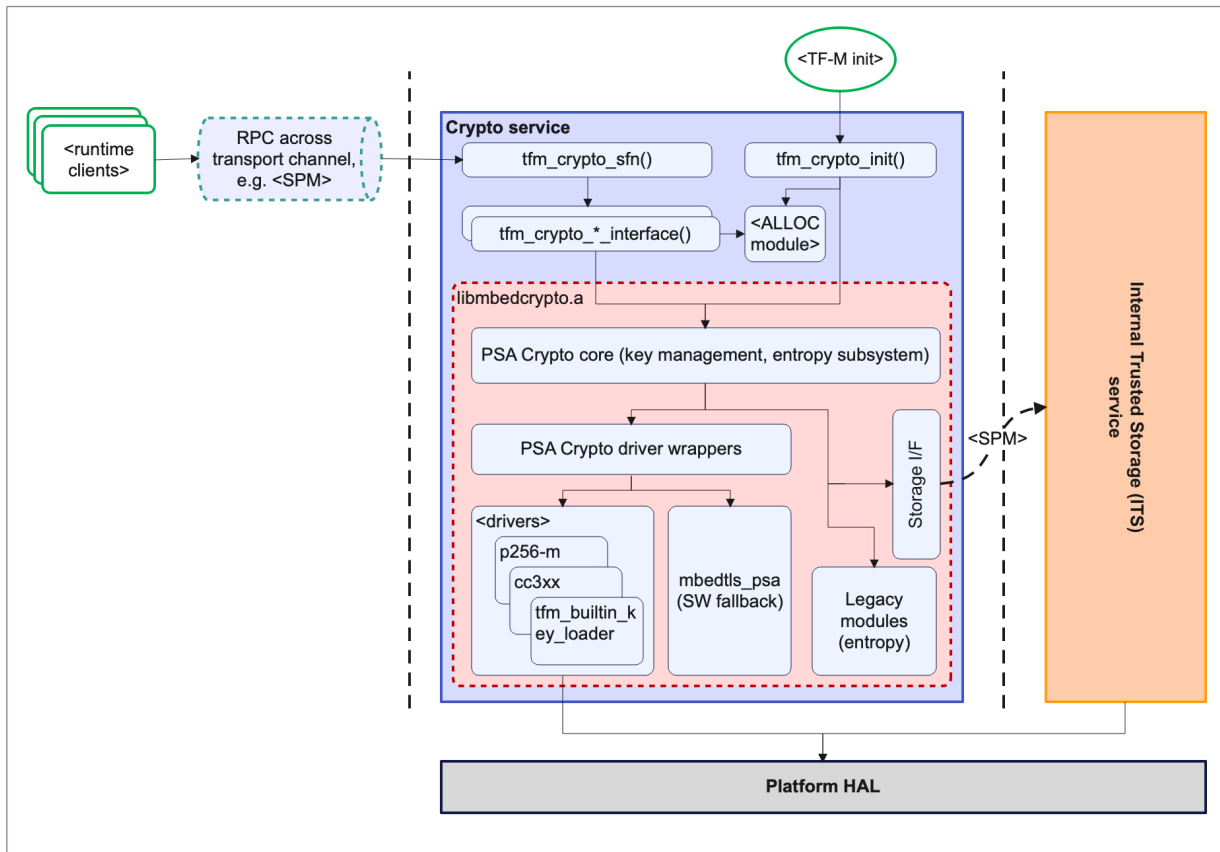


Figure 13:: Block diagram of the firmware architecture of the TF-M Crypto service. Dotted lines between services represent isolation boundaries once runtime firmware is initialized, i.e. TF-M init phase has completed. The diagram is simplified and shows only the major functional blocks, for a more detailed `libtfpsacrypto.a` architecture please refer to⁷.

Relationship between TF-PSA-Crypto and the TF-M Crypto service

TF-M Crypto as a particular configuration of TF-PSA-Crypto

Up until *TF-Mv2.0*, the TF-M Crypto service used to provide its own separate implementation of the PSA Certified Crypto APIs, i.e. it provided its own version of the implementation defined aspects of the specifications. Starting from *TF-Mv2.1*, the TF-M Crypto service fully aligns to the implementation defined by the Mbed TLS project (aligns to TF-PSA-Crypto starting from *TF-Mv2.3*), i.e. its implementation defined aspects are the same as the ones defined by TF-PSA-Crypto/Mbed TLS.

As a consequence, starting from *TF-Mv2.1* the PSA Crypto headers available in TF-M are a copy of those distributed by the TF-PSA-Crypto/Mbed TLS project. TF-M just uses them and won't accept any contribution to them, as those need to be discussed in the scope of the TF-PSA-Crypto/Mbed TLS project.

TF-M then represents just a configuration of the TF-PSA-Crypto reference implementation where the TF-M Crypto APIs are provided as a remote call across a transport channel, which might be represented by a TrustZone boundary (in Armv8.x-M systems), by a mailbox channel in heterogeneous systems, e.g. Cortex-A + Cortex-M systems, by an SPM mediated interface, e.g. partition to partition calls or, in general, through a mechanism which provides process separation between the client and the service sides of the API. In this context, the client must always define the TF-PSA-Crypto config option `MBEDTLS_PSA_CRYPTO_CLIENT`, while the service must always have `MBEDTLS_PSA_CRYPTO_SPM`,

mainly to avoid symbol clashing at link time between the library interface and the `tfm_crypto_api.c` interface. When there is a component on the service side which is able to identify the client through an ID, it is recommended to also define `MBEDTLS_PSA_CRYPTO_KEY_ID_ENCODES_OWNER` option in order to provide separation in the key space.

Usage of TF-PSA-Crypto configuration header

Crypto uses a configuration header, specified through the setting of the `TF_PSA_CRYPTO_CONFIG_FILE`, i.e. PSA Crypto configuration. In order to be able to perform header inclusion for `psa/crypto.h`, the configuration file must be visible to the compilation unit through the include hierarchy. If the macro is not defined, the fall back strategy is to include the default config file available in the TF-PSA-Crypto repo, i.e. `include/psa/crypto_config.h`, which contains a set of default values for the macros.

Usage of the default header config when using the TF-M Crypto service is highly discouraged, mainly because both on the client side and on the service side a set of options must always be defined (or undefined), as described in the previous section. TF-M provides example `_profiles_` which show the options and how they should be used on both client and service side of the integration. The symbol to enable the Crypto config `TF_PSA_CRYPTO_CONFIG_FILE` instead must be available to the unit being compiled which is including `psa/crypto.h`, i.e. passed by the build system config stage.

Hardware acceleration

The TF-M Crypto partition must handle all HW related crypto tasks, if the platform is capable of offering hardware acceleration or if a complete Secure Element is present. The main difference between the two is that a hardware accelerator does not store keys but just accelerates operations, while a Secure Element is capable of storing keys and the PSA Crypto core running on the host must interface with it to store, retrieve or use them for crypto tasks, etc.

There are currently two methods to interface an accelerator into the Crypto service, and both rely on the Crypto partition fully owning the Crypto hardware, i.e. the memory mapped IO space must be bound the Crypto partition only. Both methods are implemented through the capability of the `_backend_` library to either:

1. Provide a link time mechanism to replace pure SW implementations for algorithms with HW assisted implementations. In this case, the TF-M platform provides some additional HW abstraction through the usage of `crypto_hw_accelerator_*` APIs. This is dubbed the *_ALT* approach and will be soon to be deprecated potentially starting from the release of Mbed TLS 4.0
2. Provide a cleanly defined interface specification⁴ to describe the APIs that a driver must expose to the PSA Crypto core in order for the core to be able to offload operations to hardware. This is the preferred method for interfacing with HW.

Both solutions are currently handled at build time (either compilation or linking) by TF-PSA-Crypto. For details on how to integrate a driver please refer directly to the documentation referenced above and to the TF-PSA-Crypto repo.

Builtin keys

A particular driver using the interface described in² is the TF-M Builtin Key Loader driver⁵. The goal of the driver is to make TF-PSA-Crypto aware of *transparent builtin keys*, i.e. keys which can be read from the core (i.e. not fully opaque keys), but that are normally bound to the platform and provisioned in it, for which it would be more appropriate to treat them as standard *transparent keys*. The concept of *transparent builtin keys* is not defined in the spec so it is specifically a non standard extension added by TF-M to the TF-PSA-Crypto implementation, which might be changed between releases until a standard solution is adopted. TF-M patches TF-PSA-Crypto on the fly to enable such behaviour using patches available in `lib/ext/tf-psa-crypto`. Implementations might disable the `tfm_builtin_key_loader` and then must provide their own alternative storage location for all of the TF-M required builtin keys, e.g. by having them stored in a Secure Element with a corresponding opaque driver.

⁴ PSA Unified Driver Interface for Cryptoprocessors: <https://github.com/Mbed-TLS/TF-PSA-Crypto/blob/development/docs/proposed/psa-driver-interface.md>

⁵ TF-M Builtin Key Loader driver, normally described as *tfm_builtin_key_loader*

Service API description

The `Alloc` and `Init` modules implement public APIs which are specific to the TF-M Crypto service, and are available only internally to other components of the TF-M Crypto partition. For a detailed description of the prototypes please refer to the `tfm_crypto_api.h` header.

Table 45:: Init and Alloc modules APIs

Function	Module	Caller	Scope
<code>tfm_crypto_init</code>	Init	SPM	Called during TF-M boot for initialisation. It does modules initialisation (it initializes the <code>Alloc</code> module) and initializes the <i>backend</i> library. Being the partition enabled for the SFN model, it does not implement any IPC specific message handler, instead it relies on the SPM being able to schedule SFN partitions using the SFN dispatcher with little overhead
<code>tfm_crypto_ipc_dispatch</code>	Init	SPM	Function to handle an SFN request or to interface with the message handler when running in IPC model
<code>tfm_crypto_alloc_init</code>	Alloc	Init	Called by <code>tfm_crypto_init()</code> , it initialises the internal memory storage in the TF-M Crypto partition that the service uses to store multipart operation contexts as requested by clients.
<code>tfm_crypto_alloc</code>	Alloc	Service modules	It allocates a new operation context for a multipart operation. It returns an handle to the allocated context in secure memory.
<code>tfm_crypto_alloc_get</code>	Alloc	Service modules	It retrieves a previously allocated operation context of a multipart operation, based on the handle given as input.
<code>tfm_crypto_alloc_release</code>	Alloc	Service modules	It releases a previously allocated operation context of a multipart operation, based on the handle given as input.
<code>tfm_crypto_dispatcher</code>	*	Init	Interface functions called by the dispatcher to service PSA Crypto APIs requests

Configuration parameters

The TF-M Crypto service exposes some configuration parameters to tailor the service configuration in terms of supported functionalities and hence FLASH/RAM size to meet the requirements of different platforms and use cases. These parameters can be provided via CMake parameters during the CMake configuration step and as a configuration header to allow the configuration of the TF-PSA-Crypto library. When using Kconfig they are also exported in the Kconfig menus.

Table 46:: Configuration parameters table

Parameter	Type	Description	Default
<i>CRYPTO_E</i>	CMake build configuration parameter	Buffer used by TF-PSA-Crypto for its own allocations at runtime. This is a buffer allocated in static memory.	8096 (bytes)
<i>CRYPTO_C</i>	CMake build configuration parameter	This parameter defines the maximum number of possible concurrent operation contexts (cipher, MAC, hash and key deriv) for multipart operations, that can be allocated simultaneously at any time.	8
<i>CRYPTO_I</i>	CMake build configuration parameter	This parameter applies only to IPC model builds. In IPC model, during a Service call, input and outputs are allocated temporarily in an internal scratch buffer whose size is determined by this parameter.	5120 (bytes)
<i>CRYPTO_S</i>	CMake build configuration parameter	Defines the stack size assigned to the crypto partition in higher level of isolation configurations (L1 isolation has a common stack shared by all partitions)	6912 (bytes)
<i>CRYPTO_N</i>	CMake build configuration parameter	Uses the TF-PSA-Crypto NV seed feature to provide entropy in case there is no HW acceleration providing HW entropy	Defined for platforms which don't have CRYPTO_HW_ACCELERATOR
<i>CRYPTO_I</i>	CMake build configuration parameter	Defines the size of scratch buffers to handle input/outputs if the Memory Mapped IOVEC feature is not enabled	5120 (bytes)
<i>CRYPTO_S</i>	CMake build configuration parameter	When enabled, only the multipart, i.e. non-integrated APIs will be available in the service	Not defined (Profile default)
<i>CRYPTO_*</i>	CMake build configuration parameters	When enabled, the corresponding shim layer module and relative APIs are available in the service	Defined (Profile default)
<i>TF_PSA_C</i>	Configuration header	The TF-PSA-Crypto library can be configured to support different algorithms through the usage of a configuration header file at build time. This specifies which cryptographic mechanisms are available through the PSA API and also allows for tailoring FLASH/RAM requirements for different platforms and use cases.	lib/ext/ tf-psa-crypto/ tfpsacrypto_config/ crypto_config_default.h (Profile default)

References

SPDX-License-Identifier: BSD-3-Clause

SPDX-FileCopyrightText: Copyright The TrustedFirmware-M Contributors

10.2.7 Symmetric key algorithm based Initial Attestation

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Introduction

This document proposes a design of symmetric key algorithm based Initial Attestation in TF-M.

Symmetric key algorithm based Initial Attestation (*symmetric Initial Attestation* for short) signs and verifies Initial Attestation Token (IAT) with a symmetric cryptography signature scheme, such as HMAC. It can reduce TF-M binary size and memory footprint on ultra-constrained devices without integrating asymmetric ciphers.

This proposal follows PSA Attestation API document¹.

Note

As pointed out by PSA Attestation API², the use cases of Initial Attestation based on symmetric key algorithms can be limited due to the associated infrastructure costs for key management and operational complexities. It may also restrict the ability to interoperate with scenarios that involve third parties.

Design overview

The symmetric Initial Attestation follows the existing IAT generation sequence for Initial Attestation based on asymmetric key algorithm (*asymmetric Initial Attestation* for short).

As Profile Small design² requests, a configuration flag `SYMMETRIC_INITIAL_ATTESTATION` selects symmetric initial attestation during build.

The top-level design is shown in *Overall design diagram* below.

Symmetric Initial Attestation adds its own implementations of some steps in IAT generation in Initial Attestation secure service. More details are covered in *IAT generation in Initial Attestation secure service*.

The interfaces and procedures of Initial Attestation secure service are not affected. Refer to Initial Attestation Service Integration Guide³ for details of the implementation of Initial Attestation secure service.

Symmetric Initial Attestation invokes `t_cose` library to build up `COSE_Mac0` structure. `COSE_Mac0` support was originally added to the `t_cose` library fork in TF-M, however since `t_cose 2.0` it is part of the upstream library⁴ which is already used by TF-M too.

¹ PSA Attestation API 1.0 (ARM IHI 0085)

² Trusted Firmware-M Profile Small Design

³ Initial Attestation Service Integration Guide

⁴ MAC compute in `t_cose` library

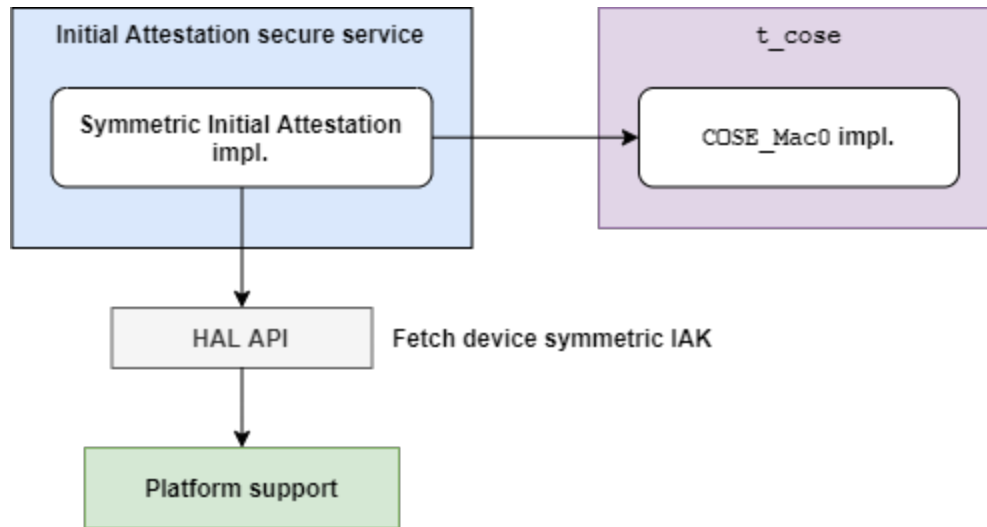


Figure 14:: Overall design diagram

Several HAL APIs are defined to fetch platform specific assets required by Symmetric Initial Attestation. For example, `tfm_plat_get_symmetric_iak()` fetches symmetric Initial Attestation Key (IAK). Those HAL APIs are summarized in *HAL APIs*.

Decoding and verification of symmetric Initial Attestation is also included in this proposal for regression test. The test suites and IAT decoding are discussed in *TF-M Test suite*.

QCBOR library and Crypto service are also invoked. But this proposal doesn't require any modification to either QCBOR or Crypto service. Therefore, descriptions of QCBOR and Crypto service are skipped in this document.

IAT generation in Initial Attestation secure service

The sequence of IAT generation of symmetric Initial Attestation is shown in *Symmetric IAT generation flow in Initial Attestation secure service* below. Note that the Register `symmetric IAK` stage is no longer required due to changes in the Crypto partition (`attest_symmetric_key.c` is now responsible only for calculating the instance ID).

In Initial Attestation secure service, symmetric Initial Attestation implements the following steps in `attest_create_token()`, which are different from those of asymmetric Initial Attestation.

- `attest_token_start()`,
- Instance ID claims,
- `attest_token_finish()`.

If `SYMMETRIC_INITIAL_ATTESTATION` is selected, symmetric Initial Attestation dedicated implementations of those steps are included in build. Otherwise, asymmetric Initial Attestation dedicated implementations are included instead. Symmetric Initial Attestation implementation resides a new file `attest_symmetric_key.c` to handle symmetric Instance ID related operations. Symmetric Initial Attestation dedicated `attest_token_start()` and `attest_token_finish()` are added in `attestation_token.c`. The details are covered in following sections.

Symmetric Instance ID

Symmetric Initial Attestation dedicated `attest_symmetric_key.c` implements the `attest_get_instance_id()` function. This function returns the Instance ID value, calculating it if it has not already been calculated. Refer to *Instance ID claim* for more details.

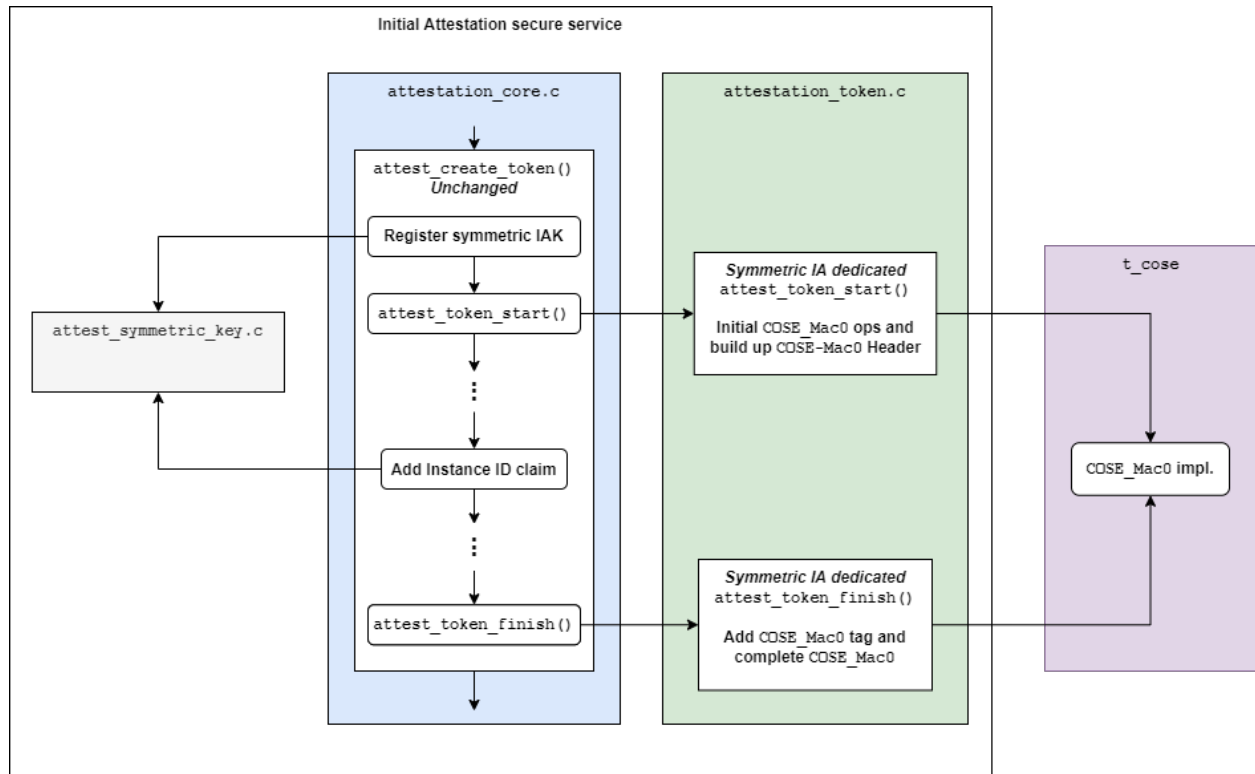


Figure 15:: Symmetric IAT generation flow in Initial Attestation secure service

Note

Only symmetric IAK for HMAC algorithm is allowed so far.

Instance ID calculation

In symmetric Initial Attestation, Instance ID is also calculated the first time it is requested. It can protect critical symmetric IAK from being frequently fetched, which increases the risk of asset disclosure.

The Instance ID value is the output of hashing symmetric IAK raw data *twice*, as requested in PSA Attestation API⁷. HMAC-SHA256 may be hard-coded as the hash algorithm of Instance ID calculation.

Note

According to RFC2104⁵, if a HMAC key is longer than the HMAC block size, the key will be first hashed. The hash output is used as the key in HMAC computation.

In current design, HMAC is used to calculate the authentication tag of COSE_Mac0. Assume that symmetric IAK is longer than HMAC block size (HMAC-SHA256 by default), the Instance ID is actually the HMAC key for COSE_Mac0 authentication tag generation, if Instance ID value is the output of hashing IAK only *once*. Therefore, attackers may request a valid IAT from device and fake malicious ones by using Instance ID to calculate valid authentication tags, to cheat others.

As a result, symmetric IAK raw data should be hashed *twice* to generate the Instance ID value.

⁵ HMAC: Keyed-Hashing for Message Authentication

The Instance ID calculation result is stored in a static buffer. Token generation process can call `attest_get_instance_id()` to fetch the data from that static buffer.

`attest_token_start()`

Symmetric Initial Attestation dedicated `attest_token_start()` initializes the COSE_Mac0 computation context and builds up the COSE_Mac0 Header.

The workflow inside `attest_token_start()` is shown in *Workflow in symmetric Initial Attestation attest_token_start()* below.

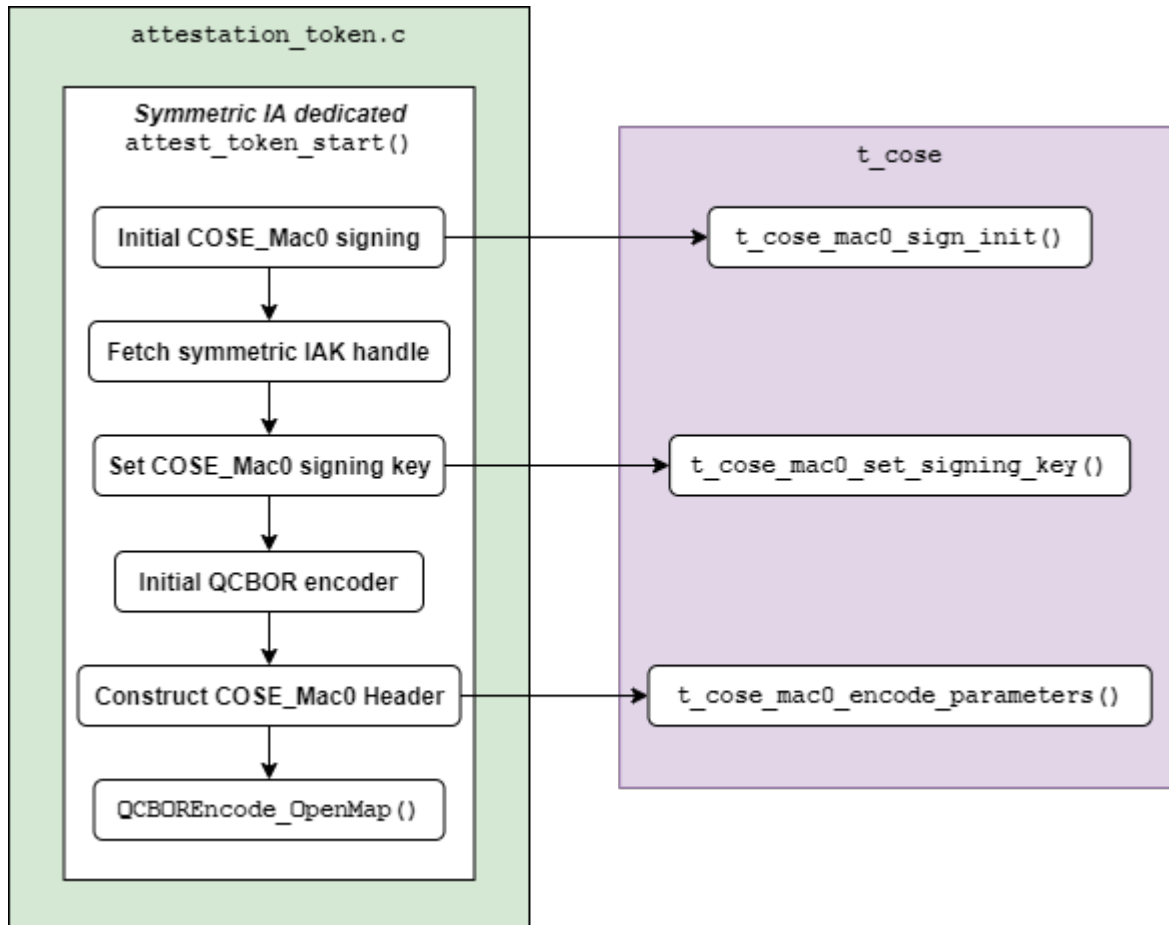


Figure 16:: Workflow in symmetric Initial Attestation `attest_token_start()`

Descriptions of each step are listed below:

1. `t_cose_mac0_sign_init()` is invoked to initialize COSE_Mac0 signing context in `t_cose`.
2. The symmetric IAK handle is set into COSE_Mac0 signing context via `t_cose_mac0_set_signing_key()`.
3. Initialize QCBOR encoder.
4. The header parameters are encoded into COSE_Mac0 structure in `t_cose_mac0_encode_parameters()`.
5. `QCBOREncode_OpenMap()` prepares for encoding the COSE_Mac0 payload, which is filled with IAT claims.

For detailed description and documentation of the COSE_Mac0 functionalities please refer to the `t_cose` repository[?].

Instance ID claim

Symmetric Initial Attestation also implements Instance ID claims in `attest_add_instance_id_claim()`. The Instance ID value is fetched via `attest_get_instance_id()`. The value has already been calculated during symmetric IAK registration. See *Instance ID calculation* for details.

The other steps are the same as those in asymmetric Initial Attestation implementation. The UEID type byte is set to 0x01.

`attest_token_finish()`

Symmetric Initial Attestation dedicated `attest_token_finish()` calls `t_cose_mac0_encode_tag()` to calculate and encode the authentication tag of `COSE_Mac0` structure. The whole COSE and CBOR encoding are completed in `attest_token_finish()`. The simplified flow in `attest_token_finish()` is shown in *Workflow in symmetric Initial Attestation `attest_token_finish()`* below.

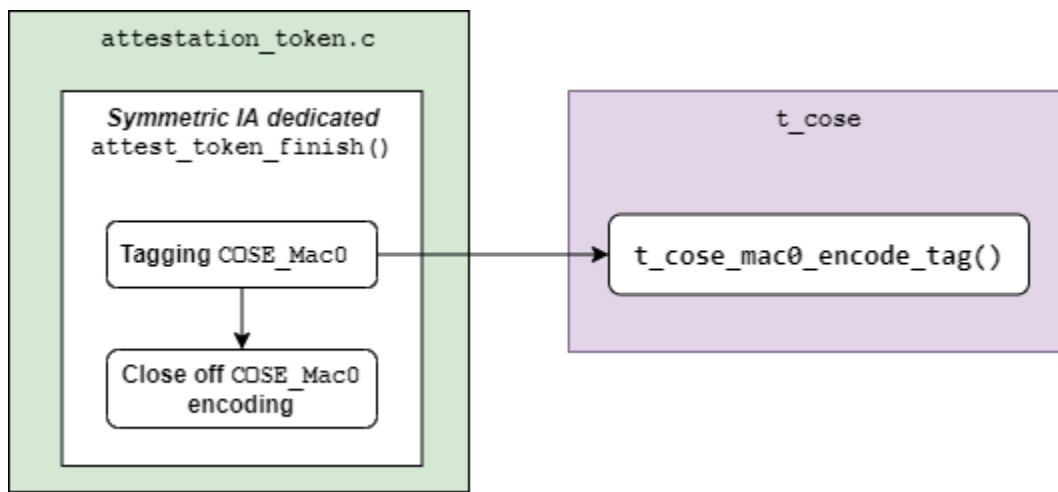


Figure 17:: Workflow in symmetric Initial Attestation `attest_token_finish()`

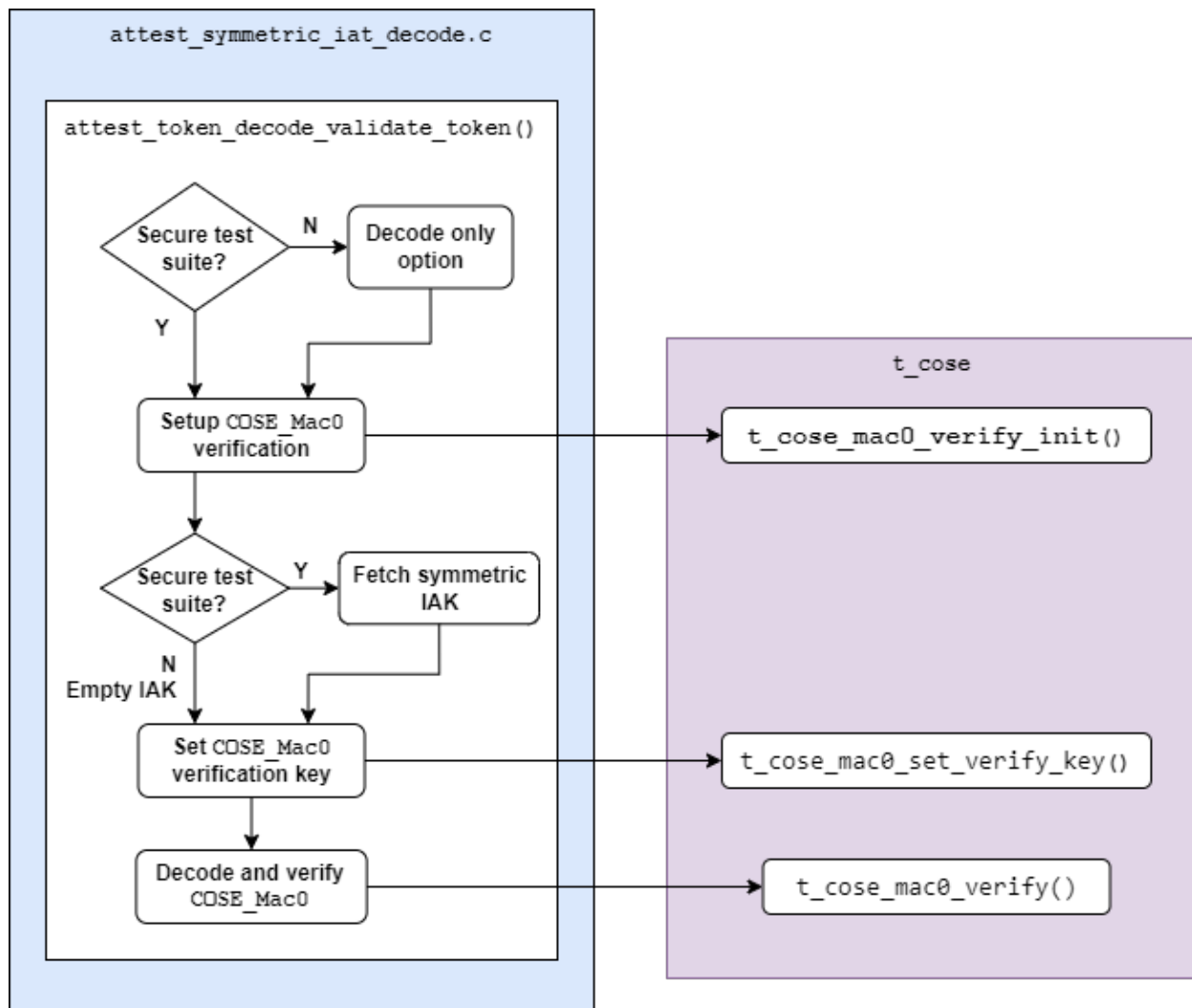
TF-M Test suite

Symmetric Initial Attestation adds dedicated non-secure and secure test suites. The test suites also follow asymmetric Initial Attestation test suites implementation but optimize the memory footprint. Symmetric Initial Attestation non-secure and secure test suites request Initial Attestation secure service to generate IATs. After IATs are generated successfully, test suites decode IATs and parse the claims. Secure test suite also verifies the authentication tag in `COSE_Mac0` structure.

Symmetric Initial Attestation implements its dedicated `attest_token_decode_validate_token()` in `attest_symmetric_iat_decoded.c` to perform IAT decoding required by test suites. If `SYMMETRIC_INITIAL_ATTESTATION` is selected, `attest_symmetric_iat_decoded.c` is included in build. Otherwise, asymmetric Initial Attestation dedicated implementations are included instead.

The workflow of symmetric Initial Attestation dedicated `attest_token_decode_validate_token()` is shown below.

If the decoding is required from secure test suite, `attest_token_decode_validate_token()` will fetch symmetric IAK to verify the authentication tag in `COSE_Mac0` structure. If the decoding is required from non-secure test suite, `attest_token_decode_validate_token()` will only decode `COSE_Mac0` by setting `T_COSE_OPT_DECODE_ONLY` option flag. Non-secure must not access the symmetric IAK.

Figure 18:: Workflow in symmetric Initial Attestation `attest_token_decode_validate_token()`

HAL APIs

HAL APIs are summarized below.

Fetch device symmetric IAK

`tfm_plat_get_symmetric_iak()` fetches device symmetric IAK.

```
enum tfm_plat_err_t tfm_plat_get_symmetric_iak(uint8_t *key_buf,
                                              size_t buf_len,
                                              size_t *key_len,
                                              psa_algorithm_t *key_alg);
```

Parameters:

<code>key_buf</code>	Buffer to store the symmetric IAK.
<code>buf_len</code>	The length of <code>key_buf</code> .
<code>key_len</code>	The length of the symmetric IAK.
<code>key_alg</code>	The key algorithm. Only HMAC SHA-256 is supported so far.

It returns error code specified in enum `tfm_plat_err_t`.

Get symmetric IAK key identifier

`attest_plat_get_symmetric_iak_id()` gets the key identifier of the symmetric IAK as the `kid` parameter in COSE Header.

Optional if device doesn't install a key identifier for symmetric IAK.

```
enum tfm_plat_err_t attest_plat_get_symmetric_iak_id(void *kid_buf,
                                                    size_t buf_len,
                                                    size_t *kid_len);
```

Parameters:

<code>kid_buf</code>	Buffer to store the IAK identifier.
<code>buf_len</code>	The length of <code>kid_buf</code> .
<code>kid_len</code>	The length of the IAK identifier.

It returns error code specified in enum `tfm_plat_err_t`.

Reference

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10.2.8 Internal Trusted Storage (ITS) Service

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Add support for block-aligned flash in Internal Trusted Storage

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Abstract

The proposal is describing a mechanism to enable the use of larger flash devices, imposing a requirement for word-aligned full-block program operations, in Trusted Firmware-M.

Requirements

- Allow page-aligned writes for up to 512 Bytes per page.
- Guarantee data integrity and power-failure reliability.
- Do not alter existing supported platform behaviour.

Current implementation

In the current ITS filesystem design, each filesystem create or write operation requires two flash blocks to be updated: first the data block and then the metadata block. Buffering is avoided as much as possible to reduce RAM requirements.

However, if the ITS_FLASH_PROGRAM_UNIT is 512 Bytes then the data will have to be stored in a temporary memory location in order to be able to write that much data in one-shot.

Proposed implementation overview

1. A new block-sized static buffer should be added to `its_flash.c` when `ITS_FLASH_PROGRAM_UNIT` is larger than currently supported.
2. Methods calling the flash API such as `its_flash_write()` or `its_flash_block_to_block_move()` will populate the buffer instead of directly programming the flash.
3. A new method `its_flash_flush()`, should be provided in order to flush the block buffer to the device.
4. `its_flash_flush()` should be called twice: Once after a data block update and once more after the metadata block update is completed.
5. The proposed design should require that the data block update is always completed before the metadata block update starts
6. Writes to the block buffer should be atomic, and guarded against corruption by data from different blocks.

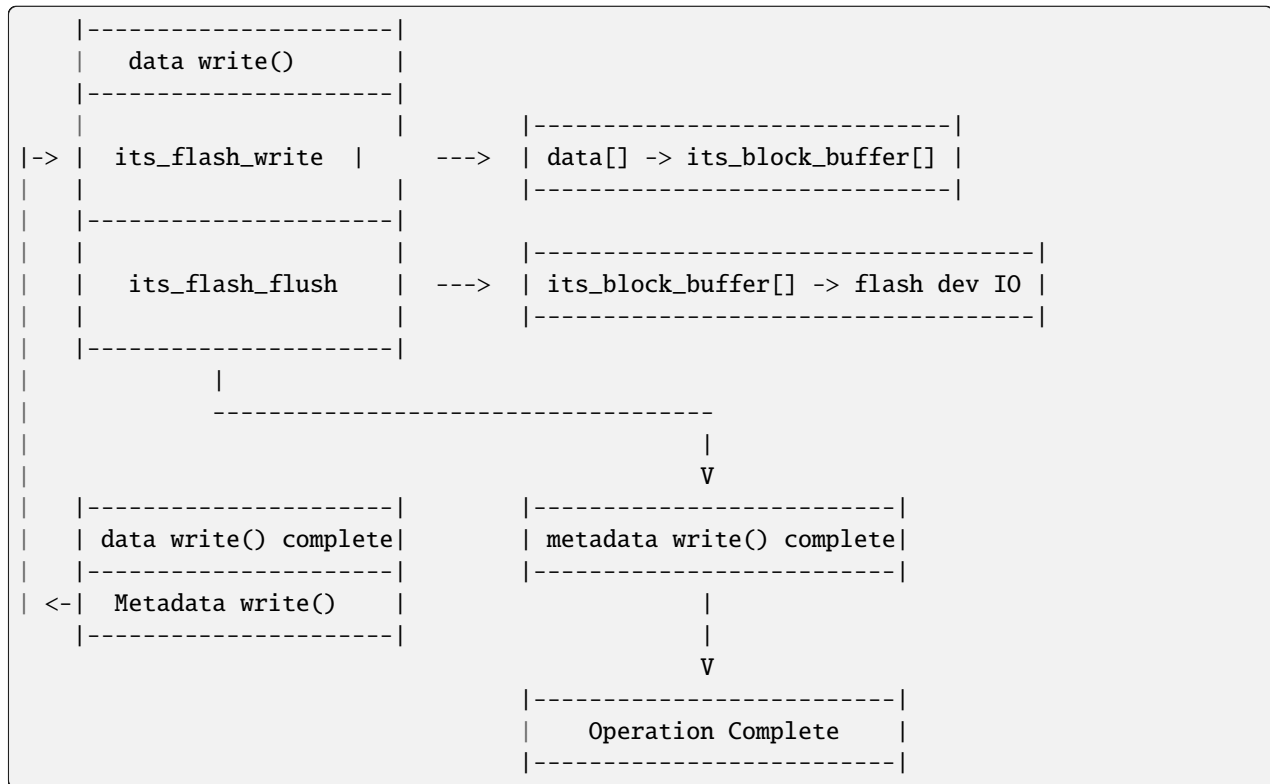
Considerations

- The proposed implementation will increase the RAM usage of ITS by the size of a block, only for platforms which require block-aligned writes.
- Currently power-failure is detected by software by incrementing an 8-bit metadata header field (`swap_count`), as the last written byte. When the proposed block-buffer is used, the block is programmed in one-shot and the order the bytes are written on the physical device, is hardware dependent.
- A set of guarantees are required by the supported flash ECC devices. The device's flash APIs should provide a mechanism to capture and raise incomplete program operations, as well as write bytes in a sequential order.

For example, if a board powers down through a 512 page program operation, the next read operation should return an error rather than read invalid data.

Functional flow diagram

The logic of the proposal is described in the following diagram



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PSA Internal Trusted Storage

PSA Internal Trusted Storage (ITS) is a PSA RoT Service for storing the most security-critical device data (e.g. cryptographic keys) in internal storage, which is trusted to provide data confidentiality and authenticity. This contrasts with PSA Protected Storage, which is an Application RoT service that allows larger data sets to be stored securely in external flash, with the option for encryption, authentication and rollback protection to protect the data-at-rest.

Current TF-M Secure Storage

Currently, the TF-M Secure Storage service implements PSA Protected Storage version 1.0-beta2. There is not yet an implementation of PSA Internal Trusted Storage in TF-M.

New TF-M service

The proposal is to implement the *PSA Internal Trusted Storage API* with the *TF-M Internal Trusted Storage service*. It can be abbreviated to *TF-M ITS service* in general and to *its* in code. This name has the advantage of making clear the correspondence between the service and the API it implements.

If this name is adopted, then it may make sense to rename the *Secure Storage service* to the *Protected Storage service* in the future to match. Then “secure storage” could refer to the two services as a collective.

The TF-M ITS service will implement PSA ITS version 1.0. It will be provided by a separate partition to Protected Storage, for a couple of reasons:

- To permit isolation between the services.
 - ITS is a PSA RoT Service, while Protected Storage is an Application RoT Service.
- To avoid circular dependencies.
 - The PSA Firmware Framework does not permit circular dependencies between partitions, which would occur if Protected Storage and ITS were provided by the same partition. Protected Storage depends on Crypto, which in turn depends on ITS.

The existing SST filesystem will be reused to provide the backend of the service, with the flash layer modified to direct storage to internal flash, rather than external.

Compared to Protected Storage, encryption, authentication and rollback protection are not required, so the SST encrypted object layer and the crypto and NV counter interfaces are not required. The rollback protection feature of the object table is also not required.

Code structure

The code structure of the service will be as follows:

TF-M repo:

interface/

- include/psa/internal_trusted_storage.h - PSA ITS API
- src/tfm_its_api.c - PSA ITS API implementation for NSPE

secure_fw/ns_callable/tfm_veneers.c - ITS veneers (auto-generated from manifest)

secure_fw/partitions/internal_trusted_storage/

- tfm_internal_trusted_storage.yaml - Partition manifest
- tfm_its_secure_api.c - PSA ITS API implementation for SPE
- tfm_its_req_mgr.c - Uniform secure functions and IPC request handlers
- tfm_internal_trusted_storage.h - TF-M ITS API (with client_id parameter)
- tfm_internal_trusted_storage.c - TF-M ITS implementation, using the flash_fs as a backend
- its_crypto_interface.h - APIs for encrypting ITS assets used by ITS implementation (optional)
- its_crypto_interface.c - Implementation for ITS encryption (optional)
- platform/ext/target/.../tfm_hal_its_encryption.c - Platform implementation for ITS encryption HAL APIs (optional)
- flash_fs/ - Filesystem
- flash/ - Flash interface

tf-m-tests repo:

test/secure_fw/suites/its/

- non_secure/psa_its_ns_interface_testsuite.c - Non-secure interface tests
- secure/psa_its_s_interface_testsuite.c - Secure interface tests

TF-M ITS implementation

The following APIs will be exposed by `tfm_internal_trusted_storage.h`:

```
psa_status_t tfm_its_init(void);

psa_status_t tfm_its_set(int32_t client_id,
                        psa_storage_uid_t uid,
                        size_t data_length,
                        const void *p_data,
                        psa_storage_create_flags_t create_flags);

psa_status_t tfm_its_get(int32_t client_id,
                        psa_storage_uid_t uid,
                        size_t data_offset,
                        size_t data_size,
                        void *p_data,
                        size_t *p_data_length);

psa_status_t tfm_its_get_info(int32_t client_id,
                             psa_storage_uid_t uid,
                             struct psa_storage_info_t *p_info);

psa_status_t tfm_its_remove(int32_t client_id,
                             psa_storage_uid_t uid);
```

That is, the TF-M ITS APIs will have the same prototypes as the PSA ITS APIs, but with the addition of a `client_id` parameter, which will be passed from the ITS request manager. A `tfm_its_init` function will also be present, which will be called at initialisation time and not exposed through a veneer or SID.

The implementation in `tfm_internal_trusted_storage.c` must validate the parameters (excepting memory references, which are validated by the SPM), translate the UID and client ID into a file ID and then make appropriate calls to the filesystem layer. It must also take care ensure that any PSA Storage flags associated with the UID are honoured.

Filesystem

The ITS filesystem will be copied and modified from the SST filesystem. The modifications required will be to rename symbols from `sst` to `its` and to update the implementation to be aligned with the latest version of the PSA Storage spec (which consists mainly of moving to the `psa_status_t` error type and using common error codes from `psa/error.h`).

The filesystem will also be modified to align the size of each file stored to the alignment requirement exposed by the flash interface, by adding appropriate padding.

The filesystem code will be de-duplicated again once the ITS service is implemented (see below).

Flash layer

The flash layer will be copied from SST, and modified to direct writes to the internal flash device. It too needs to be updated to use `psa_status_t` error types.

Platform layer

The TF-M platform layer must be updated to distinguish between the external flash device used for Protected Storage and internal flash device used for ITS. A flash region for the relevant storage service needs to be allocated in each.

On test platforms these may just be two distinct regions of the same flash device, but in general they will separate devices with their own drivers.

Detailed design considerations

Mapping UID onto file ID

The ITS APIs identify assets with 64-bit UIDs, to which the ITS service must append the 32-bit client ID of the calling partition for access control. The existing filesystem uses 32-bit file IDs to identify files, so some mapping would be required to convert between the identifiers.

SST uses the object table to do the mapping from client ID, UID pairs to file IDs, which means making an extra filesystem read/write for each get/set operation. This mapping has minimal overhead for SST though, because object table lookups are already required for rollback protection.

For ITS, no rollback protection feature is required, so there are two options:

- Keep a simplified version of the SST object table that just maps from (client ID, UID) to file ID
- Modify the filesystem to take (at least) 96-bit file IDs, in the form of a fixed-length char buffer.

The advantage of the former is that it would require no extra modification to the existing filesystem code, and the existing SST object table could be cut down for ITS. However, it would mean that every ITS request would invoke twice the number of filesystem operations, increasing latency and flash wear. The code size of the ITS partition would be increased, as would RAM usage as the table would need to be read into RAM.

The latter option would make the filesystem slightly more complex: the size of a metadata entry would be increased by 64-bits and the 96-bit fids would need to be copied and compared with `memcpy` and `memcmp` calls. On the other hand, mapping onto file IDs would incur only the cost of copying the UID and client ID values into the file ID buffer.

A third, even more general, solution would be to use arbitrary-length null-terminated strings as the file IDs. This is the standard solution in full-featured filesystems, but we do not currently require this level of complexity in secure storage.

With this in mind, the proposed option is the second.

Storing create flags

The ITS APIs provide a 32-bit `create_flags` parameter, which contains bit flags that determine the properties of the stored data. Only one flag is currently defined for ITS: `PSA_STORAGE_FLAG_WRITE_ONCE`, which prevents a UID from being modified or deleted after it is set for the first time.

There are two places that these flags could be stored: in the file data or as part of the file metadata.

For the first option, the ITS implementation would need to copy the flags into the buffer containing the data, and adjust the size accordingly, for each set operation, and the reverse for each get. Every `get_info` operation would need to read some of the file data, rather than just the metadata, implying a second flash read. A potential downside is that many of the cryptographic assets stored in ITS will be aligned to power-of-two sizes; adding an extra 32-bits would misalign the size, which may reduce flash performance or necessitate adding padding to align to the flash page size.

To implement the second option, a 32-bit `flag` field would be added to the filesystem's metadata structure, whose interpretation is defined by the user. This field would clearly be catered towards the PSA Storage APIs, even if nominally generic, and alternative filesystems may not have any such field. However, it is a more intuitive solution and would simplify both flash alignment and `get_info` operations.

Overall, it seems more beneficial to store the flags in the metadata, so this is the proposed solution.

Code sharing between Protected Storage and ITS

To de-duplicate the filesystem code used by both Protected Storage and ITS, it is proposed that Protected Storage calls ITS APIs as its backend filesystem.

Protected Storage essentially becomes an encryption, authentication and rollback protection layer on top of ITS. It makes IPC requests or secure function calls to the ITS service to do filesystem operations on its behalf.

This has a couple of advantages:

- It shrinks Protected Storage's stack size, because the filesystem and flash layer stack is only in ITS.
- It automatically solves the problem of ensuring mutual exclusion in the filesystem and flash layers when Protected Storage and ITS are called concurrently. The second request to ITS will just be made to wait by the SPM.

The disadvantage of this approach is that it will increase the latency of Protected Storage requests, due to the extra overhead associated with making a second IPC request or secure function call. It also limits Protected Storage to using only the ITS APIs, unless extra veneers are added solely for Protected Storage to use. This, for example, prevents Protected Storage from doing partial writes to file without reading and re-writing the whole file.

ITS will need to be modified to direct calls from Protected Storage to a different flash device. It can use the client ID to detect when the caller is Protected Storage, and pass down the identity of the flash device to use to the flash layer, which then calls the appropriate driver.

An open question is what to do if Protected Storage itself wants to store something in internal storage in the future (e.g. rollback counters, hash tree/table or top hash). A couple of possible solutions would be:

- Divide up the UIDs, so certain UIDs from Protected Storage refer to assets in internal storage, and others to ones in external storage.
- Use the `type` field of `psa_call` in IPC model to distinguish between internal and external storage requests.

The other option for code sharing would be for Protected Storage and ITS to directly share filesystem code, which would be placed in a shared code region. With this approach, mutual exclusion to the flash device would need to be implemented separately, as would some way of isolating static memory belonging to each partition but not the code. Because of these complications, this option has not been considered further at this time.

Encryption in ITS

The ITS can optionally be configured to encrypt the internal trusted storage data. To support encryption in ITS the target platform must provide an implementation of the APIs defined in `platform/include/tfm_hal_its_encryption.h`:

```
enum tfm_hal_status_t tfm_hal_its_aead_generate_nonce(uint8_t *nonce,
                                                    const size_t nonce_size);

enum tfm_hal_status_t tfm_hal_its_aead_encrypt(
    struct tfm_hal_its_auth_crypt_ctx *ctx,
    const uint8_t *plaintext,
    const size_t plaintext_size,
    uint8_t *ciphertext,
    const size_t ciphertext_size,
    uint8_t *tag,
    const size_t tag_size);

enum tfm_hal_status_t tfm_hal_its_aead_decrypt(
    struct tfm_hal_its_auth_crypt_ctx *ctx,
    const uint8_t *ciphertext,
    const size_t ciphertext_size,
    uint8_t *tag,
```

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```
const size_t tag_size,
uint8_t *plaintext,
const size_t plaintext_size);
```

Then encryption can be enabled by setting the build option -DITS_ENCRYPTION=ON.

The figure Figure ?? describes the encryption and decryption process happening when calling tfm_its_set and tfm_its_get.

By using an AEAD scheme, it is possible to not only encrypt the file data but also authenticate the file meta data, which include:

- File id
- File size
- File flags

The key used to perform the AEAD operation must be derived from a long-term key-derivation key and the file id, which is used as a derivation label. The long-term key-derivation key must be managed by the target platform.

There is a generic implementation of the abovementioned functions under platform/ext/common/template/tfm_hal_its_encryption.c using PSA crypto calls similar to Protected Storage solution. When used, the default NV seed template under platform/ext/common/template/crypto_nv_seed.c must be disabled, as it relies on ITS. If there is a need for NV seed usage, an ITS independent implementation is required. If NV seed is not necessary, it can be turned off by setting CRYPTO_NV_SEED=0.

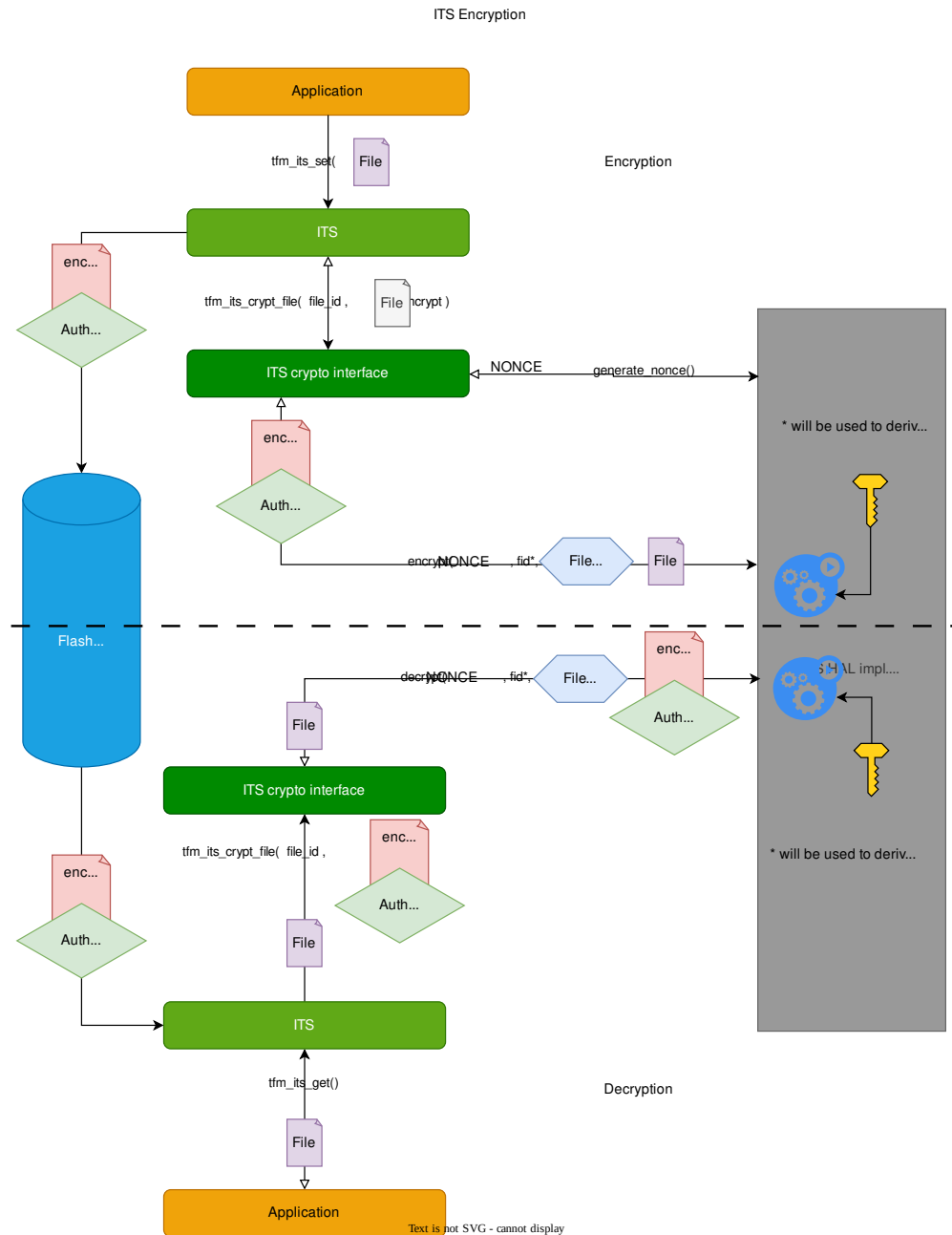
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10.2.9 Firmware Update Service

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- *Benefits Analysis on this Partition*
 - * *Implement the FWU functionality in the non-secure side*
 - * *Pros and Cons for implementing FWU APIs in secure side*
- *Reference*

Introduction of Firmware Update service

The Firmware Update(FWU) service provides the functionality of updating firmware images. It provides a standard interface for updating firmware and it is platform independent. TF-M defines a shim layer to support cooperation between bootloader and FWU service.

This partition supports the following features:

- Query the firmware store information.
- Image preparation: prepare a new firmware image in the component's firmware store.
- Image installation: install prepared firmware images on all components that have been prepared for installation.
- Image trial: manage a trial of new firmware images atomically on all components that are in TRIAL state.

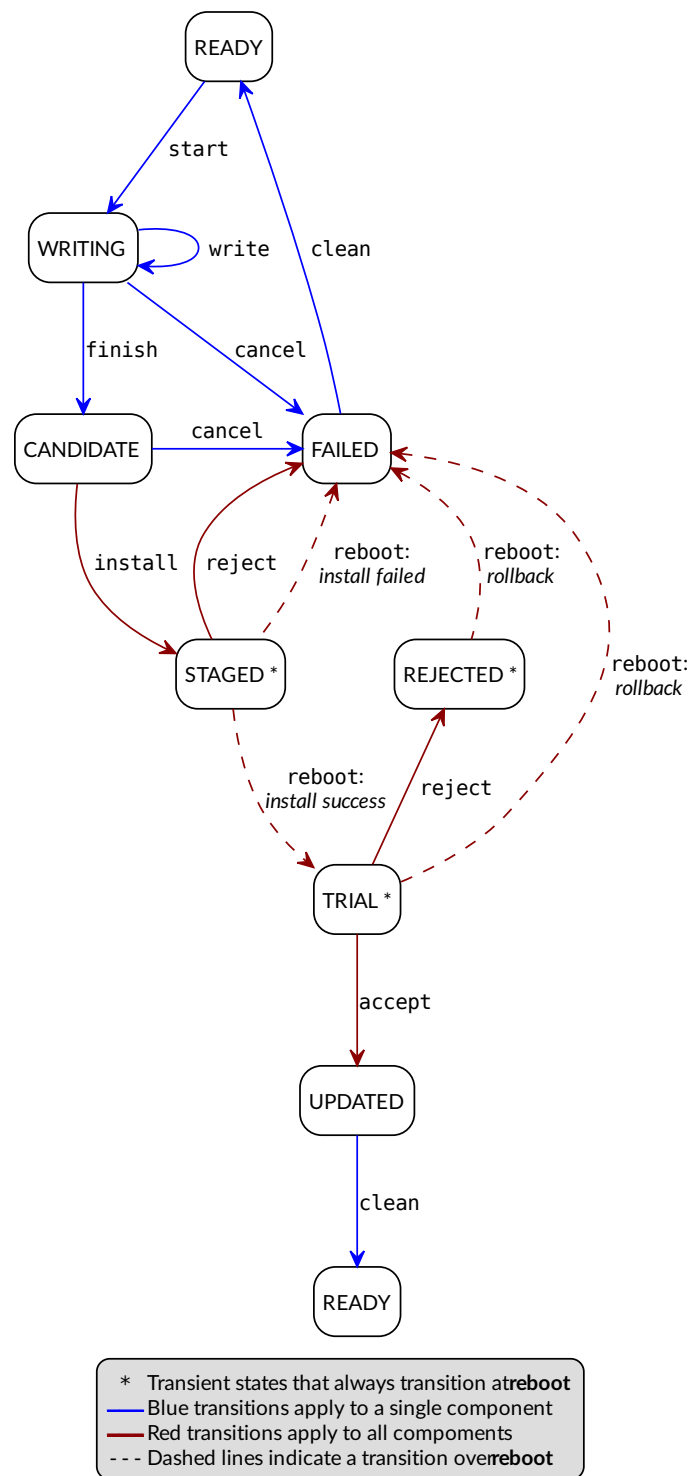
A typical flow through the component states is shown below¹.

Components

The structure of the TF-M Firmware Update service is listed below:

Component name	Description	Location
Client API interface	This module exports the client API of PSA Firmware Update to the users.	./interface/src/tfm_fwu_api.c
Manifest	The manifest file is a description of the service components.	./secure_fw/partitions/firmware_update/tfm_firmware_update.yaml
NSPE client API interface	This module exports the client API of PSA Firmware Update to the NSPE(i.e. to the applications).	./interface/src/tfm_fwu_api.c
IPC request handlers	This module handles all the secure requests in IPC model. It maintains the image state context and calls the image ID converter to achieve the firmware update functionalities.	./secure_fw/partitions/firmware_update/tfm_fwu_req_mgr.c
Shim layer between FWU and bootloader	This module provides the APIs with the functionality of operating the bootloader to cooperate with the Firmware Update service	./secure_fw/partitions/firmware_update/bootloader/tfm_bootloader_fwu_abstraction.h
Shim layer example based on MCUboot	This module is the implementation of the shim layer between FWU and bootloader based on MCUboot.	./secure_fw/partitions/firmware_update/bootloader/mcuboot/tfm_mcuboot_fwu.c

¹ PSA Firmware Update API



Service API description

This service follows the PSA Firmware Update API spec of version 1.0⁷. Please refer to Firmware Update spec for the detailed description.

Shim Layer between FWU and bootloader

The firmware update operations are achieved by calling the shim layer APIs between bootloader and FWU.

Shim layer introduction

This shim layer provides the APIs with the functionality of operating the bootloader to cooperate with the Firmware Update service. This shim layer is decoupled from bootloader implementation. Users can specify a specific bootloader by setting TFM_FWU_BOOTLOADER_LIB build configuration and adding the specific build scripts into that file. By default, the MCUboot is chosen as the bootloader.

Interfaces of the shim Layer

fwu_bootloader_init(function)

Prototype

```
psa_status_t fwu_bootloader_init(void);
```

Description

Bootloader related initialization for the firmware update. It reads some necessary shared data from the memory if needed. It initializes the flash drivers defined in FLASH_DRIVER_LIST. Platform can define FLASH_DRIVER_LIST in flash_layout.h to overload the default driver list.

Parameters

N/A

fwu_bootloader_staging_area_init(function)

Prototype

```
psa_status_t fwu_bootloader_staging_area_init(psa_fwu_component_t component,
                                              const void *manifest,
                                              size_t manifest_size);
```

Description

The component is in READY state. Prepare the staging area of the component for image download. For example, initialize the staging area, open the flash area, and so on.

Parameters

- **component**: The identifier of the target component in bootloader.
- **manifest**: A pointer to a buffer containing a detached manifest for the update. If the manifest is bundled with the firmware image, manifest must be NULL.
- **manifest_size**: Size of the manifest buffer in bytes.

fwu_bootloader_load_image(function)**Prototype**

```
psa_status_t fwu_bootloader_load_image(psa_fwu_component_t component,
                                       size_t          image_offset,
                                       const void      *block,
                                       size_t          block_size);
```

Description

Load the image into the target component.

Parameters

- **component**: The identifier of the target component in bootloader.
- **image_offset**: The offset of the image being passed into block, in bytes.
- **block**: A buffer containing a block of image data. This might be a complete image or a subset.
- **block_size**: Size of block.

fwu_bootloader_install_image(function)**Prototype**

```
psa_status_t fwu_bootloader_install_image(psa_fwu_component_t *candidates,
                                           uint8_t number);
```

Description

Check the authenticity and integrity of the image. If a reboot is required to complete the check, then mark this image as a candidate so that the next time bootloader runs it will take this image as a candidate one to boot-up. Return the error code PSA_SUCCESS_REBOOT.

Parameters

- **candidates**: A list of components in CANDIDATE state.
- **number**: Number of components in CANDIDATE state.

fwu_bootloader_mark_image_accepted(function)**Prototype**

```
psa_status_t fwu_bootloader_mark_image_accepted(const psa_fwu_component_t *trials,
                                                 uint8_t number);
```

Description

Call this API to mark the TRIAL(running) image in component as confirmed to avoid revert when next time boot-up. Usually, this API is called after the running images have been verified as valid.

Parameters

- **trials**: A list of components in TRIAL state.
- **number**: Number of components in TRIAL state.

fwu_bootloader_reject_staged_image(function)**Prototype**

```
psa_status_t fwu_bootloader_reject_staged_image(psa_fwu_component_t component);
```

Description

The component is in STAGED state. Call this API to Uninstall the staged image in the component so that this image will not be treated as a candidate next time boot-up.

Parameters

- component: The identifier of the target component in bootloader.

fwu_bootloader_reject_trial_image(function)**Prototype**

```
psa_status_t fwu_bootloader_reject_trial_image(psa_fwu_component_t component);
```

Description

The component is in TRIAL state. Mark the running image in the component as rejected.

Parameters

- component: The identifier of the target component in bootloader.

fwu_bootloader_clean_component(function)**Prototype**

```
psa_status_t fwu_bootloader_clean_component(psa_fwu_component_t component);
```

Description

The component is in FAILED or UPDATED state. Clean the staging area of the component.

Parameters

- component: The identifier of the target component in bootloader.

fwu_bootloader_get_image_info(function)**Prototype**

```
psa_status_t fwu_bootloader_get_image_info(psa_fwu_component_t component,
                                           bool query_state,
                                           bool query_impl_info,
                                           psa_fwu_component_info_t *info);
```

Description

Get the image information of the given bootloader_image_id in the staging area or the running area.

Parameters

- component: The identifier of the target component in bootloader.
- query_state: Whether query the 'state' field of psa_fwu_component_info_t.

- `query_impl_info`: Whether Query 'impl' field of `psa_fw_component_info_t`.
- `info`: Buffer containing return the component information.

Additional shared data between BL2 and SPE

An additional TLV area “image version” is added into the shared memory between BL2 and TF-M. So that the firmware update partition can get the image version. Even though the image version information is also included in the BOOT RECORD TLV area which is encoded by CBOR, adding a dedicated `image version` TLV area is preferred to avoid involving the CBOR encoder which can increase the code size. The FWU partition will read the shared data at the partition initialization.

Build configurations related to FWU partition

- `TFM_PARTITION_FIRMWARE_UPDATE` Controls whether FWU partition is enabled or not.
- `TFM_FWU_BOOTLOADER_LIB` Bootloader configure file for FWU partition.
- `TFM_CONFIG_FWU_MAX_WRITE_SIZE` The maximum permitted size for block in `psa_fw_write`, in bytes.
- `TFM_FWU_BUF_SIZE` Size of the FWU internal data transfer buffer (defaults to `TFM_CONFIG_FWU_MAX_WRITE_SIZE` if not set).
- `FWU_STACK_SIZE` The stack size of FWU Partition.
- `FWU_DEVICE_CONFIG_FILE` The device configuration file for FWU partition. The default value is the configuration file generated for MCUboot. The following macros should be defined in the configuration file:
 - `FWU_COMPONENT_NUMBER` The number of components on the device.

Note

In this design, component ID ranges from 0 to `FWU_COMPONENT_NUMBER - 1`.

- `FWU_SUPPORT_TRIAL_STATE` Whether TRIAL component state is supported.
- `TEST_NS_FWU` FWU nonsecure tests switch.
- `TEST_S_FWU` FWU secure tests switch.

Note

The running image which supports revert mechanism should be confirmed before initiating a firmware update process. For example, if the running image is built with `-DMCUBOOT_UPGRADE_STRATEGY=SWAP_USING_MOVE`, the image should be confirmed either by adding `-DMCUBOOT_CONFIRM_IMAGE=ON` build option or by calling `psa_fw_accept()` API before initiating a firmware update process. Otherwise, `PSA_ERROR_BAD_STATE` will be returned by `psa_fw_start()`.

Limitations of current implementation

Currently, the MCUboot based implementation does not record image update results like failure or success. And FWU partition does not detect failure errors in bootloader installation. If an image installation fails in the bootloader and the old image still runs after reboot, `PSA_FWU_READY` state will be returned by `psa_fw_query()` after reboot.

Currently, image download recovery after a reboot is not supported. If a reboot happens in image preparation, the downloaded image data will be ignored after the reboot.

The MCUboot based implementation only fills the data structure `psa_fwu_impl_info_t` in `psa_fwu_query()` when the component is in CANDIDATE state. Otherwise, an empty `psa_fwu_impl_info_t` is returned when the component is in other states.

Benefits Analysis on this Partition

Implement the FWU functionality in the non-secure side

The APIs listed in PSA Firmware Update API spec⁷ can also be implemented in the non-secure side.

Pros and Cons for implementing FWU APIs in secure side

Pros

- It protects the image in the passive or staging area from being tampered with by the NSPE. Otherwise, a malicious actor from NSPE can tamper the image stored in the non-secure area to break image update.
- It protects secure image information from disclosure. In some cases, the non-secure side shall not be permitted to get secure image information.
- It protects the active image from being manipulated by NSPE. Some bootloader supports testing the image. After the image is successfully installed and starts to run, the user should set the image as permanent image if the image passes the test. To achieve this, the area of the active image needs to be accessed. In this case, implementing FWU service in SPE can prevent NSPE from manipulating the active image area.
- On some devices, such as the Arm Musca-B1 board, the passive or staging area is restricted as secure access only. In this case, the FWU partition should be implemented in the secure side.

Cons

- It increases the image size of the secure image.
- It increases the execution latency and footprint. Compared to implementing FWU in NSPE directly, calling the Firmware Update APIs which are implemented in the secure side increases the execution latency and footprint.
- It can increase the attack surface of the secure runtime.

Users can decide whether to call the FWU service in TF-M directly or implement the Firmware Update APIs in the non-secure side based on the pros and cons analysis above.

Reference

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10.2.10 Protected Storage service key management

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Background

The PSA Protected Storage API requires confidentiality for external storage to be provided by:

cryptographic ciphers using device-bound keys, a tamper resistant enclosure, or an inaccessible deployment location, depending on the threat model of the deployed system.

A TBSA-M-compliant device must embed a Hardware Unique Key (HUK), which provides the root of trust (RoT) for confidentiality in the system. It must have at least 128 bits of entropy (and a 128 bit data size), and be accessible only to Trusted code or Trusted hardware that acts on behalf of Trusted code. [?]

Design description

Each time the system boots, PS will request that the Crypto service uses a key derivation function (KDF) to derive a storage key from the HUK, by referring to the builtin key handle for the HUK. The storage key could be kept in on-chip volatile memory private to the Crypto partition, or it could remain inside a secure element. Either way it will not be returned to PS.

For each call to the PSA Protected Storage APIs, PS will make requests to the Crypto service to perform AEAD encryption and/or decryption operations using the storage key (providing a fresh nonce for each encryption).

At no point will PS access the key material itself, only referring to the HUK and storage key by their handles in the Crypto service.

Key derivation

PS will make key derivation requests to the Crypto service with calls to the PSA Crypto APIs. In order to derive the storage key, the following calls are required:

```
status = psa_key_derivation_setup(&op, PSA_ALG_HKDF(PSA_ALG_SHA_256));

/* Set up a key derivation operation with HUK */
status = psa_key_derivation_input_key(&op, PSA_KEY_DERIVATION_INPUT_SECRET,
                                     TFM_BUILTIN_KEY_ID_HUK);

/* Supply the PS key label as an input to the key derivation */
status = psa_key_derivation_input_bytes(&op, PSA_KEY_DERIVATION_INPUT_INFO,
                                       key_label,
                                       key_label_len);

/* Create the storage key from the key derivation operation */
status = psa_key_derivation_output_key(&attributes, &op, &ps_key);
```

Note

TFM_BUILTIN_KEY_ID_HUK is a static key ID that is used to identify the HUK. It has an arbitrary value defined in `tfm_crypto_defs.h`

`ps_key` is a PSA Crypto key handle to a volatile key, set by the derivation operation. After the call to `psa_key_derivation_output_key`, it can be used to refer the storage key.

`key_label` can be any string that is independent of the input key material and different to the label used in any other derivation from the same input key. It prevents two different contexts from deriving the same output key from the same input key.

The key derivation function used by the crypto service to derive the storage key will be HKDF, with SHA-256 as the underlying hash function. HKDF is suitable because:

- It is simple and efficient, requiring only two HMAC operations when the length of the output key material is less than or equal to the hash length (as is the case here).
- The trade-off is that HKDF is only suitable when the input key material has at least as much entropy as required for the output key material. But this is the case here, as the HUK has 128 bits of entropy, the same as required

by PS.

- HKDF is standardised in RFC 5869 [?] and its security has been formally analysed. [?]
- It is supported by the TF-M Crypto service.

The choice of underlying hash function is fairly straightforward: it needs to be a modern standardised algorithm, considered to be secure and supported by TF-M Crypto. This narrows it down to just the SHA-2 family. Of the hash functions in the family, SHA-256 is the simplest and provides more than enough output length.

Keeping the storage key private to PS

The Crypto service derives a platform key from the HUK, using the partition ID as the input to that derivation, and that platform key is used for the key derivation by PS. This happens transparently, and to PS is indistinguishable from deriving from the HUK except that other partitions cannot derive the storage key even if they know the derivation parameters.

Key use

To encrypt and decrypt data, PS will call the PSA Crypto AEAD APIs in the same way as the current implementation, but `ps_key` will refer to the storage key, rather than the imported HUK. For each encryption operation, the following call is made (and analogously for decryption):

```
psa_aead_encrypt(ps_key, PS_CRYPT0_ALG,
                 crypto->ref.iv, PS_IV_LEN_BYTES,
                 add, add_len,
                 in, in_len,
                 out, out_size, out_len);
```

References

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10.3 Software Design Notes

10.3.1 Code sharing between independently linked XIP binaries

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Motivation

Cortex-M devices are usually constrained in terms of flash and RAM. Therefore, it is often challenging to fit bigger projects in the available memory. The PSA specifications require a device to both have a secure boot process in place at device boot-up time, and to have a partition in the SPE which provides cryptographic services at runtime. These two

entities have some overlapping functionality. Some cryptographic primitives (e.g. hash calculation and digital signature verification) are required both in the bootloader and the runtime environment. In the current TF-M code base, both firmware components use the mbed-crypto library to implement these requirements. During the build process, the mbed-crypto library is built twice, with different configurations (the bootloader requires less functionality) and then linked to the corresponding firmware component. As a result of this workflow, the same code is placed in the flash twice. For example, the code for the SHA-256 algorithm is included in MCUboot, but the exact same code is duplicated in the SPE cryptography partition. In most cases, there is no memory isolation between the bootloader and the SPE, because both are part of the PRoT code and run in the secure domain. So, in theory, the code of the common cryptographic algorithms could be reused among these firmware components. This could result in a big reduction in code footprint, because the cryptographic algorithms are usually flash hungry. Code size reduction can be a good opportunity for very constrained devices, which might need to use TF-M Profile Small anyway.

Technical challenge

Code sharing in a regular OS environment is easily achievable with dynamically linked libraries. However, this is not the case in Cortex-M systems where applications might run bare-metal, or on top of an RTOS, which usually lacks dynamic loading functionality. One major challenge to be solved in the Cortex-M space is how to share code between independently linked XIP applications that are tied to a certain memory address range to be executable and have absolute function and global data memory addresses. In this case, the code is not relocatable, and in most cases, there is no loader functionality in the system that can perform code relocation. Also, the lack of an MMU makes the address space flat, constant and not reconfigurable at runtime by privileged code.

One other difficulty is that the bootloader and the runtime use the same RAM area during execution. The runtime firmware is executed strictly after the bootloader, so normally, it can reuse the whole secure RAM area, as it would be the exclusive user. No attention needs to be paid as to where global data is placed by the linker. The bootloader does not need to retain its state. The low level startup of the runtime firmware can freely overwrite the RAM with its data without corrupting bootloader functionality. However, with code sharing between bootloader and runtime firmware, these statements are no longer true. Global variables used by the shared code must either retain their value or must be reinitialised during low level startup of the runtime firmware. The startup code is not allowed to overwrite the shared global variables with arbitrary data. The following design proposal provides a solution to these challenges.

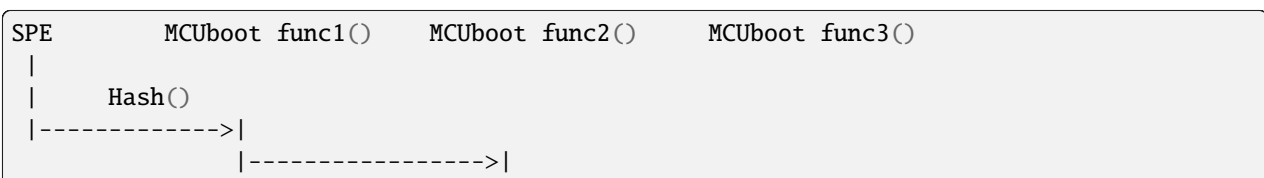
Design concept

The bootloader is sometimes implemented as ROM code (BL1) or stored in a region of the flash which is lockable, to prevent tampering. In a secure system, the bootloader is immutable code and thus implements a part of the Root of Trust anchor in the device, which is trusted implicitly. The shared code is primarily part of the bootloader, and is reused by the runtime SPE firmware at a later stage. Not all of the bootloader code is reused by the runtime SPE, only some cryptographic functions.

Simplified steps of building with code sharing enabled:

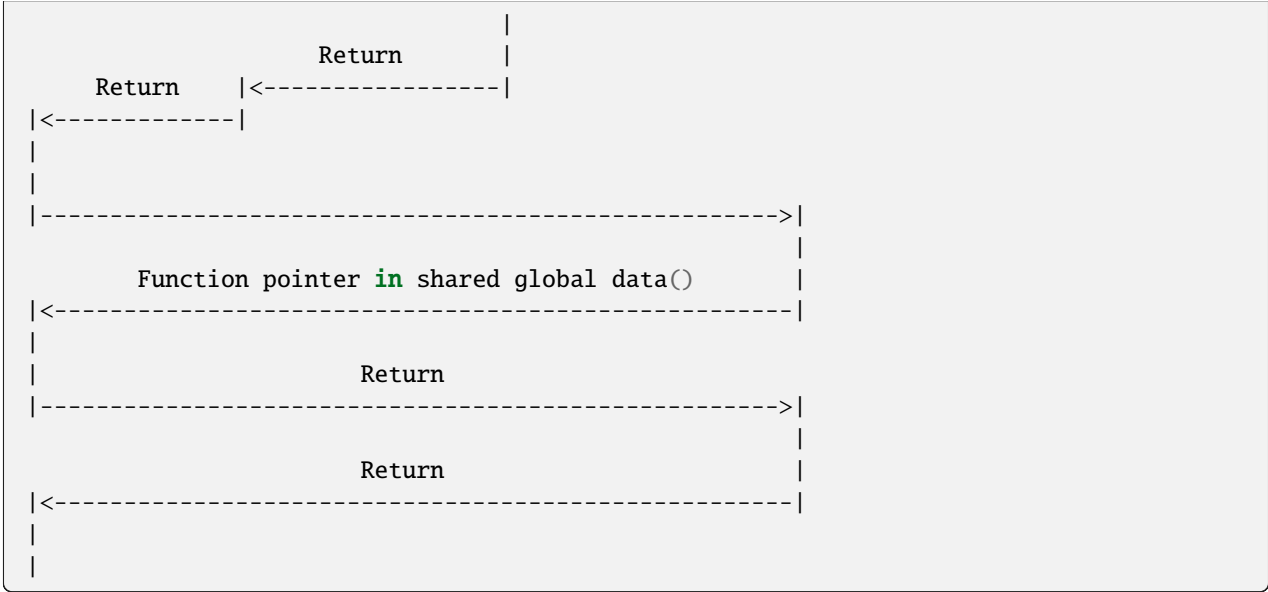
- Complete the bootloader build process to have a final image that contains the absolute addresses of the shared functions, and the global variables used by these functions.
- Extract the addresses of the functions and related global variables that are intended to be shared from the bootloader executable.
- When building runtime firmware, provide the absolute addresses of the shared symbols to the linker, so that it can pick them up, instead of instantiating them again.

The execution flow looks like this:



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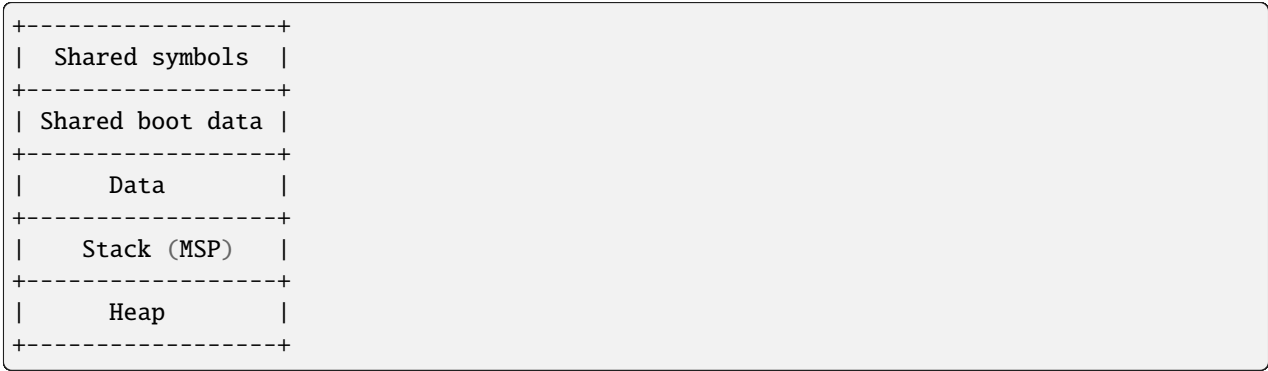


The execution flow usually returns from a shared function back to the SPE with an ordinary function return. So usually, once a shared function is called in the call path, all further functions in the call chain will be shared as well. However, this is not always the case, as it is possible for a shared function to call a non-shared function in SPE code through a global function pointer.

For shared global variables, a dedicated data section must be allocated in the linker configuration file. This area must have the same memory address in both MCUboot's and the SPE's linker files, to ensure the integrity of the variables. For simplicity's sake, this section is placed at the very beginning of the RAM area. Also, the RAM wiping functionality at the end of the secure boot flow (that is intended to remove any possible secrets from the RAM) must not clear this area. Furthermore, it must be ensured that the linker places shared globals into this data section. There are two way to achieve this:

- Put a filter pattern in the section body that matches the shared global variables.
- Mark the global variables in the source code with special attribute `__attribute__((section(<NAME_OF_SHARED_SYMBOL_SECTION>)))`

RAM memory layout in MCUboot with code sharing enabled:



RAM memory layout in SPE with code sharing enabled:



(continues on next page)

(continued from previous page)

```

+-----+
| Shared boot data |
+-----+
|   Stack (MSP)   |
+-----+
|   Stack (PSP)   |
+-----+
| Partition X Data |
+-----+
| Partition X Stack|
+-----+
|
|   .
|   .
|   .
+-----+
| Partition Z Data |
+-----+
| Partition Z Stack|
+-----+
|   PRoT Data     |
+-----+
|   Heap          |
+-----+

```

Patching Mbed TLS

In order to share some global function pointers from mbed-crypto that are related to dynamic memory allocation, their scope must be extended from private to global. This is needed because some compiler toolchain only extract the addresses of public functions and global variables, and extraction of addresses is a requirement to share them among binaries. Therefore, a short patch was created for the mbed-crypto library, which “globalises” these function pointers:

lib/ext/mbedcrypto/0002-Enable-crypto-code-sharing-between-independent-binaries.patch

The patch needs to be manually applied in the Mbed TLS repo, if code sharing is enabled. The patch has no effect on the functional behaviour of the cryptographic library, it only extends the scope of some variables.

Tools support

All the currently supported compilers provide a way to achieve the above objectives. However, there is no standard way, which means that the code sharing functionality must be implemented on a per compiler basis. The following steps are needed:

- Extraction of the addresses of all global symbols.
- The filtering out of the addresses of symbols that aren’t shared. The goal is to not need to list all the shared symbols by name. Only a simple pattern has to be provided, which matches the beginning of the symbol’s name. Matching symbols will be shared. Examples are in : *bl2/shared_symbol_template.txt*
- Provision of the addresses of shared symbols to the linker during the SPE build process.
- The resolution of symbol collisions during SPE linking. Because mbed-crypto is linked to both firmware components as a static library, the external shared symbols will conflict with the same symbols found within it. In order to prioritize the external symbol, the symbol with the same name in mbed-crypto must be marked as weak in the symbol table.

The above functionalities are implemented in the toolchain specific CMake files:

- *toolchain_ARMCLANG.cmake*
- *toolchain_GNUARM.cmake*

By the following two functions:

- *target_share_symbols()*: Extract and filter shared symbol addresses from MCUboot.
- *target_link_shared_code()*: Link shared code to the SPE and resolve symbol conflict issues.

ARMCLANG

The toolchain specific steps are:

- Extract all symbols from MCUboot: add *-symdefs* to the compiler command line
- Filter shared symbols: call CMake script *FilterSharedSymbols.cmake*
- Weaken duplicated (shared) symbols in the mbed-crypto static library that are linked to the SPE: *arm-none-eabi-objcopy*
- Link shared code to SPE: Add the filtered output of *-symdefs* to the SPE source file list.

GNUARM

The toolchain specific steps are:

- Extract all symbols from MCUboot: *arm-none-eabi-nm*
- Filter shared symbols: call CMake script: *FilterSharedSymbols.cmake*
- Strip unshared code from MCUboot: *arm-none-eabi-strip*
- Weaken duplicated (shared) symbols in the mbed-crypto static library that are linked to the SPE: *arm-none-eabi-objcopy*
- Link shared code to SPE: Add *-Wl -R <SHARED_STRIPPED_CODE.axf>* to the compiler command line

IAR

Functionality currently not implemented, but the toolchain supports doing it.

Memory footprint reduction

Build type: MinSizeRel Platform: mps2/an521 Version: TF-Mv1.2.0 + code sharing patches MCUboot image encryption support is disabled.

	ConfigDefault		ConfigProfile-M		ConfigProfile-S	
	ARM-CLANG	GNUARM	ARM-CLANG	GNUARM	ARM-CLANG	GNUARM
CODE_SHARING=OFF	122268	124572	75936	75996	50336	50224
CODE_SHARING=ON	113264	115500	70400	70336	48840	48988
Difference	9004	9072	5536	5660	1496	1236

If MCUboot image encryption support is enabled then saving could be up to ~13-15KB.

Note

Code sharing on Musca-B1 was tested only with SW only crypto, so crypto hardware acceleration must be turned off: `-DCRYPTO_HW_ACCELERATOR=OFF`

Usability considerations

Functions that only use local variables can be shared easily. However, functions that rely on global variables are a bit tricky. They can still be shared, but all global variables must be placed in the shared symbol section, to prevent overwriting and to enable the retention of their values.

Some global variables might need to be reinitialised to their original values by runtime firmware, if they have been used by the bootloader, but need to have their original value when runtime firmware starts to use them. If so, the reinitialising functionality must be implemented explicitly, because the low level startup code in the SPE does not initialise the shared variables, which means they retain their value after MCUboot stops running.

If a bug is discovered in the shared code, it cannot be fixed with a firmware upgrade, if the bootloader code is immutable. If this is the case, disabling code sharing might be a solution, as the new runtime firmware could contain the fixed code instead of relying on the unfixed shared code. However, this would increase code footprint.

API backward compatibility also can be an issue. If the API has changed in newer version of the shared code. Then new code cannot rely on the shared version. The changed code and all the other shared code where it is referenced from must be ignored and the updated version of the functions must be compiled in the SPE binary. The Mbed TLS library is API compatible with its current version (v2.24.0) since the `mbedtls-2.7.0` release (2018-02-03).

To minimise the risk of incompatibility, use the same compiler flags to build both firmware components.

The artifacts of the shared code extraction steps must be preserved so as to remain available if new SPE firmware (that relies on shared code) is built and released. Those files are necessary to know the address of shared symbols when linking the SPE.

How to use code sharing?

Considering the above, code sharing is an optional feature, which is disabled by default. It can be enabled from the command line with a compile time switch:

- `TFM_CODE_SHARING`: Set to *ON* to enable code sharing.

With the default settings, only the common part of the mbed-crypto library is shared, between MCUboot and the SPE. However, there might be other device specific code (e.g. device drivers) that could be shared. The shared cryptography code consists mainly of the SHA-256 algorithm, the *bignum* library and some RSA related functions. If image encryption support is enabled in MCUboot, then AES algorithms can be shared as well.

Sharing code between the SPE and an external project is possible, even if MCUboot isn't used as the bootloader. For example, a custom bootloader can also be built in such a way as to create the necessary artifacts to share some of its code with the SPE. The same artifacts must be created like the case of MCUboot:

- ***shared_symbols_name.txt***: Contains the name of the shared symbols. Used by the script that prevents symbol collision.
- *shared_symbols_address.txt*: Contains the type, address and name of shared symbols. Used by the linker when linking runtime SPE.
- *shared_code.axf*: GNUARM specific. The stripped version of the firmware component, only contains the shared code. It is used by the linker when linking the SPE.

Note

The artifacts of the shared code extraction steps must be preserved to be able to link them to any future SPE version.

When an external project is sharing code with the SPE, the `SHARED_CODE_PATH` compile time switch must be set to the path of the artifacts mentioned above.

Further improvements

This design focuses only on sharing the cryptography code. However, other code could be shared as well. Some possibilities:

- Flash driver
- Serial driver
- Image metadata parsing code
- etc.

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10.3.2 Hardware Abstraction Layer

Organization

Arm Limited

Contact

tf-m@lists.trustedfirmware.org

API Version

0.9

Introduction

TF-M HAL abstracts the hardware-oriented and platform specific operations on the *SPE* side and provides a set of APIs to the upper layers such as *SPM*, *RoT Service*. The *HAL* aims to cover the platform different aspects whereas common architecturally defined aspects are done generically within the common *SPE*. In some cases, although the operations are defined architecturally, it may not be possible to generalize implementations because lots of information is only known to platforms. It is more efficient to define a *HAL API* for those architectural operations as well.

Note

TBSA-M provides the hardware requirements for security purposes. *TF-M HAL* tries to reference *TBSA-M* recommendations in the interfaces from the software perspective only. Please reference *TBSA-M* for your security hardware design.

Design Goals

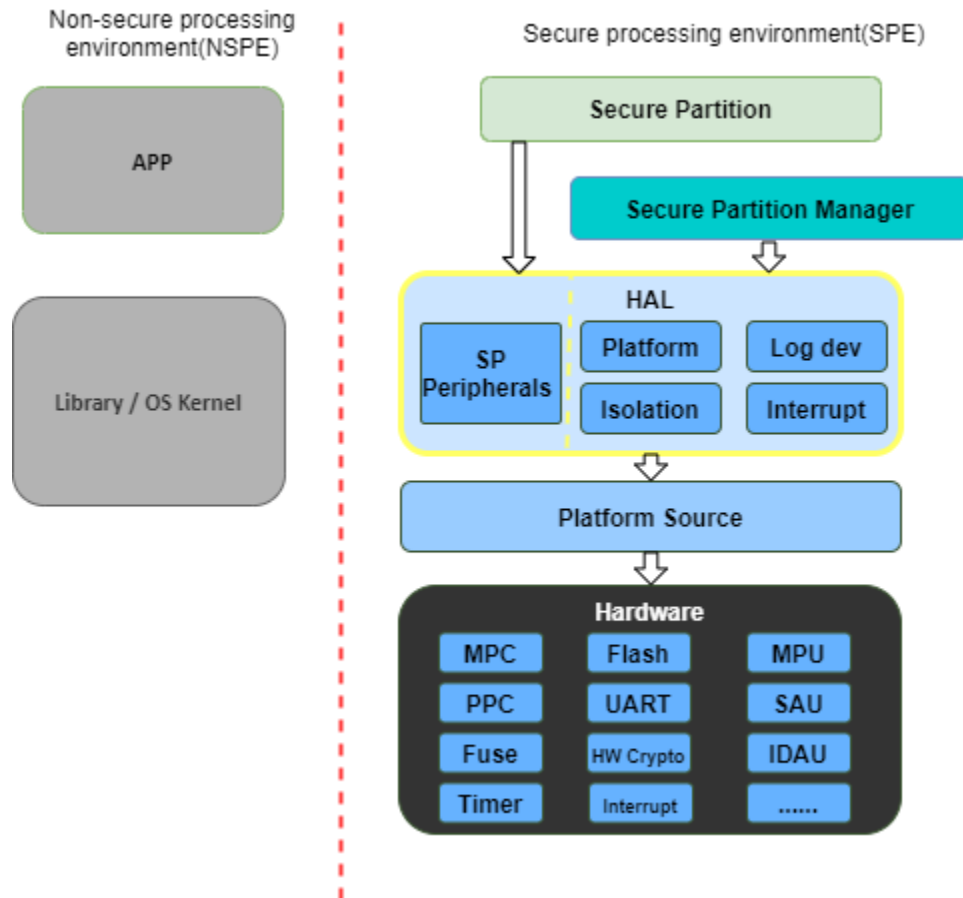
TF-M HAL is designed to simplify the integration efforts on different platforms.

TF-M HAL is designed to make it easy to use the hardware and develop the *SPM* and *RoT Service* which need to access the devices.

TF-M HAL is designed to make the structure clearer and let the *TF-M* mainly focus on *PSA* implementation.

Overview

This section provides an overview of the abstraction layer structure.



Here lists a minimal set of necessary functionalities:

- **Isolation API:** Provides the necessary isolation functionalities required by the *PSA-FF-M* and *TBSA-M*, and provides APIs to *SPM* to check the permissions of memory access.
- **Platform API:** Provides the platform initialization, platform-specific memory information, system reset, etc.
- **Log dev API:** Provides the log system functions.
- **Interrupt API:** Provides the interrupt functions.

Note

- There is a non-secure *HAL* that focuses on the mailbox operation API for Dual-core topology. For more information about it, please refer to *Mailbox Design in TF-M on Dual-core System*.
- The minimal set of *TF-M HAL* is sufficient for Secure Partitions by using customized peripheral interfaces. To provide easier portability for the Secure Partitions, a Secure Partition *HAL* is provided in this design too.
- The debug mechanisms give the external entity the corresponding right to access the system assets. *TF-M* ensures that the external entity is permitted access to those assets. Currently, *TF-M* only needs the debug authentication. The whole debug mechanism and related *HAL* will be enhanced in the future. Please refer to the *Debug authentication settings section* for more details.

Design Principles

As *TF-M* runs on resource-constrained devices, the *HAL* tries to avoid multiple level abstractions which cost more resources.

Part of the *HAL* interfaces does not focus on exact hardware operations such as power on/off or PIN manipulation. Instead, the *HAL* abstracts higher-level interfaces to reserve the implementation flexibility for the platform vendors.

The *TF-M HAL* should be easy to deprecate APIs and provide compatibilities. Any API incompatibility should be detected during building.

TF-M relies on the *HAL* APIs to be implemented correctly and trusts the *HAL* APIs. *TFM* can provide assertions to detect common programming errors but essentially no further extensive checks will be provided.

Source Files

This section describes the source file of the *TF-M HAL*, including the header and c files.

`tfm_hal_defs.h`

This header file contains the definitions of common macros and types used by all *HAL* APIs. Please refer to *Status Codes* for detailed definitions.

`tfm_hal_[module].[h/c]`

All other headers and c files are classified by the modules, such as isolation, platform, interrupt, devices, etc.

Note

There are common files in the platform folder include the implemented *HAL* APIs. The platform vendors can use them directly but need to implement all the sub APIs.

Status Codes

These are common status and error codes for all *HAL* APIs.

Types

`tfm_hal_status_t`

This is a status code to be used as the return type of *HAL* APIs.

```
enum tfm_hal_status_t {
    TFM_HAL_ERROR_MEM_FAULT = SCHAR_MIN,
    TFM_HAL_ERROR_MAX_VALUE,
    TFM_HAL_ERROR_BAD_STATE,
    TFM_HAL_ERROR_NOT_SUPPORTED,
    TFM_HAL_ERROR_INVALID_INPUT,
    TFM_HAL_ERROR_NOT_INIT,
    TFM_HAL_ERROR_GENERIC,
    TFM_HAL_SUCCESS = 0
};
```

Error Codes

Negative values indicate an error. Zero and positive values indicate success.

Here is the general list. The detailed usages for each error code are described in the API introduction sections.

TFM_HAL_SUCCESS

Status code to indicate general success.

TFM_HAL_ERROR_GENERIC

Status code to indicate an error that does not correspond to any defined failure cause.

TFM_HAL_ERROR_NOT_INIT

Status code to indicate that the module is not initialed.

TFM_HAL_ERROR_INVALID_INPUT

Status code to indicate that the input is invalid.

TFM_HAL_ERROR_NOT_SUPPORTED

Status code to indicate that the requested operation or a parameter is not supported.

TFM_HAL_ERROR_BAD_STATE

Status code to indicate that the requested action cannot be performed in the current state.

TFM_HAL_ERROR_MAX_VALUE

Status code to indicate that the current number has got the max value.

TFM_HAL_ERROR_MEM_FAULT

Status code to indicate that the memory check failed.

API Definition for TF-M SPM

This section describes the APIs defined for *TF-M SPM*.

Platform API

The platform API is a higher-level abstraction layer of the platform, other than a dedicated API set for the special hardware devices.

APIs

tfm_hal_platform_init()

Prototype

```
enum tfm_hal_status_t tfm_hal_platform_init(void)
```

Description

This API performs the platform initializations **before** the *SPM* initialization.

The initializations could include but not limited to: - Fault handlers - Reset configurations - Debug init - NVIC init

Parameter

- void - None.

Return Values

- TFM_HAL_SUCCESS - Init success.
- TFM_HAL_ERROR_GENERIC - Generic errors.

tfm_hal_system_reset()**Prototype**

```
void tfm_hal_system_reset(uint32_t sw_reset_syn_value)
```

Description

This API performs a system reset (cold reset).

The platform can uninitialize some resources before reset. The function gets passed a parameter, usually a register value, that gets used to set the system reset syndrome register before issuing the reset. This register survives resets and can be read to understand the root cause of the reset. A platform that does not provide this functionality still needs to pass TFM_PLAT_SWSYN_DEFAULT as a safe default value. A platform can also to redefine the value of TFM_PLAT_SWSYN_DEFAULT in case the default of 0x0UL does not meet its requirements.

When CONFIG_TFM_HALT_ON_CORE_PANIC is disabled this function is called to reset the system when a fatal error occurs.

Parameter

- uint32_t sw_reset_syn_value - The value of the register to be used to set the platform reset syndrome

Return Values

- void - None

Note

This API should not return.

tfm_hal_get_reset_syndrome()**Prototype**

```
void tfm_hal_get_reset_syndrome(void)
```

Description

This API performs reads the value of the reset syndrome register that has been mirrored after the last reset.

Parameter

- void - None

Return Values

- uint32_t - The value of the reset syndrome register, usually a full 32 bits register

Note

None

tfm_hal_clear_reset_syndrome_bit()**Prototype**

```
void tfm_hal_clear_reset_syndrome_bit(uint8_t bit_pos)
```

Description

This API performs clears a specific bit of the reset syndrome register passed as input.

Parameter

- uint8_t bit_pos - Bit position to clear on the reset syndrome register

Return Values

- void - None

Note

None

tfm_hal_system_halt()**Prototype**

```
void tfm_hal_system_halt(void)
```

Description

This API enters the CPU into an infinite loop.

The platform can uninitialize some resources before looping forever.

When CONFIG_TFM_HALT_ON_CORE_PANIC is enabled this function is called to halt the system when a fatal error occurs.

Parameter

- void - None

Return Values

- void - None

Note

This API should not return.

Isolation API

The *PSA-FF-M* defines three isolation levels and a memory access rule to provide diverse levels of security. The isolation API provides the functions to implement these requirements.

The Isolation API operates on boundaries. A boundary represents a set of protection settings that isolates components and domains. Below are the boundary examples in the current implementation:

- Boundaries between SPM and Secure Partitions.
- Boundaries between ARoT domain and PRoT domain.

There are two types of boundaries:

- Static boundaries: Set up when the system is initializing and persistent after the initialization. This type of boundary needs the set-up operations only.
- Partition boundaries: Keeps switching from one to another when the system is running. This type of boundary needs both set-up and switching operations.

The boundary operations are abstracted as HAL interfaces because isolation hardware can be different for platforms:

- The set-up HAL interface creates a partition boundary based on given partition information. This created boundary is bound with the partition for subsequent usage. The binding is done by storing the boundary into partition runtime data.
- The activation HAL interface activates the partition boundary to secure the execution for the partition to be switched. The target partition's information and boundary are given to the activation HAL to accomplish the operation.

The data representing the partition boundary in runtime is defined with the opaque type `uintptr_t`:

- It is required that one value represents one boundary. The different values represent different boundaries.
- The value is created by HAL implementation with its own-defined encoding scheme.

The HAL implementation defined encoding scheme can be designed for implementation convenience. For example:

- The implementation scheme can encode attribute flags into integer bits. This could help the activation HAL to extract the protection settings quickly from this encoded value, or even write to hardware registers directly in the most ideal case. The initial TF-M Isolation HAL reference implementation applies this scheme.
- The implementation scheme can reference the addresses of isolation hardware description data. This could help the activation HAL to reference the protection settings directly by pointers.

Multiple Secure Partitions can bind with the same boundary value. This flexibility is useful for specific configurations. Take Isolation Level 2 as an example, assigning PRoT and ARoT domain boundaries to respective partitions can make execution more efficient, because switching two partitions in the same domain does not need to change the activated boundary.

The boundary contains the partition's memory accessibility information, hence memory access check shall be performed based on boundary.

Memory Access Attributes

The memory access attributes are encoded as bit fields, you can logic OR them to have a combination of the attributes, for example `TFM_HAL_ACCESS_UNPRIVILEGED | TFM_HAL_ACCESS_READABLE` is unprivileged readable. The data type is `uint32_t`.

TFM_HAL_ACCESS_EXECUTABLE

The memory is executable.

```
#define TFM_HAL_ACCESS_EXECUTABLE (1UL << 0)
```

TFM_HAL_ACCESS_READABLE

The memory is readable.

```
#define TFM_HAL_ACCESS_READABLE (1UL << 1)
```

TFM_HAL_ACCESS_WRITABLE

The memory is writable.

```
#define TFM_HAL_ACCESS_WRITABLE (1UL << 2)
```

TFM_HAL_ACCESS_UNPRIVILEGED

The memory is unprivileged mode accessible.

```
#define TFM_HAL_ACCESS_UNPRIVILEGED (1UL << 3)
```

TFM_HAL_ACCESS_DEVICE

The memory is a MMIO device.

```
#define TFM_HAL_ACCESS_DEVICE (1UL << 4)
```

TFM_HAL_ACCESS_NS

The memory is accessible from *NSPE*

```
#define TFM_HAL_ACCESS_NS (1UL << 5)
```

APIs

tfm_hal_verify_static_boundaries()

Prototype

```
fh_int tfm_hal_verify_static_boundaries(void)
```

Description

This API verifies the static isolation boundaries.

Refer to the *PSA-FF-M* for the definitions of the isolation boundaries.

Parameter

- void - None

Return Values

- TFM_HAL_SUCCESS - Verification has been successful.
- TFM_HAL_ERROR_GENERIC - Verification failed.

tfm_hal_set_up_static_boundaries()

Prototype

```
enum tfm_hal_status_t tfm_hal_set_up_static_boundaries(void)
```

Description

This API sets up the static isolation boundaries which are constant throughout the system runtime.

These boundaries include:

- The boundary between the *SPE* and the *NSPE*
- The boundary to protect the SPM execution. For example, the PSA RoT isolation boundary between the PSA Root of Trust and the Application Root of Trust which is for isolation level 2 and 3 only.

Refer to the *PSA-FF-M* for the definitions of the isolation boundaries.

Parameter

- void - None

Return Values

- TFM_HAL_SUCCESS - Isolation boundaries have been set up.
- TFM_HAL_ERROR_GENERIC - Failed to set up the static boundaries.

tfm_hal_bind_boundary()

Prototype

```
enum tfm_hal_status_t tfm_hal_bind_boundary(
    const struct partition_load_info_t *p_ldinf,
    uintptr_t *p_boundary);
```

Description

This API binds partition with a platform-generated boundary. The boundary is bound by writing the generated value into `p_boundary`. And this bound boundary is used in subsequent calls to `tfm_hal_activate_boundary()` when boundary's owner partition get scheduled for running.

Parameter

- `p_ldinf` - Load information of the partition that is under loading.
- `p_boundary` - Pointer for holding a partition's boundary.

Return Values

- TFM_HAL_SUCCESS - The boundary has been bound successfully.
- TFM_HAL_ERROR_GENERIC - Failed to bind the handle.

tfm_hal_activate_boundary()

Prototype

```
enum tfm_hal_status_t tfm_hal_activate_boundary(
    const struct partition_load_info_t *p_ldinf,
    uintptr_t boundary);
```

Description

This API requires the platform to activate the boundary to ensure the given Secure Partition can run successfully.

The access permissions outside the boundary is platform-dependent.

Parameter

- `p_ldinf` - The load information of the partition that is going to be run.
- `boundary` - The boundary for the owner partition of `p_ldinf`. This value is bound in function `tfm_hal_bind_boundary`.

Return Values

- TFM_HAL_SUCCESS - the isolation boundary has been set up.
- TFM_HAL_ERROR_GENERIC - failed to set up the isolation boundary.

tfm_hal_memory_check()**Prototype**

```
tfm_hal_status_t tfm_hal_memory_check(uintptr_t boundary,
                                       uintptr_t base,
                                       size_t size,
                                       uint32_t access_type)
```

Description

This API checks if a given range of memory can be accessed with specified access types in boundary. The boundary belongs to a partition which contains asset info.

Parameter

- boundary - Boundary of a Secure Partition. Check *tfm_hal_bind_boundary* for details.
- base - The base address of the region.
- size - The size of the region.
- access_type - The memory access types to be checked between given memory and boundaries. The *Memory Access Attributes*.

Return Values

- TFM_HAL_SUCCESS - The memory region has the access permissions.
- TFM_HAL_ERROR_MEM_FAULT - The memory region does not have the access permissions.
- TFM_HAL_ERROR_INVALID_INPUT - Invalid inputs.
- TFM_HAL_ERROR_GENERIC - An error occurred.

Note

If the implementation chooses to encode a pointer as the boundary, a platform-specific pointer validation needs to be considered before referencing the content in this pointer.

tfm_hal_boundary_need_switch()**Prototype**

```
bool tfm_hal_boundary_need_switch(uintptr_t boundary_from,
                                   uintptr_t boundary_to)
```

Description

This API let the platform decide if a boundary switch is needed.

Parameter

- boundary_from - Boundary to switch from.
- boundary_to - Boundary to switch to.

Return Values

- true - A switching is needed
- false - No need for a boundary switch

tfm_hal_post_partition_init_hook()

Prototype

```
enum tfm_hal_status_t tfm_hal_post_partition_init_hook(void)
```

Description

This API let the platform to finish static isolation after all partitions have been bound. It's called by SPM right before starting scheduler. Use CONFIG_TFM_POST_PARTITION_INIT_HOOK to enable it.

Parameter

- void - None

Return Values

- TFM_HAL_SUCCESS - Booting has been successful.
- TFM_HAL_ERROR_GENERIC - Error occurred.

tfm_hal_shared_metadata_rw_enable()

Prototype

```
void tfm_hal_shared_metadata_rw_enable(void);
```

Description

This API enables Read-Write access to the shared metadata section. Available when CONFIG_TFM_PARTITION_META_DYNAMIC_ISOLATION is set to 1.

Return Values

- void - None

Note

This API is only available if CONFIG_TFM_PARTITION_META_DYNAMIC_ISOLATION is enabled.

tfm_hal_shared_metadata_rw_disable()

Prototype

```
void tfm_hal_shared_metadata_rw_disable(void);
```

Description

This API disables Read-Write access to the shared metadata section, restoring Read-Only access. Available when CONFIG_TFM_PARTITION_META_DYNAMIC_ISOLATION is set to 1.

Return Values

- void - None

Note

This API is only available if CONFIG_TFM_PARTITION_META_DYNAMIC_ISOLATION is enabled.

Log API

The log API is used by the *TF-M log system*. The log device could be uart, memory, usb, etc.

APIs

`tfm_hal_output_partition_log()`

Prototype

```
int32_t tfm_hal_output_partition_log(const unsigned char *str, uint32_t len)
```

Description

This API is called by Secure Partition to output logs.

Parameter

- `str` - The string to output.
- `len` - Length of the string in bytes.

Return Values

- Positive values - Number of bytes output.
- `TFM_HAL_ERROR_NOT_INIT` - The log device has not been initialized.
- `TFM_HAL_ERROR_INVALID_INPUT` - Invalid inputs when `str` is NULL or `len` is zero.

Note

None.

`tfm_hal_output_spm_log()`

Prototype

```
int32_t tfm_hal_output_spm_log(const unsigned char *str, uint32_t len)
```

Description

This API is called by *SPM* to output logs.

Parameter

- `str` - The string to output.
- `len` - Length of the string in bytes.

Return Values

- Positive numbers - Number of bytes output.
- `TFM_HAL_ERROR_NOT_INIT` - The log device has not been initialized.
- `TFM_HAL_ERROR_INVALID_INPUT` - Invalid inputs when `str` is NULL or `len` is zero.

Note

Please check *TF-M log system* for more information.

Interrupt APIs

The SPM HAL interrupt APIs are intended for operations on Interrupt Controllers of platforms.

APIs for control registers of interrupt sources are not in the scope of this set of APIs. Secure Partitions should define the APIs for managing interrupts with those MMIO registers.

APIs

`tfm_hal_irq_enable()`

Prototype

```
enum tfm_hal_status_t tfm_hal_irq_enable(uint32_t irq_num)
```

Description

This API enables an interrupt from the Interrupt Controller of the platform.

Parameter

- `irq_num` - the interrupt to be enabled with a number

Return Values

- `TFM_HAL_ERROR_INVALID_INPUT` - the `irq_num` exceeds The maximum supported number of external interrupts.
- `TFM_HAL_ERROR_GENERIC` - failed to enable the interrupt.
- `TFM_HAL_SUCCESS` - the interrupt is successfully enabled.

`tfm_hal_irq_disable()`

Prototype

```
enum tfm_hal_status_t tfm_hal_irq_disable(uint32_t irq_num)
```

Description

This API disables an interrupt from the Interrupt Controller of the platform.

Parameter

- `irq_num` - the interrupt to be disabled with a number

Return Values

- `TFM_HAL_ERROR_INVALID_INPUT` - the `irq_num` exceeds The maximum supported number of external interrupts.
- `TFM_HAL_ERROR_GENERIC` - failed to disable the interrupt.
- `TFM_HAL_SUCCESS` - the interrupt is successfully disabled.

`tfm_hal_irq_clear_pending()`

Prototype

```
enum tfm_hal_status_t tfm_hal_irq_clear_pending(uint32_t irq_num)
```

Description

This API clears an active and pending interrupt.

Parameter

- `irq_num` - the interrupt to be disabled with a number

Return Values

- `TFM_HAL_ERROR_INVALID_INPUT` - the `irq_num` exceeds The maximum supported number of external interrupts.
- `TFM_HAL_ERROR_GENERIC` - failed to clear the pending interrupt.
- `TFM_HAL_SUCCESS` - the pending interrupt has been cleared.

Initializations

Prototype

```
enum tfm_hal_status_t {source_symbol}_init(void *p_pt,  
                                           const struct irq_load_info_t *p_ildi)
```

The `{source_symbol}` is:

- `irq_{source}`, if the source attribute of the IRQ in Partition manifest is a number
- Lowercase of source attribute, if source is a symbolic name

Description

Each interrupt has an initialization function individually. The *SPM* calls the functions while loading the Partitions.

The following initializations are required for each interrupt:

- Setting the priority. The value must between 0 to 0x80 exclusively.
- Ensuring that the interrupt targets the Secure State.
- Saving the input parameters for future use.

Platforms can have extra initializations as well.

Parameter

- `p_pt` - pointer to Partition runtime struct of the owner Partition
- `p_ildi` - pointer to `irq_load_info_t` struct of the interrupt

Note

Please refer to the :doc: *IRQ integration guide*<tfm_secure_irq_integration_guide> for more information.

API Definition for Secure Partitions

The Secure Partition (SP) *HAL* mainly focuses on two parts:

- Peripheral devices. The peripherals accessed by the *TF-M* default Secure Partitions.
- Secure Partition abstraction support. The Secure Partition data that must be provided by the platform.

The Secure Partition abstraction support will be introduced in the peripheral API definition.

ITS and PS flash API

There are two kinds of flash:

- Internal flash. Accessed by the PSA Internal Trusted Storage (ITS) service.
- External flash. Accessed by the PSA Protected Storage (PS) service.

The ITS HAL for the internal flash device is defined in the `tfm_hal_its.h` header and the PS HAL in the `tfm_hal_ps.h` header.

Macros

Internal Trusted Storage

TFM_HAL_ITS_FLASH_DRIVER

Defines the identifier of the CMSIS Flash `ARM_DRIVER_FLASH` object to use for ITS. It must have been allocated by the platform and will be declared extern in the HAL header.

TFM_HAL_ITS_PROGRAM_UNIT

Defines the size of the ITS flash device's physical program unit (the smallest unit of data that can be individually programmed to flash). It must be equal to `TFM_HAL_ITS_FLASH_DRIVER.GetInfo()->program_unit`, but made available at compile time so that filesystem structures can be statically sized.

TFM_HAL_ITS_FLASH_AREA_ADDR

Defines the base address of the dedicated flash area for ITS.

TFM_HAL_ITS_FLASH_AREA_SIZE

Defines the size of the dedicated flash area for ITS in bytes.

TFM_HAL_ITS_SECTORS_PER_BLOCK

Defines the number of contiguous physical flash erase sectors that form a logical filesystem erase block.

Protected Storage

TFM_HAL_PS_FLASH_DRIVER

Defines the identifier of the CMSIS Flash `ARM_DRIVER_FLASH` object to use for PS. It must have been allocated by the platform and will be declared extern in the HAL header.

TFM_HAL_PS_PROGRAM_UNIT

Defines the size of the PS flash device's physical program unit (the smallest unit of data that can be individually programmed to flash). It must be equal to `TFM_HAL_PS_FLASH_DRIVER.GetInfo()->program_unit`, but made available at compile time so that filesystem structures can be statically sized.

TFM_HAL_PS_FLASH_AREA_ADDR

Defines the base address of the dedicated flash area for PS.

TFM_HAL_PS_FLASH_AREA_SIZE

Defines the size of the dedicated flash area for PS in bytes.

TFM_HAL_PS_SECTORS_PER_BLOCK

Defines the number of contiguous physical flash erase sectors that form a logical filesystem erase block.

Optional definitions

The `TFM_HAL_ITS_FLASH_AREA_ADDR`, `TFM_HAL_ITS_FLASH_AREA_SIZE` and `TFM_HAL_ITS_SECTORS_PER_BLOCK` definitions are optional. If not defined, the platform must implement `tfm_hal_its_fs_info()` instead.

Equivalently, `tfm_hal_its_ps_info()` must be implemented by the platform if `TFM_HAL_ITS_FLASH_AREA_ADDR`, `TFM_HAL_ITS_FLASH_AREA_SIZE` or `TFM_HAL_ITS_SECTORS_PER_BLOCK` is not defined.

Objects

ARM_DRIVER_FLASH

The ITS and PS HAL headers each expose a CMSIS Flash Driver instance.

```
extern ARM_DRIVER_FLASH TFM_HAL_ITS_FLASH_DRIVER
extern ARM_DRIVER_FLASH TFM_HAL_PS_FLASH_DRIVER
```

The CMSIS Flash Driver provides the flash primitives `ReadData()`, `ProgramData()` and `EraseSector()` as well as `GetInfo()` to access flash device properties such as the sector size.

Types

tfm_hal_its_fs_info_t

Struct containing information required from the platform at runtime to configure the ITS filesystem.

```
struct tfm_hal_its_fs_info_t {
    uint32_t flash_area_addr;
    size_t flash_area_size;
    uint8_t sectors_per_block;
};
```

Each attribute is described below:

- `flash_area_addr` - Location of the block of flash to use for ITS
- `flash_area_size` - Number of bytes of flash to use for ITS
- `sectors_per_block` - Number of erase sectors per logical FS block

tfm_hal_ps_fs_info_t

Struct containing information required from the platform at runtime to configure the PS filesystem.

```
struct tfm_hal_ps_fs_info_t {
    uint32_t flash_area_addr;
```

(continues on next page)

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```
size_t flash_area_size;
uint8_t sectors_per_block;
};
```

Each attribute is described below:

- `flash_area_addr` - Location of the block of flash to use for PS
- `flash_area_size` - Number of bytes of flash to use for PS
- `sectors_per_block` - Number of erase sectors per logical FS block

Functions

`tfm_hal_its_fs_info()`

Prototype

```
enum tfm_hal_status_t tfm_hal_its_fs_info(struct tfm_hal_its_fs_info_t *fs_info);
```

Description

Retrieves the filesystem configuration parameters for ITS.

Parameter

- `fs_info` - Filesystem config information

Return values

- `TFM_HAL_SUCCESS` - The operation completed successfully
- `TFM_HAL_ERROR_INVALID_INPUT` - Invalid parameter

Note

This function should ensure that the values returned do not result in a security compromise. The block of flash supplied must meet the security requirements of Internal Trusted Storage.

`tfm_hal_ps_fs_info()`

Prototype

```
enum tfm_hal_status_t tfm_hal_ps_fs_info(struct tfm_hal_ps_fs_info_t *fs_info);
```

Description

Retrieves the filesystem configuration parameters for PS.

Parameter

- `fs_info` - Filesystem config information

Return values

- `TFM_HAL_SUCCESS` - The operation completed successfully
- `TFM_HAL_ERROR_INVALID_INPUT` - Invalid parameter

Note

This function should ensure that the values returned do not result in a security compromise.

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10.3.3 Cooperative Scheduling Rules

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TF-M Scheduler - Rules

On ArmV8-M CPUs, NSPE and SPE share the same physical processing element(PE). A TF-M enabled systems need to be able to handle asynchronous events (interrupts) regardless of current security state of the PE, and that may lead to scheduling decisions. This introduces significant complexity into TF-M. To keep the integrity of (NSPE and SPE) schedulers and call paths between NSPE and SPE, following set of rules are imposed on the TF-M scheduler design.

Objectives and Requirements

1. Decoupling of scheduling decisions between NSPE and SPE
2. Efficient interrupt handling by SPE as well as NSPE
3. Reduce critical sections on the secure side to not block interrupts unnecessarily
4. Scalability to allow simplification/reduction of overheads to scale down the design for constrained devices
5. Reduce impact on NSPE software design
 - a. NSPE interrupt handling implementation should be independent
 - b. Impact on the NSPE scheduler should be minimal
 - c. No assumptions should be made about NSPE's scheduling capabilities

Scheduler Rules for context switching between SPE and NSPE

To allow coherent cooperative scheduling, following set of rules are imposed on security state changes. The switching between SPE and NSPE can be triggered in multiple ways.

Involuntary security state switch; when the software has no control over the switch:

- A NSPE interrupt takes control into NSPE from SPE
- A SPE interrupt takes control into SPE from NSPE

Voluntary security state switch; when software programmatically makes the switch:

- A NSPE exception handler returns from NSPE to pre-empted SPE context
- A SPE exception handler returns from SPE to pre-empted NSPE context
- NSPE makes a function call into SPE
- SPE returns a call from NSPE
- SPE makes a function call into NSPE (not covered in current design)
- NSPE returns a call from SPE (not covered in current design)

In order to maintain the call stack integrity across NSPE and SPE, following rules are imposed on all security state switches.

Rules for NSPE Exception handling

1. **The NSPE exception handler is allowed to trigger a NSPE context switch (regardless of security state of the preempted context.**

This is expected behaviour of any (RT)OS.

2. **The NSPE scheduler must eventually ‘restore’ the preempted (by exception) context.**

This is expected behaviour of any (RT)OS.

3. **If NSPE exception results in a NSPE context switch, SPM must be informed of the scheduling decision; this must be done BEFORE the execution of newly scheduled-in context.**

This is to ensure integrity of the call stack when SPE is ready to return a previous function call from NSPE.

Rules for SPE Exception handling

1. **All of the SPE interrupts must have higher priority than NSPE interrupts**

This is rule is primarily for simplifying the SPM design.

2. **The SPE interrupt handler is allowed to trigger a SPE context switch (regardless of security state of the pre-empted context)**

If the SPE context targeted by the interrupt is not same as current SPE context, the SPM may choose to switch the current running SPE context based on priority.

3. **SPE scheduler must treat pre-empted context as one of the SPE contexts**

- a. If the pre-empted SPE context is SP1, the TCB for SP1 should be used for saving the context. i.e. the context of SP1 should be saved before scheduling anything other secure partition.
- b. If SP1 was pre-empted by a NSPE interrupt, and subsequent NSPE execution is pre-empted by SPE exception (before NSPE scheduling decision is communicated back to SPM) – SP1 TCB must be used for saving the context. In this case SPM is not yet aware of the NSPE context switch, from SPM’s standpoint SP1 is still executing, so SPM assumes that the preempted context is SP1.
- c. If SP1 was pre-empted by a NSPE interrupt, and subsequent NSPE execution is pre-empted by SPE exception *after* NSPE scheduling decision is communicated back to SPM) - a TCB dedicated to NSPE should be used for saving the context.

When NSPE scheduler communicates the scheduling decision to SPM, SPM must save the SP1 context, if a SPE interrupt preempts the currently running NSPE context, SPM should save the context to a dedicated NSPE TCB.

- d. The SPE scheduler must eventually ‘restore’ the pre-empted context. This is an expected behaviour of any scheduler.

4. **All of the interrupts belonging to a partition must have same priority.**

This serializes ISR execution targeted for same partition.

5. **In case of nested interrupts, all of the ISRs must run to finish before any service code is allowed to run**

This is an expected behaviour of any scheduler.

6. **If the previously preempted context was a NSPE ISR context, SPE ISR handler must return to preempted NSPE context.**

This is an expected behaviour of any scheduler to return to preempted context.

Rules for NSPE to SPM function call (and NSPE scheduler)

1. Current NSPE context must have been communicated to SPM, otherwise SPM cannot guarantee NSPE function calling stack integrity.

Rules for Function Return from SPE to NSPE with result

1. **The result available on SPE side are for currently active NSPE context.**

To maintain call stack integrity, if SPE is ready to return to NSPE, it can do function return only if the SPE return path corresponds to currently active NSPE context.

2. **Last entry into secure world happened programmatically (Voluntary security state switch into SPE)**

i.e. control is voluntarily given back by NSPE, either through a function call, or a context restore via 'return to SPE from NSPE'. As opposed to a SPE interrupt bringing back the execution into SPE.

3. **The current NSPE call stack has not already been returned with SPM_IDLE.**

This rule applies if following optional feature is enabled.

Rules for Return from SPE to NSPE with SPM_IDLE

This is optional part of the design as it introduces significant complexity on both sides of the security boundary. It allows yielding of the CPU to NSPE when SPE has not CPU execution to do but it has not yet finished the previous request(s) from NSPE; i.e. SPE is waiting on arrival of a SPE interrupt.

1. **Last entry into secure world happens programmatically (Voluntary security context switch into SPE)**

i.e. control is voluntarily given back by NSPE, either through a function call, or a context restore via 'return to SPE from NSPE'. As opposed to a SPE interrupt bringing back the execution into SPE.

2. **The result for the currently active NSPE entity is not yet available, the called service is waiting (on interrupt/event).**

SPE request corresponding to currently active NSPE caller is not yet completed and is waiting on an ISR.

3. **The current NSPE call stack has not already been returned with SPM_IDLE.**

Rules for NSPE pend irq based return from SPE to NSPE

This is optional part of the design as it introduces significant complexity on both sides. This works in conjunction with [Rules for Return from SPE to NSPE with SPM_IDLE](#rules-for-return-from-spe-to-nspe-with-spm_idle). In this scenario, when SPE is ready with result for a previous call from NSPE, it raises a pended IRQ to NSPE instead of returning the function call path.

1. **The SPE has finished a NSPE request.**
2. **The corresponding NSPE context has already been returned with SPM_IDLE.**

Rules for ISR pre-emption

1. **A higher priority NSPE interrupt is allowed to preempt a lower priority NSPE ISR**
2. **A higher priority SPE interrupt is allowed to preempt a lower priority SPE ISR**
3. **A SPE interrupt is allowed to preempt NSPE ISR**
4. **A NSPE interrupt is not allowed to preempt SPE ISR**
5. **All interrupts belonging to a service must have same priority**

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10.3.4 Code Generation With Jinja2

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Generating files from templates in TF-M

Some of the files in TF-M are generated from template files. The files to be generated are listed in `tools/tfm_generated_file_list.yaml`. For each generated file `<path_to_file>/<filename>` the template file is `<path_to_file>/<filename>.template`. The templates are populated with partition information from the partition manifests. The manifests to be used for the generation are listed in `tools/tfm_manifest_list.yaml`.

Custom generator script - Current method

`tools/tfm_parse_manifest_list.py` Python script is used to generate files from the templates. This script calls the `tools/generate_from_template.py` to parse the template files, and uses `tools/keyword_substitution.py` to substitute the keychains with actual values from the manifest files.

Use Jinja2 template engine - proposal

The proposal is to eliminate the template parser and substituter scripts, and call the Jinja2 template engine library from `tools/tfm_parse_manifest_list.py` to do the substitution.

More information on jinja2: <https://jinja.palletsprojects.com/en/stable/>

Changes needed:

- `tools/tfm_parse_manifest_list.py` have to be modified to call the Jinja2 library instead of the custom scripts. The data structure required by the library is very similar to the one required by the current scripts.
- template files needs to be rewritten to the Jinja syntax: The control characters to be replaced (like `@! -> {%}`) and `for` statements to be added to iterate over the substitutions.

Improvements over the current solution

- Compatible with Python 2.7 and Python 3, while current solution only supports Python 2.7
- More advanced functionality: direct control over the list of items for a keychain, meta information on that list (like length)
- Well documented (see website)
- Jinja2 is free and open-source software, BSD licensed, just like TF-M. It is a well established software in contrast with the current proprietary solution.

Example

Below code snippet enumerates services in Secure Partitions:

```
{% for partition in partitions %}  
  {% if partition.manifest.services %}  
    {% for service in partition.manifest.services %}  
      {do something for the service}  
    {% endfor %}  
  {% endif %}  
{% endfor %}
```

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10.3.5 Fixing implicit casting for C enumeration values

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Abstract

C enumerations provide a nice way to increase readability by creating new enumerated types but the developer should take extra care when mixing enumeration and integer values. This document investigates C enumerations safety and proposes strategies on how to fix the implicit casting of the enumeration values of TF-M with other types.

C99 Standard point of view

In TF-M many implicit casts are done between integer types (`uint32_t`, `int32_t`, `size_t`, etc), enumerated types (`enum foobar`) and enumeration constants (`FOOBAR_ENUM_1`).

According to the C99 standard¹:

§6.2.5, 16: An enumeration comprises a set of named integer constant values. Each distinct enumeration constitutes a different numerated type.

§6.7.2.2, 2: The expression that defines the value of an enumeration constant shall be an integer constant expression that has a value representable as an `int`.

§6.7.2.2, 3: The identifiers in an enumerator list are declared as constants that have type `int` and may appear wherever such are permitted.

§6.7.2.2, 4: Each enumerated type shall be compatible with `char`, a signed integer type, or an unsigned integer type. The choice of type is implementation-defined, but shall be capable of representing the values of all the members of the enumeration.

From these four quotes from the C99 standard², the following conclusions can be made:

- an enumeration defines a new type and should be treated as such
- the enumeration constants must only contains value representable as an `int`
- the enumeration constants have type `int`
- the actual type of the enumeration can be between `char`, signed and unsigned `int`. The compiler chooses the type it wants among those that can represent all declared constants of the enumeration.

¹ C99 standard: <http://www.open-std.org/jtc1/sc22/WG14/www/docs/n1256.pdf>

Example:

```
enum french_cities {
    MARSEILLE,
    PARIS,
    LILLE,
    LYON
};
```

In that example, `MARSEILLE`, `PARIS`, `LILLE` and `LYON` are enumeration constants of type `int` and `enum french_cities` is a enumerated type which can be of actual type `char`, `unsigned int` or `int` (the compiler chooses!).

For these reasons, doing an implicit cast between an enumeration and another type is the same as doing an implicit cast between two different types. From a defensive programming point of view, it should be checked that the destination type can represent the values from the origin type. In this specific case it means four things for enumerations:

- it is always safe to assign an enumeration constant to an `int`, but might be better to cast to show intent.
- when casting an enumeration constant to another type, it should be checked that the constant can fit into the destination type.
- when casting from an integer type (`uint32_t`, `int32_t`, etc) to an enumeration type, it should be checked that the integer's value is one of the enumeration constants. The comparison needs to be done on the biggest type of the two so that no information is lost. C integer promotion should automatically do that for the programmer (check §6.3.1.8, 1 for the rules).
- when casting from an enumeration type to an integer type, it should be checked that the enumeration type value fits into the integer type. The value of a variable which has the type of an enumeration type is not limited to the enumeration constants of the type. An enumeration constant will always fit into an `int`.

Strategies to fix

0. Replace the enumerated type with an integer type and replace the enumeration constant with preprocessor constants.
1. Whenever possible, try to use matching types to avoid implicit casting. It happens, for example, for arithmetic operations, function calls and function returns. This strategy always have the lowest performance impact.
2. When using an enumeration constant in an arithmetic operation with another type, verify that the constant can fit in the other type and cast it.
3. When converting an integer to an enumeration type, use a conversion function to check if the integer matches an enumeration constant. To not impact performance too much, this function should be an inline function. If it does not match, use (or add) the error constant or return an error value.
4. When converting an enumeration type to an integer, use a conversion function to check that the integer type can contain the enumeration value.

Design proposal for TF-M

In TF-M, an action will be taken for all enumerated types and enumeration constants that are used for implicit casting. The goal of this proposal is to remove all implicit casting of enumeration values in TF-M.

The following enumerated types will be removed and replaced with preprocessor constants (strategy 0). These enumerated types are not used in TF-M but only the constants they declare.

- `enum spm_part_state_t`
- `enum spm_part_flag_mask_t`

- `enum tfm_partition_priority`

The following enumerated types will be kept because they are used in the prototypes of functions and are useful for debugging. Whenever possible, strategy 1 will be applied to remove implicit casting related with those enumerations but dynamic conversions will be used if the first option would create too much change in the code base.

- `enum tfm_status_e`: the return type of the following functions will be changed to return the `enum tfm_status_e` type. These functions are already returning the enumeration constants, but implicitly casted to an integer type like `int32_t`.
 - `int32_t check_address_range`
 - `int32_t has_access_to_region`
 - `int32_t tfm_core_check_sfn_parameters`
 - `int32_t tfm_start_partition`
 - `int32_t tfm_return_from_partition`
 - `int32_t tfm_check_sfn_req_integrity`
 - `int32_t tfm_core_check_sfn_req_rules`
 - `int32_t tfm_spm_sfn_request_handler`
 - `int32_t tfm_spm_sfn_request_thread_mode`
- `enum tfm_buffer_share_region_e`: the following function prototypes will be changed:
 - `tfm_spm_partition_set_share(uint32_t partition_idx, uint32_t share)` → `tfm_spm_partition_set_share(uint32_t partition_idx, enum tfm_buffer_share_region_e share)`
- `enum tfm_memory_access_e`
- `enum attest_memory_access_t`
- `enum engine_cipher_mode_t`
- `mbedtls_cipher_type_t`

The following enumerated types are used for error code values of Secure service calls. They should be kept as they are part of the interface and might be used by external parties in Non-Secure code. For the Initial Attestation service, the enumeration is defined in the PSA Attestation API specifications.

- `enum psa_attest_err_t`
- `enum psa_audit_err`
- `enum tfm_platform_err_t`

Implicit casting related with these enumerations is happening in two locations of TF-M and need conversion functions in those locations, because the types can not be changed:

- In the Non-Secure Client library, all of the Secure Service functions implicitly cast the `uint32_t` returned by `tfm_ns_lock_dispatch` to these enumerated types. Strategy 3 is needed here.
- In all of the veneer functions, there is an implicit cast from the `int32_t` value returned by the SVC request function (`tfm_core_*_request`) to these enumerated types. Strategy 3 is needed here as well. The implicit cast will eventually be removed if all of the services are using the Uniform Signatures Prototypes so that the veneer functions all return `psa_status_t` which is an `int32_t`.

If the interface of those services can be changed, these enumerations could be removed and replaced with the `psa_status_t` type to remove the implicit casting.

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10.4 FF-M Isolation

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This document analyzes the isolation rules of implementing Firmware Framework for M 1.1 (FF-M 1.1)² isolation and introduces the reference implementation in TF-M, which complies with the rules by operating the hardware and software resources.

Note

Reference the document *Glossary* for terms and abbreviations.

10.4.1 Introduction

This chapter describes the definitions from Firmware Framework for M (FF-M)¹ and analyzes the possible implementation keypoints.

Isolation Levels

There are 3 isolation levels (1-3) defined in FF-M, the greater level number has more isolation boundaries.

The definition for Isolation Level 1:

- L1.1 NPSE needs protection from nobody.
- L1.2 SPE needs protection from NSPE.

The definition for Isolation Level 2:

- L2.1 NPSE needs protection from nobody.
- L2.2 Application Root of Trust (ARoT) needs protection from NSPE.
- L2.3 PSA Root of Trust (PRoT) needs protection from NSPE and ARoT.

The definition for Isolation Level 3:

- L3.1 NPSE needs protection from nobody.
- L3.2 Secure Partition needs protection from NSPE and other Secure Partitions.
- L3.3 PSA Root of Trust (RoT) domain needs protection from NSPE and all Secure Partitions.

² Arm Firmware Framework for M 1.1 Extension (FF-M 1.1)

¹ Arm PSA Firmware Framework for M 1.0 (FF-M 1.0)

Important

A Secure Partition RoT Service is a Root of Trust Service implemented within a Secure Partition. An Application RoT Service must be implemented as a Secure Partition RoT Service. But it is implementation-defined whether a PSA RoT Service is a Secure Partition RoT Service.

Here listed several possible PSA RoT Service implementation mechanisms:

1. Implement PSA RoT Services in Secure Partitions with respective boundaries.
2. Implement PSA RoT Services in Secure Partitions, but no boundaries between these Secure Partitions as they are in the PSA RoT Domain.
3. Implement PSA RoT Services in a customized way instead of Secure Partitions, an internal library of PSA RoT domain e.g.

TF-M chooses the 2nd option to balance performance and complexity.

Isolation Rules

The essence of isolation is to protect the assets of one protection domain from being accessed from other domains. The isolation levels define where the isolation boundaries should be placed, the isolation rules define the strength of the isolation the boundaries should offer.

Note

Refer to chapter *Memory Assets* in FF-M⁷ to know asset class items. Assets are represented by memory addresses in the system memory map, which makes assets named *Memory Assets*. The often-seen asset items are ROM, RAM, and memory-mapped peripherals.

Memory Asset Class

There are 3 memory asset classes defined in FF-M:

- Code
- Constant data
- Private data

There are 7 isolation rules for protecting assets described in chapter 3.1 of FF-M 1.0⁷ and chapters 8.4 to 8.5 of FF-M 1.1⁷. The following is a summary of the rules:

- I1. (Changed in FF-M 1.1) Private data is not executable.
- I2. Only private data is writable.
- I3. If domain A needs protection from domain B, then Private data in domain A cannot be accessed by domain B.
- I4. (Optional, changed in FF-M 1.1) If domain A needs protection from domain B, then Code and Constant data in domain A, which is not part of a shared library, is not readable or executable by domain B.
- I5. (Optional, changed in FF-M 1.1) Code in a domain, which is not part of a shared library, is not executable by any other domain.
- I6. (Optional) All assets in a domain are private to that domain and cannot be accessed by any other domain, with the following exception: The domain containing the SPM can only access Private data and Constant data assets of other domains when required to implement the PSA Firmware Framework API.

- I7. (Optional, added in FF-M 1.1) Constant data is not executable.

The first 3 rules from I1 to I3 define the mandatory rules to comply with the PSA security model, while I4 to I7 are optional rules to enhance the isolation boundaries.

Important

There is a table in chapter 3.1.2 of FF-M 1.0² under I1 that lists the asset types and allowed access method. Preventing executable access to constant data costs more hardware resources, so there is an optional rule I7 created in FF-M 1.1² to aid implementations with constrained hardware resources.

Hardware Infrastructure

To implement a secure system, the hardware security framework (e.g. TrustZone or multiple-core) and their auxiliary components (e.g. SAU) are required to ensure the isolation between SPE and NSPE, as described in Trusted Base System Architecture for M (TBSA-M)³.

Important

The interface between secure and non-secure states needs to be fully enumerated and audited to prove the integrity of the secure state isolation.

Besides this SPE and NSPE isolation mechanism, the following analyzes the implementation rules to find out the hardware requirements for isolation inside SPE domains:

- I1, I2 and I7: The assets can be categorized into 3 *Memory Asset Class*, each type has the specific access rules.
- I3: The private data access from the prevented domain needs to be blocked.
- I4: All the assets access from the prevented domain needs to be blocked.
- I5: Code execution from all other domains (even the domain not prevented from) needs to be blocked.
- I6: All the assets access from all other domains (includes non-prevented domain) needs to be blocked, but, SPM is an exception, which can access the private data and constant data of the current domain.

The above items list the requirements for memory access, here are two more points:

- If the memory device or the peripheral are shared between multiple hosts (such as multiple CPU or DMA, etc.), specific hardware protection units need to be available for validating accesses to that device or peripheral.
- The MMIO range for Secure Partitions is not allowed to be overlapped, which means each partition should have exclusive memory-mapped region if they require a peripheral device. The memory-mapped region is regarded as the private data so access to this area needs to be validated.

10.4.2 Reference Implementation

This chapter describes the isolation implementation inside SPE by using the Armv8-M architecture component - Memory Protection Unit (MPU). The MPU can isolate CPU execution and data access.

Note

Previous version M-profile architecture MPU setting is similar in concept but the difference in practical register formats, which is not described in this document.

³ Arm Trusted Base System Architecture for M (TBSA-M)

The MPU protects memory assets by regions. Each region represents a memory range with specific access attributes.

Note

The maximum numbers of MPU regions are platform-specific.

The SPM is running under the privileged mode for handling access from services. The MPU region for SPM needs to be available all the time since SPM controls the MPU setting while scheduling.

Since partitions are scheduled by SPM, the MPU regions corresponding to the partitions can be configured dynamically while scheduling. Since there is only one running at a time and all others are deactivated, the SPM needs to set up necessary regions for each asset type in one partition only.

There is reusable code like the C-Runtime and RoT Service API which are same across different partitions. TF-M creates a Secure Partition Runtime Library (SPRTL) as a specific library shared by the Secure Partition. Please refer to *Secure Partition Runtime Library* for more detail.

Note

Enable SPRTL makes it hard to comply with the rules I4, I5 and I6, duplicating the library code can be one solution but it is not “shared” library anymore.

As mentioned in the last chapter, MMIO needs extra MPU regions as private data.

MPU Region Access Permission

The following content describes the memory access permission to represent the corresponding asset classes.

These access permissions are available on Armv8-M MPU:

- Privileged Read-Only (RO)
- All RO
- Privileged Read-Write (RW)
- All RW
- Execute Never (XN)

And one more Armv8.1-M access permission:

- Privileged Execute Never (PXN)

The available regions type list:

Type	Attributes	Privilege Level	Asset
P_RO	RO	Privileged	PRoT Code
P_ROXN	RO + XN	Privileged	PRoT Constant Data
P_RWXN	RW + XN	Privileged	PRoT Private Data/Peripheral
A_RO	RO	Any privilege	Partition/SPRTL Code
A_ROXN	RO + XN	Any privilege	Partition/SPRTL Constant Data
A_RWXN	RW + XN	Any privilege	Partition/SPRTL Private Data/Peripheral
A_ROPXN	RO + PXN	Any privilege	Armv8.1-M Partition Code

Example Image Layout

The secure firmware image contains components such as partitions, SPM and the shared code and data. Each component may have different class assets. There would be advantages if placing the assets from all components with the same access attributes into one same region:

- The data relocating or clearing when booting can be done in one step instead of breaking into fragments.
- Assets with statically assigned access attribute can share the same MPU region which saves regions.

Take the TF-M existing implementation image layout as an example:

Level 1 Boundaries	Level 2 Boundaries	Level 3 Boundaries	
Code (ROM)	PRoT	PRoT SPM	Code
	Code	PRoT Service	Code
		Partition 1	Code
	ARoT	Partition N	Code
	Code	SPRTL	Code
Check [4] for more details between Code and Constant Data.			
Constant Data (ROM)	PRoT	PRoT SPM	Constant Data
	Constant	PRoT Service	Constant Data
	Data	Partition 1	Constant Data
	ARoT	Partition N	Constant Data
	Constant	SPRTL	Constant Data
	Data		
Private Data (RAM)	PRoT	PRoT SPM	Private Data
	Private	PRoT Service	Private Data
	Data	Partition 1	Private Data
	ARoT	Partition N	Private Data
	Private	SPRTL	Private Data
	Data		

Note

1. Multiple binaries image implementation could also reference this layout if its hardware protection unit can cover the exact boundaries mentioned above.

- Private data includes both initialized and zero-initialized (ZI) sections. Check chapter 3.1.1 of FF-M² for the details.
- This diagram shows the boundaries but not orders. The order of regions inside one upper region can be adjusted freely.
- As described in the important of *Memory Asset Class*, the setting between Code and Constant Data can be skipped if the executable access method is not applied to constant data. In this case, the groups of Code and Constant Data can be combined or even mixed – but the boundary between PRoT and ARoT are still required under level higher than 1.

Example Region Numbers under Isolation Level 3

The following table lists the required regions to comply with the rules for implementing isolation level 3. The level 1 and level 2 can be exported by simplifying the items in level 3 table.

Important

The table described below is trying to be shared between all supported platforms in Trusted Firmware-M. It is obvious that some platforms have special characteristics. In that case, the specific layout table for a particular platform can be totally redesigned but need to fulfil the isolation level requirements.

- Only the running partition's assets are covered since the inactive partitions do not need regions.
- X indicates the existence of this region can't comply with the rule.
- An ATTR + n represent extra n regions are necessary.
- The table assumes each rule is in addition to the rules in previous columns (but the FF-M permits any combination of optional rules).

The following table lists the required regions to comply with the rules:

Region Purpose	I1 I2 I3 I7	I4	I5	I6
PRoT SPM Code	A_RO	P_RO	P_RO	P_RO
PRoT Service Code				A_ROPXN
Active Partition Code		A_RO	A_ROPXN	
SPRTL Code		A_RO	A_RO	X
PRoT SPM RO	A_ROXN	P_ROXN	P_ROXN	P_ROXN
PRoT Service RO				A_ROXN
Active Partition RO		A_ROXN	A_ROXN	
SPRTL RO		A_ROXN	A_ROXN	X
PRoT SPM RW	P_RWXN	P_RWXN	P_RWXN	P_RWXN
PRoT Service RW				A_RWXN
Active Partition RW	A_RWXN	A_RWXN	A_RWXN	
SPRTL RW [5]	A_RWXN + 1	A_RWXN + 1	A_RWXN + 1	X
Partition Periph	A_RWXN + n	A_RWXN + n	A_RWXN + n	A_RWXN + n
Total Numbers	[1]	[2]	[3]	[4]

Note

- Total number = A_RO + A_ROXN + P_RWXN + (2 + n)A_RWXN = 5 + n, where n is the maximum number of peripherals needed by one partition.

2. Total number = $[1] + P_RO + A_RO + P_ROXN + A_ROXN = 9 + n$ (or $6 + n$ without SPRTL).
3. Total number = $[2] = 9 + n$, and Armv8.1-M PXN is required.
4. Total number = $P_RO + A_ROPXN + P_ROXN + A_ROXN + P_RWXN + (1 + n)A_RWXN = 6 + n$, but SPRTL cannot be used, and Armv8.1-M PXN is required. To comply with this rule, the PSA RoT Services need to be implemented as Secure Partitions.
5. This data belongs to SPRTL RW but it is set as Read-Only and only SPM can update this region with the active partition's metadata for implementing functions with owner SP's context, such as heap functions. This region can be skipped if there is no metadata required (such as no heap functionalities required).

The memory-mapped regions for peripherals have different memory access attributes in general, they are standalone regions in the MPU even though their attributes covers 'A_RWXN'.

TF-M provides the reference linker scripts `tfm_isolation_s.sct.template`, `tfm_isolation_s.ld.template`, and `tfm_isolation_s.icf.template` for the Armclang, GCC, and IAR compilers in the `platform/ext/common` directory. These linker scripts arrange the SPE's memory by attribute to aid the implementation of the isolation rules.

The reference isolation HAL for Armv8-M and Armv8.1-M platforms in `platform/ext/common/tfm_hal_isolation_v8m.c` uses the memory position tags defined by the linker scripts to set up the MPU region base and limit addresses. It implements isolation rules I1, I2, I3 and I7 for isolation levels 1 to 3. It partially implements rule I4 by statically partitioning code into privileged and unprivileged read-only regions but collecting all constant data into a single unprivileged, read-only, execute-never region, which provides a good level of protection against ROP attacks by limiting the amount of executable memory. On Armv8.1-M targets, it statically configures all Application RoT Partition code as PXN, which is sufficient to satisfy rule I5 in isolation level 2, but not in isolation level 3, where it would require dynamically mapping only the active partition's code region.

TF-M platforms may replace the reference linker scripts and/or isolation HAL with a customized implementation, but it must meet at least isolation rules I1, I2 and I3 to comply with the PSA security model.

Default access rules

Hardware protection components MAY have the capability to collect regions not explicitly configured in static or runtime settings, and then apply default access rules to these regions. Furthermore, one default rule can be applied to multiple non-contiguous regions which makes them share a common boundary. This operation sets up a standalone 'region' the same as other explicitly configured regions. And it doesn't affect the analysis summary above - just be aware that some regions listed in the table MAY not be explicitly configured.

Take the MPU as an example, MPU can assign a default privileged access attribute to the regions (e.g. SPM and PRoT regions) not explicitly configured. This feature can reduce required MPU regions and ease the programming because regions can be put non-address-contiguous and skip the explicit configuration.

Important

When this default access rules mechanism is applied, the implicitly configured regions must be reviewed to ensure the isolation boundaries are set correctly.

Interfaces

The isolation implementation is based on the HAL framework. The SPM relies on the HAL API to perform the necessary isolation related operations.

The requirement the software need to do are these:

- Create enough isolation protection at the early stage of system booting, just need to focus on the SPM domain.

- Create an isolation domain between secure and non-secure before the jump to the non-secure world.
- Create an isolation domain for each Secure Partition after the Secure Partition is loaded and before jumping to its entry point. The isolation domain should cover all the assets of the Secure Partition, include all its memory, interrupts, and peripherals.
- Switch isolation domains when scheduling different Secure Partitions.
- It is also a requirement that the platform needs to help to check if the caller of the PSA APIs is permitted to access some memory ranges.

The design document *TF-M Hardware Abstraction Layer* gives a detail design, include the platform initialization, isolation interfaces. Please refer to it for more detail.

10.4.3 References

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10.5 TF-M builtin keys

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10.5.1 Introduction

When TF-M is integrated on a platform, the platform itself can provide several keys for usage which are bound to the device instead of being owned by a specific secure partition. When those keys are readable by the secure processing environment, the platform must provide a function to load these keys in the HAL layer, either loading them from OTP or another platform specific location or subsystem. These keys are henceforth referred to as “builtin keys”, and might include (but are not limited to) the following:

1. The Hardware Unique Key (HUK)
2. The Initial Attestation Key (IAK)

The `tfm_builtin_key_loader` component implements a mechanism to discover those keys and make them available to the PSA Crypto APIs for usage. Note that if a key is stored in a subsystem which can't be read by the secure processing environment, a full opaque driver¹ must be used to be able to use those keys through the PSA Crypto APIs. This document focuses only on the case where the keys can be read by the SPE (i.e. running TF-M), so the driver applies only on those cases, which can be considered as “_transparent_ builtin keys”.

In TF-M's legacy solution, the IAK is loaded by the attestation service as a volatile key, which requires some key-loading logic to be implemented by that partition. The HUK is not loaded in the crypto service, and is instead used by an implementation of a TF-M specific KDF algorithm which then loads the key and invokes TF-PSA-Crypto directly. Both solutions are far from ideal as they require an effort to load (and duplicate) the keys per each user, or require a dedicated code in TF-M that directly interacts with the HAL layer and invokes functions from the crypto library bypassing the PSA Crypto interface. The aim of the `tfm_builtin_key_loader` driver is to provide a uniform interface to use the builtin keys available in a platform.

¹ PSA cryptoprocessor driver interface: <https://github.com/Mbed-TLS/TF-PSA-Crypto/blob/development/docs/proposed/psa-driver-interface.md>

Implementing a driver to deal with builtin keys allows to expand the legacy solution for dealing with this type of keys in several ways. For example, it avoids the need for partitions to implement dedicated mechanisms for probing the HAL layer for builtin keys and load them as volatile. It removes the need to have implementation specific call flows for deriving keys from the HUK (as that requirement now is fulfilled by the driver itself transparently). It allows uniform access to the keys also from the NS world (in any case, subject to the policy dictated by the platform). Correctly abstracts away details of the key handling mechanism from TF-M partitions into the PSA Crypto core key management subsystem.

10.5.2 PSA builtin keys

The PSA Cryptographic API provides a mechanism for accessing keys that are stored in platform-specific locations (often hardware accelerators or OTP). A builtin key is assigned a specific `key_id`, i.e. a handle, which is hardcoded at build time and must be selected in the range `[MBEDTLS_PSA_KEY_ID_BUILTIN_MIN, MBEDTLS_PSA_KEY_ID_BUILTIN_MAX]`. A user of the PSA Crypto API can reference those keys directly by using these handles. It is up to the platform to specify policy restrictions for specific users of the keys, e.g. an NS entity, or a secure partition. The PSA Crypto core will then check those policies to grant or deny access to the builtin key for that user.

10.5.3 PSA cryptoprocessor driver API

The PSA specification allows the PSA Crypto APIs to defer their operation to an accelerator driver, through a mechanism described in the PSA cryptoprocessor driver interface specification². This specification defines the concept of PSA Crypto core, i.e. “An implementation of the PSA Cryptography API is composed of a core and zero or more drivers”. The PSA specification also has the concept of storage locations for keys, through the type `psa_key_location_t`, which is used to access keys that don’t have local storage². This is leveraged mainly by opaque drivers mainly that use keys for which the key material is not readable by the PSA crypto core layer.

TF-M defines a software driver called `tfm_builtin_key_loader` that provides no acceleration but just defines a dedicated key location defined through the `TFM_BUILTIN_KEY_LOADER_KEY_LOCATION` define. By resorting to the entry points provided by this driver, the PSA Crypto core slot management subsystem can access keys stored in the underlying platform, validate key usage policies and allow PSA Crypto APIs to uniformly access builtin keys using the same call flows used with traditional local-storage keys.

This is implemented by hooking the entry points defined by the driver into the `library/psa_crypto_driver_wrappers.c` file provided by the PSA Crypto core. This is currently done manually but eventually could be just autogenerated by parsing a description of the driver entry points in the JSON format. These entry points are:

1. `tfm_builtin_key_loader_init`
2. `tfm_builtin_key_loader_get_key_buffer_size`
3. `tfm_builtin_key_loader_get_builtin_key`

The call flow for the entry points from the driver wrapper layer is as follows:

1. During the driver initialisation phase started by the PSA Crypto Core init, the `tfm_builtin_key_loader_init` function is called to probe the platform HAL and retrieve the builtin keys. Those keys are described by the types defined in the HAL layer interface header `tfm_plat_crypto_keys.h`. In particular global tables describing key properties and user policies must be implemented by each platform and are retrieved by the two accessor functions `tfm_plat_builtin_key_get_desc_table_ptr()` and `tfm_plat_builtin_key_get_policy_table_ptr()`. The keys are loaded from the platform in secure RAM in the TF-M Crypto partition with associated metadata. It’s worth to note that the keys are loaded through callback functions which the platform lists in the key descriptor table.

² Definition of `psa_key_location_t` type in the PSA spec: https://arm-software.github.io/psa-api/crypto/1.1/api/keys/lifetimes.html#c.psa_key_location_t

2. Once the TF-M Crypto service is initialised, at runtime it might receive a request through an API call to use one of the builtin key IDs. Those IDs are described in the `tfm_builtin_key_ids.h` header. Platforms can override the default values providing their own header.
3. When the PSA Crypto core in the `psa_get_and_lock_key_slot()` function checks that the `key_id` being requested is in the builtin range region, it probes the platform through the function `MBEDTLS_PSA_PLATFORM_GET_BUILTIN_KEY` which must be implemented by the TF-M Crypto service library abstraction layer. This function just checks the key descriptor table from the HAL to understand if such `key_id` is available in the platform and what are the corresponding slot and lifetime values associated to it. The lifetime contains the location of the key, which determines which driver is responsible for dealing with it (for the `tfm_builtin_key_loader` driver, that is `TFM_BUILTIN_KEY_LOADER_KEY_LOCATION`), while the slot value maps the key to the slot in the driver internal storage.
4. At this point, the PSA Crypto core knows that the key exists and is bound to the location associated to the driver. By calling into the driver wrappers layer is then able to retrieve the key attributes stored in the platform for that key ID, and the size required to allocate in its internal slot management system in order to load the key material in the core. This is done by calling `tfm_builtin_key_loader_get_builtin_key` just with a valid key attributes pointer (and nothing else), to retrieve the attributes. Once the attributes are available, the required size is retrieved through the driver wrapper by calling `tfm_builtin_key_loader_get_key_buffer_size`.
5. At this stage, the slot management subsystem calls again into the driver wrapper layer through `tfm_builtin_key_loader_get_builtin_key` with a valid buffer to hold the key material returned by the `tfm_builtin_key_loader` driver. When loading the key, the user requiring that `key_id` is validated by the driver code against the policies defined by the platform. If the policies match, the builtin key material and meta-data is loaded and is used like a transparent key available to the PSA Crypto core slot management subsystem.

10.5.4 Technical details

Builtin key IDs and overriding

TF-M builtin key IDs are defined in `interface/include/crypto_keys/tfm_builtin_key_ids.h` through the enum `tfm_key_id_builtin_t`. They are allocated inside the range that PSA specifies for the builtin keys, i.e. between `MBEDTLS_PSA_KEY_ID_BUILTIN_MIN` and `MBEDTLS_PSA_KEY_ID_BUILTIN_MAX`. A platform can specify extra builtin key IDs by setting the `PLATFORM_DEFAULT_CRYPTOKEYS` variable to OFF, creating the header `tfm_builtin_key_ids.h`, and specifying new keys and IDs.

Builtin key access control

A builtin key is accessible by all callers since the `key_id` associated to it is public information. Access to the keys must be mediated, which is done by matching the user requesting the `key_id` against the policies available for that user on that particular key in the policy table. If no policies are specified for a specific combination of user and `key_id`, the usage flags in the key attributes will be all set to zeros, meaning the key will be unusable for any operation for that particular user.

Multi-partition key derivation

The HUK is used for key derivation by any secure partition or NS caller that requires keys that are bound to a particular context. For example, Protected Storage derives keys uniquely for each user of the service which are used to encrypt user files. In order to provide HUK derivation to every secure partition / NS caller, it must be ensured that no service that utilises HUK derivation can derive the same key as another service (simply by using the same inputs for the KDF APIs, i.e. accessing the same base key for derivation).

This is accomplished by deriving a further “platform key” for each builtin key that has `PSA_KEY_USAGE_DERIVE` set in its attributes. These platform keys are derived from the builtin key, using the partition ID as a KDF input, and can then be used for safely for further derivations by the user, without risks to derive the same keys as other users. This is enforced directly by the `tfm_builtin_key_loader` driver.

Note

If the NS client ID feature is disabled, all NS callers share a partition ID of -1, and therefore will share a platform key and be therefore be able to derive the same keys as other NS callers.

For keys that are not exposed outside the device, this is transparent to the service that is using the key derivation, as they have no access to the builtin key material and cannot distinguish between keys derived directly from it and keys derived from the platform key. For some builtin keys, deriving platform keys is not acceptable, as the key is used outside the device (i.e. the IAK public key is used to verify attestation tokens) so the actual builtin key is used.

The decision has been taken to derive platform keys for any key that can be used for key derivation (PSA_KEY_USAGE_DERIVE), and not derive platform keys otherwise. For builtin keys that do not derive platform keys but are directly used, care must be taken with access control where multiple partitions have access to the same raw key material.

TF-PSA-Crypto transparent builtin keys

TF-PSA-Crypto does not natively support transparent builtin keys (transparent keys are keys where the key material is directly accessible by the PSA Crypto core), so some modifications had to be made. Opaque keyslots have the same basic structure as standard transparent key slots, and can be passed to the functions usually reserved for transparent keys, though this is a private implementation detail of the TF-PSA-Crypto library and is not specified in the driver interface. Therefore, the only modification required currently is to allow keys that have the location `TFM_BUILTIN_KEY_LOADER_KEY_LOCATION` to be passed to the functions that usually accept transparent keys only, i.e. with the location `PSA_KEY_LOCATION_LOCAL_STORAGE`. This is due to the fact that the standard assumption of the PSA Crypto core is that, if a driver that provides an additional location, will also provide dedicated cryptographic mechanisms to act on those keys, but this is not the case of the `tfm_builtin_key_loader`, as it just provides a mechanism to load keys (which act as a transparent key with local storage, once loaded), but TF-PSA-Crypto does not support such “transparent builtin key” concept. Note that the modifications on TF-PSA-Crypto are relying on non standard implementation details hence this particular integration can change between releases³.

10.5.5 Using Opaque PSA crypto accelerators with TF-M

For platforms which have a cryptographic accelerator which has a corresponding Opaque PSA crypto accelerator driver, the TF-M builtin key loader driver can be disabled using the `-DCRYPTO_TFM_BUILTIN_KEYS_DRIVER=OFF` cmake option. The platform can then redefine the HAL function `tfm_plat_builtin_key_get_desc_table_ptr` to point to a table where the location and slot number of the keys corresponds instead to the opaque driver. The PSA driver wrapper will then route the calls into the opaque driver, with no other changes needed. If the description table is altered but the builtin key loader driver is not disabled, it is possible to mix software builtin keys with keys stored in opaque accelerators on a per-key level. Note that because the key policy enforcement via `tfm_plat_builtin_key_get_desc_table_ptr` is currently applied by the builtin key loader driver, other opaque drivers must apply either this policy or their own policy (Though this may be changed in future).

10.5.6 References

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³ Interface for platform keys: <https://github.com/ARM-software/psa-crypto-api/issues/550> (private)

10.6 Log system design document

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10.6.1 Background

In current TF-M log system, the SPM and Secure partitions share the same log APIs and implementations. While TF-M is keep evolving, the requirements for the log system has changed:

- Log level is required for both SPM and SP sides to output message in different scenarios.
- SPM only needs simple log format such as hex and string, while SP needs rich formatting.
- Distinctions on log output between SPM and SP are required.

A new log system is needed to separate the SPM and Secure partitions and to meet their different requirements.

10.6.2 Design

To allow customizable configurations, the log interfaces are defined as macros. The macros are easy to be forwarded or even empty. When SPM trying to output message and a value, it relies on a wrapper function, and finally output the formatted message by the HAL API.

The design principles of TF-M log system:

- Configurable log levels.
- Separated SPM and SP log implementations.
- Platforms provide log HAL implementations.

SPM Log System

Level Control

Three log levels for SPM log system are defined:

- LOG_LEVEL_NONE
- LOG_LEVEL_ERROR
- LOG_LEVEL_NOTICE
- LOG_LEVEL_WARNING
- LOG_LEVEL_INFO
- LOG_LEVEL_VERBOSE

Then a macro TFM_SPM_LOG_LEVEL is defined as an indicator, it should be equal to one of the six log levels.

API Definition

The following APIs output the string with optional format specifiers. The `_RAW` forms of the macros, output the string directly, without adding a prefix identifying the log level of the print:

```
VERBOSE_RAW(...); VERBOSE(...);
INFO_RAW(...); INFO(...);
WARN(...); WARN_RAW(...);
NOTICE(...); NOTICE_RAW(...);
ERROR(...); ERROR_RAW(...);
```

Here is a table about the effective APIs with different SPM log level.

	DEBUG	INFO	ERROR	NONE
VERBOSE_RAW VERBOSE	Yes	No	No	No
INFO_RAW INFO	Yes	Yes	No	No
NOTICE_RAW NOTICE	Yes	Yes	No	No
WARN_RAW WARN	Yes	Yes	No	No
ERROR_RAW ERROR	Yes	Yes	Yes	No

Partition Log System

Partition log outputting required rich formatting in particular cases. There is a customized print inside `TFM(tfm_log_unpriv)`, and it is wrapped as macro.

Level Control

Three log levels for partition log system are defined:

- `LOG_LEVEL_NONE`
- `LOG_LEVEL_ERROR`
- `LOG_LEVEL_NOTICE`
- `LOG_LEVEL_WARNING`
- `LOG_LEVEL_INFO`
- `LOG_LEVEL_VERBOSE`

Then a macro `TFM_PARTITION_LOG_LEVEL` is defined as an indicator. It should be equal to one of the six log levels and it is an overall setting for all partitions.

Log Format

Compared to SPM, SP log API supports formatting. Similar to `printf`, these log APIs use a format outputting to output various type of data:

```
%d - decimal signed integer
%u - decimal unsigned integer
%x - hex(hexadecimal)
%s - string
```

API Definition

Define partition log APIs:

```
VERBOSE_UNPRIV_RAW(...); VERBOSE_UNPRIV(...);  
INFO_UNPRIV_RAW(...); INFO_UNPRIV(...);  
WARN_UNPRIV(...); WARN_UNPRIV_RAW(...);  
NOTICE_UNPRIV(...); NOTICE_UNPRIV_RAW(...);  
ERROR_UNPRIV(...); ERROR_UNPRIV_RAW(...);
```

Here is a table about the effective APIs with different partition log level.

	DEBUG	INFO	ERROR	NONE
VERBOSE_UNPRIV_RAW VERBOSE_UNPRIV	Yes	No	No	No
INFO_UNPRIV_RAW INFO_UNPRIV	Yes	Yes	No	No
NOTICE_UNPRIV_RAW NOTICE_UNPRIV	Yes	Yes	No	No
WARN_UNPRIV_RAW WARN_UNPRIV	Yes	Yes	No	No
ERROR_UNPRIV_RAW ERROR_UNPRIV	Yes	Yes	Yes	No

HAL API

Please refers to the HAL design document.

10.6.3 Log Devices

In most of the cases, a serial device could be used as a log device. And in other particular cases, a memory-based log device could be applied as well. These log device interfaces are abstracted into HAL APIs.

Note

It is not recommended to reuse the same HAL for both SPM and SP log outputting especially when SPM and SP run under different privileged level, which makes them have a different information confidential level. Unless:

- The SPM log outputting would be disabled as silence in the release version.

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10.7 Physical attack mitigation in Trusted Firmware-M

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10.7.1 Requirements

PSA Certified Level 3 Lightweight Protection Profile¹ requires protection against physical attacks. This includes protection against manipulation of the hardware and any data, undetected manipulation of memory contents, physical probing on the chip's surface. The RoT detects or prevents its operation outside the normal operating conditions (such as voltage, clock frequency, temperature, or external energy fields) where reliability and secure operation has not been proven or tested.

Note

Mitigation against certain level of physical attacks is a mandatory requirement for PSA Level 3 certification. The *TF-M countermeasures against physical attacks* discussed below doesn't provide mitigation against all the physical attacks considered in scope for PSA L3 certification. Please check the Protection Profile document for an exhaustive list of requirements.

10.7.2 Physical attacks

The goal of physical attacks is to alter the expected behavior of a circuit. This can be achieved by changing the device's normal operating conditions to untested operating conditions. As a result a hazard might be triggered on the circuit level, whose impact is unpredictable in advance but its effect can be observed. With frequent attempts, a weak point of the system could be identified and the attacker could gain access to the entire device. There is a wide variety of physical attacks, the following is not a comprehensive list rather just give a taste of the possibilities:

- Inject a glitch into the device power supply or clock line.
- Operate the device outside its temperature range: cool down or warm it up.
- Shoot the chip with an electromagnetic field. This can be done by passing current through a small coil close to the chip surface, no physical contact or modification of the PCB (soldering) is necessary.
- Point a laser beam on the chip surface. It could flip bits in memory or a register, but precise knowledge of chip layout and design is necessary.

The required equipment and cost of these attacks varies. There are commercial products to perform such attacks. Furthermore, they are shipped with a scripting environment, good documentation, and a lot of examples. In general, there is a ton of videos, research paper and blogs about fault injection attacks. As a result the threshold, that even non-proficient can successfully perform such attack, gets lower over time.

10.7.3 Effects of physical attacks in hardware and in software execution

The change in the behavior of the hardware and software cannot be seen in advance when performing a physical attack. On circuit-level they manifest in bit faults. These bit faults can cause varied effects in the behavior of the device micro-architecture:

- Instruction decoding pipeline is flushed.
- Altering instructions when decoding.
- Altering data when fetching or storing.
- Altering register content, and the program counter.
- Flip bits in register or memory.

These phenomena happen at random and cannot be observed directly but the effect can be traced in software execution. On the software level the following can happen:

- A few instructions are skipped. This can lead to taking different branch than normal.

¹ PSA Certified Level 3 Lightweight Protection Profile

- Corrupted CPU register or data fetch could alter the result of a comparison instruction. Or change the value returned from a function.
- Corrupted data store could alter the config of peripherals.
- Very precise attacks with laser can flip bits in any register or in memory.

This is a complex domain. Faults are not well-understood. Different fault models exist but all of them target a specific aspect of fault injection. One of the most common and probably the easily applicable fault model is the instruction skip.

10.7.4 Mitigation against physical attacks

The applicability of these attacks highly depends on the device. Some devices are more sensitive than others. Protection is possible at hardware and software levels as well.

On the hardware level, there are chip design principles and system IPs that are resistant to fault injection attacks. These can make it harder to perform a successful attack and as a result the chip might reset or erase sensitive content. The device maker needs to consider what level of physical attack is in scope and choose a SoC accordingly.

On top of hardware-level protection, a secondary protection layer can be implemented in software. This approach is known as “defence in depth”.

Neither hardware nor software level protection is perfect because both can be bypassed. The combination of them provides the maximum level of protection. However, even when both are in place, it is not certain that they provide 100% protection against physical attacks. The best of what is to be achievable to harden the system to increase the cost of a successful attack (in terms of time and equipment), thereby making it non profitable to perform them.

Software countermeasures against physical attacks

There are practical coding techniques which can be applied to harden software against fault injection attacks. They significantly decrease the probability of a successful attack:

- Control flow monitor

To catch malicious modification of the expected control flow. When an important portion of a program is executed, a flow monitor counter is incremented. The program moves to the next stage only if the accumulated flow monitor counter is equal to an expected value.

- Default failure

The return value variable should always contain a value indicating failure. Changing its value to success is done only under one protected flow (preferably protected by double checks).

- Complex constant

It is hard to change a memory region or register to a pre-defined value, but usual boolean values (0 or 1) are easier to manipulate.

- Redundant variables and condition checks

To make branch condition attack harder it is recommended to check the relevant condition twice (it is better to have a random delay between the two comparisons).

- Random delay

Successful fault injection attacks require very precise timing. Adding random delay to the code execution makes the timing of an attack much harder.

- Loop integrity check

To avoid to skip critical loop iterations. It can weaken the cryptographic algorithms. After a loop has executed, check the loop counter whether it indeed has the expected value.

- Duplicated execution

Execute a critical step multiple times to prevent fault injection from skipping the step. To mitigate multiple consecutive fault injections, random delay can be inserted between duplicated executions.

These techniques should be applied in a thoughtful way. If it is applied everywhere then it can result in messy code that makes the maintenance harder. Code must be analysed and sensitive parts and critical call path must be identified. Furthermore, these techniques increase the overall code size which might be an issue on the constrained devices.

Currently, compilers are not providing any support to implement these countermeasures automatically. On the contrary, they can eliminate the protection code during optimization. As a result, the C level protection does not add any guarantee about the final behavior of the system. The effectiveness of these protections highly depends on the actual compiler and the optimization level. The compiled assembly code must be visually inspected and tested to make sure that proper countermeasures are in-place and perform as expected.

10.7.5 TF-M Threat Model against physical attacks

Physical attack target

A malicious actor performs physical attack against TF-M to retrieve assets from device. These assets can be sensitive data, credentials, crypto keys. These assets are protected in TF-M by proper isolation.

For example, a malicious actor can perform the following attacks:

- Reopen the debug port or hinder the closure of it then connect to the device with a debugger and dump memory.
- Bypass secure boot to replace authentic firmware with a malicious image. Then arbitrary memory can be read.
- Assuming that secure boot cannot be bypassed then an attacker can try to hinder the setup of the memory isolation hardware by TF-M *Secure Partition Manager* (SPM) and manage to execute the non-secure image in secure state. If this is achieved then still an exploitable vulnerability is needed in the non-secure code which can be used to inject and execute arbitrary code to read the assets.
- Device might contain unsigned binary blob next to the official firmware. This can be any data, not necessarily code. If an attacker manages to replace this data with arbitrary content (e.g. a NOP slide leading to a malicious code) then they can try to manipulate the program counter to jump to this area before setting up the memory isolation.

Assumptions on attacker capability

It is assumed that the attacker owns the following capabilities to perform physical attack against devices protected by TF-M.

- Has physical access to the device.
- Able to access external memory, read and possibly tamper it.
- Able to load arbitrary candidate images for firmware upgrade.
- Able to manage that bootloader tries to upgrade the arbitrary image from staging area.
- Able to inject faults on hardware level (voltage or power glitch, EM pulse, etc.) to the system.
- Precise timing of fault injection is possible once or a few times, but in general the more intervention is required for a successful attack the harder will be to succeed.

It is out of the scope of TF-M mitigation if an attacker is able to directly tamper or disclose the assets. It is assumed that an attacker has the following technical limitations.

- No knowledge of the image signing key. Not able to sign an arbitrary image.
- Not able to directly access to the chip through debug port.
- Not able to directly access internal memory.

- No knowledge of the layout of the die or the memory arrangement of the secure code, so precise attack against specific registers or memory addresses are out of scope.

Physical attack scenarios against TF-M

Based on the analysis above, a malicious actor may perform physical attacks against critical operations in *SPE* workflow and critical modules in TF-M, to indirectly gain unauthenticated accesses to assets.

Those critical operations and modules either directly access the assets or protect the assets from disclosure. Those operations and modules can include:

- Image validation in bootloader
- Isolation management in TF-M, including platform specific configuration
- Cryptographic operations
- TF-M Secure Storage operations
- PSA client permission check in TF-M

The detailed scenarios are discussed in following sections.

Physical attacks against bootloader

Physical attacks may bypass secure image validation in bootloader and a malicious image can be installed.

The countermeasures is bootloader specific implementation and out of the scope of this document. TF-M relies on MCUboot by default. MCUboot has already implemented countermeasures against fault injection attacks³.

Physical attacks against TF-M SPM

TF-M SPM initializes and manages the isolation configuration. It also performs permission check against secure service requests from PSA clients.

Static isolation configuration

It is TF-M SPM's responsibility to build up isolation during the initialization phase. If this is missed or not done correctly then it might be possible for non-secure code to access some secure memory area or an external device can access assets in the device through a debug port.

Therefore, hindering the setup of memory or peripheral isolation hardware is an obvious candidate for physical attacks. The initialization phase has a constant time execution (like the previous boot-up state), therefore the timing of the attack is simpler, compared to cases when secure and non-secure runtime firmware is up-and-running for a while and IRQs make timing unpredictable.

Some examples of attacking isolation configuration are shown in the list below.

- Hinder the setting of security regions. Try to execute non-secure code as secure.
- Manipulate the setting of secure regions, try to extend the non-secure regions to cover a memory area which otherwise is intended to be secure area.
- Hinder the setting of isolation boundary. In this case vulnerable ARoT code has access to all memory.
- Manipulate peripheral configuration to give access to non-secure code to a peripheral which is intended to be secure.

³ [MCUboot fault injection mitigation](#)

PSA client permission checks

TF-M SPM performs several permission checks against secure service requests from a PSA client, such as:

- Check whether the PSA client is a non-secure client or a secure client

NS client's PSA client ID is negative. NS client is not allowed to directly access secure areas. A malicious actor can inject faults when TF-M SPM authenticates a NS client. It may manipulate TF-M to accept it as a secure client and allow the NS client to access assets.

- Memory access checks

TF-M SPM checks whether the request has correct permission to access a secure memory area. A malicious actor can inject faults when TF-M SPM checks memory access permission. It may skip critical check steps or corrupt the check result. Thereby a malicious service request may pass TF-M memory access check and accesses assets which it is not allowed to.

The physical attacks mentioned above relies on the a malicious NS application or a vulnerable RoT service to start a malicious secure service request to access the assets. The malicious actor has to be aware of the accurate timing of dealing with the malicious request in TF-M SPM. The timing can be affected by other clients and interrupts. It should be more difficult than pure fault injection.

Dynamic isolation boundary configuration

Physical attack may affect the isolation boundary setting during TF-M context switch, especially in Isolation Level 3. For example:

- A fault injection may cause TF-M SPM to skip clear privileged state before switching in an ARoT service.
- A fault injection may cause TF-M SPM to skip updating MPU regions and therefore the next RoT service may access assets belonging to a previous one.

However, it is much more difficult to find out the accurate timing of TF-M context switch, compared to other scenarios in TF-M SPM. It also requires a vulnerable RoT service to access assets after fault injection.

Physical attacks against TF-M Crypto service

Since crypto operations are done by TF-PSA-Crypto library or by a custom crypto accelerator engine and its related software driver stack, the analysis of physical attacks against crypto operations is out-of-scope for this document. However, in general the same requirements are applicable for the crypto, to be compliant with PSA Level 3 certification. That is, it must be resistant against physical attacks. So crypto software and hardware must be hardened against side-channel and physical attacks.

Physical attacks against Secure Storage

Physical attacks against Internal Trusted Storage

Based on the assumption in *Assumptions on attacker capability*, a malicious actor is unable to directly retrieve assets via physical attacks against *Internal Trusted Storage* (ITS).

Instead, a malicious actor can inject faults into isolation configuration of ITS area in TF-M SPM to gain the access to assets stored in ITS. Refer to *Physical attacks against TF-M SPM* for details.

Physical attacks against Protected Storage

Based on the assumption in *Assumptions on attacker capability*, a malicious actor can be able to directly access external storage device. Therefore *Protected Storage* (PS) shall enable encryption and authentication by default to detect tampering with the content in external storage device.

A malicious actor can also inject faults into isolation configuration of PS and external storage device peripherals in TF-M SPM to gain the access to assets stored in PS. Refer to *Physical attacks against TF-M SPM* for details.

It is out of the scope of TF-M to fully prevent malicious actors from directly tampering with or retrieving content stored in external storage devices.

Physical attacks against platform specific implementation

Platform specific implementation includes critical TF-M HAL implementations. A malicious actor can perform physical attack against those platform specific implementations to bypass the countermeasures in TF-M common code.

Platform early initialization

TFM provides a HAL API for platforms to perform HW initialization before SPM initialization starts. The system integrator is responsible to implement this API on a particular SoC and harden it against physical attacks:

```
enum tfm_hal_status_t tfm_hal_platform_init(void);
```

The API can have several initializations on different modules. The system integrator can choose to even harden some of these initializations functions within this platform init API. One of the examples is the debug access setting.

Debug access setting

TFM configures debug access according to device lifecycle and accessible debug certificates. In general, TF-M locks down the debug port if the device is in secure production state. The system integrator can put the settings into an API and harden it against physical attacks.

Platform specific isolation configuration

TFM SPM exposes a HAL API for static and dynamic isolation configuration. The system integrator is responsible to implement these API on a particular SoC and harden it against physical attacks.

```
enum tfm_hal_status_t tfm_hal_set_up_static_boundaries(void);  
enum tfm_hal_status_t tfm_hal_bind_boundary(const struct partition_load_info_t *p_ldinf,  
                                             uintptr_t *p_boundary);
```

Memory access check

TFM SPM exposes a HAL API for platform specific memory access check. The system integrator is responsible to implement this API on a particular SoC and harden it against physical attacks.

```
tfm_hal_status_t tfm_hal_memory_check(uintptr_t boundary,  
                                       uintptr_t base,  
                                       size_t size,  
                                       uint32_t access_type);
```

10.7.6 TF-M countermeasures against physical attacks

This section propose a design of software countermeasures against physical attacks.

Fault injection hardening library

There is no open-source library which implements generic mitigation techniques listed in *Software countermeasures against physical attacks*. TF-M project implements a portion of these techniques. TF-M software countermeasures are implemented as a small library Fault Injection Hardening (FIH) in TF-M code base. A similar library was first introduced and tested in the MCUboot project (version 1.7.0)² which TF-M relies on.

The FIH library is put under TF-M lib/fih/.

The implementation of the different techniques was assigned to fault injection protection profiles. Four profiles (OFF, LOW, MEDIUM, HIGH) were introduced to fit better to the device capability (memory size, TRNG availability) and to protection requirements mandated by the device threat model. Fault injection protection profile is configurable at compile-time, default value: OFF.

Countermeasure profiles and corresponding techniques are listed in the table below.

Countermeasure	Profile LOW	Profile MEDIUM	Profile HIGH	Comments
Control flow monitor	Y	Y	Y	
Failure loop hardening	Y	Y	Y	
Complex constant		Y	Y	
Redundant variables and checks		Y	Y	
Random delay			Y	Implemented, but depends on HW capability

Similar to MCUboot, four profiles are supported. It can be configured at build time by setting (default is OFF):

```
-DTFM_FIH_PROFILE=<OFF, LOW, MEDIUM, HIGH>
```

How to use FIH library

As analyzed in *TF-M Threat Model against physical attacks*, this section focuses on integrating FIH library in TF-M SPM to mitigate physical attacks.

- Identify critical function call path which is mandatory for configuring isolation or debug access. Change their return types to FIH_RET_TYPE and make them return with FIH_RET. Then call them with FIH_CALL. These macros are providing the extra checking functionality (control flow monitor, redundant checks and variables, random delay, complex constant) according to the profile settings. More details about usage can be found here: `trusted-firmware-m/lib/fih/inc/fih.h`

Take simplified TF-M SPM initialization flow as an example:

```
main()
|
|--> tfm_core_init()
|
|           |--> tfm_hal_set_up_static_boundaries()
|           |
|           |           |--> platform specific isolation impl.
|           |
|           |--> tfm_hal_platform_init()
|           |
|           |--> platform specific init
```

(continues on next page)

² MCUboot project - fault injection hardening

(continued from previous page)

```

|
|--> During each partition initialization
|
|       |--> tfm_hal_bind_boundary()
|
|               |--> platform specific peripheral isolation impl.

```

- Might make the important setting of peripheral config register redundant and verify them to match expectations before continue.
- Implements an extra verification function which checks the critical hardware config before secure code switches to non-secure. Proposed API for this purpose:

```

fih_int tfm_hal_verify_static_boundaries(void);

```

This function is intended to be called just after the static boundaries are set up and is responsible for checking all critical hardware configurations. The goal is to catch if something is missed and act according to system policy. The introduction of one more checking point requires one more intervention with precise timing. The system integrator is responsible to implement this API on a particular SoC and harden it against physical attacks. Make sure that all platform dependent security feature is properly configured.

- The most powerful mitigation technique is to add random delay to the code execution. This makes the timing of the attack much harder. However it requires an entropy source. It is recommended to use the HIGH profile when hardware support is available. There is a porting API layer to fetch random numbers in FIH library:

```

void fih_delay_init(void);
uint8_t fih_delay_random(void);

```

- Similar countermeasures can be implemented in critical steps in platform specific implementation.

Take memory isolation settings on AN521 platform as an example. The following hardware components are responsible for memory isolation in a SoC, which is based on SSE-200 subsystem. System integrators must examine the chip specific memory isolation solution, identify the key components and harden the configuration of those. This list just serves as an example here for easier understanding:

- Implementation Defined Attribution Unit (IDAU): Implementation defined, it can be a static config or dynamic. Contains the default security access permissions of the memory map.
- SAU: The main module in the CPU to determine the security settings of the memory.
- MPC: External module from the CPU point of view. It protects the non security aware memories from unauthenticated access. Having a properly configured MPC significantly increases the security of the system.
- PPC: External module from the CPU point of view. Protects the non security aware peripherals from unauthenticated access.
- MPU: Protects memory from unprivileged access. ARoT code has only a restricted access in secure domain. It mitigates that a vulnerable or malicious ARoT partition can access to device assets.

The following AN521 specific isolation configuration functions shall be hardened against physical attacks.

```

sau_and_idau_cfg()
mpc_init_cfg()
ppc_init_cfg()

```

Some platform specific implementation rely on platform standard device driver libraries. It can become much more difficult to maintain drivers if the standard libraries are modified with FIH library. Platform specific imple-

mentation can implement duplicated execution and redundant variables/ condition check when calling platform standard device driver libraries according to usage scenarios.

Impact on memory footprint

The addition of protection code against physical attacks increases the memory footprint. The actual increase depends on the selected profile and where the mitigation code is added.

Attack experiment with SPM

The goal is to bypass the setting of memory isolation hardware with simulated instruction skips in fast model execution (FVP_MPS2_AEMv8M) in order to execute the regular non-secure test code in secure state. This is done by identifying the configuration steps which must be bypassed to make this happen. The instruction skip simulation is achieved by breakpoints and manual manipulation of the program counter. The following steps are done on AN521 target, but this can be different on another target:

- Bypass the configuration of isolation HW: SAU, MPC.
- Bypass the setting of the PSP limit register. Otherwise, a stack overflow exception will happen. Because the secure PSP will be overwritten by the address of the non-secure stack and on this particular target the non-secure stack is on lower address than the value in the secure PSP_LIMIT register.
- Avoid the clearing of the least significant bit in the non-secure entry point, where BLXNS/BXNS is jumping to non-secure code. Having the least significant bit cleared indicates to the hardware to switch security state.

The previous steps are enough to execute the non-secure Reset_Handler() in secure state. Usually, RTOS is executing on the non-secure side. In order to properly boot it up further steps are needed:

- Set the S_VTOR system register to point the address of the NS Vector table. Code is executed in secure state therefore when an IRQ hit then the handler address is fetched from the table pointed by S_VTOR register. RTOS usually do an SVC call at start-up. If S_VTOR is not modified then SPM's SVC handler will be executed.
- TBC: RTX osKernelStart still failing.

The bottom line is that in order to execute the regular non-secure code in secure state the attacker need to interfere with the execution flow at many places. Successful attack can be made even harder by adding the described mitigation techniques and some random delays.

10.7.7 Reference

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10.8 Memory-mapped IOVECs

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This document describes briefly the Memory-mapped IOVECs at higher isolation levels (isolation levels 2 and 3). The MM-IOVEC is fully described in FF-M v1.1 [?].

10.8.1 Introduction

Key terms:

- **IL1**: Lowest isolation level
- **IL2 / IL3**: Higher isolation levels
- **SSM** - Secure Shared Memory: refers to a block of secure memory that is available to RoTs but is not private to any single RoT.

FF-M v1.1 introduced MM-IOVEC functionality. Its initial implementation in TF-M restricted the capability to isolation level 1. The build system displays a warning banner when MM-IOVEC is enabled.

10.8.2 Problem statement

The current MM-IOVEC mechanism is restricted to partitions operating at IL1. This limitation prevents efficient transfer of large payloads for partitions at higher isolation levels.

The objective is to enable MM-IOVEC support for IL2/IL3 partitions while preserving isolation guarantees and without affecting security properties.

Reasons to extend support to higher isolation levels are:

- Process large payloads that do not fit in the client buffers
- Enable multiple actors to process and forward the ‘original’ vectors

10.8.3 Proposed solution

Remove the `tfm_invalid_config` check for `TFM_ISOLATION_LEVEL GREATER 1` and `PSA_FRAMEWORK_HAS_MM_IOVEC`. This change enables MM-IOVEC for IL2 and IL3.

No changes to the SPM implementation are required. However additional responsibilities are introduced for relevant partitions.

This solution enables:

- mapping and passing of Secure Shared Memory, for example a mmio flash
- mapping and passing NSPE buffers

This document uses the Generic TF-M Threat Model to assess the impact of the solution.

Targeted Threat analysis

This section presents a simplified text-based data flow sequence (DF) to help analyse whether boundaries, data classification and actors need to change.

- DF-1..DF-5 - Data provided by NSPE

DF-1 NS sends a request to SPE with iovecs (already covered)

DF-2 - 1st partition

- the partition maps the input/output vectors
- processes the data
- temporarily copies the NS vector pointers in own memory/stack and forward those to another partition via `psa_call()`
- unmap the vectors

DF-3 - 2nd partition + the partition maps the input/output vectors + processes the data + unmap the vectors

DF-4 - 1st partition + the partition maps the input/output vectors + processes the data + unmap the vectors

DF-5 NS resumes with the processed data in the output vectors

- DF-6..DF-10 - Data provided by SPE

DF-6 NS sends a request to SPE WITHOUT iovecs (already covered)

DF-7 - 1st partition

- the partition process the request, prepares the SSM references
- processes the data
- temporarily copies the SSM pointers into vectors in own memory/stack and forward those to another partition via `psa_call()`
- unmap the vectors

DF-8 (equal to DF-3) - 2nd partition

- the partition maps the input/output vectors
- processes the data
- unmap the vectors

DF-9 - 1st partition

- the partition maps the input/output vectors
- processes the data
- unmap the vectors

DF-10 NS resumes with the processed data in the output vectors

We analyze if with the DF above the threats scenario is worsened. We assume IL3. The NSPE to SPE interactions are already covered by the Generic TF-M Threat Model. We limit the analysis for the MM-IOVEC mapping/unmapping between the partitions.

DF-1, DF-6 Already covered in Generic TF-M Threat Model

DF-2, DF-3, DF-4, DF-7, DF-8, DF-9 Not covered in the Generic TF-M Threat Model because of the definitions of trusted boundaries (see section https://trustedfirmware-m.readthedocs.io/en/latest/security/threat_models/generic_threat_model.html#data-flows-inside-spe-informative)

Spoofing The caller partition may try to access and use another's partition identity

Not a threat. The caller partition's identity is assigned and checked by SPM and cannot be directly provided.

Tampering The caller partition may try to change or modify persistent data

Not a threat. We trust the partition to not behave as an attacker (see FF-M).

Also, the caller partition can already map/unmap vectors from NS or from SSM by "design (of the memory)".

Repudiation The caller partition may try to perform prohibited operations in a system that lacks the ability to trace the operations.

Not a threat. We trust the partition to behave responsibly.

Information disclosure The caller partition may try to access a file or data it was not granted access to.

Not a threat. We trust the partition to not behave as an attacker (see FF-M).

Allowing the mapping/unmapping does not change the isolation rules. Thus if unprivilege code tries to access priv or another unpriv data, the system raises a Fault. Also, the caller partition can already map/unmap vectors from NS or from SSM by “construction”.

DoS Not covered in the threat model

Out of scope as per TF-M Generic Threat Model.

Elevation of privilege The caller partition may try to gain privileged access to the data

Not a threat. We trust the partition to not behave as an attacker.

Interaction with the isolation model analysis

According to FF-Mv1.1, in an implementation with IL2, IL3, MM-IOVEC provides a mechanism that may conflict with the isolation rules.

A. Possible conflict

With IL3 a Secure Partition is not permitted to access another Secure Partition’s Private data. MM-IOVEC may provide a mechanism for one Secure Partition to access the other’s Private data.

A. Mitigation

SPM implemented in TF-M ensures that:

- the mapped vectors fulfill the requirements for the callee partition by testing the vectors with `cmse_check_address_range` when mapping/unmapping
- the execution environment for ARoT provided by SPM is such that a ARoT cannot access data belonging to other ARoTs. This is enforced regardless the MM-IOVEC support (`PSA_FRAMEWORK_HAS_MM_IOVEC`)

B. Possible conflict

In a system that implements isolation rule I6 (see [FF-M] §3.1.5), only the SPM is permitted to access memory in another protection domain when required. MM-IOVEC can provide access from a Secure Partition directly to client memory.

B. Mitigation

Isolation rule I6 is not implemented.

C. Possible imprecise mapping

The mapping mechanism can provide more access than is strictly required.

C. Mitigation

We must take into account the hardware limitations that can impact mapping precision. Any client of the API must take steps to mitigate the MPU granularity, for example by aligning the buffer and buffer size in memory. It is IMPLEMENTATION DEFINED how such imprecise mappings are handled by the implementation. The decision to permit imprecise mappings depends on the security requirements for the system, and the nature of the additional access.

An implementation that provides MM-IOVEC functionality must document its behavior when an input or output vector mapping is imprecise.

The services designed and implemented to handle such payloads may need to provide platform hooks to correctly set the MPU alignment for the SSM. Alternatively, the MPU can be set once at init stage to correctly bound the region limits.

10.8.4 Discussion

The FF-M architecture assumes that RoT partitions do not behave maliciously and are implemented correctly. Through secure boot, such RoTs are authenticated.

Based on the above, there is no need to introduce new trust boundaries between partitions' domains.

The assets protection does not change when the mapping/unmapping is performed, because the isolation boundaries are enforced (take place) when SPM changes the running partition and is independent from the MM-IOVEC mapping.

Accessing NS buffers with IL3 is not any different from IL1 since the isolation domain b/w NS and the PRoT-ARoT does not change.

Accessing non-private shared secure data does not conflict with the isolation domains, that is, there's no direct data sharing b/w partitions.

SPM ensures that the vectors can be accessed by the partitions according to their boundaries settings, irrespective of the `PSA_FRAMEWORK_HAS_MM_IOVEC` capability.

When mapping/unmapping is required, additional memory checks are performed to ensure the calling partition can safely access the given vectors.

The non-private shared secure regions need to have the correct attributes for the mapping partitions to be consumed.

10.8.5 Conclusion

Based on the analysis and discussion above, we don't consider that an update in the Threat Model is required.

Relaxing the usage of MM-IOVEC to higher isolation level does not change the threat landscape and does not violate the isolation rules.

Build options and configs

Unchanged

Remarks

PRoT and ARoT must not access vectors after unmapping.

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CONTRIBUTION GUIDELINES

11.1 Contributing Process

Contributions to the TF-M project need to follow the process below.

Note

Please contact TF-M mailing list tf-m@lists.trustedfirmware.org for any question.

- It is recommended to subscribe to TF-M mailing list via [this page](#).
- Refer to the *Roadmap* or send a mail to the TF-M mailing list tf-m@lists.trustedfirmware.org to get the latest status and plan of TF-M.
- Follow *Design Proposal Guideline* to propose your design.
- Follow guidelines below to prepare the patch:
 - Clone the TF-M code on your own machine from [TF-M git repository](#).
 - Follow the TF-M getting started, *Build Instructions* and *Coding Guide* for the TF-M project.
 - Make your changes in logical chunks to help reviewers. Each commit should be a separate review and either work properly or be squashed after the review and before merging.
 - Follow *Documentation Contribution Guidelines* to update documentation in docs folder if needed.
 - Test your changes and add details to the commit description.
 - The code is accepted under *Developer Certificate of Origin (DCO)*. Use `git commit -s` to add a Signed-off-by trailer at the end of the commit log message. See [git-commit](#) for details.
 - Ensure that each changed file has the correct copyright and license information. Files that entirely consist of contributions to this project should have a copyright notice and BSD-3-Clause SPDX license identifier of the form

```
SPDX-License-Identifier: BSD-3-Clause
```

```
SPDX-FileCopyrightText: Copyright The TrustedFirmware-M Contributors
```

to reflect the community ownership of the project. Individual copyrights can be traced back through the corresponding commits. For more details please refer to [Linux Foundation guidelines on copyright notices](#) for OSS projects. Patches that contain changes to imported Third Party IP files should retain their original copyright and license notices. If changes are made to the imported files, then add an additional `SPDX-FileCopyrightText` tag line as shown above.

The original form of the project copyright note as

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is still appropriate but shall be avoided in the new files.

- Add a [Change-Id](#) to the commit message, which can be generated any way you like (e.g. from the SHA of the commit). It is suggested to clone repositories with commit-msg hook. The commit-msg hook attaches Change-Id automatically. Take [trusted-firmware-m](#) as an example.
 - Submit your patch for review. Refer to [Uploading Changes](#) for details of uploading patch.
 - Add relevant *code owner(s)* for reviewing the patch.
 - You may be asked to provide further details or make additional changes.
 - You can discuss further with code owner(s) and maintainer(s) directly via TF-M mailing list tf-m@lists.trustedfirmware.org if necessary.
 - If multiple patches are linked in a chain then code owners and maintainers should review and merge individual patches when they are ready, rather than waiting for the entire chain to be reviewed. If multiple patches are intended to be merged together then authors shall use topics or explicitly mention that in the commit message.
 - Click Allow-CI +1 button on Gerrit page to run CI to validate your patch. Your patch shall pass CI successfully before being merged. Code owner(s) and maintainer(s) may ask for additional test.
 - Once the change is approved by code owners, the patch will be merged by the maintainer.
-

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11.2 Code Review Guideline

The purpose of this document is to clarify design items to be reviewed during the code review process.

Please contact *maintainers* or write an e-mail thread on the TF-M mailing list tf-m@lists.trustedfirmware.org for any questions.

11.2.1 List of the guidelines

The prerequisites before going to the review stage:

- Read the *Contributing Process* to know basic concepts.
- Read the *Source Structure* for structure related reference.

The review guidelines consist of these items:

- *Exceptions.*
- *Non-source items.*
- *Namespace.*
- *Assembler code.*
- *Include.*
- *Auto-doc.*
- *Commit Message.*

Note

Each guideline item is assigned with a unique symbol in the format Rx.y, while x and y are numeric symbols. These symbols are created for easy referencing during the code review process.

Exceptions

Files under listed folders are fully or partially imported from 3rd party projects, these files follow the original project defined guidelines and styles:

Important

- R1.1 ext and its subfolders.

Non-source items

For all non-source files such as build system files or configuration files:

Important

- R2.1 Follow the existing style while making changes.

Namespace

TF-M assign the namespace to files and symbols for an easier code reading. The symbol here includes functions, variables, types and MACROs. The prerequisites:

Important

- R3.1 Follow the documents or specifications if they propose namespaces ('psa_' for PSA defined items E.g.,). Ask the contributor to tell the documents or specifications if the reviewer is not sure about a namespace.
- R3.2 Assign TF-M specific namespace 'tfm_' or 'TFM_' for symbols implementing TF-M specific features. Ask the contributor to clarify the purpose of the patch if the reviewer is not sure about the namespace.

For the sources out of the above prerequisites (R3.1 and R3.2):

Important

- R3.3 Do not assign a namespace for source files.
- R3.4 Assigning a namespace for interface symbols is recommended. The namespace needs to be expressed in one word to be followed by other words into a phrase can represent the functionalities implemented. One word naming is allowed if the namespace can represent the functionality perfectly.
- R3.5 Assign a namespace for private symbols is NOT recommended.
- R3.6 Do not assign characters out of alphabets as the leading character.

Examples:

```

/* R3.1 FILE: s/spm/core/psa_client.c */

/* R3.2 FILE: s/spm/core/tfm_secure_context.c */

/* R3.4 FILE: s/spm/core/spm.c, 'spm\_ ' as the namespace */
void spm_init(void);

/* R3.5 FILE: s/spm/core/main.c */
static void init_functions(void);

/* R3.6 Not permitted: */
/* static uint32_t __count; */

```

Assembler code

Important

- R4.1 Pure assembler sources or inline assembler code are required to be put under the platform-independent or architecture-independent folders. The logic folders should not contain any assembler code, referring to external MACRO wrapped assembler code is allowed. Here is one example of the logic folder:
 - ‘secure_fw/spm’.

Examples:

```

/*
 * R4.1 The following MACRO is allowed to be referenced under
 * 'secure_fw/spm'
 */
#define SVC(code) __asm volatile("svc %0", ::"I"(code))

```

Include

This chapter describes the placement of the headers and including. There are two types of headers: The **interface** headers contain symbols to be shared between modules and the **private** headers contain symbols only for internal usage.

Important

- R5.1 Put the **interface** header of one module in the **include** folder under the root of this module. Deeper sub-folders can not have **include** folders, which means only one **include** is allowed for one module.
- R5.2 Creating sub-folders under **include** to represent the more granular scope of the interfaces is allowed.
- R5.3 **private** header can be put at the same place with the implementation sources for the private symbols contained in the header. It also can be put at the place where the sources need it. The latter is useful when some “private header” contains abstracted interfaces, but these interfaces are not public interfaces so it won’t be put under “include” folder.
- R5.4 Use `<>` when including public headers.
- R5.5 Use `“”` when including private headers.
- R5.6 The module’s **include** folder needs to be added into referencing module’s header searching path.

- R5.7 The module's `include` folder and the root folder needs to be added into its own header searching path and apply a hierarchy including with folder name.
- R5.8 Path hierarchy including is allowed since there are sub-folders under `include` folder and the module folder.
- R5.9 The including statement group order: the beginning group contains toolchain headers, then follows the public headers group and finally the private headers group.
- R5.10 The including statement order inside a group: Compare the include statement as strings and sort by the string comparison result.
- R5.11 The header for the referenced symbol or definition must be included even this header is included inside one of the existing included headers. This improves portability in case the existing header implementation changed.

Examples:

```
/*
 * The structure:
 *   module_a/include/func1.h
 *   module_a/include/func2/fmain.h
 *   module_a/func1.c
 *   module_a/func2/fmain.c
 *   module_b/include/funcx.h
 *   module_b/include/funcy/fmain.h
 *   module_b/funcx.c
 *   module_b/funcxi.h
 *   module_b/funcy/fmain.c
 *   module_b/funcy/fsub.c
 *   module_b/funcy/fsub.h
 * Here takes module_b/funcx.c as example:
 */
#include <func1.h>
#include <func2/fmain.h>
#include <funcx.h>
#include "funcxi.h"
#include "funcy/fsub.h"
```

Auto-doc

Auto document system such as doxygen is useful for describing interfaces. While it would be a development burden since the details are described in the design documents already. The guidelines for auto-doc:

Important

- R6.1 Headers and sources under these folders need to apply auto-doc style comments: `*include`.
- R6.2 Developers decide the comment style for sources out of listed folders.

Commit Message

TF-M has the requirements on commit message:

Important

- R7.1 Assign correct topic for a patch. Check the following table.

Topic	Justification
Boot	<i>bl2/*</i> or <i>bl1/*</i>
Build	For build system related purpose.
Docs	All *.rst changes.
Dualcpu	Dual-cpu related changes.
HAL	Generic HAL interface/implementation changes.
Interface	Interface changes, either Non-source and secure.
Pack	For packing purpose.
Platform	Generic platform related changes under <i>platform/*</i> .
Platform Name	Specific platform changes.
Partition	Multiple partition related changes.
Partition Name	Specific partition related changes.
Service	Multiple service related changes.
Service Name	Specific service related changes.
SPM	<i>secure_fw/spm/*</i>
SPRTL	<i>secure-fw/partitions/lib/runtime/*</i>
Tool	<i>tools</i> folder or <i>tf-m-tools</i> repo

Note

Ideally, one topic should cover one specific type of changes. For crossing topic changes, check the main part of the change and use the main part related topic as patch topic. If there is no suitable topics to cover the change, contact the community for an update.

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Trusted Firmware-M (TF-M) is an open govcommunity project. All contributions are ultimately merged by the maintainers listed below. Technical ownership of most parts of the codebase falls on the code owners listed below. An acknowledgement from these code owners is required before the maintainers merge a contribution.

More details may be found in the [Project Maintenance Process](#) document.

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11.6 Issue tracking

To trace TF-M issues and to maintain LTS versions transparently, all issues found after an official release and breaking TF-M functionality, i.e. bugs in either common code or platform code, or security vulnerability post public disclosure shall be registered in TF-M Github [Issue tracker](#) . The identified issues shall be addressed in the **main** development branch to ensure normal progress. Additionally, these fixes will be backported to the active Long-Term Support (LTS) branches and will be included in the upcoming LTS release.

Warning

Security-related issues should be addressed through the *Security Disclosure* and recorded in the [Issue tracker](#) **only** after public disclosure.

When reporting a new issue please cover:

1. **Summary:** Provide a concise overview of the issue. What problem are you encountering?
2. **Technical Description:** Explain the issue thoroughly. Include relevant logs or screenshots.
3. **Build and Execution Environment:**
 - The hardware platform
 - Build toolchain with versions

4. **Reproduction Steps:** Describe how to reproduce the issue step by step. If possible, provide sample code or configuration settings.
5. **Proposed Fix (Optional):** If you have ideas on how to address the issue, share them. It's not mandatory, but it can be helpful.

11.6.1 Is it a bug or security vulnerability?

A security vulnerability refers to a flaw that an attacker can exploit to gain unauthorized access to system secrets, manipulate data, or perform actions beyond the intended functionality. However, it's important to note that defects that cause system crashes or lead to a Denial of Service (DoS) state are considered bugs rather than security vulnerabilities. When faced with uncertainty in classifying a new defect, it is wise to use caution and consider it as a potential safety issue.

As implied in the *Contributing Process* maintainers reserve the right to decide on what's acceptable to be backported to LTS branches in case of any divergence.

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11.7 Yet another coding standard :)

Warning

Every rule has an exception so if you disagree or dislike then reach out!

The coding style of TF-M project is based on [Linux coding style](#) but there are updates for domain specific conventions as listed below.

TF-M also reuses code from other SW projects, e.g. CMSIS_5, which means some areas of code may have different styles. We use common sense approach and new code may inherit coding style from external projects but it needs to remain within clear scope.

The guidance below is provided as a help. It isn't meant to be a definitive list.

As implied in the *contributing guide* maintainers have the right to decide on what's acceptable in case of any divergence.

Warning

Text files do not fall within these rules as they may require different formatting.

11.7.1 Consistent style

The code needs to be consistent with itself, so if existing code in the file violates listed coding style rules then it is better to follow existing style in the file and not break consistency by following the rules listed here.

You may need to add a comment in the commit description to clarify this.

List of rules

- Use 4 spaces indentation. No tabs.
- Use `lower_case_with_underscore` for filenames, variable and function names.
- Use standard types e.g. `uint32_t`, `uint16_t`, `uint8_t`, `int32_t` etc.

- Use `uintptr_t` type when representing addresses as numbers.
- Use `static` for all private functions and variables.
- Use `void` argument if your function doesn't contain any argument.
- Use C90 `/* */` for comments rather than C99 `//`
- No trailing spaces in code.
- Keep up to 100 characters per line but 140 chars can be accepted exceptionally.
- Open curly brace `{` at the same `if/else/while/for/switch` statement line.
- Use curly braces `{ }` for one line statement bodies also.
- Put open curly brace in the line below the function header.
- Order function parameters so that input params are before output params.
- Declare local variables at the beginning of a scope.
- Define macros in all caps i.e. `CAPITAL_WITH_UNDERSCORE`.
- Use typedefs **ONLY** for function prototype declarations.
- Type definitions in `lower_case_with_underscore` ended by `_t`.
- Do not use typedef for other constructs as it obfuscates code.
- Do not use bitfields.
- Use `static` for all global variables to reduce the variable scope.
- Use enumeration for error codes to keep the code readable.
- Use descriptive variable and functions names.
- Put the correct license header at the beginning of the file.
- Keep the files (`.h/.c`) self-contained, i.e. put include dependencies in the file.
- Put appropriate header define guard in `.h` files: `__HEADER_NAME__`. Any divergence from this rules should be clearly documented.
- In a `.c` file, `#include` its own header file first.
- Document all the public functions in the header file only.
- Document all the private functions in the source file only.
- Add “extern C “ definition for C++ support in the header files.
- In the switch statement, aligned cases with the `switch` keyword.
- For enums, use upper case letters with digits and underscore only.
- Do not code while eating.

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11.8 Documentation Contribution Guidelines

The Trusted Firmware-M project uses [Sphinx](#) to generate the Official Documentation from [reStructuredText](#) *.rst* source files,

The aim is to align as much as possible with the official [Python Documentation Guidelines](#) while keeping the consistency with the already existing files.

The guidance below is provided as a help. It is not meant to be a definitive list.

11.8.1 Overview

The following short-list provides a quick summary of the rules.

- If the patch modifies a present file, the file's style should be followed
- If creating a new file, *integration guide* can be used as a reference.
- When a new style is to be expressed, consult the [Python Documentation Guidelines](#)

11.8.2 List of rules

The following rules should be considered:

1. H1 document title headers should be expressed by # with over-line (rows on top and bottom) of the text
2. Only ONE H1 header should allowed per document, ideally placed on top of the page.
3. H2 headers should be expressed by * with over-line
4. H2 header's text should be UNIQUE in per document basis
5. H3 headers should be expressed by an underline of '='
6. H4 headers should be expressed by an underline of '-'
7. H3 and H4 headers have no limitation about naming. They can have similar names on the same document, as long as they have different parents.
8. H5 headers should be expressed by an underline of '^'
9. H5 headers will be rendered in document body but not in menus on the navigation bar
10. Do not use more than 5 levels of heading
11. When writing guides, which are expected to be able to be readable by command line tools, it would be best practice to add long complicated tables, and UML diagrams in the bottom of the page and using internal references(auto-label)
12. No special formatting should be allowed in Headers (code, underline, strike-through etc)
13. Long URLs and external references should be placed at the bottom of the page and referenced in the body of the document
14. New introduced terms and abbreviations should be added to Glossary and directly linked by the *:term:* notation across all documents using it.

11.8.3 Platform Documentation

The Documentation Build system provides an interface with the platform directory allowing maintainers to bundle platform specific documentation. Platforms are grouped by vendor. **This behaviour needs to be explicitly enabled for each vendor's space** by providing the *<vendor>/index.rst* (responsible for generating the Platform Index File, as seen linked in *TF-M Platforms*) contents entry for the corresponding vendor's space in *TF-M Platforms*. The format

and structure of this entry is not strictly defined, and allows flexible control of vendor's and platform's documentation space. Follow the *Platform Documentation* document for more details.

11.8.4 Common Use Cases

The section below describes with examples, a rule compliant implementation for most common documentation elements.

Headers

```
#####  
Document Title (H1)  
#####  
  
*****  
Chapter Title (H2)  
*****  
  
Chapter Section (H3)  
=====
```

```
Chapter Sub-Section (H4)  
-----  
  
Subsection of Chapter sub-section (H5)  
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
```

Code Blocks

The recommendation for code content, is to use the explicit code-block directive, ideally with a defined lexer for the language the block contains.

A list of compatible lexers can be found at [Pygments Lexers](#)

```
.. code-block:: bash

    ls
    pwd

.. code-block:: doscon

    dir

.. code-block:: c

    static struct rn_object_t;

.. code-block:: python3

    print("Hello TF-M")
```

reStructuredText supports implicit code-blocks by indenting a section of text, surrounded by new lines. While this formatting is allowed, it should be avoided if possible.

The quick brown fox jumps over the lazy dog

```
ls
pwd
```

Note

Mixing two different code-block formats in the same document will break the whole document's rendering. When editing an existing document, please follow the existing format.

New documents should always use the explicit format.

Tables

For adding new tables the `table::` notation should be used.

```
.. table:: Components table
:widths: auto
```

```
+-----+-----+-----+
| **Title A** | **Title B** | **Title C** |
+=====+=====+=====+
| Text A      | Text B      | Text C      |
+-----+-----+-----+
```

While the equivalent simple table code will render correctly in the output, it will not be added to the index (So it cannot be referenced if needed)

```
+-----+-----+-----+
| **Title A** | **Title B** | **Title C** |
+=====+=====+=====+
| Text A      | Text B      | Text C      |
+-----+-----+-----+
```

Other types of tables such as list-tables and csv-tables are also permitted, as seen on [/getting_started/tfm_getting_started](#) and [/releases/1.0](#)

External Links

External links should be placed in the bottom of a document.

The quick brown fox jumps over the lazy dog according to ``Link_A`_`

```
.. _Link_A: https://www.aaa.org
.. _Link_B: https://www.bbb.org
```

```
-----
```

```
*Copyright (c) XYZ *
```

Creating in-line links is permitted, but should be avoided if possible. It should be only used for consistency purposes or for a small amount of short links.

```
The quick brown fox jumps over the lazy dog according to `Link_A <https://www.aaa.org>`_
```

If for the purposes of content, a link is to be referenced by multiple labels, internal linking is the recommended approach.

```
The quick brown fox jumps over the lazy dog according to `Link_A_New`_
```

```
.. _Link_A: https://www.aaa.org
.. _Link_A_New: `Link_A`_
```

```
-----
```

```
*Copyright (c) XYZ *
```

Document Links

A document included in the documentation can be referenced by the *doc:* notation

```
:doc:`integration guide </integration_guide/tfm_integration_guide>`
```

The path is relative to the root of the Trusted Firmware-M code.

Trusted Firmware-M project is spread among multiple repositories: Trusted Firmware-M, TF-M Tests, TF-M Tools and TF-M Extras. Every repository has its own documentation, and since v2.0.0, they can be found under *Links*.

Using *Intersphinx*, it is now possible to use cross-reference roles from Sphinx to reference documentation from different projects (repositories), like TF-M Tests. Referencing documentation using the *doc:* notation is preferred and helps to avoid broken cross-references if the link of the document changes.

For example, to get this: Adding TF-M Regression Test Suite; the reStructuredText would look like this:

```
:doc:`Adding TF-M Regression Test Suite <TF-M-Tests:tfm_test_suites_addition>`
```

As can be seen, it is quite similar to cross-referencing in Sphinx, except the path to the document is preceded with the external project name. For TF-M Tests, it is TF-M-Tests. The names of other projects configured to be referenced using *Intersphinx* can be seen in the *conf.py* file under *intersphinx_mapping*.

Reference specific section of a document

In order to reference a specific section of a document, up to level 4 headers (if they are included in the index), the *ref:* keyword can be used

```
:ref:`docs/getting_started/tfm_getting_started:Tool & Dependency overview`
```

This can also be used to quickly scroll to the specific section of the current document. This technique can be used to add complex table in the bottom of a document and create clickable quick access references to it for improved user experience.

Glossary term

For technical terms and abbreviations, the recommended guidance is to add an entry to the *Glossary of terms and abbreviations* and refer to it, using the *term:* directive

```
HAL
```

```
Hardware Abstraction Layer
```

```
    Interface to abstract hardware-oriented operations and provides a set of
    APIs to the upper layers.
```

(continues on next page)

(continued from previous page)

.....

As described in the design document `:term:`HAL`` abstracts the
hardware-oriented and platform specific

.....

Note

The “:term:” directive does not work when used in special formatting. Using `*:term:HAL*` **will not link to the glossary term**.

References

1. Sphinx
2. reStructuredText
3. Python Documentation Guidelines
4. Pygments Lexers
5. Intersphinx

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11.9 Design proposal guideline

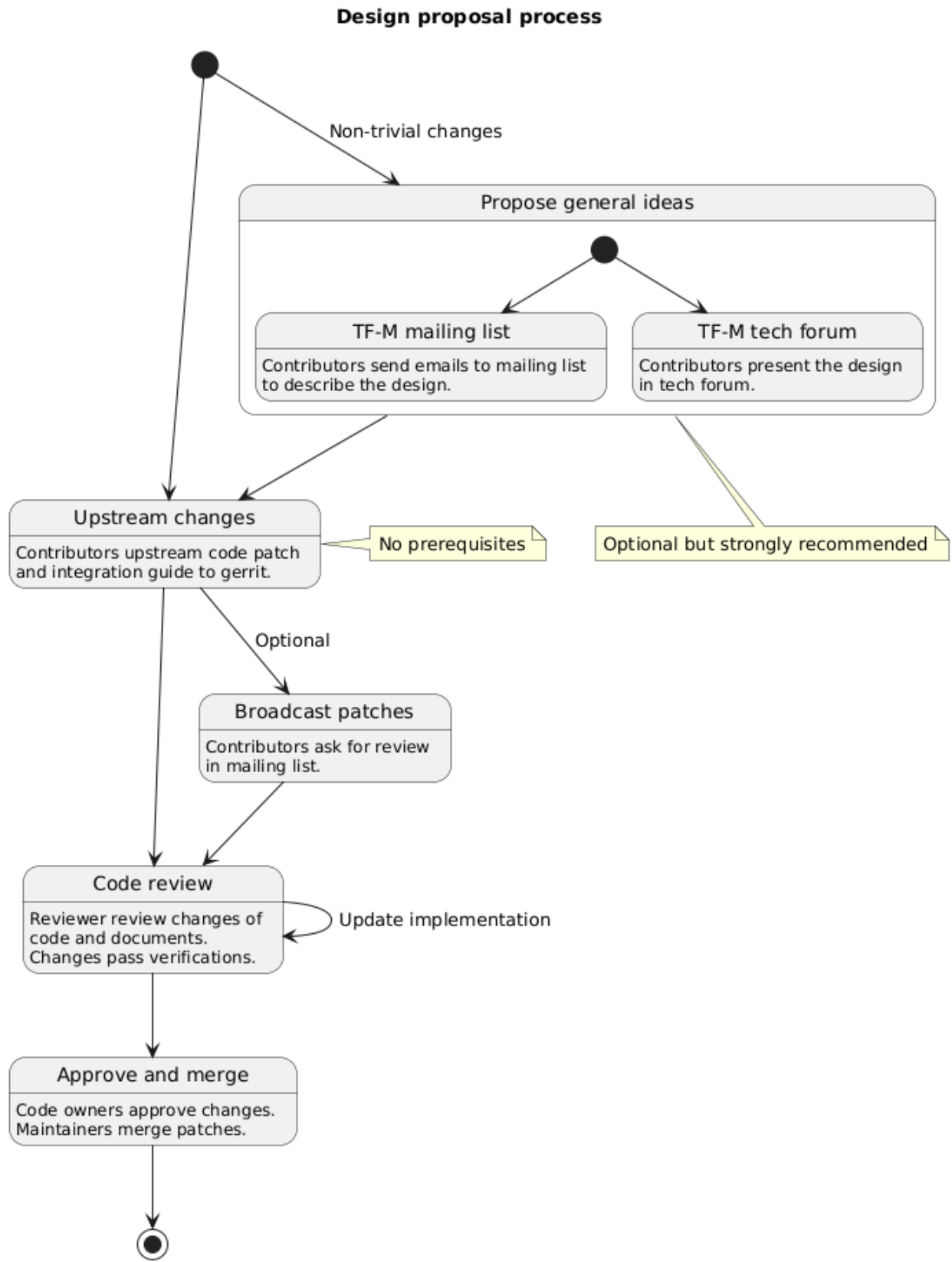
The design proposal guideline specifies the steps to propose and upload design proposals to TF-M. Those steps are lightweight and flexible to make sure that contributors can focus more on actual code implementation and iteration.

The guideline encourages developers to share design proposal via TF-M mailing list¹ and TF-M technical forum (tech forum)². The design details can be discussed via code reviews of actual implementations.

Typical steps are shown as the diagram below.

¹ TF-M mailing list

² TF-M technical forum



11.9.1 Discussion in TF-M mailing list and tech forum

It is **highly recommended** to propose and discuss designs in TF-M mailing list or TF-M tech forum, before or while the code implementation is under review.

It is efficient and flexible to directly discuss design proposal via TF-M mailing list and TF-M tech forum. Contributors can receive quick and broad feedback from TF-M community.

Although it is optional to present the ideas in mailing list or tech forum, it will help reviewers understand the design much better and expedite the code review process.

11.9.2 Code review of details

It is straightforward and convenient for contributors and reviewers to deliberate over design and implementation details via code review.

Contributors can implement their design proposal and upstream the patch set to TF-M [gerrit](#)³ for code review. For non-trivial changes or new major features, it is **strongly suggested** to propose the design to TF-M mailing list and tech forum in advance.

Contributors don't have to wait for any approvals before upstreaming patches, even if the changes are non-trivial. No formal design document in advance is required anymore.

The review process is the same as the general one⁴, with some specific requirements:

- Contributors can send an email to TF-M mailing list to ask for review.
- If it requires additional reviewers besides code owners and maintainers, contributors shall add the specific reviewers in the review list.
- Authors shall clearly specify the design purpose and briefly describe the implementation in the commit message.
- Authors shall put essential comments and notes in code for the code changes.

Code owners and maintainers may require contributors to further verify the implementation besides normal per-patch CI test. Contributors shall provide the verification results as requested.

11.9.3 Integration guide and manual

Contributors can create an integration guide or a user manual to describe how to integrate the new features related to the design proposal.

Contributors shall update the corresponding documents if the design changes existing implementation.

Reference

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11.10 MISRA-C Standard

11.10.1 Introduction

The project checks for MISRA-C violations and cautions in the code base by running the Eclair MISRA-C compliance tool as part of our CI process. The results for the daily job can be found at [TF-M Eclair daily CI pipeline](#).

The project, is not fully MISRA-C compliant. However, it aims to be MISRA-C-friendly by:

³ [TF-M gerrit](#)

⁴ [Contributing process](#)

- Avoiding the introduction mandatory MISRA-C violations and cautions.
- Running regular checks on the code base.
- Analysing and documenting violations and cautions prompted by the Eclair MISRA-C static analyser. This can be useful for downstream consumer looking for MISRA-C compliance.

While the project's intention is to keep violations and cautions to a minimum, it will not block new features, bug fixes, or improvements even if they introduce difficult-to-resolve cautions or violations. Currently, the project is not equipped with necessary fast-feedback-loop tooling to inform contributors about the MISRA-C violations/cautions they introduce. The project welcomes suggestions and contributions to better align with the MISRA-C standard.

The *coverity_check.h* library provides helper macros to document MISRA-C static-analysis findings in a consistent, reviewable way (line-level deviations, block deviations, and false positives).

In the common framework code, a limited set of MISRA-C rules are intentionally suppressed where the design requires it (for example, PSA APIs using *void ** and platform/arch interfaces). The current suppressions focus on rules 2.2, 11.3, 11.5, 11.6, and 17.9, and they are always accompanied by a justification comment.

Note

In this document, *Caution* refers to a [caution report](#) generated by Eclair MISRA-C static analyser.

11.10.2 Update schedule

The analysis table for mandatory rules will be updated with every release. Currently, the table is up-to-date for release v2.2.0.

11.10.3 Assessment of mandatory MISRA-C rules violation and cautions

The following table presents an analysis of the violations and cautions of mandatory MISRA-C:2012 rules report by Eclair.

The reported violations cannot be easily resolved with the current code structure and are discussed further in the table. The project is not actively prioritizing their resolution; however, as mentioned earlier, it welcomes suggestions and contributions from the community.

The community should proactively address any cautions that can be reasonably resolved. However, if resolving a caution does not positively affect code quality and readability, it may be safely ignored.

MISRA-C Mandatory	Error Type	Rule definition	Explanation
Rule 9.1	Caution	The value of an object with automatic storage duration shall not be read before it has been set	Cautions are due to the objects not being explicitly initialised during declaration. In some of the cases the address of the object is passed to a function to retrieve required value which gives the appearance that the object is used uninitialized.
Rule 17.4	Violation	All exit paths from a function with non-void return type shall have an explicit 'return' statement with an expression	All violations are exactly the same case of having a naked function with no C return statement. This cannot be resolved due to the body of the function being written in assembly.
Rule 21.17	Caution	Use of the string handling functions from <code><string.h></code> shall not result in accesses beyond the bounds of the objects referenced by their pointer parameters	Currently seen with <code>tfm_arch_init_context</code> . It is used to clear the memory (stack) to be used to store program context (as part of the context initialisation). Remember, stack grows downwards - hence, pointer is decremented in the implementation.
Rule 21.18	Caution	The <code>size_t</code> argument passed to any function in <code><string.h></code> shall have an appropriate value	Functions in TF-M use <code><string.h></code> functions with buffers/objects passed in by the caller. The functions suspected by Eclair have been double checked for verifying size of buffer/object before using them.

11.10.4 List of required MISRA-C rules violation and cautions

The list of required MISRA-C rules violations can be found in [TF-M Eclair daily CI pipeline](#) -> Last Successful Artefacts -> index.html -> Report by issue strictness (Mandatory/Required/Advisory) (violations)

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11.11 Python Scripts and Modules

11.11.1 Overview

This document describes the guidelines for adding Python scripts and modules to TF-M. TF-M supports multiple platforms, each potentially requiring platform-specific scripts or modules. Scripts and modules are organized as [Namespace Packages](#).

11.11.2 Adding Platform-Specific Modules

Suppose a module is located at `path/to/module/` and its package name is `foo`:

- Add the module path to `pyproject.toml`:

```
[tool.setuptools.package-dir]
foo = "path/to/module/"
```

- Register the package in `pyproject.toml`:

```
[tool.setuptools]
packages = ["foo"]
```

- Install the module:

```
pip install .
```

Or, for editable installation:

```
source .venv/bin/activate
pip install -e .
```

- Import the installed module in other Python scripts/modules:

```
import foo.<MODULE_NAME> as <MODULE_SHORT_NAME>
```

11.11.3 Adding Platform-Specific Scripts

- Every script should define a `main()` entry point.
- Add the script's entry point to `pyproject.toml`. The script executable name must be prefixed with the platform name.

For example, for a script `bar` in package `foo`:

- Register the script in `pyproject.toml`:

```
[project.scripts]
foo_bar = "foo.bar:main"
```

- Install scripts in a virtual environment:

```
# Subsequent edits made to scripts after install **WONT** be reflected immediately,
↔and will need to be installed again
source .venv/bin/activate
pip install .
```

Or, for editable installation:

```
# Edits made to the scripts will be reflected immediately
source .venv/bin/activate
pip install -e .
```

- Run the script by calling the executable:

```
(.venv) ± foo_bar
foo_bar help text
```

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- [ITS-INTEGRATE] [ITS integration guide](#)
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- [a] PS service is enabled by default. Platforms without off-chip storage devices can turn off TFM_PARTITION_PROTECTED_STORAGE to disable PS service. See *Protected Storage Secure Partition* for details.

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